

Quick Reference Card

Challenger 605

-RETURN QRC AFTER USE-

① APU

APU FIRE

1. APU FIRE PUSH.....PRESS IN

After 5 seconds and APU FIRE stays on:

2. APU BOTTLE ARMED.....PRESS
and hold to discharge.

Proceed to APU 01-1

② AVIONICS

Uncommanded Yaw Motion

1. Controls.....ASSUME MANUAL CONTROL
and counter airplane motion using handwheel inputs.
2. YAW DAMPER, DISC.....PRESS

Proceed to AVNC 02-23

Unreliable Airspeed

1. ATS DISC switch.....DEPRESS
2. Autopilot.....DISCONNECT
3. FDs.....DESELECT
4. Airplane Altitude.....MAINTAIN A SAFE ALTITUDE

If Climb or Go-Around is required:

5. Pitch/N₁ 15 DEGREES/TO FROM SL TO 15,000 FT
10 DEGREES/CLB ABOVE 15,000 FT
6. Airplane Configuration..... FLAPS 20/GEAR UP
FOR GO AROUND
FLAPS 0/GEAR UP ONCE CLIMBING

⚠ CAUTION: Respect stall warning/stick shaker and stick pusher.

Proceed to AVNC 02-25

④ ELECTRICAL

EMER PWR ONLY

OR

Loss of All Normal Electrical Power

If ADG did not deploy:

1. ADG manual deploy handle.....PULL
and leave extended.
2. AC ESS BUS.....CHECK POWERED

When ADG power is established:

3. STAB TRIM, CH 2.....PRESS TO ENGAGE
4. Engine Instruments.....VERIFY
N₁, N₂ & ITT.

Proceed to ELEC 04-1

⑤ ENGINE

L (R) ENGINE FIRE (In Flight)

OR

Severe Engine Damage (In Flight)

After takeoff and at a safe altitude:

Affected engine:

9. ATS DISCPRESS
10. Thrust leverCONFIRM and IDLE
11. Thrust leverCONFIRM and SHUT OFF
12. ENG FIRE PUSHPRESS IN
13. BOOST PUMPPRESS OUT

L (R) ENGINE FIRE stays on (after 10 seconds):

14. BOTTLE ARMEDPRESS
and hold to discharge.

L (R) ENGINE FIRE stays on (after additional 30 seconds):

15. Other BOTTLE ARMEDPRESS
and hold to discharge.

Proceed to ENG 05-2

L (R) ENGINE FIRE (On Ground)

OR

Severe Engine Damage (On Ground)

1. Parking brakeSET

Affected engine:

2. Thrust leverSHUT OFF
3. ENG FIRE PUSHPRESS IN
4. Both BOOST PUMPSPRESS OUT

L (R) ENGINE FIRE stays on (after 10 seconds):

5. BOTTLE 1 and BOTTLE 2 ARMEDPRESS
and hold to discharge.

Proceed to ENG 05-3

ENGINE

1. ATS DISCPRESS

Affected engine:

2. Thrust leverCONFIRM and RETARD
until engine responds and ITT is below limit range.

Proceed to ENG 05-4

L (R) JETPIPE OVHT

1. ATS DISCPRESS

Affected engine:

2. Thrust leverCONFIRM and RETARD
slowly until message goes out.

Proceed to ENG 05-7

L (R) REV UNLOCKED

In flight, on affected engine:

3. Thrust leverCONFIRM and IDLE
4. ATS DISCPRESS
5. THRUST REVERSER, EMER STOWPRESS IN
L (R) REV UNLOCKED will stay on.
6. Airspeed200 KIAS
maximum.

Proceed to ENG 05-13

⑤ ENGINE (cont'd)

Double Engine Failure

1. CONT IGNITIONPRESS IN
2. Airspeed240 KIAS
minimum.

Proceed to ENG 05-16

Post-Shutdown Engine Tail Pipe Fire

Affected engine:

1. Dry motorACCOMPLISH
until ITT is reduced below 150°C or starter limit.

Proceed to ENG 05-42

Uncommanded Acceleration

1. ATS DISCPRESS

On ground, affected engine:

2. Thrust leverCONFIRM and IDLE
then SHUT OFF.

In flight, affected engine:

3. Thrust leverCONFIRM and IDLE
Check engine response.

Proceed to ENG 05-44

⑥ ENVIRONMENTAL / BLEED AIR

CABIN ALT

1. Oxygen masksDON
Set 100%.
2. Crew communicationESTABLISH
3. Both PASS SIGNSON
4. Passenger oxygenDEPLOY
5. DescentINITIATE
to 10,000 feet or lowest safe altitude, whichever is higher.
6. Thrust leversIDLE
7. ATS DISCPRESS
8. FLIGHT SPOILERMAX

Proceed to ENVR 06-4

Emergency Descent

1. Oxygen masksDON
Set 100%.
2. Crew communicationESTABLISH
3. Both PASS SIGNSON
4. Passenger oxygenDEPLOY
5. DescentINITIATE
to 10,000 feet or lowest safe altitude, whichever is higher.
6. Thrust leversIDLE
7. ATS DISCPRESS
8. FLIGHT SPOILERMAX

Proceed to ENVR 06-12

⑦ FLIGHT CONTROLS

Stabilizer Trim Runaway

1. ControlsASSUME MANUAL CONTROL
and override runaway.
2. Pilot or copilot's STAB TRIM DISCPRESS, HOLD
and RELEASE

Proceed to FCTL 07-21

⑧ FORCED LDG / EVACUATION

Emergency Evacuation

Pilot:

1. PARKING BRAKE SET
2. Evacuation COMMAND
3. GND SPOILERS DISARM
4. FLIGHT SPOILER RETRACT
5. Thrust levers SHUT OFF
6. APU, LH and RH ENG FIRE PUSH PRESS IN
7. BATT MASTER OFF

to prevent CVR erasure.

Copilot: On EVACUATION command

8. ATC NOTIFY
of condition and intention to evacuate.
9. Both PACKs PRESS OUT
10. EMER DEPRESS PRESS IN
11. EMER LTS ON

Proceed to F-LDG 08-1

Ditching or Forced Landing Imminent

1. Both PACKs PRESS OUT
2. EMER DEPRESS PRESS IN
prior to 200 feet AGL.

Proceed to F-LDG 08-2

⑩ GEAR / BRAKES

MLG BAY OVHT

1. Airspeed 197 KIAS
maximum.
2. LDG GEAR DN

Proceed to GEAR 10-4

Asymmetric Braking / Loss of Braking

If loss of directional control or braking is observed:

1. Wheel brakes RELEASE
momentarily.
2. ANTI-SKID OFF
3. Wheel brakes RE-APPLY
as required.

Proceed to GEAR 10-14

⑬ OTHER

Rejected Takeoff

Simultaneously:

1. Takeoff DISCONTINUE
immediately.
2. Thrust levers IDLE
3. ATS DISC PRESS
4. Wheel brakes APPLY MAX BRAKING
until safe stop.
5. FLIGHT SPOILER MAX
6. Operating thrust reverser(s) APPLY MAX
consistent with directional control.

Proceed to OTHER 13-2

Stall Recovery

1. Autopilot DISCONNECT
2. Pitch attitude LOWER NOSE
3. ATS DISC PRESS
4. Thrust levers MAX POWER
5. Roll attitude WINGS LEVEL
6. FLIGHT SPOILER RETRACT

Proceed to OTHER 13-3

Challenger 605

Quick Reference Handbook (Imperial Version)

Volume 2

Non-normal procedures

MODEL	A/C EFFECTIVITY
CL-600-2B16	5701 to 6049
FIRST ISSUED: Oct 05/2006	REVISION 67: Nov 22/2022
QUICK REFERENCE CARD REVISION 64: Jun 24/2022	

This handbook contains suggested procedures only for quick reference. These procedures in no way supersede current procedures outlined in the approved Airplane Flight Manual and any revisions thereto. In case of conflict, the Airplane Flight Manual takes precedence.

CHALLENGER 605* (MODEL CL-600-2B16)
QUICK REFERENCE HANDBOOK (IMPERIAL VERSION) –
VOLUME 2 – CH 605 QRH

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To ease the ordering process, the following Document
Identification Number should be used: **CH 605 QRH-I**

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This is revision **67** to the Non-normal procedures, PSP 605–15. To bring this manual up to date remove old pages and insert revised pages as detailed below:

	REMOVE	INSERT
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	02–42	02–42

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NOTE

Record the date you insert this revision in the Record of Revisions at the front of the manual.

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CHECKLIST PREFACE

Purpose of the QRH Vol. 2

The QRH Vol. 2 contains procedures that are associated with Non-Normal operations. It serves a tactical purpose providing the crew with rapid access to critical information at time of need, and has been optimized for efficient flight compartment operation. It also contains other pertinent information (visual symbols, system schematics...), aimed at supporting the crew decision making process and understanding of the procedures in relation to the situation being encountered.

How to use the manual

Two types of procedures are presented in the QRH Vol. 2; EICAS message-based procedures, and Non-EICAS types of procedures.

Warning messages – Are associated with emergency conditions that require immediate crew awareness and rapid corrective or compensatory action as soon as the situation safely permits.

Caution messages – Are associated with serious system malfunctions that require immediate crew awareness and subsequent action when time and conditions permit.

Advisory messages – Are associated with a loss of system redundancy that requires crew awareness and where subsequent action may be needed.

Status procedures – Reflect the state of a particular system, still under normal condition but operated in an alternate configuration or condition.

Non-EICAS procedures – Can be associated with either emergency conditions when immediate crew awareness is self-evident, or with potentially serious conditions when the crew may have to recognize a problem and take appropriate action based on airmanship system knowledge and indirect information.

Master Index

To help the crew find procedures a Master Index is located under the INDEX tab.

Procedures in the master index are segregated by color level. Within each level, procedures are placed in alphabetic order. Reference letters are displayed using the letter of the sub-list.

To help the crew find procedures using the Master Index, when applicable some procedures may be cross-indexed under multiple letters. For example a generator failure may be associated with either the left or right generator which would trigger the associated **L AOA HEAT** or **R AOA HEAT** caution message, therefore this message will be indexed at three locations in the Master Index Caution message list; under **L** as **L AOA HEAT** under **R** as **R AOA HEAT** and under **A** as **AOA HEAT**, all leading to the same page location.

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CHECKLIST PREFACE (CONT'D)

Section Organizational Structure

System related tabs are organized in alphabetical order and numbered 1 to 14. Each marks the beginning of a new system and contains the following elements:

- Topic covered page,
- Table of contents (TOC),
- Procedure and information related to the specified systems,
- Related system schematics,
- Power supply and circuit breaker summary.

The INDEX tab located at the beginning of the manual and the GO/NO–GO tab located at the end of the manual are not numbered.

Procedure Navigation Flow

Titles for EICAS based procedure titles are presented in black capital letters over a colored background matching the level of EICAS message. For example:

MLG BAY OVHT

APU GEN OVLD

By contrast, Non–EICAS procedure titles use white letters against a black background, capitalizing each word. For example:

ADG Inadvertent Deployment

Status messages do not have associated procedures. They are listed in a table with an expanded description.

All checklist steps including conditional statements are in continuous numerical sequence.

Affirmation statements and conditional statement answers are not numbered.

Some procedures contain steps that may be different depending on service bulletin configuration or serial number effectivity. Operators should therefore ensure that crews are aware of the equipment configuration specific to their airplane.

An arrowhead symbol (▼) at the bottom of the page indicates that the procedure continues on to the next page.

For procedures that carry over to multiple pages, the title is repeated at the top of the continuing page with the word (Cont'd).

CHECKLIST PREFACE (CONT'D)

Section Abbreviations

The following table lists the section titles and their abbreviations used in this manual.

Section Name	Abbreviated Name	Section Name	Abbreviated Name
Auxiliary Power Unit	APU	Forced Landing / Evacuation	F-LDG
Avionics	AVNC	Fuel	FUEL
Doors	DOOR	Gear / Brakes	GEAR
Electrical	ELEC	Hydraulics	HYD
Engine	ENG	Ice Protection	ICE
Environmental / Bleed Air	ENVR	Other Procedures	OTHER
Flight Controls	FCTL	Smoke / Fire	SMK

Conditional Statements and Answers

Conditional statements are system state or situation related questions which may have 2 or more possible responses. They are written in **bold**, numbered and end with a colon (:).

Each answer leads the crew to another step (e.g. Go to (8)) or informs of the end of the procedure (e.g. **End of procedure.**).

Answer to conditional questions follow immediately below the conditional questions and states the options such as “YES”, “NO”, “ATS” or “Windmill”, etc. These conditional answers are in **bold** and are indented to the right with an arrow.

Immediate Action Items / Quick Reference Card (QRC)

Procedures that contain Immediate Action Items, are shown within a box with a red-dashed border, and are also provided on the Quick Reference Card (QRC). The QRC can be used by the crew to more efficiently perform the Immediate Action Items without relying solely on memory, thus reducing the possibility of omission or commission errors.

The QRC contains only the Immediate Action Items from the QRH, it does not contain the entire procedure. On the QRC, each list of Immediate Action Items is followed by either:

- The page location reference to the equivalent checklist in the QRH Vol 2,
- The reference to the Emergency Evacuation procedure on the back cover,
- If the procedure in the QRH leads to another procedure, the QRC will lead to the page location reference to this following procedure,
- The end of the procedure.

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CHECKLIST PREFACE (CONT'D)

Immediate Action Items / Quick Reference Card (QRC) (Cont'd)

When the crew uses the QRC to accomplish the Immediate Action Items, these items are considered already performed and reviewed, and the crew can continue the procedure with the QRH at the next step that follows the boxed items.

To help the crew locate where to continue following the use of the QRC, a visual marker is located directly below the Immediate Action Items box.

➡ FROM Quick Reference Card ⬅

EICAS Alerted Procedures

When an EICAS message triggers an aural alert, the actual aural alert message is indicated between quotation marks in uppercase letters and accompanied by a speaker symbol.

Assumptions, Airmanship and Crew Coordination

Procedures in this manual are designed to provide safe and effective mitigation to Non-Normal situation but cannot account for all possible scenarios. Therefore, elements that are considered basic pilot knowledge or airmanship will normally not be included in these procedures. It is assumed that a pilot using this manual:

- Is proficient in using the English language (International Civil Aviation Organization level 4 or higher),
- Has already acquired the level of airmanship and decision making skills associated with a competent type-rated pilot (instrument rated, multi-crew and multi-engine experience),
- Is already familiar with normal pilot technique common to similar airplane types.
- Is using an appropriate level of automation in relation to phase of flight and / or related failure.

Procedures contained herein also assume that actions listed below are accomplished by the crew:

1. Master Warning/Master Caution PBAs are reset,
2. Emergency level procedures have been properly accomplished,
3. Relevant Normal phase of flight procedures have been properly accomplished,
4. Caution level procedures are completed as required,
5. General assessment of airplane systems status is completed and,
6. Obvious corrective action (if any) is taken for crew awareness items.

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CHECKLIST PREFACE (CONT'D)

Landing Expressions

“Land immediately at the nearest suitable airport:” Land the airplane without delay at the nearest airport where a safe approach/landing is reasonably assured considering the severity of the emergency, weather conditions, field facility, ambient lighting, and aircraft gross weight.

“Land at the nearest suitable airport:” Extending the flight beyond the nearest suitable airport is not recommended. However, the duration of flight and landing location are at the discretion of the pilot in control.

Condition / Objective Statements

The “Condition” statement provides a brief description of known symptoms associated with the problem.

The “Objective” statement provides guidance as to what is the desired outcome of the procedure.

Prospective Cues

Thin arrows are used for actions that are prospective cues e.g.”→ If wing anti-ice is required”. Prospective cues are statements within the checklist that define the conditions in which a delayed step may have to be performed should these conditions be encountered at a later time. These cues help the pilot remember the delayed action if the set conditions are later encountered. The pilot performs the action items only if meeting the condition. If not, the item may be skipped and performed later if the condition arise.

Notes, Cautions and Warnings

Specific items requiring emphasis are expanded upon and ranked in increasing order of importance in the form of NOTE, a CAUTION or a WARNING.

NOTE: Information that provides special emphasis on aspects of a procedure, and may be safety related.

⚠CAUTION: Information that is considered important and may negatively impact the safe outcome of the procedure or lead to adverse effects or damage if not considered by the crew.

⚠WARNING: Information that is considered critical for the safe outcome of the procedure and may result in injury or loss of life if not followed.

End of procedure / – COMPLETE–

“End of procedure” following a YES or NO answer to a conditional statement. This means that the pilot has completed all necessary steps in the checklist and no subsequent actions are required.

“–COMPLETE–” This means that the pilot has completed all necessary steps in the checklist and no subsequent actions are required.

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ABBREVIATIONS

The following abbreviations are used throughout the manual. Some abbreviations also appear in lower case letters. Abbreviations having very limited usage are explained in the chapters where they are used.

A

A/C	Air-Conditioning, Aircraft
ACPC	Alternating Current Power Center
ACARS	Aircraft Communication Addressing and Reporting System
ADS-B	Automatic Dependent Surveillance-Broadcast
ADS-C	Automatic Dependent Surveillance-Contract
A-ICE, A/ICE	Anti-ice
AC	Alternating Current
ACP	Audio Control Panel
ADC	Air Data Computer
ADI	Attitude Director Indicator
AFCS	Automatic Flight Control System
AFM	Airplane Flight Manual
AFN	ATS Facilities Notification
AGL	Above Ground Level
AIL	Aileron
ALT	Altitude, Altimeter
AOA	Angle of Attack
AOC	Airline Operational Communications
AP	Autopilot
APM	Aircraft Personality Module
APPR	Approach Mode
APU	Auxiliary Power Unit
ASCB	Avionics Standard Communications Bus
ATC	Air Traffic Control
AT or A/T	Autothrottle
ATS	Air Turbine Start
ATS	Auto Thrust/Throttle System
ATS	Air Traffic Services
ATT	Attitude
AUTH	Authority
AUTO	Automatic
AUX	Auxiliary
AV	Avionics
AVAIL	Available

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ABBREVIATIONS (CONT'D)

B

BARO	Barometric
BATT	Battery
BC	Back Course
BCU	Brake Control Unit
BLTS	Belts
BRK	Brake
BTMS	Brake Temperature Monitoring System

C

C	Center, Caution, Cabin
CAA	Civil Aviation Authority (UK)
CAB	Cabin
CAI	Cowl Anti-ice
CAIV	Cowl Anti-ice Valve
CAS	Calibrated Air Speed
CAS	Crew Alerting System
CAT	Category
CAT 2	Category 2
CB, cb	Circuit Breaker
CBP	Circuit Breaker Panel
CBW	Cross-Bleed Wing
CCBP	Cockpit Circuit Breaker Panel
CCP	Cursor Control Panel
CDI	Course Deviation Indicator
CDL	Configuration Deviation List
CDU	Control Display Unit
CG, C of G	Centre of Gravity
CH	Chapter, Channel
CHGR	Charger
CLSD	Closed
Cm	Centimeters
COM	Communication
COMP	Compressor, Comparator
CONT'D	Continued
CONFIG	Configuration
CONT	Control, Continuous, Contactor, Controller
COMPUTR	Computer
COOL	Cooling, cooler
CPDLC	Controller Pilot Data Link Communications
CPLT	Copilot
CTA	Centro Tecnico Aeroespacial (Brazil)

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ABBREVIATIONS (CONT'D)

CTP	Control Tuning Panel
CTR	Center
CTRL	Control
CVR	Cockpit Voice Recorder

D

DB	Database
DC	Direct Current
DCPC	Direct Current Power Center
DCU	Data Concentration Unit
DEPRESS	Depressurization
DEGRAD	Degraded
DIR	Direct
DISC	Disconnect
DISCH	Discharge
DISP	Dispatch, Display
DMC	DCU Module Cabinet
DME	Distance Measuring Equipment
DN	Down
DOT	Department of Transport (Canada)
DSK	Data Select Knob
DSM	Digital Switching Module
DU	Display Unit

E

EASA	European Aviation Safety Agency
ECS	Environmental Control System
EDP	Engine Driven Pump/Engine Primary Hydraulic Pump
EEC	Electronic Engine Controller
EFIS	Electronic Flight Instrument System
EFVS	Enhanced Flight Vision System
EGNOS	European Geostationary Navigation Overlay Service
EGT	Exhaust Gas Temperature
EICAS	Engine Indication and Crew Alerting System
ELEC	Electrical
ELEV	Elevator, Elevation
ELT	Emergency Locator Transmitter
EMER(G)	Emergency
EMS	Electrical Management System
ENG	Engine
ENGAGE	Engagement

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ABBREVIATIONS (CONT'D)

EPR	Engine Pressure Ratio
ESS	Essential
EVS	Enhanced Vision System
EXT	External

F

F/CTL	Flight Controls
FAA	Federal Aviation Administration (USA)
FAF	Final Approach Fix
FAIL	Failure
FADEC	Full Authority Digital Engine Control
FANS	Future Air Navigation System
FCU	Flight Control Unit
FD	Flight Director
FGC	Flight Guidance Computer
FIREX	Fire Extinguisher
FL	Flight Level
FLC	Flight Level Change (airspeed hold mode)
FLT	Flight
FMS	Flight Management System
FOD	Foreign Object Damage
FPM	Feet Per Minute
FRTT	Fuel Return To Tank
FT	Feet, Foot
FUSE	Fuselage
FWD	Forward

G

GA	Go-Around
GAL	Gallon
GEN	Generator
GLD	Ground Lift Dump
GMT	Greenwich Mean Time
GND	Ground
GNSS	Global Navigation Satellite System
GPM	Gallons Per Minute
GPS	Global Positioning System
GPWS	Ground Proximity Warning System
GR	Gear
GRAV	Gravity
GS	Ground Speed
GW	Gross Weight

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ABBREVIATIONS (CONT'D)

H

HDD	Head Down Display
HDG	Heading
HEAT	Heater
HF	High Frequency (3 – 30 mHz)
Hg	Mercury
HI	High
HP	High Pressure
hPa	HectoPascals
HSI	Horizontal Situation Indicator
HT	Heater
HUD	Head-up Display
HYD	Hydraulic
Hz	Hertz

I

IAF	Initial Approach Fix
IAS	Indicated Air Speed
ICAO	International Civil Aviation Organization
ISI	Integrated Standby Instrument
IFR	Instrument Flight Rules
IGN	Ignition
ILS	Instrument Landing System
IMC	Instrument Meteorological Conditions
IMS	Information Management System
IN	Inch, Inches
INIT	Initialize
IN Hg	Inches of Mercury
INBD	Inboard
INTERCOMM	Intercommunication
IND	Indication, Indicator
INOP	Inoperative
IPC	Integrated Processing Cabinet
IRS	Inertial Reference System
IRU	Inertial Reference Unit
ISA	International Standard Atmosphere
ISO	International Standards Organization
ISOL	Isolation, Isolated
ITT	Inter Turbine Temperature (°C), Engine
INVAL	Invalid

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ABBREVIATIONS (CONT'D)**J**

JAA Joint Aviation Authorities

K

KTS Knots
kg Kilogram
kHz KiloHertz
KIAS Knots Indicated Airspeed
kPa KiloPascals
kW(s) KiloWatt(s)
kVA KiloVolt–Ampere

L

L Left, Landing
LAV Lavatory
LAT Latitude
LB, Lb Pound
LDG Landing
LDS Lightning Detection System
LEV Level
LFL Landing Field Length
L/G Landing Gear
LH Left Hand
LNAV Lateral Navigation
LO Low
LOC ILS Localizer
LP Low Pressure
LR Long Range
LRU Line Replaceable Unit
LTS Lights
LWD Left Wing Down

M

M Mach Number
m Meter
MAC Mean Aerodynamic Chord
MAN Manual
MCT Maximum Continuous Thrust
MEL Minimum Equipment List
MFW Multifunction Window
MHz MegaHertz

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ABBREVIATIONS (CONT'D)

MI	Miles
MI	Mach Number Indicated
MIC	Microphone
MILS	.001 of an inch
MIN	Minimum
MISC	Miscellaneous
MISCMP	Miscompare
MISMTCH	Mismatch
MKP	Multifunctional Keyboard Panel
MKR	Marker
MLG	Main Landing Gear
MLW	Maximum Landing Weight
MMEL	Master Minimum Equipment List
M _{MO}	Maximum Operating Speed in Mach Number
MNPS	Minimal Navigational Performance Standards
MOCA	Minimum Obstruction Clearance Altitude
MON	Monitor
MPH	Miles Per Hour
MSG	Message
MTO	Maximum Takeoff
MTOW	Maximum Takeoff Weight
MRW	Maximum Ramp Weight
MX	Maximum
MZFW	Maximum Zero Fuel Weight

Misc

%	Percent
&	and
°C	Degrees Centigrade
°F	Degrees Fahrenheit

N

N/A	Not applicable
N ₁	Low Pressure Rotor
N ₂	High Pressure Rotor
NAT	North Atlantic Tracks
NAV	Navigation
ND	Nose Down
NDU	Navigational Display Unit
NEG	Negative
NEUT	Neutral
NLG	Nose Landing Gear
NM	Nautical Miles

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ABBREVIATIONS (CONT'D)

No.	Number
NORM	Normal
NOSE	Nose Wheel
NU	Nose Up

O

OAT	Outside Air Temperature
OUTBD	Outboard
OEI	One Engine Inoperative
OEW	Operating Empty Weight
OK	Okay
OMNI	Omnidirectional
OUTBD	Outboard
OVBD	Overboard
OVRD	Override
OVHT	Overheat
OVL	Overload
OXY, O ₂	Oxygen

P

PA	Passenger Announcement
PASS	Passenger
PACK	Air-Conditioning Unit
PAX	Passenger
PBE	Protective Breathing Equipment (Smoke Hood)
PCU	Power Control Unit
PF	Pilot Flying
PFD	Primary Flight Display
PITOT	Pitot-Static
PLT	Pilot
PNF	Pilot Not Flying
P, PRESS	Pressure, Pressurization
PROTECT	Protection
PRI	Primary
PSEU	Proximity Sensor Electronics Unit
PSI	Pounds Per Square Inch
PSID	Pounds Per Square Inch Differential
PSIG	Pounds Per Square Inch Gauge
PRV	Pressure Regulating Valve
PWR	Power

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ABBREVIATIONS (CONT'D)

Q

QAK	Quick Access Keys
QFE	Local Station Pressure
QNE	ISA, Barometric Pressure
QNH	Altimeter Setting
QTY	Quantity

R

R	Right
RA	Radar Altitude, Resolution Advisory
RAI	Registro Aeronautico Italiano (Italy)
RAT	Ram Air Turbine
R/D	Refuel / Defuel
RECIRC	Recirculation
REF(s)	Reference(s)
REV	Reverse
RH	Right Hand
RMI	Radio Magnetic Indicator
RNP	Required Navigation Performance
RNP AR	Required Navigation Performance Approval Required
ROTR	Record of Temporary Revisions
RPM, rpm	Revolutions Per Minute
RSP	Reversion Switch Panel
RTA	Required Time of Arrival
RTO	Rejected Takeoff
RUD	Rudder
RVR	Runway Visual Range
RVSM	Reduced Vertical Separation Minimum
RWD	Right Wing Down
RWY	Runway
RA	Resolution Advisory

S

SAAU	State Aviation Administration of Ukraine
SAT	Static Air Temperature
SAV	Starter Air Valve
SB	Service Bulletin
SBAS	Space Based Augmentation System
SCHED	Schedule
SELCAL	Selective Calling
SERV	Service

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ABBREVIATIONS (CONT'D)

SG	Signal Generator
SMKG	Smoking
SOV	Shutoff Valve
SPC	Stall Protection Computer
SPD	Speed
SPDA	Secondary Power Distribution Assembly
SPLR(S)	Spoiler(s)
STAB	Stabilizer
STAT	Status
STBY	Standby
STEER	Steering
SVS	Synthetic Vision System
SYNC	Synchronization
SYS	System

T

TAT	Total Air Temperature
TAWS	Terrain Awareness Warning System
TCAS	Traffic Alert and Collision Avoidance System
TEMP	Temperature
TOFL	Takeoff Field Length
TOLD	Takeoff and Landing Data
TRU	Transformer Rectifier Unit
TA	Traffic Advisory
TD	Traffic Display

U

UNLKD	Unlocked
US	United States
USB	Universal Serial Bus
USG	United States Gallons

V

V_	V-Speed (See DEFINITIONS Airspeeds in this section)
Vspeeds	Departure and arrival performance
V	Volt
V/S	Vertical Speed
VERT	Vertical
VFR	Visual Flight Rules
VHF	Very High Frequency (30 – 300 mHz)
VIB	Vibration
VLV(s)	Valve(s)

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ABBREVIATIONS (CONT'D)

VMC	Visual Meteorological Conditions
VNAV	Vertical Navigation Mode
Vol	Volume
VOLT	Voltage
VOR	VHF Omnidirectional Range Station
VS	Vertical Speed Mode
VSD	Vertical Situation Display
VSİ	Vertical Speed Indicator

W

WARN	Warning
WAAS	Wide Area Augmentation System
WAI	Wing Anti-Ice
WAIV	Wing Anti-Ice Valve
WF	Runway Length Limited Weight
WG	Wing
WOW	Weight-On-Wheels
WPT(s)	Waypoint(s)
WS	Second Segment Limited Weight
WSHLD	Windshield
WX	Weather
WXR	Weather Radar

X

XFER	Transfer
XDUCER	Transducer
XBLEED	Cross Bleed
XFEED	Cross Feed

Y

YD	Yaw Damper
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DEFINITIONS

The following definitions are for terms commonly used throughout the manual. Items not commonly used will be defined in the respective sections where they apply.

Airspeeds

CAS	Calibrated airspeed – Indicated airspeed corrected for static source position error.
EAS	Equivalent airspeed – Calibrated airspeed corrected for compressibility.
IAS	Indicated airspeed – Airspeed indicator reading corrected for instrument error. Values in this manual assume zero instrument error.
M	Indicated Mach number – Machmeter reading corrected for instrument error.
M_T	True Mach number – Machmeter reading corrected for both instrument and static source position errors.
TAS	True airspeed – Equivalent airspeed corrected for density.
V_1	Takeoff decision speed (formerly designated as Critical engine failure recognition speed) – The speed used as a reference at which, due to engine failure or other causes, the pilot may elect to stop or continue the takeoff.
V_{1MBE}	Maximum V_1 for brake energy – The maximum speed on the ground from which a stop can be accomplished within the energy capabilities of the brakes.
V_{1MCG}	Minimum V_1 limited by control on the ground – The takeoff decision speed following an engine failure at V_{MCG} . V_1 may not be less than V_{1MCG} .
V_2	Takeoff safety speed – Target climb speed to be attained at or before a height of 35 feet above the runway during a continued takeoff, following an engine failure.
V_A	Design maneuvering speed – The maximum speed at which application of full available aileron, rudder or elevator will not overstress the airplane.
V_{AC}	Approach Climb Speed – The target climb speed to be attained after initiation of a Go-Around and selection of the approach climb configuration following an engine failure.

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DEFINITIONS (CONT'D)

Airspeeds (Cont'd)

V_{ATE}	Autothrottle automatic engagement speed – The speed at which the Autothrottle will automatically engage. Corresponds nominally to $1.2 V_S$ and displayed on the Airspeed tape by an amber line located above the low speed cue indication.
V_{EF}	Critical engine failure speed – The speed at which, if the critical engine fails, the engine failure is recognized at V_1 .
V_{FE}	Maximum flap extended speed – is the highest speed permissible with the wing flaps in a prescribed extended position.
V_{FTO}	Final takeoff climb speed – The speed attained at the end of the flight path acceleration segment during a continued takeoff following an engine failure, and is the climb speed scheduled for the final takeoff climb.
V_{LC}	Landing Climb Speed – The target climb speed to be attained after initiation of a Go-Around while in the landing configuration with all engines operating.
V_{LE}	Maximum landing gear extended speed – The maximum speed at which the airplane can be safely flown with the landing gear extended.
V_{LO}	Maximum landing gear operating speed – The maximum speed at which the landing gear can be safely extended or retracted.
V_{MO}/M_{MO}	Maximum Operating Limit Speed / Mach Number – The maximum operating limit speed (airspeed or Mach number) is the speed that may not be deliberately exceeded in any regime of flight (climb, cruise, or descent) unless a higher speed is authorized for flight test or pilot training operations.
V_{MCA}	Minimum control speed, air – Minimum flight speed at which the airplane is controllable when the critical engine suddenly becomes inoperative, with the remaining engine at maximum takeoff thrust.

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DEFINITIONS (CONT'D)

Airspeeds (Cont'd)

V_{MCG}	Minimum control speed, ground – Minimum speed on the ground at which control can be maintained and the takeoff continued using aerodynamic controls alone, when the critical engine suddenly becomes inoperative, with the remaining engine at maximum takeoff thrust.
V_{MCL}	Minimum control speed during landing approach – Minimum flight speed at which the airplane is controllable when the critical engine suddenly becomes inoperative, with the remaining engine at maximum takeoff thrust.
V_{MD}	Minimum Drag Speed – The speed that provides the maximum lift/drag ratio.
V_R	Rotation speed – Speed at which rotation is initiated during takeoff.
V_{REF}	Approach speed – The landing reference speed at a height of 50 feet above the runway threshold in the normal landing configuration.
V_S	Stalling speed.

Temperature

ISA	International Standard Atmosphere.
OAT	Outside air temperature – The free air temperature, obtained either from in-flight temperature indications or ground meteorological sources.
SAT	Static air temperature – Outside air (ambient) temperature as computed by the Air Data Computer and presented on the SAT indicator.
TAT	Total air temperature – Static air temperature plus adiabatic compression (ram) rise.

Distances

Takeoff Field Length (Dry and wet runways)	The takeoff field length is the longer of: a. The takeoff distance (see below). b. The accelerate-stop distance (see below).
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DEFINITIONS (CONT'D)**Distances (Cont'd)**

Takeoff Distance (Dry runways)	The takeoff distance on a dry runway is the longer of: - The distance from the start of the takeoff roll to the point where the airplane attains a height of 35 feet above the takeoff surface, with a failure of the critical engine at V_{EF}. 115% of the distance from the start of the takeoff roll to the point at which the airplane attains a height of 35 feet above the takeoff surface, with all engines operating.
Takeoff Distance (Wet runways)	The takeoff distance on a wet runway is the longer of: - The takeoff distance on a dry runway. - The distance from the start of the takeoff roll to the point where the airplane attains a height of 15 feet above the takeoff surface, with a failure of the critical engine at V_{EF}. 115% of the distance from the start of the takeoff roll to the point at which the airplane attains a height of 35 feet above the takeoff surface, with all engines operating.

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DEFINITIONS (CONT'D)

Distances (Cont'd)

Accelerate – Stop Distance (Dry runways)	The accelerate – stop distance on a dry runway is the longer of subpara a. and subpara b. as defined below: a. The sum of the distances necessary to: 1) Accelerate the airplane from a standing start to V_{EF} with all engines operating; 2) Accelerate the airplane from V_{EF} to V_1 assuming the critical engine fails at V_{EF} ; and 3) Come to a full stop from the point reached at the end of the acceleration period prescribed in subpara a. 2) above, assuming that the pilot does not apply any means of retarding the airplane until that point is reached; plus 4) A distance equivalent to 2 seconds at constant V_1 . b. The sum of the distances necessary to: 1) Accelerate the airplane from a standing start to V_1 with all engines operating; 2) Come to a full stop at the end of the period prescribed in subpara b. 1) above, assuming that the pilot does not apply any means of retarding the airplane until that point is reached; plus 3) A distance equivalent to 2 seconds at constant V_1 .
Accelerate – Stop Distance (Wet runways)	The accelerate – stop distance on a wet runway is the longer of: a. The accelerate – stop distance on a dry runway. b. The accelerate – stop distance determined as per the accelerate – stop distance on a dry runway, except that the corresponding wet runway values of V_{EF} and V_1 are used.

DEFINITIONS (CONT'D)**Distances (Cont'd)**

Takeoff Run (Dry runways)	The takeoff run on a dry runway is the longer of: - The distance from the start of the takeoff roll to the mid point between lift-off and the point at which the airplane attains a height of 35 feet above the takeoff surface, with a failure of the critical engine at V_{EF}. 115% of the distance from the start of the takeoff roll to the mid point between lift-off and the point at which the airplane attains a height of 35 feet above the takeoff surface, with all engines operating.
Takeoff Run (Wet runways)	The takeoff run on a wet runway is equal to the takeoff distance on a wet runway.
Clearway	An area beyond the runway, not less than 500 feet wide, centrally located about the extended centerline of the runway, and under the control of the airport authorities. The clearway is expressed in terms of a clearway plane, extending from the end of the runway with an upward slope not exceeding 1.25% above which no object nor any terrain protrudes. However, threshold lights may protrude above the plane if their height above the end of the runway is 26 inches or less and if they are located to each side of the runway.
Stopway	An area beyond the takeoff runway at least as wide as the runway and centered upon the extended centerline of the runway, able to support the airplane during an aborted takeoff without causing structural damage to the airplane, and designated by the airport authorities for use in decelerating the airplane during an aborted takeoff.
Actual Landing Distance	Actual landing distance is the distance from a 50 feet height at V_{REF} with the flaps in the normal landing configuration, to a full stop on a smooth, dry, level, hard-surfaced runway and ISA temperature.

DEFINITIONS (CONT'D)

Distances (Cont'd)

Landing Field Length	The performance charts in this chapter include factors for operational rules which require the use of 60% of the available runway in determining landing field length requirements. The required landing field length on a dry runway is the actual landing distance divided by 0.6. The landing field length required under wet runway conditions is equal to 115% of the dry runway landing field length.
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Takeoff Path

Takeoff Path	The takeoff path begins from a standing start and ends at 1500 feet above the takeoff surface or at the point where transition from takeoff to enroute configuration is completed, whichever is higher.
Takeoff Flight Path	The takeoff flight path begins at the end of the takeoff distance and at a height of 35 feet above the takeoff surface, and ends at 1500 feet above the takeoff surface, or at the point where transition from takeoff to enroute configuration is completed, whichever is higher. This is also known as the Gross Takeoff Flight Path.
Net Takeoff Flight Path	The net takeoff flight path is the gross takeoff flight path diminished by 0.8% climb gradient capability (or equivalent reduction in acceleration along that part of the takeoff flight path at which the airplane is accelerated in level flight). The net takeoff flight path must clear all obstacles in the takeoff area by at least 35 feet vertically.
Gross Level-off Height	Actual (gross) geometric height above takeoff surface for the flight acceleration segment.

Climb Gradient

Climb Gradient	The ratio of the change in height, during a portion of a climb, to the horizontal distance traversed in the same time interval.
Gross Gradient	Climb gradient expressed as a percentage ratio, obtained using the following formula: $(\text{Change in height} \div \text{Horizontal distance travelled}) \times 100$ The gradients shown on the charts are based on true (not pressure) rates of climb.

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DEFINITIONS (CONT'D)

Climb Gradient (Cont'd)

Net Gradient	The gross gradient reduced by the required margins.
First Segment	The first segment starts from the 35 foot height and extends to the point where the landing gear is fully retracted, at a constant V_2 speed and flaps in the takeoff position.
Second Segment	The second segment starts at the point where the landing gear is fully retracted up to at least 400 feet above the runway, flown at V_2 speed and flaps in the takeoff position.
Acceleration Segment	The acceleration segment is the part of the takeoff flight path that begins at the end of the second segment and extends horizontally over the distance required to retract flaps to zero, and accelerate to the final takeoff climb speed.
Final Segment	The final segment starts from the end of the acceleration segment and extends to the end of the takeoff flight path, flown at the final segment climb speed, flaps up.

Other

Centre of gravity (c.g.)	The point at which the airplane would balance if suspended. The c.g. distance from the reference datum is found by dividing the total moment by the total weight of the airplane.
GW	Gross weight – The maximum weight to which the airplane is certificated.
MLW	Maximum landing weight – The maximum airplane weight approved for the landing touchdown.
Max OWE	Maximum operating weight, empty – The maximum weight for a loaded airplane which includes the flight and cabin crew, galley loading and forward / aft stowage items.
Min OWE	Minimum operating weight, empty – The minimum weight for a loaded airplane which includes minimum crew, no galley loading and no stowage items.
Min FW	Minimum Flight Weight – The minimum flight weight is the minimum weight empty with 45 minutes of fuel reserve.

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DEFINITIONS (CONT'D)

Other (Cont'd)

MRW	Maximum ramp weight – The maximum airplane weight approved for taxiing.
MTOW	Maximum takeoff weight – The maximum airplane weight approved for the start of takeoff.
MZFW	Maximum zero fuel weight – The Max OWE plus passengers and baggage.

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APU FIRE



Fire Bell



Condition: APU fire is detected.

Objective: Extinguish fire and land as soon as possible.

- (1) APU FIRE PUSH PRESS IN
After 5 seconds and APU FIRE stays on:
- (2) APU BOTTLE ARMED PRESS
and hold to discharge.

➡ FROM Quick Reference Card ⬅

- (3) **APU FIRE** stays on during flight:
- ➡ YES – Go to (4)
 - ➡ NO – Go to (7)
- (4) **APU FIRE** stays on during flight:
- (5) Land at the nearest suitable airport.
- (6) Go to (8)
- (7) **APU FIRE** goes out during flight:
- (8) **APU** is supplying bleed air for air-conditioning:
- ➡ YES – Go to (9)
 - ➡ NO – Go to (12)
- (9) **APU** is supplying bleed air for air-conditioning:
- (10) PACKS TRANSFER
to engine bleeds.
- (11) Go to (13)
- (12) **APU** is NOT supplying bleed air for air-conditioning:
- (13) APU SHUTDOWN
- (14) Do not restart APU.

– COMPLETE –

APU OVERSPEED

“APU”

Condition: APU has auto shutdown due to an overspeed.

Objective: Shut off the APU and prevent further use.

(1) APU is supplying bleed air for air-conditioning:

➡ **YES** – Go to (2)

➡ **NO** – Go to (5)

(2) APU is supplying bleed air for air-conditioning:

(3) PACKS TRANSFER
to engine bleeds.

(4) Go to (6)

(5) APU is NOT supplying bleed air for air conditioning:

(6) APU SHUTDOWN

(7) Do not restart APU.

– COMPLETE –

APU OVERTEMP

“APU”

Condition: APU has auto shutdown due to an overtemp.

Objective: Shut off the APU and prevent further use.

(1) APU is supplying bleed air for air-conditioning:

➡ **YES** – Go to (2)

➡ **NO** – Go to (5)

(2) APU is supplying bleed air for air-conditioning:

(3) PACKS TRANSFER
to engine bleeds.

(4) Go to (6)

(5) APU is NOT supplying bleed air for air conditioning:

(6) APU SHUTDOWN

(7) Do not restart APU.

– COMPLETE –

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APU LCV FAIL



APU LCV

Condition: Load control valve failed open when commanded closed.

Objective: Isolate left 10th stage bleed system to prevent damage.

- I (1) L 10TH STAGE bleed.....PRESS OUT
- (2) 10TH STAGE ISOLPRESS OUT

→ If above 40,000 feet:

- I • L COWL ANTI-ICEPRESS IN
- (3) L PACK.....PRESS OUT

– COMPLETE –

APU NEG-G SOV

Condition: Valve is not in the commanded position (opened or closed).

Objective: Prevent feeding fuel to the APU.

(1) **APU FIRE PUSH is pressed in:**

➡ **YES** – Go to (2)

➡ **NO** – Go to (5)

(2) **APU FIRE PUSH is pressed in:**

(3) L BOOST PUMPPRESS OUT

Left engine:

(4) In-Flight Engine
Failure/ShutdownACCOMPLISH

(See Page ENG 05–10)

– COMPLETE –

(5) **APU FIRE PUSH is pressed OUT (no APU fire condition):**

(6) Land at the nearest suitable airport.

– COMPLETE –

Non-normal procedures PSP 605–15 – QRH–I Vol. 2	APU
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APU OIL PRESS

Condition: APU has auto shutdown due to a low oil pressure.

Objective: Shut off the APU and prevent further use.

(1) APU is supplying bleed air for air-conditioning:

➡ **YES** – Go to (2)

➡ **NO** – Go to (5)

(2) APU is supplying bleed air for air-conditioning:

(3) PACKS TRANSFER
to engine bleeds.

(4) Go to (6)

(5) APU is NOT supplying bleed air for air conditioning:

(6) APU SHUTDOWN

(7) Do not restart the APU.

– COMPLETE –

APU OIL TEMP

Condition: APU has auto shutdown due to a high oil temperature.

Objective: Shut off the APU and prevent further use.

(1) APU is supplying bleed air for air-conditioning:

➡ **YES** – Go to (2)

➡ **NO** – Go to (5)

(2) APU is supplying bleed air for air-conditioning:

(3) PACKS TRANSFER
to engine bleeds.

(4) Go to (6)

(5) APU is NOT supplying bleed air for air conditioning:

(6) APU SHUTDOWN

(7) Do not restart the APU.

– COMPLETE –

APU PUMP



PWR FUEL

Condition: APU fuel pump has failed. Fuel feed is from left engine feed line.

Objective: Shut off the APU, unless the APU generator is essential.

(1) Identify phase of flight:

➡ **On ground** – Go to (2)

➡ **In flight** – Go to (4)

(2) On ground:

(3) APU SHUTDOWN

– COMPLETE –

(4) In flight:

(5) PACKS TRANSFER
to engine bleeds.

(6) APU generator is essential:

➡ **YES** – Go to (7)

➡ **NO** – Go to (10)

(7) APU generator is essential:

(8) APU indications MONITOR

(9) Land at the nearest suitable airport.

▲ CAUTION: APU adapter oil will no longer be cooled. Adapter oil will overheat causing APU to auto shutdown.

– COMPLETE –

(10) APU generator is NOT essential:

(11) APU SHUTDOWN

– COMPLETE –

APU SOV

Condition: Valve is not in commanded position (opened or closed).

Objective: Shut off the APU, unless APU generator is essential.

(1) APU FIRE PUSH is pressed in:

➡ **YES** – Go to (2)

➡ **NO** – Go to (10)

(2) APU FIRE PUSH is pressed in:

(3) APU is supplying bleed air for air-conditioning:

➡ **YES** – Go to (4)

➡ **NO** – Go to (7)

(4) APU is supplying bleed air for air-conditioning:

(5) PACKS TRANSFER
to engine bleeds.

(6) Go to (8)

(7) APU is NOT supplying bleed air for air-conditioning:

(8) APU SHUTDOWN

(9) Land at the nearest suitable airport.

– COMPLETE –

(10) APU FIRE PUSH is pressed OUT (no APU fire condition):

(11) APU is operating:

➡ **YES** – Go to (12)

➡ **NO** – No further action required.

NOTE: The APU is NOT available.

– COMPLETE –

(12) APU is operating:

(13) PACKS TRANSFER
to engine bleeds.

(14) APU generator is essential:

➡ **YES** – Go to (15)

➡ **NO** – Go to (18)

(15) APU generator is essential:

(16) APU indications MONITOR



Challenger 605	APU 01-7
	REV 50, Mar 25/19

APU SOV (Cont'd)

(17) Land at the nearest suitable airport.

▲ CAUTION: APU adapter oil will no longer be cooled. Adapter oil will overheat causing APU to auto-shutdown.

– COMPLETE –

(18) APU generator is NOT essential:

(19) APU SHUTDOWN

– COMPLETE –

TRAINING PURPOSES ONLY

Challenger 605	APU 01-8
	REV 49, Nov 19/18

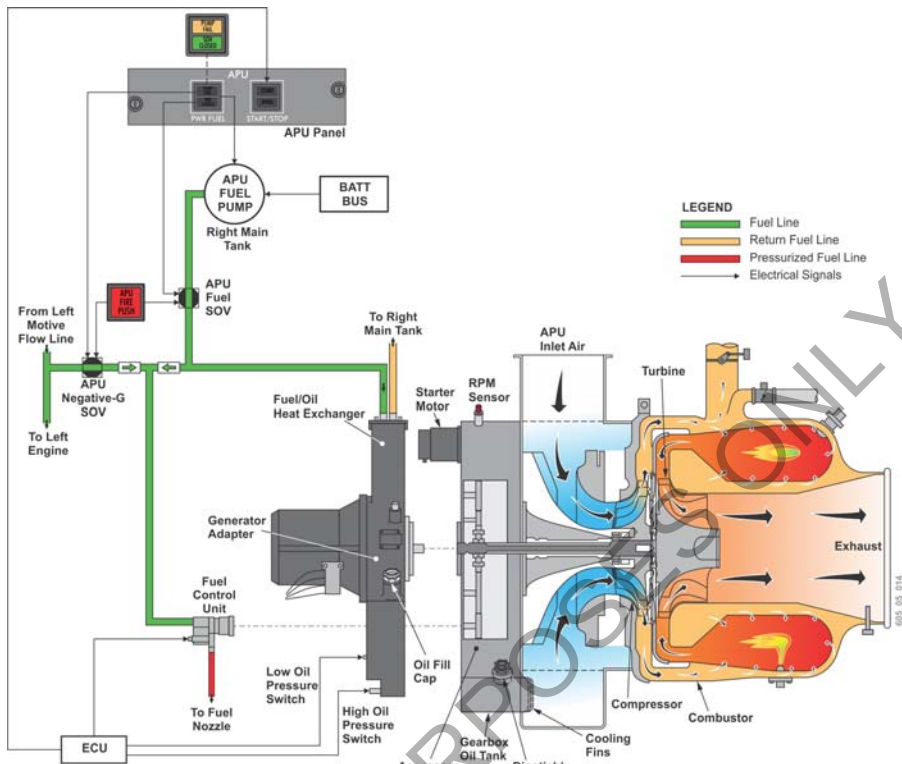
Advisory Messages

APU ADVISORY MESSAGES	
APU SOV CLSD	APU fuel feed and negative-g shutoff valves are confirmed closed after APU fire push switch is pushed, during an APU fire condition.

TRAINING PURPOSES ONLY

SCHEMATICS

Auxiliary Power Unit



Challenger 605	APU 01-10
	REV 49, Nov 19/18

Power Supply and Circuit Breaker Summary

CB NAME	BUS	CB LOCATION
APU CONTROL	BATT BUS	CBP1-L01
APU ECU PRIM	BATT BUS	CBP5-A04
APU ECU SEC	APU BATT DIR	CBP5-B09
APU START CONT	BATT BUS	CBP1-L03

TRAINING PURPOSES ONLY

Challenger 605	AVNC TOC-1
	REV 49, Nov 19/18

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CONFIG AP	AVNC 02-1

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AP HOLDING LWD (RWD) (NOSE DOWN) (NOSE UP)	AVNC 02-2
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EFIS COMP INOP	AVNC 02-4
EFIS MISCOMP	AVNC 02-5
EICAS COMP INOP	AVNC 02-6
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Challenger 605	AVNC TOC-2
	REV 59, Jun 02/21

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POWER SUPPLY AND CIRCUIT BREAKER SUMMARY

I **Power Supply and Circuit Breaker Summary**AVNC 02–47

AFCFS MSGS FAIL

AP

(PFD annunciation - flashing)

Condition: No valid communications between IAPS and AFCS.

Objective: Disconnect autopilot and assume manual control.

- (1) AutopilotDISCONNECT
using AP/SP DISC switch on control wheel or AP DISC
switch on flight control panel.

NOTE: Do not use the autopilot when **AFCS MSGS FAIL** is on.

- (2) Manual control.....RESUME

- COMPLETE -

CONFIG AP



“CONFIG AUTOPILOT”

Condition: Autopilot is engaged during takeoff.

Objective: Discontinue takeoff.

- (1) TakeoffREJECT
(2) Airplane configuration.....CONFIRM
Check disengagement of the autopilot.

(3) Erroneous **CONFIG AP** and aural occur:

- ➔ **YES** – Go to (4)
- ➔ **NO** – No further action required. **End of procedure.**

(4) Erroneous CONFIG AP and aural occur:

(5) **WOW OUTPUT** ProcedureACCOMPLISH

(See Page GEAR 10–12)

- COMPLETE -

Challenger 605	AVNC 02-2
	REV 49, Nov 19/18

ADS-B OUT FAIL

- (1) Contact ATC on radio.

- COMPLETE -

ADS-B OUT FAULT

- (1) MFD or CDU, ATC/TCAS
CONTROL menu.....SELECT
alternate transponder.

- COMPLETE -

AP HOLDING LWD (RWD) (NOSE DOWN) (NOSE UP)

Condition: Autopilot is holding control force in the direction indicated.

Objective: Correct the mistrim condition.

- (1) Autopilot operationMONITOR
Retrim as required.

⚠WARNING: An abrupt change in control force, or an out of trim situation may be experienced when disconnecting the autopilot.

SEAT BLTS should be selected ON.

- (2) **AP HOLDING LWD (RWD) (NOSE DOWN) (NOSE UP) stays on:**

➡ **YES** – Go to (3)

➡ **NO** – No further action required. **End of procedure.**

- (3) **AP HOLDING LWD (RWD) (NOSE DOWN) (NOSE UP) stays on:**

- (4) Flight controls HOLD FIRMLY

- (5) Autopilot DISCONNECT

- (6) AFCS SEL 1 (2).....SELECT
operative channel.

- (7) FDs and autopilot ENGAGE
and monitor operation. Retrim as required.

- (8) Autopilot ENGAGE
and monitor operation. Retrim as required.



Challenger 605	AVNC 02-3
	REV 49, Nov 19/18

AP HOLDING LWD (RWD) (NOSE DOWN) (NOSE UP) (Cont'd)

- (9) **AP HOLDING LWD (RWD) (NOSE DOWN) (NOSE UP) stays on:**
- ➔ **YES** – Go to (10)
 - ➔ **NO** – No further action required. **End of procedure.**
- (10) **AP HOLDING LWD (RWD) (NOSE DOWN) (NOSE UP) stays on:**
- (11) Flight controls HOLD FIRMLY
- (12) Autopilot DISCONNECT
Assume manual control.

– COMPLETE –

AP PITCH TRIM

- Condition:** Autopilot pitch trim function has failed or unable to keep up with trim change.
- Objective:** Assume manual control.
- ⚠WARNING:** An abrupt change in control force, or an out of trim situation may be experienced when disconnecting the autopilot.
- SEAT BLTS should be selected ON.
- (1) Flight controls HOLD FIRMLY
- (2) Autopilot DISCONNECT
Retrim as required.
- (3) AFCS SEL 1 (2) SELECT
operative channel.
- NOTE:** AFCS mode will have to be reselected.
- (4) FDs and autopilot ENGAGE
and monitor operation. Retrim as required.
- (5) **AP PITCH TRIM stays on:**
- ➔ **YES** – Go to (6)
 - ➔ **NO** – No further action required. **End of procedure.**
- (6) **AP PITCH TRIM stays on:**
- (7) Flight controls HOLD FIRMLY
- (8) Autopilot DISCONNECT
Assume manual control.

– COMPLETE –

Challenger 605	AVNC 02-4
	REV 49, Nov 19/18

EFIS COMP INOP

Condition: EFIS comparator unable to determine differences between input information.

Objective: Monitor instrument accuracy.

NOTE: If **EFIS COMP INOP** is on during an LPV approach with indications of an air data or attitude/heading failure, refer to ADC Failure (See Page AVNC 02–10) or IRS Failure (See Page AVNC 02–15).

- (1) Affected flight instrumentsMONITOR
- (2) Integrated Standby Instrument CROSS CHECK

NOTE: If any of the following occurs, refer to Unreliable Airspeed (See Page AVNC 02–24)

- Pitch attitude, thrust setting or external noise not consistent with indicated airspeed
- Large airspeed differences between PFDs and/or ISI
- Loss of multiple airspeed indications.

– COMPLETE –

Challenger 605	AVNC 02-5
	REV 49, Nov 19/18

EFIS MISCOMP

Condition: There is a mismatch between the onside and cross-side data.

Objective: Determine and revert to accurate source.

(1) Select appropriate flag(s):

- ➡ **Accompanied by ALT and/or IAS** – Go to (2)
- ➡ **Accompanied by ATT, ROL or PIT** – Go to (5)
- ➡ **Accompanied by HDG** – Go to (8)
- ➡ **Accompanied by RA** – Go to (15)
- ➡ **Accompanied by GS or LOC** – Go to (17)

(2) Accompanied by ALT and/or IAS:

- (3) Flight instruments and Integrated Standby Instrument **CROSS CHECK**
- (4) ADC source selector **SELECT**
to reliable side.

NOTE: If any of the following occurs, refer to Unreliable Airspeed (See Page AVNC 02–24):

- Pitch attitude, thrust setting or external noise not consistent with indicated airspeed.
- Large airspeed differences between PFDs and/or ISI.
- Loss of multiple airspeed indication.

– COMPLETE –

(5) Accompanied by ATT, ROL or PIT:

- (6) Flight instruments and Integrated Standby Instrument **CROSS CHECK**
- (7) IRS source selector **SELECT**
to reliable alternate source.

– COMPLETE –

(8) Accompanied by HDG:

(9) IRS 1 (2) (3) IN ATT status message is on:

- ➡ **YES** – Go to (10)
- ➡ **NO** – Go to (12)

(10) IRS 1 (2) (3) IN ATT status message is on:

- (11) FMS **SET IRS HDG**
Enter present airplane heading on the IRS CONTROL page.

– COMPLETE –



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Challenger 605	AVNC 02-6
	REV 49, Nov 19/18

EFIS MISCOMP (Cont'd)

(12) IRS 1 (2) (3) IN ATT status message is out:

(13) Flight instruments and Integrated
Standby Instrument CROSS CHECK

(14) IRS source selectorSELECT
to reliable alternate source.

– COMPLETE –

(15) Accompanied by RA:

(16) Radio Altimeter Failure ProcedureACCOMPLISH
(See Page AVNC 02–18)

– COMPLETE –

(17) Accompanied by GS or LOC:

If not visual (IMC) during approach:

(18) Go-around INITIATE

– COMPLETE –

EICAS COMP INOP

Condition: EICAS comparator inoperative (N_1 , N_2 or ITT data comparison not active).

Objective: Determine reliable engine data source.

(1) MFD displays (both)EICAS PAGE

(2) Engine and instrument
parameters, messages CROSS CHECK

(3) MFD, EICAS pageUSE RELIABLE SIDE

(4) Affected engine and/or system MAINTAIN
within limits.

– COMPLETE –

EVS HEAT FAIL

(1) EVS imageMONITOR
for degradation.

– COMPLETE –

Challenger 605	AVNC 02-7
	REV 49, Nov 19/18

EVS HEAT OVHT

- (1) EVS imageMONITOR
for degradation.

– COMPLETE –

PFD X-TALK FAIL

Condition: Unable to share information between PFDs.

Objective: Determine reliable source of information.

- (1) **PFD is blank:**

➡ **YES** – Go to (2)

➡ **NO** – Go to (4)

- (2) **PFD is blank:**

- (3) Primary Flight Display (PFD)

Failure ProcedureACCOMPLISH

(See Page AVNC 02–17)

– COMPLETE –

- (4) **PFD is still active:**

- (5) Instruments CROSS CHECK

NOTE: Verify that PFD reference values, speed and Vspeed bugs, and selected altitude are set on both DCPs or are correctly synchronized between both PFDs.

– COMPLETE –

XPDR 1 (2) FAIL

Condition: Single transponder failure.

- (1) MFD or CDU, ATC/TCAS

CONTROLSELECT
alternate transponder.

NOTE: TCAS and ADS–B OUT will stop functioning until alternate transponder is activated.

– COMPLETE –

Challenger 605	AVNC 02-8
	REV 51, Apr 05/19

XPDR 1 & 2 FAIL

Condition: Both transponders have failed.

- (1) Contact ATC on radio.

– COMPLETE –

YAW DAMPER

Condition: Both yaw damper channels are disengaged or failed.

Objective: Re-engage at least one yaw damper.

- (1) YD1 and YD2 ENGAGE

- (2) **YAW DAMPER** stays on:

➡ **YES** – Go to (3)

➡ **NO** – Go to (7)

- (3) **YAW DAMPER** stays on:

- (4) YAW DAMPER, DISC PRESS

- (5) FLAPS AS REQUIRED
Land with flaps at 20 degrees, commensurate with the controllability of the airplane.

- (6) Go to (8)

- (7) **YAW DAMPER** goes out:

- (8) Landing with FLAPS at 20 degrees:

➡ **YES** – Go to (9)

➡ **NO** – No further action required. **End of procedure.**

- (9) **Prior to landing:**

- (10) TAWS WARNING, FLAPS OFF PRESS IN

- (11) TAWS WARNING, RAAS OFF
(if installed) PRESS IN

- (12) Approach speed $V_{REF} + 14$
minimum.

- (13) ATS DISC PRESS
prior to 100 feet AGL.

⚠CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.



Challenger 605	AVNC 02-9
	REV 49, Nov 19/18

YAW DAMPER (Cont'd)

(14) Actual landing distance INCREASE
as applicable by factor given below.

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

NOTE: Select the runway available with minimum crosswind.

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

TRAINING PURPOSES ONLY

Challenger 605	AVNC 02-10
	REV 62, Nov 24/21

ADC Failure

Condition: Erratic or failed air data information. May be accompanied by **ALT**, **AOA**, **IAS**, **MACH** and/or **VS** flags.

Objective: Transfer to reliable side.

- Airplanes **incorporating** Service Bulletin:
 - SB 605-34-019: Navigation, Installation of Proline 21 Advanced Upgrade.

NOTE: If FMS 3 is reverted in, use of VNAV guidance is prohibited following reversion.

(1) Choose applicable:

- ➔ **Pilot's ADC has failed** – Go to (2)
- ➔ **Copilot's ADC has failed** – Go to (5)

(2) Pilot's ADC has failed:

(3) ADC source selectorSELECT
To copilot's side (ADC 2).

Check PFDs indicate single air data source selected (ADC2).

(4) Go to (7)

– COMPLETE –

(5) Copilot's ADC has failed:

(6) ADC source selectorSELECT
To pilot's side (ADC 1).

Check PFDs indicate single air data source selected (ADC1).

(7) N₁ settingDESELECT
from EICAS.

NOTE: Data from the CDU can be used if crossed checked with the AFM. TGT thrust mode may be used to set the N₁ setting on EICAS after confirmation of the data.

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Avionics
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Challenger 605	AVNC 02-11
	REV 49, Nov 19/18

Autopilot Failure



CAVALRY CHARGE



(PFD annunciation - flashing)

Condition: Uncommanded autopilot disengagement. Accompanied by AP flag and continuous aural.

Objective: Select alternate AFCS system.

- (1) AutopilotDISCONNECT
using AP/SP DISC switch on control wheel or AP DISC switch on flight control panel.
- (2) AFCS SEL 1 (2) (centre pedestal).....SELECT
operative channel.
- (3) Autopilot RE-ENGAGE
- (4) **Autopilot warning comes on again:**
 - ➔ **YES** – Go to (5)
 - ➔ **NO** – No further action required. **End of procedure.**
- (5) **Autopilot warning comes on again:**
- (6) AutopilotDISCONNECT
- (7) Manual control RESUME

– COMPLETE –

Cursor Control Panel (CCP) Failure

Condition: Unable to control pilot or copilot CCP functions.

Objective: Operate in degraded mode.

- (1) **EICAS displayed on same side MFD as failed CCP:**
 - ➔ **YES** – No further action required. **End of procedure.**
 - ➔ **NO** – Go to (2)
- (2) **EICAS displayed on cross side MFD:**
- (3) EICAS BOTH PFD'S

– COMPLETE –

Challenger 605	AVNC 02-12
	REV 62, Nov 24/21

DCU Failure or DCU Aural Failure

Condition: Failure or fault of the indicated Data Concentrator Unit. Associated with **DCU 1 (2) INOP** or **DCU 1 (2) AURAL INOP**.

- (1) Affected AUDIO WARNING DISABLE
- (2) Other AUDIO WARNING NORMAL

– COMPLETE –

Display Control Panel (DCP) Failure

Condition: Unable to control pilot or copilot DCP functions. Baro, course and HDG knobs may be inoperative.

Objective: Transfer to operative DCP side.

- (1) Flight control panel, XFRSELECT side corresponding to operative DCP.

NOTE: Display features may be controlled using the CCP and the PFD submenu on the MFD.

– COMPLETE –

Dual GPS Failure – Oceanic and Remote Ops

Condition: Loss of both GPS channels. Accompanied by IRS ONLY, and/or FMS DR or DR EXCEEDS 5 MIN messages.

Objective: Attempt to regain GPS function.

- (1) GPS 1 and GPS 2 circuit breakers (1N3 and 2H11)PULL and RESET.

(2) Message stays on after approximately 60 seconds:

- ➡ YES – Go to (3)
- ➡ NO – No further action required. **End of procedure.**

(3) Message stays on after approximately 60 seconds:

- (4) Notify controlling agency.

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Avionics
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Challenger 605	AVNC 02-13
	REV 49, Nov 19/18

Dual HF Transmission Failure

Condition: No HF transmission. Amber Frequency tuning areas on HF 1 and HF 2.

Objective: Attempt to regain HF communication capability.

- (1) HF COMM 2 circuit breakers (2L8, 2L9 and 2L10)PULL
- (2) HF 1ATTEMPT communication.
- (3) **Transmission successful:**
 - ➡ **YES** – No further action required. **End of procedure.**
 - ➡ **NO** – Go to (4)
- (4) **Transmission NOT successful:**
- (5) HF COMM 2 circuit breakers (2L8, 2L9 and 2L10)RESET
- (6) HF COMM 1 circuit breakers (1K5, 1K6 and 1K7)PULL
- (7) HF 2ATTEMPT communication.

– COMPLETE –

Flight Management System CDU/FMC Failure

Condition: No data available on CDU or unresponsive CDU.

Objective: Isolate affected FMS.

NOTE: If a red FMS flag is displayed during an LPV approach with indications of an air data or attitude/heading failure, refer to ADC Failure (See Page AVNC 02–10) or IRS Failure (See Page AVNC 02–15).

- (1) FMS 3 (if installed)SET
to failed FMS position.
- (2) CDU 1 (2) TUNE INHIBPUSH
to affected side.
- (3) **Only one FMS is operative:**
 - ➡ **YES** – Go to (4)
 - ➡ **NO** – No further action required. **End of procedure.**
- (4) **Only one FMS is operative:**
- (5) N₁ Thrust TargetDESELECT
on FMS.

- NOTE:**
1. Data from the CDU can be used if cross checked with the AFM. TGT thrust mode may be used to set the N₁ setting on EICAS after confirmation of the data.
 2. The input parameters on the THRUST LIMIT, TAKEOFF REF and APPROACH REF CDU pages must be re-confirmed by the pilot following an FMS 3 reversion (if installed), an FMS SYNC command or an FMS cold start.

- Airplanes **incorporating** Service Bulletin:
- SB 605–34–019: Navigation, Installation of Proline 21 Advanced Upgrade.

The input parameters on the THRUST LIMIT, TAKEOFF REF and APPROACH REF CDU pages must be re-confirmed by the pilot following an FMS 3 reversion (if installed) or an FMS cold start.

– COMPLETE –

Challenger 605	AVNC 02-15
	REV 49, Nov 19/18

FMS V Speed Failure

VSPD

(1) V Speeds.....SET

- NOTE:**
1. Set V speeds manually through the PFD REFS to clear the Flag. Use the AFM to determine V speeds.
 2. Do not use the FMS takeoff and approach performance calculation function.

– COMPLETE –

FSU Failure

Condition: IFIS electronic charts, MFD graphical weather and enhanced map overlays are inoperative.

Objective: Operate in degraded mode.

(1) **FSU INOP** displayed on both MFDs:

- ➡ **YES** – Go to (2)
- ➡ **NO** – Go to (4)

(2) **FSU INOP** displayed on both MFDs:

(3) Use alternate source for electronic charts.

– COMPLETE –

(4) **FSU INOP** displayed on only one MFD :

(5) EICAS..... BOTH PFD'S

NOTE: Electronic charts can be displayed on the unaffected MFD.

– COMPLETE –

IRS Failure

Condition: IRS has failed or is providing erratic attitude and/or heading data. May be accompanied by **ATT** and/or **HDG** flags.

Objective: Transfer to operative IRS.

NOTE: If an IRS is in ATT mode, a heading drift of up to 15 degrees/hr can occur. The FMS heading on the IRS CONTROL page should be updated periodically from the best available alternate navigation source.



Challenger 605	AVNC 02-16
	REV 49, Nov 19/18

IRS Failure (Cont'd)

(1) Choose applicable:

- ➡ **Pilot's IRS is failed** – Go to (2)
- ➡ **Copilot's IRS is failed** – Go to (4)
- ➡ **Both IRS failed** – Go to (6)

(2) Pilot's IRS is failed:

- (3) IRS source selectorSELECT
to an operative IRS.

NOTE: 1. If IRS 3 is not installed, pilot's flight director is inoperative, and FD 1 FAIL status message comes on.
2. PFDs indicate IRS 2 single source selected (if IRS 3 is installed and selected, PFD 1 indicates IRS 3 source selected).

– COMPLETE –

(4) Copilot's IRS is failed:

- (5) IRS source selectorSELECT
to an operative IRS.

NOTE: 1. If IRS 3 is not installed, copilot's flight director is inoperative, and FD 2 FAIL status message comes on.
2. PFDs indicate IRS 1 single source selected (if IRS 3 is installed and selected, PFD 2 indicates IRS 3 source selected).

– COMPLETE –

(6) Both IRS failed:

- (7) ISI and standby compassMONITOR
(8) Land at the nearest suitable airport.

– COMPLETE –

Loss of EICAS

Condition: EICAS data is not available on selected MFD.

If display blanks, the system automatically transfers the EICAS primary page data to the operational MFD.

- (1) EICAS..... BOTH PFD'S

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Avionics
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Challenger 605	AVNC 02-17
	REV 49, Nov 19/18

Multifunction Display (MFD) Failure

Condition: MFD is intermittent or goes blank.

Objective: Operate in reversion mode.

On the reversionary panel:

- (1) L (R) Displays PFD REV
- (2) EICAS BOTH PFD'S

- Airplanes **incorporating** Service Bulletin:
 - SB 605-34-019: Navigation, Installation of Proline 21 Advanced Upgrade.
- (3) MFD TUNE INHIB PRESS IN

– COMPLETE –

Primary Flight Display (PFD) Failure

Condition: PFD is intermittent or goes blank.

Objective: Operate in reversion mode.

On the reversionary panel:

- (1) L (R) Displays MFD REV
- (2) EICAS BOTH PFD'S

– COMPLETE –

RAAS Advisory Unexpected

Condition: Advisory or aural is in conflict with expectations.

(1) Identify phase of flight:

- ➡ **During low speed ground operations** – Go to (2)
- ➡ **In flight or during high speed ground operations** – Go to (6)

(2) During low speed ground operations:

- (3) Stop.
- (4) Verify position.
- (5) Go to (8)

(6) In flight or during high speed ground operations:

- (7) Verify position and/or distance remaining or distance available.



Challenger 605	AVNC 02-18
	REV 49, Nov 19/18

RAAS Advisory Unexpected (Cont'd)

(8) Contact ATC for assistance if necessary.

NOTE: If aural alerts become distracting or interfere with communications between flight crew or ATC, press in the TAWS WARNING RAAS switch/light to select it OFF.

– COMPLETE –

RAAS Inoperative

Condition: Runway awareness system inoperative. Associated with the **RAAS FAIL** status message.

(1) TAWS WARNING, RAAS OFF.....PRESS IN

– COMPLETE –

RAAS Not Available

Condition: Insufficient information for runway awareness to function. Associated with the **RAAS NOT AVAILABLE** status message.

(1) TAWS WARNING, RAAS OFF.....PRESS IN

– COMPLETE –

Radio Altimeter Failure

Condition: False Radio Altimeter height and/or RA fail flag. May be accompanied with **EFIS COMP INOP** / **EFIS MISCOMP**.

Objective: Review affected systems.

(1) Choose applicable:

- ➡ **Pilot’s radio altimeter is failed** – Go to (2)
- ➡ **Copilot’s radio altimeter is failed** – Go to (6)

(2) **Pilot’s radio altimeter is failed:**

(3) ATS DISC..... PRESS
prior to 100 feet AGL.

⚠CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

(4) Thrust levers..... POSITION MANUALLY



Challenger 605	AVNC 02-19
	REV 49, Nov 19/18

Radio Altimeter Failure (Cont'd)

(5) Go to (7)

(6) Copilot's radio altimeter is failed:

(7) Review operational notes below:

- NOTE:**
- Do not rely on affected radio altitude displays.
 - TCAS, TAWS and RAAS (if installed) and windshear are inoperative. TCAS FAIL message on PFD or MFD, **TAWS BASIC FAIL**, **RAAS FAIL** and **TAWS WINDSHEAR FAIL** on.

- Airplanes **incorporating** Service Bulletin:
 - SB 605-34-005: Radio Altimeter System, Installation of a second Radio Altimeter.*

If only one radio altimeter has failed, TCAS, TAWS and RAAS and windshear remain operational.

- For an intermittent or undetected radio altimeter error, the following may be incorrect or inactive:
 - Displayed radio altitudes
 - LOC/GS comparators
 - Excessive deviation alerts
 - Expanded lateral deviation scale
 - Approach minimums annunciation.
- For an intermittent or undetected radio altimeter error, ATS RETARD mode may occur early or not occur. Refer to ATS Radio Altimeter Input Failure. (See Page ENG 05-34) .
- For an intermittent or undetected radio altimeter error, localizer and/or glideslope AP/FD commands could be adversely affected.

– COMPLETE –

Radio Tuning Failure

Condition: Unable to tune radios. Amber Frequencies in tuning areas.

Objective: Inhibit affected tuning controls

- NOTE:**
- Note all frequencies being used prior to starting procedure.
 - During procedure, continue tuning and attempting communication even if frequency is displayed in amber.



Challenger 605	AVNC 02-20
	REV 49, Nov 19/18

Radio Tuning Failure (Cont'd)

(1) CDU 1 TUNE INHIBPRESS IN

If unable to restore radio tuning on CDU 2 or MFD:

(2) CDU 1 TUNE INHIBPRESS OUT

(3) CDU 2 TUNE INHIBPRESS IN

If unable to restore radio tuning on CDU 1 or MFD:

(4) CDU 1 TUNE INHIBPRESS IN

If unable to restore radio tuning on MFD:

(5) CDU 1 TUNE INHIB and CDU 2
TUNE INHIBPRESS OUT

(6) MFD TUNE INHIBPRESS IN

If unable to restore COM tuning on at least one side:

(7) CDU 1 TUNE INHIB and CDU 2
TUNE INHIBPRESS IN
to force COM 1 to 121.5.

- NOTE:**
1. To go straight to the emergency frequency (121.50 MHz), press in CDU 1 TUNE INHIB, CDU 2 TUNE INHIB and MFD TUNE INHIB.
 2. If 8.33 KHz channel spacing was previously tuned on VHF 1, the emergency frequency (121.50) may be displayed in amber. The amber display does not prevent VHF from operating normally on this frequency.

– COMPLETE –

Single Flight Director Failed

FD 1 FAIL

or

FD 2 FAIL



PFD
annunciation

Condition: Indicated flight director is inoperative. May be accompanied by **FD1 (2) FAIL**.

Objective: Use alternate autopilot system.

- NOTE:**
1. If a red FD flag is displayed during an LPV approach with indications of an air data or attitude/heading failure, refer to ADC Failure (See Page AVNC 02-10) or IRS Failure (See Page AVNC 02-15) .
 2. Anticipate elevator out of trim situation when disconnecting the autopilot.

- (1) Flight controls HOLD FIRMLY
- (2) Autopilot DISCONNECT
- (3) AFCS SEL 1 (2) SELECT
operative channel.

NOTE: AFCS mode will have to be reselected.

- (4) FDs and autopilot ENGAGE
and monitor operation. Retrim as required.

(5) Choose applicable:

- ➡ Red FD flag displayed on both PFDs – Go to (6)
- ➡ Sensor failure indicated (red line across affected flight director mode annunciator) – Go to (8)
- ➡ No failure indication – No further action required. **End of procedure.**

(6) Red FD flag displayed on both PFDs:

- (7) Flight control panel XFR SELECT
to operable side.

– COMPLETE –

(8) Sensor failure indicated:

- (9) Affected mode switch DESELECT

– COMPLETE –

Challenger 605	AVNC 02-22
	REV 49, Nov 19/18

Synthetic Vision System Runway Failure

- (1) SVS OFF
prior to takeoff and prior to passing FAF.

– COMPLETE –

Synthetic Vision System Failure

- (1) SVS OFF

– COMPLETE –

TCAS Failure

Condition: Loss of TCAS functionality accompanied by PFD/MFD
TCAS FAIL.

Objective: Discontinue use of TCAS.

- (1) MFD or CDU tuning, ATC/TCAS
mode ALT ON
to select TCAS off.

– COMPLETE –

TCAS VSI Resolution Advisory Failure

NO VSI RA

Condition: No TCAS Resolution Advisory on the VSI.

Objective: Operate in degraded mode.

- (1) MFD or CDU tuning, ATC/TCAS
mode TA ONLY

– COMPLETE –

Challenger 605	AVNC 02-23
	REV 51, Apr 05/19

Uncommanded Yaw Motion

Condition: Abnormal/uncommanded change in the yaw axis (“kick” or oscillation).

Objective: Assume manual control.

- (1) Controls ASSUME MANUAL CONTROL
and counter airplane motion using handwheel inputs.

(2) YAW DAMPER, DISC PRESS

➡ FROM Quick Reference Card ⬅

(3) Uncommanded motion continues:

- ➡ **YES** – Go to (4)
- ➡ **NO** – Go to (9)

(4) Uncommanded motion continues:

(5) Airspeed 250 KIAS maximum.

(6) Land at the nearest suitable airport.

NOTE: Select the runway available with minimum crosswind.

(7) FLAPS AS REQUIRED
Land with flaps at 20 degrees, commensurate with the controllability of the airplane.

(8) Go to (10)

(9) Uncommanded motion stops:

(10) Landing with FLAPS at 20 degrees:

- ➡ **YES** – Go to (11)
- ➡ **NO** – No further action required. **End of procedure.**

(11) Landing with FLAPS at 20 degrees:

Prior to landing:

(12) TAWS WARNING, FLAPS OFF PRESS IN

(13) TAWS WARNING, RAAS OFF (if installed) PRESS IN

(14) Approach speed $V_{REF} + 14$ minimum.

(15) ATS DISC PRESS
prior to 100 feet AGL.

▲ CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

(16) Actual landing distance INCREASE
as applicable by factor given below:



Challenger 605	AVNC 02-24
	REV 59, Jun 02/21

Uncommanded Yaw Motion (Cont'd)

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

Unreliable Airspeed

Condition: Loss or intermittent or large difference between airspeed sources. May be accompanied by numerous flags and comparator messages.

Objective: Use pitch and thrust references for airspeed control.

Indication:

- Pitch attitude, thrust setting or external noise not consistent with indicated airspeed.
- Loss of multiple airspeed indications.
- Multiple airspeed indications remain constant regardless of pitch or thrust adjustments.

Unreliable Airspeed (Cont'd)

- Airspeed difference between primary flight displays is more than 10 KIAS and/or more than 15 KIAS between any primary flight display and ISI and no reliable airspeed source can be determined.
- EFIS MISCOMP** with an amber IAS flag and/or **EFIS COMP INOP**

NOTE:

- If doubt exists about the reliability of aircraft speed indications, aircraft control must be established and maintained by the use of pitch and thrust settings.
- Autopilot may not be available.
- No crew action is required for the following messages as they are covered within:
 - Caution messages: **EFIS MISCOMP** with an amber IAS flag or **EFIS COMP INOP**.

- ATS DISC switch DEPRESS
- Autopilot DISCONNECT
- FDs DESELECT
- Airplane Altitude MAINTAIN A SAFE ALTITUDE

If Climb or Go-Around is required:

- Pitch/N₁ 15 DEGREES/TO FROM SL TO 15,000 FT
10 DEGREES/CLB ABOVE 15,000 FT
- Airplane Configuration FLAPS 20/ GEAR UP FOR GO-AROUND
Flaps 0 / GEAR UP ONCE CLIMBING

▲ CAUTION: Respect stall warning/stick shaker and stick pusher.

➡ FROM Quick Reference Card ⬅

At desired altitude:

- Autopilot or Flight Director (if required and available) ENGAGE BASIC PTCH/ROLL

NOTE:

- Using the FD/autopilot in PTCH, ALT, HDG and ROLL modes will help reduce workload.
- If FD/autopilot engaged in PTCH mode, one click of PTCH wheel results in 0.5 degrees of pitch change.

▲ CAUTION:

- Do not use any FD/autopilot SPEED mode.
- Do not use FLC mode or ATS.



Unreliable Airspeed (Cont'd)

(8) Pitch/N₁SET
as per Table A.

NOTE: Leave current environmental conditions if they are considered to be a reason for unreliable airspeed. Refer to Tables B or C as required for climb or descent.

(9) ANTI-ICE, LH and RH COWL
switchesAS REQUIRED

(10) ANTI-ICE, WING switchAS REQUIRED

- NOTE:**
1. Autopilot/yaw damper should only be engaged once N₁ is set.
 2. Cross check AOA after setting pitch attitude.
 3. Pitch/Thrust settings will give approximate level flight, and may result in slight climb or descent. Adjust pitch to stop climb/descent.
 4. Disregard low speed cue, flight path vector and airspeed trend vector as these may be erroneous.
 5. If overspeed warning sounds, select the affected AUDIO WARNING switch to mute aural and disregard.
 6. A nuisance **NOSE DOOR OPEN** warning message can be disregarded.
 7. Altitude, Mode C and TCAS may be in error by up to 600 feet. Vertical speed indications may also be unreliable.
 8. Groundspeed indication may be unreliable.
 9. FMS cruise time and fuel predictions may be incorrect.
 10. Mach trim may be inoperative or operate erratically. Disengage Mach trim if erratic stabilizer motion is observed.

Evaluate indicated airspeed sources in level flight as follows:

- ▲ CAUTION:** Airspeed should be considered reliable only if the following two conditions both exist:
1. Airspeed indication is consistent with pitch, thrust and airspeed range from Table A, and
 2. Airspeed indication changes and speed trend vector are consistent with pitch and thrust adjustments.

(11) ADC source selectorSELECT TO NORM

(12) Both PFDs and ISI COMPARE AIRSPEED
to expected speed range from Table A.



Unreliable Airspeed (Cont'd)

- (13) Determine which of the following conditions apply:
- If no indicated airspeed is considered reliable from Table A, proceed to step (14).
 - If one or more airspeed source is considered reliable from Table A, proceed to step (18).

NO RELIABLE AIRSPEED SOURCE

- (14) Airspeed Indications DISREGARD
- (15) Pitch/N₁ SET
using Table A, B, C, D 1, D 2, D 3, or E as appropriate for
phase of flight.
- (16) Land at the nearest suitable airport.
- (17) When ready to start descent, proceed to step (23)

ONE OR MORE AIRSPEED SOURCE IS CONSIDERED RELIABLE

- (18) ADC source selector CONFIRM NORM
- (19) Reliable airspeed source MONITOR WITH CAUTION
- (20) Pitch/N₁ SET
using Table A, B, C, D 1, D 2, D 3 or E as appropriate for
phase of flight and continue to monitor reliable airspeed
source and airspeed range.

- ▲ CAUTION:** 1. Disregard airspeed indication if at any time it becomes unreliable.
2. ADC source selector must remain in NORM position.

- (21) Land at the nearest suitable airport.
- (22) When ready to start descent, proceed to step (23)

Recovery to an airfield

- (23) Descent INITIATE WHEN READY

- ▲ CAUTION:** To avoid significant flap overspeed and flap damage:
1. Transition from FLAPS 0 to FLAPS 30 must be accomplished in level flight (Table D 1, D 2 or D 3).
 2. Select FLAPS 45 at the start of final descent on the glideslope/glidepath (Table E).

- NOTE:** 1. Plan to intercept a long final to allow time to re-configure for approach/landing and set pitch/N₁.
2. Plan to recover to an airfield with an ILS, LNAV or VNAV under Visual Meteorological Conditions if possible.



Unreliable Airspeed (Cont'd)

(25) Actual landing distance INCREASE

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.25 (25%)	1.20 (20%)

NOTE: In the event of a go-around, refer to the immediate actions at the beginning of this procedure.

- NOTE:**
- Pitch/Thrust settings will give approximate level flight, and may result in slight climb or descent. Interpolation is acceptable.
 - Adjust pitch to stop climb/descent.
 - If pitch is increased, REDUCE expected speed range by delta KIAS/delta Mach.
 - If pitch is decreased, INCREASE expected speed range by delta KIAS/delta Mach.

TABLE A - LEVEL FLIGHT - FLAP 0											
			WEIGHT 1,000 LB (1,000 KG)						EXPECTED SPD RANGE		EXPECTED SPD CHANGE FOR 0.5° CHANGE IN PITCH ΔKIAS/ ΔMACH
			26 (11,8)	30 (13,6)	35 (15,9)	40 (18,1)	45 (20,4)	48 (21,8)	KIAS	MACH	
ALTITUDE (1,000 ft)	40	Pitch (°)	1.0	1.5	2.0	2.5			220 to 240	.71 to .82	5/0.01
		N ₁ (%)	83.0	84.6	87.2	89.9					
		AoA	0.4	0.4	0.5	0.5					
	35	Pitch (°)	1.5	2.0	2.5	2.5	3.0		240 to 260	.69 to .80	10/0.03
		N ₁ (%)	81.8	82.7	83.9	85.7	87.9				
		AoA	0.4	0.4	0.4	0.4	0.5				
	30	Pitch (°)	1.5	2.0	2.5	2.5	3.0	3.0	250 to 275		10
		N ₁ (%)	79.9	80.4	81.3	82.3	83.8	84.7			
		AoA	0.3	0.4	0.4	0.4	0.4	0.4			
	25	Pitch (°)	1.5	2.0	2.5	2.5	3.0	3.0	250 to 275		10
		N ₁ (%)	76.3	76.7	77.5	78.3	79.5	80.3			
		AoA	0.4	0.4	0.4	0.4	0.5	0.5			
	20	Pitch (°)	1.5	2.0	2.5	3.0	3.0	3.5	250 to 275		10
		N ₁ (%)	72.9	73.3	74.0	74.9	75.9	76.7			
		AoA	0.4	0.4	0.4	0.4	0.5	0.5			
If any indicated speed is outside the expected indicated airspeed range, it should be considered UNRELIABLE.											



Unreliable Airspeed (Cont'd)

- NOTE:**
1. Pitch/Thrust settings will give approximate level flight, and may result in slight climb or descent. Interpolation is acceptable.
 2. Adjust pitch to stop climb/descent.
 3. If pitch is increased, REDUCE expected speed range by delta KIAS/delta Mach.
 4. If pitch is decreased, INCREASE expected speed range by delta KIAS/delta Mach.

TABLE A - LEVEL FLIGHT - FLAP 0

			WEIGHT 1,000 LB (1,000 KG)						EXPECTED SPD RANGE		EXPECTED SPD CHANGE FOR 0.5° CHANGE IN PITCH ΔKIAS/ ΔMACH
			26 (11,8)	30 (13,6)	35 (15,9)	40 (18,1)	45 (20,4)	48 (21,8)	KIAS	MACH	
ALTITUDE (1,000 ft)	15	Pitch (°)	1.5	2.0	2.5	3.0	3.0	3.5	250 to 275		10
		N ₁ (%)	70.5	70.8	71.5	72.3	73.3	74.0			
		AoA	0.4	0.4	0.4	0.4	0.5	0.5			
	10	Pitch (°)	2.0	2.5	3.5	4.0	4.5	5.0	215 to 230		5
		N ₁ (%)	58.0	58.9	60.2	61.7	63.1	63.9			
		AoA	0.4	0.4	0.5	0.5	0.6	0.6			
	5	Pitch (°)	2.5	3.0	3.5	4.5	5.0	5.5	210 to 225		5
		N ₁ (%)	54.8	55.7	57.0	58.3	59.7	60.5			
		AoA	0.4	0.4	0.4	0.5	0.5	0.5			
	0	Pitch (°)	2.5	3.0	3.5	4.5	5.0	5.5	210 to 225		5
		N ₁ (%)	51.9	52.7	53.9	55.2	56.5	57.3			
		AoA	0.3	0.4	0.4	0.5	0.5	0.5			

If any indicated speed is outside the expected indicated airspeed range, it should be considered UNRELIABLE.



Unreliable Airspeed (Cont'd)

- NOTE:**
1. Use FD in PTCH, ALT, HDG or ROLL modes only.
 2. If FD/autopilot engaged in PTCH mode, one click of PTCH wheel results in 0.5 degrees of pitch change.

- CAUTION:**
1. Do not use any AP/FD SPEED mode.
 2. Do not use FLC mode or ATS.

Table B – Climb – FLAPS 0					
ALT (ft)	Pitch (°)	N ₁ (%)	AoA (°)	Expected SPD Range	
				KIAS	MACH
0	13.5	TO	0.3	225 to 300	
5000	13.0	TO	0.3	215 to 300	
10000	11.5	TO	0.4	215 to 315	
15000	10.0	CLB	0.4	225 to 310	
20000	8.0	CLB	0.4	230 to 295	
25000	6.0	CLB	0.4	255 to 305	
30000	4.0	CLB	0.4	265 to 290	.70 to .76
35000	3.5	CLB	0.4	245 to 265	.73 to .78



Unreliable Airspeed (Cont'd)

- NOTE:**
1. Use FD in PTCH, ALT, HDG or ROLL modes only.
 2. If FD/autopilot engaged in PTCH mode, one click of PTCH wheel results in 0.5 degrees of pitch change.

- CAUTION:**
1. Do not use any AP/FD SPEED mode.
 2. Do not use FLC mode or ATS.

Table C – Descent – FLAPS 0

ALT (ft)	Pitch (°)	N ₁ (%)	AoA (°)	Expected SPD Range	
				KIAS	MACH
40000	–1.5	IDLE	0.4	230 to 255	.75 to .83
35000	–1.5	IDLE	0.4	235 to 280	.69 to .82
30000	–2.0	IDLE	0.4	235 to 305	.62 to .79
25000	–2.0	IDLE	0.4	230 to 305	
20000	–2.0	IDLE	0.4	230 to 305	
15000	–1.0	IDLE	0.4	215 to 280	
10000	–1.0	IDLE	0.4	220 to 275	
5000	–1.0	IDLE	0.4	205 to 280	
0	–1.0	IDLE	0.4	205 to 280	

- NOTE:** With anti-ice ON, increase N₁ if required to obtain the minimum N₂ of 78%.
- For an N₁ increase of 5%, add 0.5° to the pitch from Table C.
 - For an N₁ increase of 10%, add 1° to the pitch from Table C.
 - For an N₁ increase of 15%, add 2° to the pitch from Table C.



Unreliable Airspeed (Cont'd)

- NOTE:**
- Pitch/Thrust settings will give approximate level flight, and may result in slight climb or descent. Interpolation is acceptable.
 - Adjust pitch to stop climb/descent.
 - If pitch is increased, **REDUCE** expected speed range by delta KIAS.
 - If pitch is decreased, **INCREASE** expected speed range by delta KIAS.
 - Use FD in PTCH, ALT, HDG or ROLL modes only.
 - If FD/autopilot engaged in PTCH mode, one click of PTCH wheel results in 0.5 degrees of pitch change.

- CAUTION:**
- Do not use any AP/FD SPEED mode.
 - Do not use FLC mode or ATS.
- To avoid significant flap overspeed and flap damage:
- Transition from FLAPS 0 to FLAPS 30 must be accomplished in level flight.
 - Select FLAPS 45 at the start of final descent on the glideslope/glidepath.

TABLE D 1 - APPROACH - LEVEL FLIGHT - FLAPS 0 - GEAR UP										
			WEIGHT 1,000 LB (1,000 KG)						EXPECTED SPD RANGE KIAS	EXPECTED ΔKIAS PER 0.5° ΔPITCH
			26 (11,8)	30 (13,6)	35 (15,9)	40 (18,1)	45 (20,4)	48 (21,8)		
ALTITUDE (1,000 ft)	15	Pitch (°)	2.0	2.5	3.5	4.0	4.5	5.0	215 to 230	5
		N ₁ (%)	61.1	62.2	63.6	65.3	67.0	68.0		
		AoA	0.4	0.4	0.5	0.6	0.6	0.6		
	10	Pitch (°)	2.0	2.5	3.5	4.0	4.5	5.0	215 to 230	5
		N ₁ (%)	58.0	58.9	60.2	61.7	63.1	63.9		
		AoA	0.4	0.4	0.5	0.5	0.6	0.6		
	5	Pitch (°)	2.5	3.0	3.5	4.5	5.0	5.5	210 to 225	5
		N ₁ (%)	54.8	55.7	57.0	58.3	59.7	60.5		
		AoA	0.4	0.4	0.4	0.5	0.5	0.5		
	0	Pitch (°)	2.5	3.0	3.5	4.5	5.0	5.5	210 to 225	5
		N ₁ (%)	51.9	52.7	53.9	55.2	56.5	57.3		
		AoA	0.3	0.4	0.4	0.5	0.5	0.5		

Unreliable Airspeed (Cont'd)

- NOTE:**
- Pitch/Thrust settings will give approximate level flight, and may result in slight climb or descent. Interpolation is acceptable.
 - Adjust pitch to stop climb/descent.
 - If pitch is increased, **REDUCE** expected speed range by delta KIAS.
 - If pitch is decreased, **INCREASE** expected speed range by delta KIAS.
 - Use FD in PTCH, ALT, HDG or ROLL modes only.
 - If FD/autopilot engaged in PTCH mode, one click of PTCH wheel results in 0.5 degrees of pitch change.

- CAUTION:**
- Do not use any AP/FD SPEED mode.
 - Do not use FLC mode or ATS.
- To avoid significant flap overspeed and flap damage:
- Transition from FLAPS 0 to FLAPS 30 must be accomplished in level flight.
 - Select FLAPS 45 at the start of final descent on the glideslope/glidepath.

TABLE D 2 - APPROACH - LEVEL FLIGHT - FLAPS 20 - GEAR UP

			WEIGHT 1,000 LB (1,000 KG)						EXPECTED SPD RANGE KIAS	EXPECTED ΔKIAS PER 0.5° ΔPITCH
			26 (11,8)	30 (13,6)	35 (15,9)	40 (18,1)	45 (20,4)	48 (21,8)		
ALTITUDE (1,000 ft)	15	Pitch (°)	0.5	1.0	2.0	3.0	4.0	4.5	165 to 185	5
		N ₁ (%)	66.6	67.6	68.9	70.4	72.3	73.4		
		AoA	0.3	0.4	0.4	0.5	0.5	0.5		
	10	Pitch (°)	0.5	1.0	2.0	3.0	4.0	4.5	165 to 185	5
		N ₁ (%)	63.1	64.0	65.2	66.6	68.4	69.4		
		AoA	0.3	0.3	0.4	0.5	0.5	0.5		
	5	Pitch (°)	0.5	1.0	2.0	3.0	4.0	4.5	165 to 185	5
		N ₁ (%)	59.7	60.5	61.7	63.0	64.6	65.6		
		AoA	0.3	0.3	0.4	0.5	0.5	0.5		
	0	Pitch (°)	0.5	1.0	2.0	3.0	4.0	4.5	165 to 185	5
		N ₁ (%)	56.4	57.2	58.4	59.6	61.2	62.1		
		AoA	0.3	0.3	0.4	0.5	0.5	0.5		

Unreliable Airspeed (Cont'd)

- NOTE:**
- Pitch/Thrust settings will give approximate level flight, and may result in slight climb or descent. Interpolation is acceptable.
 - Adjust pitch to stop climb/descent.
 - If pitch is increased, REDUCE expected speed range by delta KIAS.
 - If pitch is decreased, INCREASE expected speed range by delta KIAS.
 - Use FD in PTCH, ALT, HDG or ROLL modes only.
 - If FD/autopilot engaged in PTCH mode, one click of PTCH wheel results in 0.5 degrees of pitch change.

- CAUTION:**
- Do not use any AP/FD SPEED mode.
 - Do not use FLC mode or ATS.
- To avoid significant flap overspeed and flap damage:
- Transition from FLAPS 0 to FLAPS 30 must be accomplished in level flight.
 - Select FLAPS 45 at the start of final descent on the glideslope/glidepath.

TABLE D 3 - APPROACH - LEVEL FLIGHT - FLAPS 30 - GEAR DOWN										
			WEIGHT 1,000 LB (1,000 KG)						EXPECTED SPD RANGE KIAS	EXPECTED ΔKIAS PER 0.5° ΔPITCH
			26 (11,8)	30 (13,6)	35 (15,9)	40 (18,1)	45 (20,4)	48 (21,8)		
ALTITUDE (1,000 ft)	15	Pitch (°)	-2.0	-1.0	0.5	1.5	3.0	3.5	160 to 175	5
		N ₁ (%)	77.1	78.0	79.0	80.1	81.3	82.2		
		AoA	0.3	0.4	0.4	0.5	0.6	0.6		
	10	Pitch (°)	-2.0	-1.0	0.5	1.5	3.0	3.5	160 to 175	5
		N ₁ (%)	73.1	73.9	74.8	75.9	77.0	77.8		
		AoA	0.3	0.4	0.4	0.5	0.6	0.6		
	5	Pitch (°)	-2.0	-1.0	0.5	1.5	3.0	3.5	160 to 175	5
		N ₁ (%)	69.1	69.9	70.9	72.0	73.0	73.7		
		AoA	0.3	0.4	0.4	0.5	0.6	0.6		
	0	Pitch (°)	-2.0	-1.0	0.5	1.5	3.0	3.5	160 to 175	5
		N ₁ (%)	65.5	66.3	67.2	68.2	69.2	69.9		
		AoA	0.3	0.4	0.4	0.5	0.6	0.6		



Unreliable Airspeed (Cont'd)

- NOTE:**
1. Set N_1 to maintain AoA.
 2. Adjust pitch as necessary to maintain required glideslope guidance/glidepath.

▲ CAUTION: Select FLAPS 45 at the start of final descent on the glideslope/glidepath.

TABLE E - APPROACH - FLAPS 45 - GEAR DN - 3 DEGREE G/S

ALT (FT)		WEIGHT KG (LB)						EXPECTED SPD RANGE KIAS
		26 (11,8)	30 (13,6)	35 (15,9)	40 (18,1)	45 (20,4)	48 (21,8)	
15000	Pitch (°)	-3.0	-3.0	-3.0	-3.0	-3.0	-3.0	V _{REF} +/- 10
	N_1 (%)	65.5	67.7	71.2	74.9	77.3	76.7	
	AoA	0.5	0.5	0.5	0.5	0.5	0.5	
10000	Pitch (°)	-2.5	-2.5	-2.5	-2.5	-2.5	-2.5	
	N_1 (%)	60.8	62.5	66.3	69.8	72.8	74.6	
	AoA	0.5	0.5	0.5	0.5	0.5	0.5	
5000	Pitch (°)	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0	
	N_1 (%)	55.9	57.9	61.4	65.0	67.9	69.4	
	AoA	0.6	0.6	0.6	0.6	0.6	0.6	
0	Pitch (°)	-2.0	-1.0	-1.0	-1.0	-1.0	-1.0	
	N_1 (%)	52.5	53.4	56.7	59.9	62.8	64.2	
	AoA	0.6	0.6	0.6	0.6	0.6	0.6	

- COMPLETE -

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Advisory Messages

AVIONICS ADVISORY MESSAGES	
DATALINK ACARS / DATALINK MESSAGE	DATALINK has received a message.
DATALINK ATN CPDLC	DATALINK has received an ATN CPDLC message.
FANS CPDLC	FANS has received an CPDLC message.
FDR EVENT	Flight data recorder event marker.
IRS 3 ALIGNING (if installed)	IRS 3 is in align mode. Do not taxi.
SELCAL DATALINK	SELCAL has been activated by ground station on the DATALINK.
SELCAL HF 1 (2)	SELCAL has been activated by ground station on the respective high frequency radio.
SELCAL VHF 1 (2) (3)	SELCAL has been activated by ground station on the respective very high frequency radio.
XPDR IN STBY	The selected transponder is operational and in standby.

Status Messages

AVIONICS STATUS MESSAGES

AFCS 1 (2) INOP	Respective autopilot system is inoperative.
CAS MISCOMP	Any warning, caution, or aural miscompare exists for more than 20 seconds.
CVR FAIL	Cockpit voice recorder failure.
DATALINK FAIL	Datalink failure.
DISPLAY 1/A FAIL	FANS failure.
DCU 1 (2) AURAL INOP	Respective data concentrator unit internal fault affecting audio warning function. See Page AVNC 02-12
DCU 1 (2) INOP	Data concentrator unit 1 (2) failure. See Page AVNC 02-12
FD 1 (2) FAIL	Respective flight director is inoperative. See Page AVNC 02-21
FDR FAIL	Flight data recorder failure.
HGS FAIL	Head-up Guidance System is inoperative.
IAPS DEGRADED	Integrated avionics processing system (IAPS) degraded, failure of any of IAPS busses on the ground.
IAPS OVERTEMP	IAPS overtemperature.
IRS 1 (2) (3) IN ATT	IRS operating in attitude mode.
IRS 1 (2) (3) POWER FAULT	IRS operating on back-up power.
IRS 3 INOP	IRS 3 inoperative (if installed).
MFD X-TALK FAIL	The MFDs have lost the ability to cross-talk information. Cross-side radio tuning from the MFD may be inop.
RAAS INOPERATIVE	Runway awareness system inoperative. See Page AVNC 02-18
RAAS NOT AVAILABLE	Insufficient information for runway awareness to function. See Page AVNC 02-18
RDC FAIL	Remote data concentrator failure.
TAWS BASIC FAIL	TAWS modes 1-6 are failed.
TAWS FAIL	All TAWS functions have failed.

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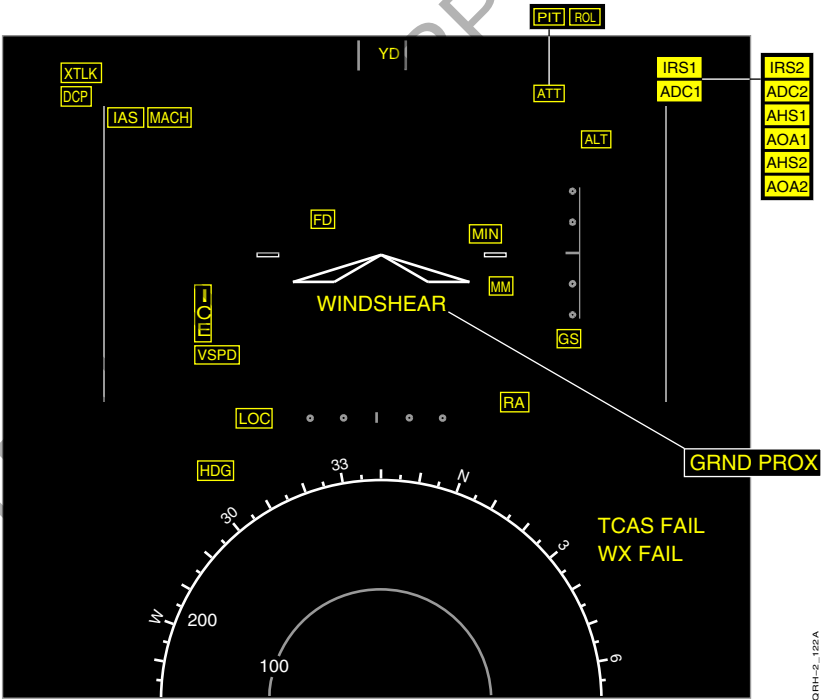
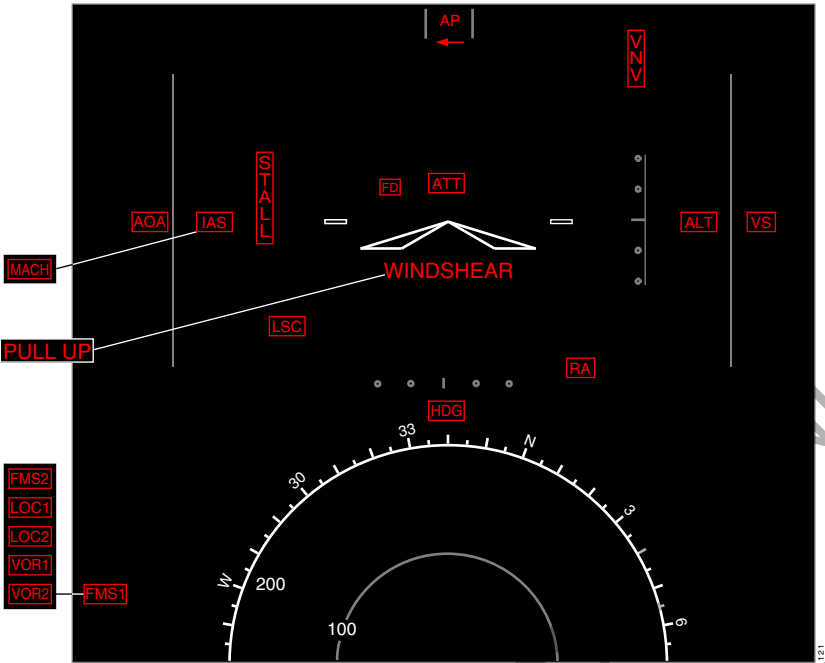
Status Messages (Cont'd)

TAWS TERR FAIL	Terrain awareness inoperative.
TAWS TERR NOT AVAIL	Terrain database not available for current location.
TAWS WINDSHEAR FAIL	Windshear detection has failed.
TSS FAN FAIL	TSS cooling fan is inoperative.
XPDR 1 (2) FAIL	The backup (not selected) transponder has failed.
YD 1 (2) INOP	Respective yaw damper is not engaged or is failed; other channel engaged/not failed.

TRAINING PURPOSES ONLY

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Display Annunciations



Display Annunciations (Cont'd)

FLAG NAME	DESCRIPTION	ASSOCIATED PROCEDURE
ALT	Altitude data from the ADC is missing or invalid.	Refer to ADC Failure procedure See AVNC 02-10
AOA	No valid AOA data is available.	Not applicable.
AP	Autopilot is automatically disconnected because data from the FGC is missing.	Refer to Autopilot Failure procedure See AVNC 02-11
ATT	Attitude data from the selected IRS is missing or invalid.	Refer to IRS Failure procedure See AVNC 02-15
FD (FGC failure)	FD command bars are removed for a detected fault in the FGC.	Refer to Single Flight Director Failed procedure See AVNC 02-21
FD (signal loss)	Signal(s) required by the selected mode (for example, LOC for approach) are invalid, the FD command bars are removed.	Refer to Single Flight Director Failed procedure See AVNC 02-21
FMS	Data from the active NAV source is missing or invalid.	Refer to Flight Management System CDU/ FMC Failure procedure See AVNC 02-14
GS	LOC is the active NAV source and vertical deviation data from the active NAV source is missing or invalid. Normal indication on a LOC only approach (no ILS GS available).	Check active NAV source.
HDG	Heading data from the IRS is missing or invalid.	Refer to IRS Failure procedure See AVNC 02-15
IAS	Airspeed data from the ADC is missing or invalid.	Refer to ADC Failure procedure See AVNC 02-10
LOC	LOC Data from the active NAV source is missing or invalid.	Check active NAV source.



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Display Annunciations (Cont'd)

FLAG NAME	DESCRIPTION	ASSOCIATED PROCEDURE
LSC	No stall warning speed comparison is available. Low Speed Cue will not show.	Not applicable.
MACH	Mach data from the ADC is missing or invalid when the airspeed reaches or exceeds 0.99 Mach.	Refer to ADC Failure procedure See AVNC 02-10
RA	RA Radio altitude data is missing or invalid.	Refer to Radio Altimeter Failure procedure See AVNC 02-18
STALL	Stall condition is detected by the Stall Protection Computer.	Refer to Stall Recovery procedure See OTHER 13-3
VNV	VNAV data from the FMS is missing or invalid.	Check VNAV source
VS	Vertical speed data from the ADC is missing or invalid.	Refer to ADC Failure procedure See AVNC 02-10

FLAG NAME	DESCRIPTION	ASSOCIATED PROCEDURE
ADC	Identifies which air data computer ADC is used for reversion.	Not applicable.
ALT	Difference between the two pressure altitude values is not within limits.	Refer to EFIS MISCOMP procedure See AVNC 02-5
ALTS	Altitude select mode – aborted capture.	Not applicable.
AOA	No valid AOA data is available.	Not applicable.
AP	Autopilot is engaged, with FD/AP synchronization in progress.	Not applicable.
ATT	Difference between the pilot-side and copilot-side attitude data source (both pitch and roll) is not within limits.	Refer to EFIS MISCOMP procedure See AVNC 02-5



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Display Annunciations (Cont'd)

FLAG NAME	DESCRIPTION	ASSOCIATED PROCEDURE
CCP	CCP is reported as failed.	Refer to Cursor Control Panel (CCP) Failure procedure See AVNC 02-11
CHECK DATABASE STATUS	A database is expired or effectivity cannot be determined due to the loss of system time. This message will show while on the ground on every MFD format until it is cleared or acknowledged by viewing the DATABASE EFFECTIVITY page.	Go to the DATABASE EFFECTIVITY page to review and acknowledge the message.
CONFIG ERROR	FSU reports that an electronic activation key (EAK) within the system cannot be recognized by the FSU, or there is a file server application (FSA) software mismatch .	Not applicable.
DCP	DCP is reported as failed.	Refer to Display Control Panel (DCP) Failure procedure See AVNC 02-12
FD	Difference between the two FD commands is not within limits.	Not applicable.
FMS	Displayed lateral deviations from each PFD not within limits during an LPV approach.	The LPV approach cannot be continued. Select a different type of approach.
FSU INOP	Fault is detected in the FSU (File Server Unit).	Not applicable.
GS	Glideslope compare function is active and the difference between the pilot-side and copilot-side heading glideslope deviation is not within limits.	Refer to EFIS MISCOMP procedure See AVNC 02-5



Display Annunciations (Cont'd)

FLAG NAME	DESCRIPTION	ASSOCIATED PROCEDURE
HDG	Difference between pilot-side and copilot-side heading values is greater than limits (6 degrees).	Refer to EFIS MISCOMP procedure See AVNC 02-5
IAS	Difference between pilot-side and copilot-side indicated airspeed values is greater than limits (10 knots).	Refer to EFIS MISCOMP procedure See AVNC 02-5
ICE	Either the left or right ice detection system detects ice and the associated-ice system is selected OFF.	Refer to ICE procedure See ICE 12-7
IRS	Identifies which IRS is used for reversion.	Not applicable.
LOC	Localizer compare function is active and the difference between the pilot-side and copilot-side localizer deviations is not within limits.	Refer to EFIS MISCOMP procedure See AVNC 02-5
LX FAIL	Lightning detection system data is failed.	Not applicable.
MACH	Difference between the pilot-side and copilot-side Mach values is greater than limits (0.03 Mach).	Refer to EFIS MISCOMP procedure and follow IAS flag guidance See AVNC 02-5
MAP	Fault is detected in data required to display the map. The MAP symbology is removed.	Not applicable.
NO VSI RA	TCAS vertical resolution advisory information cannot be provided, or vertical speed is not in view on the PFD.	Refer to TCAS VSI Resolution Advisory Fail procedure See AVNC 02-22
PIT	Difference between the pilot-side and copilot-side pitch attitude data is not within limits.	Refer to EFIS MISCOMP procedure See AVNC 02-5



Display Annunciations (Cont'd)

FLAG NAME	DESCRIPTION	ASSOCIATED PROCEDURE
RA	Radio altitude compare function is active and the difference between the pilot-side and copilot-side radio altitude supplied to the on-side and cross-side is not within limits.	Refer to EFIS MISCOMP procedure See AVNC 02-5
ROL	Difference between the pilot-side and copilot-side data is not within limits.	Refer to EFIS MISCOMP procedure See AVNC 02-5
TCAS FAIL	TCAS system is failed.	Refer to TCAS Failure procedure See AVNC 02-22
TERRAIN FAIL	TAWS system is failed. Message is removed if terrain overlay is deselected.	Not applicable.
TERRAIN NOT AVAILABLE	TAWS terrain data is not available but not failed. Graphical terrain data is removed from the display.	Not applicable.
T-1.5A	Disagreement between the tilt value commanded and the actual tilt value from the radar.	Not applicable.
VNV	Displayed vertical deviations from each PFD not within limits during an LPV approach.	The LPV approach cannot be continued. Select a different type of approach.
VSPD	VSPEEDs from the two active FMS differs by 2 knots or greater, or if the PFD received VSPEED data from only one of the two active FMS.	Refer to FMS V Speed Failure procedure See AVNC 02-15
WX FAIL	Weather radar information is not available and radar overlay selected.	Not applicable.



Display Annunciations (Cont'd)

FLAG NAME	DESCRIPTION	ASSOCIATED PROCEDURE
WX FAULT	Weather Radar display may be degraded in a manner that is not obvious to the crew. Some faults may cause the Weather Radar to stop displaying reflectively information.	Not applicable.
XTLK	PFD or MFD has lost communication with any other PFD or MFD. Removed when display is deselected by reversion mode.	Not applicable.

– COMPLETE –

Power Supply and Circuit Breaker Summary

CB NAME	BUS	CB LOCATION
ADC 1	DC ESS BUS	CBP4-C06
ADC 2	DC BUS 2	CBP2-K06
ADC STBY	BATT BUS	CBP1-P06
AFCS 2	DC BUS 1	CBP1-K04
AOA L	AC ESS BUS	CBP3-B05
AOA R	AC BUS 2	CBP2-A13
AUDIO C/PLT	BATT BUS	CBP1-Q07
AUDIO PLT	BATT BUS	CBP1-Q06
AUDIO PLT C/PLT	DC ESS BUS	CBP4-D06
AUX AOA	AC BUS 1	CBP1-C12
CCP 1	BATT BUS	CBP2-Q02
CCP 2	DC BUS 2	CBP2-L03
CREW FORCE SYST	DC BUS 1	CBP1-F09
CVR	APU BATT DIR	CBP5-B10
CVR CONT	DC ESS BUS	CBP4-D01
DATA LINK	DC BUS 1	CBP1-H09
DCP 1	DC ESS BUS	CBP4-D01
DCP 2	DC BUS 2	CBP2-L04
DISPLAY FAN	AC BUS 1	CBP1-C05
DISPLAY FAN CONT	DC BUS 1	CBP1-F01
DME 1	DC BUS 1	CBP1-H12
DME 2	DC BUS 2	CBP2-H12
EICAS DCU 1 CH A	BATT BUS	CBP2-Q09
EICAS DCU 1 CH A	DC ESS BUS	CBP4-C12
EICAS DCU 1 CH B	BATT BUS	CBP2-Q10
EICAS DCU 1 CH B	DC ESS BUS	CBP4-C13
EICAS DCU 2	MAIN BATT DIR	CBP6-B02
EICAS DCU 2 CH A	BATT BUS	CBP2-Q11
EICAS DCU 2 CH B	BATT BUS	CBP2-Q12
EICAS L LDU	DC BUS 1	CBP1-H05
EICAS R LDU	BATT BUS	CBP2-Q03
EVS CONTROL	DC BUS 1	CBP1-K03
EVS HEATER FAIRING	AC BUS 1	CBP1-D05
EVS HEATER WINDOW	AC BUS 1	CBP1-D07

Power Supply and Circuit Breaker Summary (Cont'd)

CB NAME	BUS	CB LOCATION
EVS HUD	DC BUS 1	CBP1-K01
EVS SENSOR	DC BUS 1	CBP1-K02
FDR CONT/ RDC	BATT BUS	CBP1-Q03
FDR PWR	AC ESS BUS	CBP3-C03
FMS CDU 1	BATT BUS	CBP1-M03
FMS CDU 2	DC BUS 2	CBP2-H09
FMS CDU 3	DC ESS BUS	CBP4-D09
FMS DATA LOAD	DC BUS 1	CBP1-H10
FSU	BATT BUS	CBP1-P11
GPS 1	BATT BUS	CBP1-N03
GPS 2	BATT BUS	CBP2-H11
HANDSETS/AEROCOM	DC BUS 1	CBP1-G04
HF COMM 1CPL	DC BUS 1	CBP1-K07
HF COMM 1 TX/RX	DC BUS 1	CBP1-K05
HF COMM 1 TX/RX	DC BUS 1	CBP1-K06
HF COMM 2 CPL	DC BUS 2	CBP2-L10
HF COMM 2 TX/RX	DC BUS 2	CBP2-L08
HF COMM 2 TX/RX	DC BUS 2	CBP2-L09
IAPS FAN L	BATT BUS	CBP1-Q05
IAPS FAN R	DC ESS BUS	CBP4-D02
IAPS L AFCS 1 MDC	BATT BUS	CBP1-Q04
IAPS L AFCS 2 FMS 1	BATT BUS	CBP1-M02
IAPS R AFCS 1 FMS 3	DC ESS BUS	CBP4-D03
IAPS R AFCS 2 FMS 2	DC BUS 2	CBP2-K03
IRIDIUM SATCOM	DC BUS 1	CBP1-G05
IRU 1	DC ESS BUS	CBP4-D11
IRU 1 SEC	BATT BUS	CBP1-Q08
IRU 2	DC BUS 2	CBP2-H10
IRU 2 SEC	BATT BUS	CBP1-Q09
IRU 3	DC BUS 1	CBP1-H13
IRU 3 SEC	BATT BUS	CBP1-Q10
ISI	DC ESS BUS	CBP4-D07
LDS	DC BUS 2	CBP2-H06

Power Supply and Circuit Breaker Summary (Cont'd)

CB NAME	BUS	CB LOCATION
MFD 1	BATT BUS	CBP2-Q04
MFD 1 HTR	DC BUS 1	CBP1-H04
MFD 2	DC BUS 2	CBP2-L02
MFD 2 HTR	DC BUS 2	CBP2-L06
PBA PWR 1	BATT BUS	CBP2-Q06
PBA PWR 1	DC BUS 1	CBP1-H06
PBA PWR 2	BATT BUS	CBP2-Q07
PBA PWR 2	DC BUS 1	CBP1-H07
PFD 1	DC ESS BUS	CBP4-C01
PFD 1 HTR	DC BUS 1	CBP1-H03
PFD 2	DC BUS 2	CBP2-L01
PFD 2 HTR	DC BUS 2	CBP2-L05
RAD ALT 1	DC BUS 1	CBP1-H14
RAD ALT 2	DC BUS 2	CBP2-H14
RUD TRIM	DC BUS 1	CBP1-F05
SATCOM – SRT	AC BUS 2	CBP2-B10
SATCOM ANT	DC BUS 2	CBP2-H13
SELCAL	DC BUS 1	CBP1-H01
TAWS	AC BUS 1	CBP1-B13
TCAS	AC ESS BUS	CBP3-C05
VHF COMM 1	BATT BUS	CBP1-Q02
VHF COMM 2	DC BUS 2	CBP2-H02
VHF COMM 3	DC ESS BUS	CBP4-D08
VHF NAV 1	DC ESS BUS	CBP4-D13
VHF NAV 2	DC BUS 2	CBP2-H15
WEATHER RDR R/T	DC BUS 1	CBP1-H08
XM RADIO	DC BUS 2	CBP2-H08
XM WEATHER	DC BUS 2	CBP2-H07
XPDR 1	DC ESS BUS	CBP4-H04
XPDR 2	DC BUS 2	CBP2-H04

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Challenger 605	DOOR TOC-1
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POWER SUPPLY AND CIRCUIT BREAKER SUMMARY

Power Supply and Circuit Breaker Summary	DOOR 03-8
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TRAINING PURPOSES ONLY

Challenger 605	DOOR TOC-2
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Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Doors
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Challenger 605	DOOR 03-1
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PASSENGER DOOR



“DOOR”

Condition: Two or more of the passenger door/handle sensors indicate open.

Objective: Reduce cabin pressure and descend to a safe altitude.

NOTE: **PASSENGER DOOR** comes on when two or more of the six door/handle sensors indicate that the door is open.

(1) Identify phase of flight:

➡ **On ground** – Go to (2)

➡ **In flight** – Go to (4)

(2) On ground:

(3) Door.....CHECK
correctly secured.

– COMPLETE –

(4) In flight:

(5) EMER DEPRESS.....PRESS IN

▲ CAUTION: Do not attempt to repressurize the airplane.

(6) Land at the nearest suitable airport.

(7) Emergency Descent ProcedureACCOMPLISH
(See Page ENVR 06–12)

– COMPLETE –

BAGGAGE DOOR

Condition: Baggage door is not properly latched.

Objective: Monitor cabin pressure and control manually if required.

(1) SEAT BLTS sign ON

(2) Cabin pressure CHECK

(3) Cabin pressure normal:

➡ **YES** – No further action required. **End of procedure.**

➡ **NO** – Go to (4)

(4) Cabin pressure NOT normal:

Switch to manual pressurization control:

(5) PRESS CONTMANUAL



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Challenger 605	DOOR 03-2
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BAGGAGE DOOR (Cont'd)

- (6) MAN ALT AS REQUIRED
to maintain cabin altitude at 8,000 feet.

NOTE: Select and hold to “UP” to increase cabin altitude; “DN” to reduce cabin altitude.

- (7) MAN RATE AS REQUIRED
adjust for crew and pax comfort.

When reaching target cabin altitude:

- (8) MAN ALT MID POS/CENTRE

NOTE: The cargo bay door is a plug type door, therefore, full cabin depressurization is not recommended.

- (9) Descent INITIATE
to 10,000 feet or lowest safe altitude, whichever is higher.

- (10) Land at the nearest suitable airport.

At 10,000 feet or lowest safe altitude:

- (11) MAN ALT HOLD DN
to decrease cabin altitude to landing field elevation.

- (12) MAN RATE AS REQUIRED
adjust for crew and pax comfort.

When reaching target cabin altitude:

- (13) MAN ALT MID POS/CENTRE

– COMPLETE –

PAX DOOR LATCH

Condition: One of the passenger door latches is unlocked.

Objective: Monitor cabin pressure.

- (1) SEAT BLTS sign ON

- (2) Cabin pressure CHECK

- (3) **Cabin pressure normal:**

➡ **YES** – No further action required. **End of procedure.**

➡ **NO** – Go to (4)

- (4) **Cabin pressure NOT normal:**

Switch to manual pressurization control:

- (5) PRESS CONT MANUAL



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Challenger 605	DOOR 03-3
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PAX DOOR LATCH (Cont'd)

(6) MAN ALT AS REQUIRED
to maintain cabin altitude at 8,000 feet.

NOTE: Select and hold to “UP” to increase cabin altitude; “DN” to reduce cabin altitude.

(7) MAN RATE AS REQUIRED
adjust for crew and pax comfort.

When reaching target altitude:

(8) MAN ALT MID POS/CENTRE

(9) Descent INITIATE
to 10,000 feet or lowest safe altitude, whichever is higher.

(10) Land at the nearest suitable airport.

At 10,000 feet or lowest safe altitude:

(11) MAN RATE HOLD DN
to decrease cabin altitude to landing field elevation.

(12) MAN RATE AS REQUIRED
adjust for crew and pax comfort.

When reaching target cabin altitude:

(13) MAN ALT MID POS/CENTRE

– COMPLETE –

PAX DOOR OUT HNDL

Condition: The passenger door outer handle is not stowed.

Objective: Monitor cabin pressure.

(1) SEAT BLTS sign ON

(2) Cabin pressure CHECK

(3) **Choose applicable:**

➡ **Cabin pressure is normal** – Go to (4)

➡ **Cabin pressure is NOT normal** – Go to (15)

(4) **Cabin pressure is normal:**

(5) Passenger door CHECK
visually, and ensure that 4 roll-over catch green witness
marks are correctly aligned.

(6) **All 4 witness marks are aligned:**

➡ **YES** – Go to (7)

➡ **NO** – Go to (13)



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Challenger 605	DOOR 03-4
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PAX DOOR OUT HNDL (Cont'd)

(7) All 4 witness marks are aligned:

(8) Passenger door, inner T-handlePULL
to stow outer handle.

When the handle is locked:

(9) Passenger door, inner T-handlePUSH IN
to stow inner T-handle, then close inner T-handle access
cover.

(10) PAX DOOR OUT HNDL stays on:

➡ **YES** – Go to (11)

➡ **NO** – No further action required. **End of procedure.**

(11) PAX DOOR OUT HNDL stays on:

(12) Land at the nearest suitable airport.

– COMPLETE –

(13) Not all 4 witness marks are aligned:

Switch to manual pressurization control:

(14) Go to (15)

– COMPLETE –

(15) Cabin pressure is NOT normal:

Switch to manual pressurization control:

(16) PRESS CONTMANUAL

(17) MAN ALTAS REQUIRED
to maintain cabin altitude at 8,000 feet.

NOTE: Select and hold to “UP” to increase cabin altitude; “DN” to reduce cabin altitude.

(18) MAN RATEAS REQUIRED
adjust for crew and pax comfort.

When reaching target cabin altitude:

(19) MAN ALTMID POS/CENTRE

(20) DescentINITIATE
to 10,000 feet or lowest safe altitude, whichever is higher.

(21) Land at the nearest suitable airport.

At 10,000 feet or lowest safe altitude:

(22) MAN ALTHOLD DN
to decrease cabin altitude to landing field elevation.

(23) MAN RATEAS REQUIRED
adjust for crew and pax comfort.



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Doors
--	-------

Challenger 605	DOOR 03-5
	REV 49, Nov 19/18

PAX DOOR OUT HNDL (Cont'd)

When reaching target cabin altitude:

(24) MAN ALTMID POS/CENTRE

– COMPLETE –

PAX DOOR STOW

Condition: Passenger door inner handle is not stowed.

Objective: Monitor cabin pressure.

(1) SEAT BLTS sign ON

(2) Cabin pressure CHECK

(3) Cabin pressure normal:

➡ **YES** – Go to (4)

➡ **NO** – Go to (8)

(4) Cabin pressure normal:

(5) Passenger door, inner handle STOW
Stow inner handle by pushing it down into the latched position.

(6) **PAX DOOR STOW** persists:

(7) No further action required.

– COMPLETE –

(8) Cabin pressure NOT normal:

Switch to manual pressurization control:

(9) PRESS CONT MANUAL

(10) MAN ALT AS REQUIRED
to maintain cabin altitude at 8,000 feet.

NOTE: Select and hold to “UP” to increase cabin altitude; “DN” to reduce cabin altitude.

(11) MAN RATE AS REQUIRED
adjust for crew and pax comfort.

When reaching target cabin altitude:

(12) MAN ALTMID POS/CENTRE

(13) Descent INITIATE
to 10,000 feet or lowest safe altitude, whichever is higher.

(14) Land at the nearest suitable airport.

At 10,000 feet or lowest safe altitude:



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Doors
---	--------------

Challenger 605	DOOR 03-6
	REV 49, Nov 19/18

PAX DOOR STOW (Cont'd)

- (15) MAN ALT HOLD DN
to decrease cabin altitude to landing field elevation.
- (16) MAN RATE AS REQUIRED
adjust for crew and pax comfort.

When reaching target cabin altitude:

- (17) MAN ALT MID POS/CENTRE

– COMPLETE –

TRAINING PURPOSES ONLY

Challenger 605	DOOR 03-7
	REV 49, Nov 19/18

Advisory Messages

DOORS ADVISORY MESSAGES	
PAX DOOR CLOSED	PAX door latches locked, handle stowed and airplane is on the ground.

Status Messages

DOORS STATUS MESSAGES	
EXTERNAL DOOR OPEN	All external doors not closed and latched.

TRAINING PURPOSES ONLY

Challenger 605	DOOR 03-8
	REV 49, Nov 19/18

Power Supply and Circuit Breaker Summary

CB NAME	BUS	CB LOCATION
DOOR CONT	DC ESS BUS	CBP4-B04
DOOR WARN	BATT BUS	CBP2-N03
DOOR WARN	DC ESS BUS	CBP4-B03

TRAINING PURPOSES ONLY

Challenger 605	ELEC TOC-1
	REV 49, Nov 19/18

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SCHEMATICS

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POWER SUPPLY AND CIRCUIT BREAKER

SUMMARY

Power Supply and Circuit Breaker Summary	ELEC 04-23
--	------------

Challenger 605	ELEC TOC-2
	REV 49, Nov 19/18

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TRAINING PURPOSES ONLY

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Electrical
---	-------------------

Challenger 605	ELEC 04-1
	REV 63, Mar 14/22

EMER PWR ONLY

or

Loss of All Normal Electrical Power

- Condition:** No power is available in AC bus 1 and 2, or essential bus is unpowered with no generators on-line.
- Objective:** Operate with degraded systems and attempt to regain normal power.

If ADG did not deploy:

(1) ADG manual deploy handle PULL
and leave extended.

(2) AC ESS BUS CHECK POWERED

When ADG power is established:

(3) STAB TRIM, CH 2 PRESS TO ENGAGE

(4) Engine Instruments VERIFY
N₁, N₂ & ITT.

➡ **FROM Quick Reference Card** ⬅

- (5) **Both engines confirmed failed:**
- ➡ **YES** – Go to (6)
- ➡ **NO** – Go to (8)
- (6) **Both engines confirmed failed:**
- (7) Double Engine Failure Procedure ACCOMPLISH
(See Page ENG 05-16)
- COMPLETE –**
- (8) **At least one engine still operating:**
- (9) R PACK PRESS OUT
- (10) Thrust levers POSITION MANUALLY
- (11) Cabin altitude MONITOR
- (12) Descent AS REQUIRED
- (13) GEN 1 and GEN 2 OFF/RESET
then ON.
- (14) **Any generator comes on-line:**
- ➡ **YES** – Go to (15)
- ➡ **NO** – Go to (17)
- (15) **Any generator comes on-line:**
- Re-establish normal power:**



Challenger 605	ELEC 04-2
	REV 49, Nov 19/18

EMER PWR ONLY (Cont'd)

or

Loss of All Normal Electrical Power (Cont'd)

(16) Go to (24)

(17) No generator comes on-line:

Below 20,000 feet:

(18) APU is available:

➡ YES – Go to (19)

➡ NO – Go to (32)

(19) APU is available:

(20) APU START

(21) APU GEN ON

(22) APU generator comes on-line:

➡ YES – Go to (23)

➡ NO – Go to (30)

(23) APU generator comes on-line:

(24) ADG manual deploy handle STOW

(25) ADG PWR TXFR OVERRIDE PRESS

(26) HYDRAULIC pump 3B AUTO/ON

(27) CAS messages REVIEW

(28) Affected systems RESTORE

(29) Altimeters CROSS CHECK

– COMPLETE –

(30) No APU generator comes on-line:

(31) Go to (32)

– COMPLETE –

(32) APU is NOT available:

(33) Descent CONTINUE
to 10,000 feet or lowest safe altitude, if required.

→ If wing anti-ice is on/required:

- WING ANTI-ICE STBY
- Leave icing conditions.

(34) CAS messages REVIEW



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Electrical
--	------------

Challenger 605	ELEC 04-3
	REV 49, Nov 19/18

EMER PWR ONLY (Cont'd)

or

Loss of All Normal Electrical Power (Cont'd)

The following significant systems are not available when on emergency power only:

- Yaw damper 2 and AFCS 2,
- Ground spoilers,
- Flaps channel 2,
- Stabilizer trim channel 1,
- Mach trim, aileron and rudder trim,
- Hydraulic pumps 1B, 2B & 3A,
- DME, FMS 2,
- Copilot's PFD, MFD, navigation & communication systems,
- Right pack,
- Right probe heaters, right AOA vane heater and right ice detector,
- Nose wheel steering system, anti-skid system,
- Both windshield heaters, right window heater,
- Copilot's instrument lights, NAV lights, and L & R landing lights,
- SPS altitude compensation, and
- Auto throttle system.

- NOTE:**
1. With Mach trim not available, do not exceed 250 KIAS (0.7 M), unless the autopilot is engaged and functioning normally.
 2. Without probe and AOA vane heaters, do not rely on stall protection system, ISI for altitude and airspeed, and cabin differential pressure on EICAS.
 3. Flaps will operate (extend or retract) at half speed.

(35) Land at the nearest suitable airport.

- NOTE:**
1. For flight planning, a fuel burn correction of 5% should be used when the ADG is deployed.
 2. If the ADG is not operating properly, and the airplane is on battery power only, then all electrical power may be lost after 30 minutes.
 3. Hydraulic pump 3B will be inoperative when on battery power only.

Prior to landing:

(36) ADG manual deploy handlePULL

(37) Actual landing distance INCREASE
as applicable by factor given below:



Challenger 605	ELEC 04-4
	REV 63, Mar 14/22

EMER PWR ONLY (Cont'd)

or

Loss of All Normal Electrical Power (Cont'd)

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.50 (150%)	1.85 (85%)

⚠ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

(38) HYD 3 LO PRESS is on:

- ➡ **YES** – Go to (39)
- ➡ **NO** – No further action required. **End of procedure.**

(39) HYD 3 LO PRESS is on:

(40) Landing Gear Manual Extension Procedure.....ACCOMPLISH

I (See Page GEAR 10–16)

– COMPLETE –

AC 1 AUTOXFER



Condition: AC bus auto transfer inhibited due to an overcurrent condition on AC bus 1.

Objective: Review inoperative systems.

(1) AC bus 1 is powered:

- ➡ **YES** – No further action required. **End of procedure.**
- ➡ **NO** – Go to (2)

(2) AC bus 1 is NOT powered (GEN 1 OFF and AC BUS 1 are on) :

(3) BUS INOP LIST..... REVIEW

(See Page ELEC 04–15)

– COMPLETE –

Challenger 605	ELEC 04-5
	REV 49, Nov 19/18

AC 2 AUTOXFER



Condition: AC bus auto transfer inhibited due to an overcurrent condition on AC bus 2.

Objective: Review inoperative systems.

(1) AC bus 2 is powered:

- ➡ **YES** – No further action required. **End of procedure.**
- ➡ **NO** – Go to (2)

(2) AC bus 2 is NOT powered (GEN 2 OFF and AC BUS 2 are on) :

- (3) BUS INOP LIST** REVIEW
(See Page ELEC 04–15)

– COMPLETE –

AC BUS 1

Condition: AC bus 1 is unpowered.

Objective: Attempt to reset.

- (1) GEN 1** OFF/RESET then ON.

(2) AC BUS 1 stays on:

- ➡ **YES** – Go to (3)
- ➡ **NO** – No further action required. **End of procedure.**

(3) AC BUS 1 stays on:

- (4)** Leave icing conditions.
- (5) BUS INOP LIST** REVIEW
(See Page ELEC 04–15)

NOTE:

1. Do not rely on low speed cue or Normalized AOA displayed on both PFDs.
2. Do not rely on air temperature indications.

– COMPLETE –

Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Electrical
---	-------------------

Challenger 605	ELEC 04-6
	REV 49, Nov 19/18

AC BUS 2

Condition: AC bus 2 is unpowered.

Objective: Attempt to reset.

(1) GEN 2..... OFF/RESET
then ON.

(2) **AC BUS 2 stays on:**

➡ **YES** – Go to (3)

➡ **NO** – No further action required. **End of procedure.**

(3) **AC BUS 2 stays on:**

(4) HYDRAULIC pump 3B ON

(5) ADC.....SELECT 1

(6) Leave icing conditions.

(7) BUS INOP LIST..... REVIEW

(See Page ELEC 04–15)

NOTE:

1. Do not rely on stall protection system.
2. Do not rely on ISI for altitude and airspeed.
3. Do not rely on cabin differential pressure on EICAS.

– COMPLETE –

AC ESS BUS

Condition: AC essential bus is unpowered.

Objective: Transfer to alternate source.

(1) AC ESS XFER.....PRESS IN

(2) **AC ESS BUS stays on:**

➡ **YES** – Go to (3)

➡ **NO** – No further action required. **End of procedure.**

(3) **AC ESS BUS stays on:**

(4) Leave icing conditions.

(5) BUS INOP LIST..... REVIEW

(See Page ELEC 04–15)

NOTE:

1. Do not rely on stall protection system.
2. Do not rely on ISI for altitude and airspeed.
3. Do not rely on cabin differential pressure on EICAS.

– COMPLETE –

Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Electrical
--	------------

Challenger 605	ELEC 04-7
	REV 49, Nov 19/18

APU BATTERY OFF

Condition: APU battery is not available.

Objective: Review inoperative systems.

NOTE: **APU BATTERY OFF** may appear momentarily during APU start.

- (1) BATT MASTERCHECK ON
 - (2) APU battery voltageMONITOR
 - (3) BUS INOP LIST REVIEW
- (See Page ELEC 04–15)

– COMPLETE –

APU GEN OFF

Condition: APU generator is offline with the APU running.

Objective: Attempt to reset.

- (1) APU GEN OFF/RESET
then ON.
 - (2) **APU GEN OFF** stays on:
 - ➡ YES – Go to (3)
 - ➡ NO – Go to (6)
 - (3) **APU GEN OFF** stays on:
 - (4) APU GEN OFF
 - (5) BUS INOP LIST REVIEW
- (See Page ELEC 04–15)

– COMPLETE –

- (6) **APU GEN OFF** goes out:
- (7) All AC loadsMONITOR

– COMPLETE –

Challenger 605	ELEC 04-8
	REV 49, Nov 19/18

APU GEN OVLD

Condition: Excessive electrical load on APU generator.

Objective: Reduce electrical load as required.

- (1) GEN load(s).....MONITOR
- (2) AC loads..... REDUCE
as necessary.

– COMPLETE –

BATTERY BUS

Condition: Battery bus is unpowered.

Objective: Review inoperative systems.

- (1) Land at the nearest suitable airport.
- (2) BUS INOP LIST..... REVIEW
(See Page ELEC 04–15)

– COMPLETE –

DC BUS 1

Condition: DC bus 1 is unpowered.

Objective: Attempt to repower bus.

- (1) MAIN BUS TIEPRESS IN
- (2) **DC BUS 1 stays on:**
 - ➡ **YES** – Go to (3)
 - ➡ **NO** – No further action required. **End of procedure.**
- (3) **DC BUS 1 stays on:**
- (4) Leave icing conditions.

NOTE: After reviewing the BUS INOP LIST, return to this checklist to complete the remaining steps.

- (5) BUS INOP LIST..... REVIEW
(See Page ELEC 04–15)
- (6) ATS DISC..... PRESS
- (7) Thrust levers..... POSITION MANUALLY

– COMPLETE –

Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Electrical
--	------------

Challenger 605	ELEC 04-9
	REV 51, Apr 05/19

DC BUS 2

Condition: DC bus 2 is unpowered.

Objective: Attempt to repower bus.

(1) MAIN BUS TIEPRESS IN

(2) **DC BUS 2 stays on:**

➡ **YES** – Go to (3)

➡ **NO** – No further action required. **End of procedure.**

(3) **DC BUS 2 stays on:**

NOTE: After reviewing the BUS INOP LIST, return to this checklist to complete the remaining steps.

(4) BUS INOP LIST REVIEW

(See Page ELEC 04–15)

NOTE: MACH TRIM is inoperative.

(5) Leave icing conditions.

(6) Approach speed $V_{REF} + 10$
minimum.

(7) Actual landing distance INCREASE
by applicable factor shown below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.40 (40%)	1.30 (30%)

I (8) STAB TRIM PRESS TO ENGAGE

NOTE: 1. With at least one STAB TRIM channel engaged, manually trim airplane as required.

2. With MACH TRIM inoperative, do not exceed 250 KIAS (0.7 M) unless the autopilot is engaged and functioning normally.

– COMPLETE –

Challenger 605	ELEC 04-10
	REV 52, Jun 20/19

DC EMER BUS

Condition: DC emergency bus is unpowered.

Objective: Review inoperative systems.

(1) Affected airplanes systems REVIEW

NOTE: Engine and APU fire extinguishing systems are inoperative:

- Firex bottles are inoperative,
- APU fuel feed shutoff valves will not close, and
- Engine hydraulic shutoff valves will not close.

(2) Land at the nearest suitable airport.

– COMPLETE –

DC ESS BUS

Condition: DC essential bus is unpowered.

Objective: Review inoperative systems.

(1) Leave icing conditions.

I (2) STAB TRIM PRESS TO ENGAGE

NOTE: 1. With at least one STAB TRIM channel engaged, manually trim airplane as required.

2. MACH TRIM is inoperative. Do not exceed 250 KIAS (0.7 M) unless the autopilot is engaged and functioning normally.

(3) L DISPLAYS SELECT
MFD REV.

(4) AFCS SEL SELECT
AFCS 2.

(5) YD 2 PRESS IN

NOTE: YD 1 will be inoperative. If YD 2 does not re-engage, follow procedure for **YAW DAMPER** failure.

(6) BUS INOP LIST REVIEW

(See Page ELEC 04-15)

(7) Land at the nearest suitable airport.

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Electrical
--	------------

Challenger 605	ELEC 04-11
	REV 49, Nov 19/18

GEN 1 OFF

Condition: Generator 1 is offline.

Objective: Attempt to reset.

(1) GEN 1 OFF/RESET
then ON.

(2) **GEN 1 OFF** stays on:

➡ **YES** – Go to (3)

➡ **NO** – Go to (7)

(3) **GEN 1 OFF** stays on:

(4) GEN 1 OFF

NOTE: 1. With generator 1 shut down, hydraulic pump 2B is load shed.

2. During single generator operations, AC utility busses are shed, unless on ground with flaps at 0 degrees.

(5) APU (if available) START
when at 20,000 feet or less.

(6) APU GEN ON

– COMPLETE –

(7) **GEN 1 OFF** goes out:

(8) AC voltage, load and frequency MONITOR

– COMPLETE –

GEN 2 OFF

Condition: Generator 2 is offline.

Objective: Attempt to reset.

(1) GEN 2 OFF/RESET
then ON.

(2) **GEN 2 OFF** stays on:

➡ **YES** – Go to (3)

➡ **NO** – Go to (8)



Challenger 605	ELEC 04-12
	REV 49, Nov 19/18

GEN 2 OFF (Cont'd)

(3) **GEN 2 OFF stays on:**

(4) GEN 2..... OFF

NOTE: 1. With generator 2 shut down, hydraulic pump 1B is load shed.

2. During single generator operations, AC utility busses are shed, unless on ground with flaps at 0 degrees.

(5) APU (if available).....START
when at 20,000 feet or less.

(6) APU GEN ON

(7) Altimeters CROSS CHECK

– COMPLETE –

(8) **GEN 2 OFF goes out:**

(9) AC voltage, load and frequencyMONITOR

(10) Altimeters CROSS-CHECK

– COMPLETE –

GEN 1 (2) OVLD

Condition: Excessive electrical load on indicated generator.

Objective: Reduce electrical load.

(1) AC loadsMONITOR
and load shed, as necessary.

(2) BUS INOP LIST REVIEW

(See Page ELEC 04–15)

– COMPLETE –

MAIN BATT OFF

Condition: Main battery is not available.

Objective: Review inoperative systems.

(1) BATT MASTERCHECK ON

(2) **MAIN BATT OFF stays on:**

➡ **YES** – Go to (3)

➡ **NO** – No further action required. **End of procedure.**



Challenger 605	ELEC 04-13
	REV 49, Nov 19/18

MAIN BATT OFF (Cont'd)

- (3) **MAIN BATT OFF** stays on:
- (4) Main battery voltageMONITOR
- (5) Affected airplane systems REVIEW

NOTE: 1. Back-up power to DCU 2 is not available;
2. During generator switching, **ENG MISCOMP** is on.

– COMPLETE –

ADG Inadvertent Deployment

Condition: Uncommanded deployment of the air driven generator.

Objective: Re-establish full electrical power and review operational penalties.

- NOTE:** 1. The AC essential bus and hydraulic pump No. 3B can be transferred back to either main generator, or the APU generator, if operating, by pressing the PWR TXFR OVERRIDE switch on the ADG control panel after the ADG manual deploy handle is re-stowed.
2. For flight planning, a fuel burn correction of 5% should be used when the ADG is deployed.

To re-establish normal operations:

- (1) ADG manual deploy handleSTOW
- (2) ADG PWR TXFR OVERRIDE PRESS
- (3) HYDRAULIC pump 3BAUTO
- (4) Altimeters CROSS CHECK

– COMPLETE –

Challenger 605	ELEC 04-14
	REV 49, Nov 19/18

Status Messages

ELECTRICAL STATUS MESSAGES	
AC ESS ALTN	AC essential bus is powered by AC bus 2.
APU BATT CHGR FAIL	APU battery charger failure or APU battery is not charging.
ESS TRU 1 (2) FAIL	Respective essential transformer rectifier unit (TRU) output is less than 18VDC and AC essential bus is not failed.
MAIN BATT CHGR FAIL	Main battery charger failure or main battery is not charging.

SCHEMATICS

BUS INOP LIST

NOTE: The bus inoperative list covers systems that are important to the approach and landing phases of flight. The list covers items that are NOT available. The list does not cover all electrically controlled and/or powered systems and is not a complete check of all circuit breaker panels to determine affected systems.

SYSTEM	SUB-SYSTEM	AC BUS 1	AC BUS 2	DC BUS 1	DC BUS 2	AC ESS BUS	DC ESS BUS	DC BATT BUS	APU BATT DIR BUS	MAIN BATT DIR BUS
ENGINE	IGN					A inop		B inop		
	START							L & R inop		
	OIL PRESS						R inop	L inop	BYPASS INDIC inop	
	THRUST REV						inop			

SCHEMATICS (CONT'D)

BUS INOP LIST (Cont'd)

SYSTEM	SUB-SYSTEM	AC BUS 1	AC BUS 2	DC BUS 1	DC BUS 2	AC ESS BUS	DC ESS BUS	DC BATT BUS	APU BATT DIR BUS	MAIN BATT DIR BUS
NAV/FT INST	PFD				2 inop		ISI, 1 inop			
	MFD				2 inop			1 inop		
	CCP/DCP				CCP 2, DCP 2 inop		DCP 1 inop	CCP 1 inop		
	ADC				2 inop		1 inop	STBY inop		
	IRS			♦ 3*	♦ 2		♦ 1	♦ 1, 2, 3*		
	CLK						inop			STBY PWR inop
	RAD ALT			1 inop	2 inop					
	NAV RCVR				2 inop		1 inop			
	DME			1 inop	2 inop					
	ATC XPDR				2 inop		1 inop			
	TCAS					inop				
	FMS				CDU 2 inop		CDU 3* inop	CDU 1 inop		
	GPS				2 inop			1 inop		
	TAWS	inop								
	WXR			inop	XM* inop					

LEGEND:

♦ – Indicates that the applicable system is powered by a second bus. The failure of the two marked busses (♦) will render the applicable system inoperative.

SCHEMATICS (CONT'D)

BUS INOP LIST (Cont'd)

* – If applicable (optional equipment).

SYSTEM	SUB-SYSTEM	AC BUS 1	AC BUS 2	DC BUS 1	DC BUS 2	AC ESS BUS	DC ESS BUS	DC BATT BUS	ADG BUS	DC EMER BUS
AIR COND. PRESS COOL	PRESS CONT				CPAM 2 inop		CPAM 1, EMER DE-PRESS inop	AUTO inop		
	PACK				R inop		L inop			
	RAM AIR							SOV inop		
	10TH-STAGE BLEED				R inop		L inop, ISOL inop			
	LEAK DET					BOTH inop				
	AVIONIC COOLING	DISPLAY FAN inop								
FUEL	PUMPS				R inop			L & APU inop.		
	SOV's									L, R & APU inop
	XFLOW							inop		
	TAIL TANK DUMP							SOV inop		
	TAIL TANK XFER	PRIM PUMP inop	SEC PUMP inop	AUTO XFER, PRIM PUMP CONT inop	SEC PUMP CONT inop			inop		
	GRAV XFLOW							inop		

SCHEMATICS (CONT'D)

BUS INOP LIST (Cont'd)

SYSTEM	SUB-SYSTEM	AC BUS 1	AC BUS 2	DC BUS 1	DC BUS 2	AC ESS BUS	DC ESS BUS	DC BATT BUS	ADG BUS	DC EMER BUS
HYD	PUMPS	2B inop, ♦ 3B	1B, 3A inop	2B CONT inop	1B, 3A CONT inop			3B CONT inop	♦ 3B	
	GAUGE			2 inop	1 inop			3 inop		
	EDP SOV									L, R inop
ICE/RAIN	COWL A/ICE							L, R inop		
	WING A/ICE			AUTO 1 inop	AUTO 2 inop		MAN 1 inop	MAN 2 inop		
	14TH-STAGE BLEED			L & ISOL inop	R inop					
	PROBE HEAT	TAT, AUX AOA inop	R AOA, R PITOT, R STATIC inop		ADS HTR R CONT inop	L AOA, L & STBY PITOT, L STATIC inop	ADS HTR L CONT inop			
	WSHLD	L inop	R inop	L CONT inop	R CONT inop					
	WINDOW		R inop		R CONT inop	L inop	L CONT inop			
	ICE DET		2 inop			1 inop				

Challenger 605	ELEC 04-19
	REV 49, Nov 19/18

SCHEMATICS (CONT'D)

BUS INOP LIST (Cont'd)

SYSTEM	SUB-SYSTEM	AC BUS 1	AC BUS 2	DC BUS 1	DC BUS 2	AC ESS BUS	DC ESS BUS	DC BATT BUS	MAIN BATT DIR BUS
LDG GEAR	BRAKES							MLG OVHT DET inop	
	ANTI-SKID			♦	♦				
	NWS			♦	♦				
	PSEU			♦ LGC DOOR 1, DOOR 2 ♦ SPLR 1	♦ LGC ♦ SPLR 2		DOOR WARN inop	♦ LGC DOOR1, DOOR2 ♦ SPLR 1 ♦ SPLR 2 NGDR CONT inop	
	WOW			♦ 1	♦ 2			♦ 1 & 2	
LIGHT	LAND/TAXI	L inop	R inop	PULSE inop					
	EXT	NAV inop		REAR, A. COLL inop	BCN, WING, A. COLL inop				
	INSTR			CHART HOLDER inop			PLT, CPLT CB PNL inop	PED, OVHD MAP LT 1, 2 inop	
EICAS	DCU 1						♦ A & B	♦ A & B	
	DCU 2							♦ A & B	♦ A & B
	LDU			L inop				R inop	
AUTO FLT	IAPS			♦ L	♦ R		♦ R	♦ L	
APU								♦ ECU	♦ ECU

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Electrical
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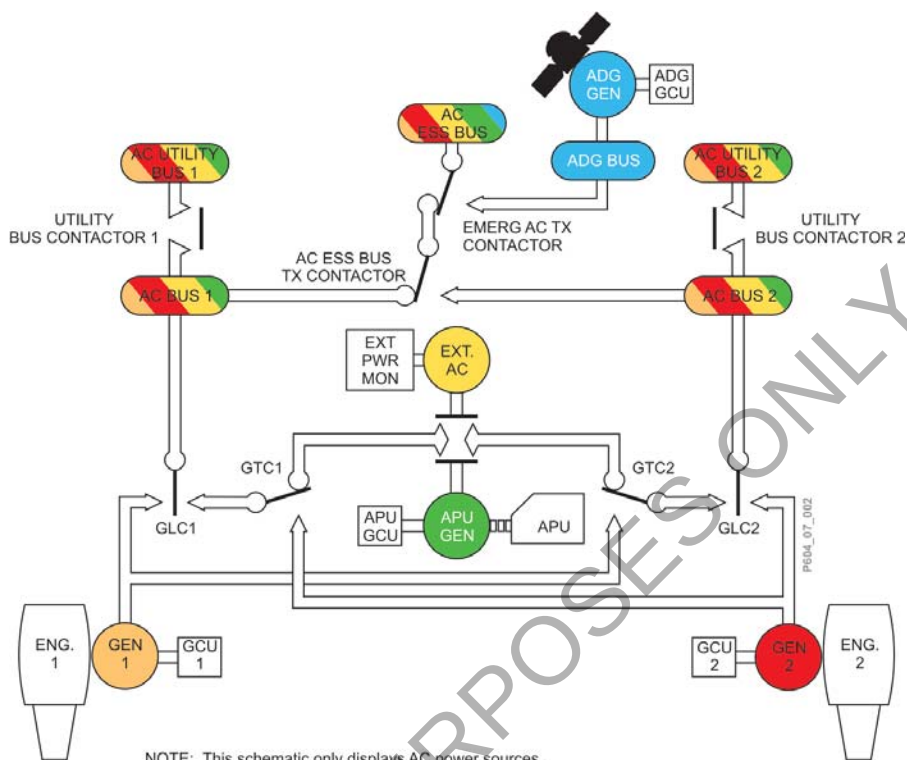
SCHEMATICS (CONT'D)**BUS INOP LIST (Cont'd)**

SYSTEM	SUB-SYSTEM	AC BUS 1	AC BUS 2	DC BUS 1	DC BUS 2	AC ESS BUS	DC ESS BUS	DC BATT BUS	MAIN BATT DIR BUS
COMM	VHF				2 inop		3* inop	1 inop	
	AUDIO CONT						♦ PLT, C/PLT	♦ PLT, C/PLT	
	HF			1 inop	2 inop				
	SELCAL			inop					

SYSTEM	SUB-SYSTEM	AC BUS 1	AC BUS 2	DC BUS 1	DC BUS 2	AC ESS BUS	DC ESS BUS	DC BATT BUS	DC EMER BUS
ELEC	GCU							1, 2, 3 inop	
	ADG							DEPLOY inop	
OXY	INDIC							QTY inop	
FIREX	LOOPS							BOTH inop	
	ENG BTL								BOTH inop
	APU BTL								inop
FLT CONT	FLAPS		PDU 2 inop		CH 2 inop	PDU 1 inop	CH 1 inop		
	TRIMS			RUD inop	AIL inop				
	STAB TRIM		HSTA CH 1 inop		HSTCU CH 1 inop	HSTA CH 2 inop	HSTCU CH 2 inop		
	STALL PROT						R CH inop	L CH, STICK PUSH inop	
	GROUND SPOILERS SOLENOID VALVE			PWR 1 inop	PWR 2 inop				

SCHEMATICS (CONT'D)

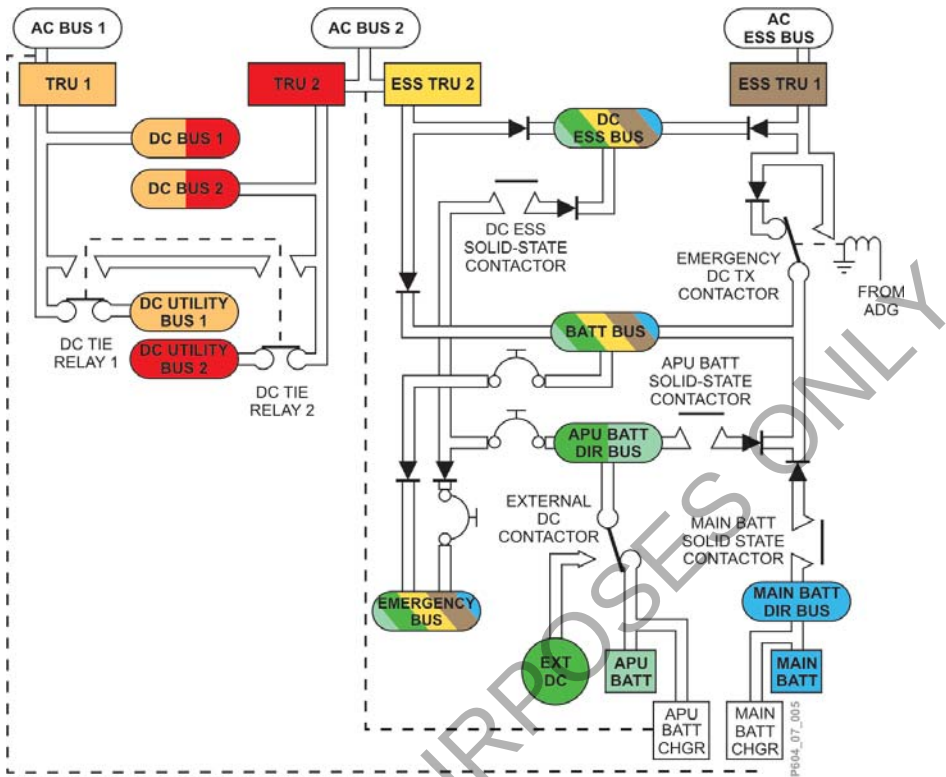
AC Electrical System



NOTE: This schematic only displays AC power sources.
Disregard the position of the GLCs and Generator Transfer Controllers (GTCs).

SCHEMATICS (CONT'D)

DC Electrical System



NOTE: This schematic only displays DC power sources. Disregard the positions of all contactors and ties.

Challenger 605	ELEC 04-23
	REV 49, Nov 19/18

Power Supply and Circuit Breaker Summary

CB NAME	BUS	CB LOCATION
115 VAC ESS FEED	AC BUS 1	CBP1-B05
115 VAC ESS FEED	AC BUS 2	CBP2-B08
AC ESS FEED	ADG BUS	CBP3-A08
AC ESS PWR CONT	MAIN BATT DIR	CBP6-A04
ADG AUTO	MAIN BATT DIR	CBP3-B05
ADG DEPLOY CONT AUTO	BATT BUS	CBP2-N06
ADG DEPLOY CONT MAN	BATT BUS	CBP2-N07
ADG DEPLOY SENSE	AC BUS 1	CBP1-C10
ADG DEPLOY SENSE	AC BUS 2	CBP2-C10
ADG HEATER	AC BUS 2	CBP2-C11
ADG MAN	MAIN BATT DIR	CBP6-B04
APU BATT CHGR	AC BUS 2	CBP2-C02
APU BATT CHGR O/P	APU BATT DIR	CBP5-B13
APU BATT CONT	APU BATT DIR	CBP5-B02
APU BATT FEED	DC EMERG BUS	CBP1-S07
BATT BUS POWER SENSE	DC ESS & BATT BUS CTRL	CBP2-M03
CONTACTOR LOGIC – ESS TRUS 1	DC ESS & BATT BUS CTRL	CBP2-M01
CONTACTOR LOGIC – ESS TRUS 2	DC ESS & BATT BUS CTRL	CBP2-M02
EMERG BUS FEED	BATT BUS	CBP1-M01
ESS AC XFER CONT	BATT BUS	CBP1-Q01
ESS TRU 1	AC ESS BUS	CBP3-A02
ESS TRU 2	AC BUS 2	CBP2-A08
ESS TRU SENSE 1	DC ESS	CBP4-A01
ESS TRU SENSE 2	DC ESS	CBP4-A02
ESS/EMER BUS FEED	APU BATT DIR	CBP5-B01
EXT AC PWR CONT	APU BATT DIR	CBP5-B12
FEED	AC UTILITY BUS 1	CBP1-D14
FEED	AC UTILITY BUS 2	CBP2-D14
FEED	DC BUS 1 CONTROL	CBP1-E08

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Electrical
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Power Supply and Circuit Breaker Summary (Cont'd)

CB NAME	BUS	CB LOCATION
FEED	DC BUS 2 CONTROL	CBP2-E08
FEED	DC UTILITY BUS 1	CBP1-E07
FEED	DC UTILITY BUS 2	CBP2-E07
FEED 1 – BATT BUS	DC ESS & BATT BUS CONTROL	CBP2-M07
FEED 1 – DC ESS	DC ESS & BATT BUS CONTROL	CBP2-M08
FEED 2 – BATT BUS	DC ESS & BATT BUS CONTROL	CBP2-M09
FEED 2 – DC ESS	DC ESS & BATT BUS CONTROL	CBP2-M10
FREQ CONV	AC UTILITY BUS 1	CBP1-D11
GALLEY FEED 1	DC UTILITY BUS 1	CBP1-J03
GALLEY FEED 1	DC UTILITY BUS 2	CBP2-J01
GALLEY FEED 2	DC UTILITY BUS 1	CBP1-J04
GALLEY FEED 2	DC UTILITY BUS 2	CBP2-J02
GCU 1	BATT BUS	CBP1-P01
GCU 2	BATT BUS	CBP1-P02
GCU 3	BATT BUS	CBP1-P03
MAIN BATT CHGR	AC BUS 1	CBP1-C02
MAIN BATT CHGR O/P	MAIN BATT DIR	CBP6-B06
MAIN BATT CONT	MAIN BATT DIR	CBP6-A03
MAIN BATT PWR SENSE	MAIN BATT DIR	CBP6-A01
MAIN BATT PWR SENSE REF	MAIN BATT DIR	CBP6-A02
PWR SENSE	AC UTILITY BUS 1	CBP1-D09
PWR SENSE	AC UTILITY BUS 2	CBP2-D09
PWR SENSE	DC BUS 1	CBP1-E12

Power Supply and Circuit Breaker Summary (Cont'd)

CB NAME	BUS	CB LOCATION
PWR SENSE	DC BUS 2	CBP2-E12
PWR SENSE	DC EMERG BUS	CBP1-R07
PWR SENSE	DC ESS BUS	CBP4-A03
PWR SENSE	DC UTILITY BUS 1	CBP1-E06
PWR SENSE	DC UTILITY BUS 2	CBP2-E06
RCCB CONT – APU BATT	DC ESS & BATT BUS CONTROL	CBP2-M05
RCCB CONT – DC ESS	DC ESS & BATT BUS CONTROL	CBP2-M06
RCCB CONT – MAIN BATT	DC ESS & BATT BUS CONTROL	CBP2-M04
TIE & UTIL	DC BUS 1 CONTROL	CBP1-E09
TIE & UTIL	DC BUS 2 CONTROL	CBP2-E09
TRU 1	AC BUS 1	CBP1-B02
TRU 1 SENSE	DC BUS 1 CONTROL	CBP1-E10
TRU 2	AC BUS 2	CBP2-B02
TRU 2 SENSE	DC BUS 2 CONTROL	CBP2-E10
UTIL BUS CONT	DC BUS 1	CBP1-E11
UTIL BUS CONT	DC BUS 2	CBP2-E11
UTILITY BUS 1 FEED	DC UTILITY BUS 1	CBP1-E07
UTILITY BUS PWR SENSE 2	DC UTILITY BUS 2	CBP2-E06
VOLTS & FREQ IND	ADG BUS	CBP3-A13

Challenger 605	ELEC 04-26
	REV 49, Nov 19/18

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TRAINING PURPOSES ONLY

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Electrical
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SUMMARY**

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Challenger 605	ENG 05-1
	REV 49, Nov 19/18

L (R) ENGINE FIRE (In Flight)

or

Severe Engine Damage (In Flight)



Fire Bell



OR



Condition: Indicated engine on fire (in flight).

Objective: Monitor and shutdown if necessary.

(1) **Identify phase of flight:**

➡ During takeoff after achieving V_1 – Go to (2)

➡ In flight – Go to (8)

(2) **During takeoff after achieving V_1 :**

(3) TakeoffCONTINUE

(4) AirplaneROTATE

When positive rate of climb is achieved:

(5) Landing gearRETRACT

(6) AirspeedMAINTAIN V_2

NOTE: If engine failure occurs above V_2 , maintain airspeed at current value (not more than $V_2 + 10$).



Challenger 605	ENG 05-2
	REV 51, Apr 05/19

L (R) ENGINE FIRE (In Flight) (Cont'd)

or

Severe Engine Damage (In Flight) (Cont'd)

(7) Go to (8)

(8) In flight:

After takeoff and at a safe altitude:

Affected engine:

(9) ATS DISC PRESS

(10) Thrust lever CONFIRM AND IDLE

(11) Thrust lever CONFIRM AND SHUT OFF

(12) ENG FIRE PUSH PRESS IN

(13) BOOST PUMP PRESS OUT

L (R) ENGINE FIRE stays on (after 10 seconds):

(14) BOTTLE ARMED PRESS
and hold to discharge.

L (R) ENGINE FIRE stays on (after additional 30 seconds):

(15) Other BOTTLE ARMED PRESS
and hold to discharge.

FROM Quick Reference Card

AFTER TAKEOFF CHECK (if applicable):

(16) Wing anti-ice system selected on:

➡ **YES** – Go to (17)

➡ **NO** – Go to (20)

(17) Wing anti-ice system selected on:

(18) 14TH STAGE ISOLPRESS IN

(19) Go to (21)

(20) Wing anti-ice NOT selected on:

(21) Both THRUST REVERSERS OFF

At not less than 400 feet AGL or the engine-out level off height:

(22) ThrustSET
to maximum continuous.

(23) AirspeedACCELERATE
to V_{FTO} .

(24) FLAPSUP
at a speed of not less than $V_{FTO}-5$.

(25) ClimbINITIATE

(26) CAS messages REVIEW



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
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Challenger 605	ENG 05-3
	REV 51, Apr 05/19

L (R) ENGINE FIRE (In Flight) (Cont'd)

or

Severe Engine Damage (In Flight) (Cont'd)

At a safe altitude:

- (27) In-Flight Engine
Failure/ShutdownACCOMPLISH
(See Page ENG 05-10)

NOTE: Do not operate the APU if engine rotor burst damage is suspected.

– COMPLETE –

L (R) ENGINE FIRE (On Ground)

or

Severe Engine Damage (On Ground)



Fire Bell



OR



Condition: Indicated engine on fire (on ground).

Objective: Secure indicated engine and extinguish fire.

- (1) Parking brakeSET

Affected engine:

- (2) Thrust leverSHUT OFF

- (3) ENG FIRE PUSHPRESS IN

- (4) Both BOOST PUMPsPRESS OUT

L (R) ENGINE FIRE stays on (after 10 seconds):

- (5) BOTTLE 1 and BOTTLE 2
ARMEDPRESS
and hold to discharge.

➡ **FROM Quick Reference Card** ⬅

- (6) Emergency Evacuation ProcedureACCOMPLISH
(See Page F-LDG 08-1)

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
--	--------

Challenger 605	ENG 05-4
	REV 49, Nov 19/18

ENGINE

Condition: N_1 , N_2 or ITT limits exceeded.

Objective: Prevent further exceedences. Shutdown if necessary.

(1) ATS DISC PRESS

Affected engine:

(2) Thrust lever CONFIRM AND RETARD
until engine responds and ITT is below limit range.

➡ FROM Quick Reference Card ⬅

(3) Engine limits continue to be exceeded:

➡ YES – Go to (4)

➡ NO – Go to (6)

(4) Engine limits continue to be exceeded:

(5) In-Flight Engine
Failure/Shutdown ACCOMPLISH
(See Page ENG 05-10)

– COMPLETE –

(6) Engine returns to normal parameters:

If N_1 or N_2 were exceeded:

(7) Both thrust levers RETARD
to below 79% N_2 .

(8) Both ENG SPEED OFF
to prevent asymmetric thrust.

NOTE: The APR system is inoperative when the ENG SPEED switches are selected OFF.

(9) Both thrust levers ADVANCE

NOTE: Thrust lever controls are more sensitive to changes.

(10) Engine accelerates to desired power:

➡ YES – Go to (11)

➡ NO – Go to (13)

(11) Engine accelerates to desired power:

(12) Engine indications MONITOR

NOTE: Exercise caution to prevent overspeed during thrust reverser use, when operating with ENG SPEED switches OFF.

– COMPLETE –



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
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Challenger 605	ENG 05-5
	REV 52, Jun 20/19

ENGINE (Cont'd)

(13) Engine not accelerating as desired:

- (14) In-Flight Engine
 Failure/ShutdownACCOMPLISH
 (See Page ENG 05-10)

– COMPLETE –

L (R) ENGINE OIL PRESS

or

Engine Oil Pressure Low



“ENGINE OIL”

Condition: Indicated engine oil pressure is low.

Objective: Monitor and shutdown if necessary.

- (1) ATS DISC PRESS

Affected engine:

- (2) Thrust lever CONFIRM AND IDLE
 (3) Oil pressure CHECK

- (4) **L (R) ENG OIL PRESS** is on AND
 affected engine oil pressure is less than 25 psi:

- ➡ YES – Go to (5)
 ➡ NO – Go to (7)

- (5) **L (R) ENG OIL PRESS** is on AND
 affected engine oil pressure is less than 25 psi:

- (6) In-Flight Engine
 Failure/ShutdownACCOMPLISH
 (See Page ENG 05-10)

– COMPLETE –

- (7) **L (R) ENG OIL PRESS** is on and associated engine oil
 pressure is normal OR **L (R) ENG OIL PRESS** is out and
 affected engine oil pressure is less than 25 psi:

Affected engine:

- (8) Thrust leverADJUST
 as required.
 (9) Engine indicationsMONITOR

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
--	--------

Challenger 605	ENG 05-6
	REV 49, Nov 19/18

L (R) ENGINE START SOV

Condition: Indicated starter remains engaged above 55% N₂ for more than 15 seconds.

Objective: Attempt to disengage starter.

(1) Identify phase of flight:

➡ **On ground** – Go to (2)

➡ **In flight** – Go to (12)

(2) On ground:

(3) Affected ENG STOP PRESS

(4) Starter cutout confirmed:

➡ **YES** – No further action required. **End of procedure.**

➡ **NO** – Go to (5)

(5) Starter does NOT cutout:

(6) Affected engine SHUTDOWN

(7) Affected 10TH STAGE bleed PRESS OUT

(8) APU LCV PRESS OUT

(9) Engine is still motoring:

➡ **YES** – Go to (10)

➡ **NO** – No further action required. **End of procedure.**

(10) Engine is still motoring:

(11) Other engine SHUTDOWN

– COMPLETE –

(12) In flight:

(13) Affected ENG STOP PRESS

(14) IGNITION A (B) PRESS OUT

(15) Affected ENG START circuit
breaker (L: 1L10) (R: 1L9) PULL

(16) Engine is still motoring:

➡ **YES** – Go to (17)

➡ **NO** – No further action required. **End of procedure.**

(17) Engine is still motoring:

(18) Affected 10 ST B/AIR SOV circuit
breaker (L: 4C11) (R: 2F11) PULL

(19) Land at the nearest suitable airport.

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
--	--------

Challenger 605	ENG 05-7
	REV 49, Nov 19/18

L (R) JETPIPE OVHT



“JETPIPE OVERHEAT”

Condition: Overheat detected in the indicated jetpipe and/or pylon area.

Objective: Attempt to control and shutdown if necessary.

(1) ATS DISC PRESS

Affected engine:
(2) Thrust lever CONFIRM AND RETARD
slowly until message goes out.

➤

FROM Quick Reference Card

◀

- (3) L (R) JETPIPE OVHT stays on:

➡ YES – Go to (4)

➡ NO – No further action required. End of procedure.

(4) L (R) JETPIPE OVHT stays on:

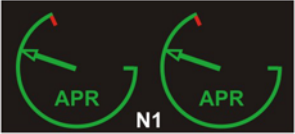
(5) In-Flight Engine Failure/ShutdownACCOMPLISH

(See Page ENG 05–10)

– COMPLETE –

Challenger 605	ENG 05-8
	REV 53, Sep 17/19

APR CMD SET



Condition: APR inadvertently activated on both engines.

Objective: Adjust to normal range (in flight).

(1) Identify phase of flight:

- ➡ **On ground** – Do not take off. **End of procedure.**
- ➡ **In flight** – Go to (2)

(2) In flight:

- (3) Thrust levers.....ADJUST to within normal range.
- (4) APR..... OFF
- (5) Engine performanceMONITOR during approach.

– COMPLETE –

Challenger 605	ENG 05-9
	REV 53, Sep 17/19

APR INOP

Condition: APR is failed or is not armed.

Objective: Initially select off and monitor.

(1) APR OFF

(2) Both ENG SPEEDCHECK ON

(3) Engine indications are normal:

➡ **YES** – Go to (4)

➡ **NO** – Go to (10)

(4) Engine indications are normal:

(5) APRARM

(6) APR INOP stays on:

➡ **YES** – Go to (7)

➡ **NO** – No further action required. **End of procedure.**

(7) APR INOP stays on:

(8) APR OFF

(9) Go to (11)

(10) Engine indications are NOT as expected:

(11) At a safe altitude:

(12) ATS DISC PRESS

(13) Both thrust leversRETARD
to below 79% N_2 .

(14) Both ENG SPEED OFF

(15) Thrust levers ADVANCE
as required.

- NOTE:**
1. Thrust lever controls are more sensitive to changes.
 2. Due to the possibility of N_1 overspeed, monitor the N_1 if more than idle reverse thrust is used after landing. Do not exceed 90% N_1 reverse thrust.
 3. Use APR-off performance. Refer to the Airplane Flight Manual, Chapter 6; PERFORMANCE.

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
--	--------

Challenger 605	ENG 05-10
	REV 49, Nov 19/18

L (R) ENGINE

or

In-Flight Engine Failure/Shutdown

Condition: One engine inoperative in-flight.

Objective: Configure the airplane for single engine flight.

Accomplish an engine shutdown only when flight conditions permit.

Affected engine:

- (1) Thrust lever CONFIRM AND IDLE
- (2) Thrust lever CONFIRM AND SHUT OFF

→ If **TAIL TANK TRANSFER OVRD** is pressed in:

- TAIL TANK TRANSFER,
MANUAL OVRD PRESS OUT

→ If **anti-ice system** is on/required:

- 14TH STAGE ISOL PRESS IN
- Affected COWL ANTI-ICE PRESS OUT

(3) **Choose applicable:**

- ➡ **HYD 1 (2) HI TEMP is out** – Go to (4)
- ➡ **HYD 1 (2) HI TEMP is on** – Go to (7)

(4) **HYD 1 (2) HI TEMP is out:**

- (5) Applicable HYDRAULIC pump ON
- (6) Go to (8)

(7) **HYD 1 (2) HI TEMP is on:**

- (8) Affected PACK PRESS OUT
- (9) Affected 10TH STAGE bleed PRESS OUT
- (10) Affected GEN OFF

Below 20,000 feet:

(11) **APU is available:**

- ➡ **YES** – Go to (12)
- ➡ **NO** – Go to (18)

(12) **APU is available:**

- (13) APU START
- (14) APU GEN ON



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
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Challenger 605	ENG 05-11
	REV 49, Nov 19/18

L (R) ENGINE (Cont'd)

or

In-Flight Engine Failure/Shutdown (Cont'd)

(15) PACKs..... TRANSFER
to APU bleed air.

NOTE: Bleed air extraction from the APU is not permitted above 15,000 feet.

(16) Affected PACKPRESS IN

(17) Go to (19)

(18) APU is NOT available:

(19) Fuel systemCHECK
and monitor fuel quantity and balance.

(20) Affected BOOST PUMPPRESS OUT

(21) Leave icing conditions.

NOTE: Icing conditions exist in flight at a TAT of 10°C (50°F) or colder, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or colder.

(22) Engine damage suspected or intentional shutdown:

➔ **YES** – Go to (23)

➔ **NO** – Go to (27)

(23) Engine damage suspected or intentional shutdown:

(24) MFD or CDU, ATC/TCAS mode..... TA ONLY

(25) Land at the nearest suitable airport.

(26) Single Engine Approach and
Landing ProcedureACCOMPLISH

(See Page ENG 05-29)

– COMPLETE –

(27) Engine relight is possible:

NOTE: 1. Do not attempt to relight an engine that is suspected to be not serviceable.

2. Relight engine using starter-assisted cross bleed whenever possible.

(28) Select applicable procedure:

- Cross Bleed Relight/Starter Assisted(See Page ENG 05-25)
- Windmill Relight (See Page ENG 05-27)

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
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Challenger 605	ENG 05-12
	REV 50, Mar 25/19

L (R) ENG MISCOMP

Condition: Engine parameters do not agree with each other.

Objective: Determine reliable engine indication source.

- (1) Multifunction Flight Displays (both).....EICAS PAGE
- (2) Affected engine's parameters
(either left or right) CROSS CHECK
to find disagreement.
- (3) Affected engine MAINTAIN WITHIN LIMITS
using reliable EICAS page.

- COMPLETE -

L (R) REV UNSAFE

Condition: Indicated thrust reverser received a deploy command while not armed.

Objective: Do not operate indicated thrust reverser.

- I (1) Affected thrust reverser OFF

- ⚠ CAUTION:**
1. Do not arm affected thrust reverser during flight.
 2. Use remaining thrust reverser carefully upon landing.

- COMPLETE -

Challenger 605	ENG 05-13
	REV 59, Jun 02/21

L (R) REV UNLOCKED



L or R THRUST REVERSER

Condition: Reverser movement detected without a deploy command.

Objective: Attempt to re-stow indicated reverser. Shutdown engine if necessary.

(1) Identify phase of flight:

- ➡ **In flight** – Go to (2)
- ➡ **On ground** – Go to (44)

(2) In flight:

Affected engine:

- (3) Thrust lever CONFIRM AND IDLE
- (4) ATS DISC PRESS
- (5) THRUST REVERSER, EMER STOW PRESS IN
- L (R) REV UNLOCKED** will stay on.
- (6) Airspeed 200 KIAS maximum.

➡ **FROM Quick Reference Card** ⬅

- (7) WING ANTI-ICE OFF
- (8) Affected COWL ANTI-ICE PRESS OUT

(9) Thrust reverser stows:

- ➡ **YES** – Go to (10)
- ➡ **NO** – Go to (19)

I (10) Thrust reverser stows:

⚠ CAUTION: Limit airspeed to not more than 200 KIAS.

- (11) Thrust levers AS REQUIRED
- (12) Both THRUST REVERSERS OFF

⚠ CAUTION: Do not arm thrust reverser.

- (13) Land at the nearest suitable airport.



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
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Challenger 605	ENG 05-14
	REV 49, Nov 19/18

L (R) REV UNLOCKED (Cont'd)

→ If anti-ice is on/required:

- WING ANTI-ICE NORM
- Affected COWL ANTI-ICE PRESS IN

Prior to landing:

- (14) TAWS WARNING, FLAPS OFF PRESS IN
- (15) TAWS WARNING, RAAS OFF (if installed) PRESS IN
- (16) Landing flaps 20 DEGREES
- (17) Approach speed $V_{REF} + 22$ minimum.
- (18) Actual landing distance INCREASE by a factor of 1.70 (70%).

– COMPLETE –

(19) Thrust reverser does NOT stow:

NOTE: Indications are thrust lever locked at idle, and continued airplane buffet & yaw.

Affected engine:

- (20) Thrust lever CONFIRM AND SHUT OFF

▲ CAUTION: Do not relight engine.

- (21) Land at the nearest suitable airport.
- (22) TAIL TANK TRANSFER, MANUAL OVRD PRESS OUT
- (23) Affected BOOST PUMP PRESS OUT
- (24) Fuel system CHECK and monitor fuel quantity and balance.

→ If anti-ice is on/required:

- 14TH STAGE ISOL PRESS IN
- WING ANTI-ICE NORM
 - Leave icing conditions (if possible).

- (25) Applicable HYDRAULIC pump ON
- (26) Affected PACK PRESS OUT
- (27) Affected 10TH STAGE bleed PRESS OUT
- (28) Affected GEN OFF
- (29) Altimeters CROSS-CHECK



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
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Challenger 605	ENG 05-15
	REV 49, Nov 19/18

L (R) REV UNLOCKED (Cont'd)

At 20,000 feet or less:

(30) APU is available:

- ➡ **YES** – Go to (31)
- ➡ **NO** – Go to (38)

(31) APU is available:

- (32) APU START
- (33) APU GEN ON
- (34) Altimeters CROSS CHECK
- (35) PACKs TRANSFER
to APU bleed air.
- (36) Affected PACK PRESS IN
- (37) Go to (39)

(38) APU is NOT available:

Prior to landing:

- (39) TAWS WARNING, FLAPS OFF PRESS IN
- (40) TAWS WARNING, RAAS OFF (if
installed) PRESS IN
- (41) Landing flaps 20 DEGREES
- (42) Approach speed $V_{REF}+22$
minimum.
- (43) Actual landing distance INCREASE
by a factor 1.70 (70%).

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

(44) On ground:

▲ CAUTION: If **L** or **R REV UNLOCKED** is on during ground operations, ensure that the affected thrust lever is at IDLE. Do not cycle the thrust reverser(s). Maintenance inspection is required.

– COMPLETE –

Double Engine Failure

Condition: Both L ENGINE and R ENGINE are on or evidence of loss of thrust on both engines.

Objective: Attempt to relight at least one engine.

- (1) CONT IGNITIONPRESS IN

(2) Airspeed 240 KIAS minimum.

FROM Quick Reference Card

- (3) Engine instruments.....VERIFY
N₁, N₂ and ITT.

If engines continue to run-down:

- (4) Thrust levers (both)SHUT OFF
- (5) ADG manual deploy handlePULL
and leave extended.
- (6) AC ESS BUS.....CHECK POWERED

When ADG power is established:

- (7) STAB TRIM, CH 2PRESS TO ENGAGE

(8) Oxygen masks (if required)DON
Set to 100%.

(9) Crew communication.....ESTABLISH

(10) Both PASS SIGNS ON

(11) APU (if available).....START
when at 20,000 feet or less.

(12) APU GEN ON

(13) Both PACKsPRESS OUT

(14) WING ANTI-ICE OFF

(15) Both COWL ANTI-ICEsPRESS OUT

Challenger 605	ENG 05-17
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Double Engine Failure (Cont'd)

⚠WARNING: To avoid thermal seizure, increase airspeed as required to maintain at least 4% N₂ indicated. If the N₂ is allowed to drop to 0%, the engine may thermally seize and may not rotate even at higher airspeeds or with starter engagement.

- NOTE:**
- Between 21,000 feet and 10,000 feet, a minimum of 12% N₂ is necessary for a windmill relight.
 - At 10,000 feet or less, a minimum of 10% N₂ is necessary for a windmill relight.
 - Acceleration to V_{MO} is recommended to attain the necessary N₂ for a windmill relight.
 - The altitude loss when accelerating from 240 KIAS to V_{MO}, could be greater than 5,000 feet.
 - A push-over to as steep as 15 degrees nose down may be required.

(16) Choose applicable:

- ➡ **Windmill relight is possible** – Go to (17)
- ➡ **APU bleed air relight is possible** – Go to (46)
- ➡ **No relight possible** – Go to (79)

(17) Windmill relight is possible:

(18) Glide airspeed INCREASE
to V_{MO}.

⚠CAUTION: The resulting high rate of descent (up to 7,800 fpm) will cause a significant reduction of available gliding distance.

(19) Descent INITIATE
to 21,000 feet or less, at V_{MO}.

(20) Both BOOST PUMPS CONFIRM PRESSED IN

At 21,000 feet or less, attempt to start both engines at the same time.

(21) When ITT is 90°C or colder and N₂ is 12% or greater (between 21,000 feet and 10,000 feet) or N₂ is 10% or greater (10,000 feet or less):

(22) CONT IGNITION CONFIRM PRESSED IN

(23) Thrust levers (both) IDLE

(24) Engine indications MONITOR
engine acceleration, ITT and oil pressure.



Challenger 605	ENG 05-18
	REV 49, Nov 19/18

Double Engine Failure (Cont'd)

**(25) At least one engine relights within 25 seconds
(stable IDLE within 2 minutes):**

- ➡ **YES** – Go to (26)
- ➡ **NO** – Go to (39)

**(26) At least one engine relights within 25 seconds
(stable IDLE within 2 minutes):**

(27) Thrust lever(s) AS REQUIRED

Operating engine(s):

(28) Applicable GEN(s) OFF/RESET
then ON.

(29) Applicable PACK(s) PRESS IN

NOTE: Use only one air-conditioning pack during single engine operations, unless the APU is the bleed source.

(30) CONT IGNITION PRESS OUT

Re-establish normal power:

(31) ADG manual deploy handle STOW

(32) ADG PWR TXFR OVERRIDE PRESS

(33) CAS messages REVIEW

(34) Affected systems RESTORE

(35) Altimeters CROSS-CHECK

(36) Only one engine is operating:

- ➡ **YES** – Go to (37)
- ➡ **NO** – No further action required. **End of procedure.**

(37) Only one engine is operating:

(38) In-Flight Engine
Failure/Shutdown ACCOMPLISH

(See Page ENG 05–10)

– COMPLETE –

(39) No engine relights within 25 seconds:

(40) Both thrust levers SHUT OFF

(41) Another windmill relight attempt is possible:

- ➡ **YES** – Go to (42)
- ➡ **NO** – Go to (46)

(42) Another windmill relight attempt is possible:

(43) Airspeed MAINTAIN V_{MO}



Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Engine
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Challenger 605	ENG 05-19
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Double Engine Failure (Cont'd)

- (44) Wait 30 seconds, then repeat windmill relight procedure.
- (45) Go to (21)
- (46) APU bleed air relight is possible:**
- (47) Airspeed MAINTAIN
240 KIAS minimum.
- (48) Both BOOST PUMPs CONFIRM PRESSED IN
- (49) 10TH STAGE bleeds PRESS OUT
- (50) APU LCV PRESS IN
- (51) Attempt to start one engine at a time.**
- At 15,000 feet or less:**
- (52) CONT IGNITION CONFIRM PRESSED IN
- (53) L (R) ENG START PRESS
- When ITT is 90°C or colder and N₂ reaches 28%:**
- (54) Thrust lever IDLE
- (55) Engine indications MONITOR
engine acceleration, ITT and oil pressure.
- (56) Engine relights within 25 seconds (stable IDLE within 2 minutes):**
- ➡ **YES** – Go to (57)
- ➡ **NO** – Go to (71)
- (57) Engine relights within 25 seconds (stable IDLE within 2 minutes):**
- (58) Thrust lever AS REQUIRED
- Operating engine:**
- (59) Applicable GEN OFF/RESET
then ON.
- (60) APU LCV PRESS OUT
- (61) Applicable 10TH STAGE bleed PRESS IN
- (62) Applicable PACK PRESS IN
- NOTE:** Use only one air-conditioning pack during single engine operations, unless the APU is the bleed source.
- (63) CONT IGNITION PRESS OUT
- Re-establish normal power:**
- (64) ADG manual deploy handle STOW
- (65) ADG PWR TXFR OVERRIDE PRESS



Double Engine Failure (Cont'd)

- (66) CAS messages..... REVIEW
- (67) Affected systems RESTORE
- (68) Altimeters CROSS CHECK
- (69) Other engine..... START
using Cross Bleed or Windmill Relight procedures.

NOTE: If only one engine is operating after trying to restart using the Cross Bleed Relight (Starter Assisted) or the Windmill Relight procedure, proceed to In-Flight Engine Failure/Shutdown procedure to secure the inoperative engine (See Page ENG 05–10).

- (70) Select applicable procedure:
 - Cross Bleed Relight/Starter Assisted (See Page ENG 05–25)
 - Windmill Relight(See Page ENG 05–27)

(71) Engine does NOT relight within 25 seconds:

- (72) Affected thrust leverSHUT OFF
- (73) Affected ENG STOP PRESS

(74) Another APU bleed air relight attempt is possible:

- ➡ **YES** – Go to (75)
- ➡ **NO** – Go to (79)

(75) Another APU bleed air relight attempt is possible:

- (76) Airspeed MAINTAIN
240 KIAS minimum.
- (77) Attempt relight on other engine.
- (78) Go to (51)

(79) No relight possible:

All Engine Out Procedure:

- (80) CONT IGNITIONPRESS OUT
- (81) AirspeedESTABLISH
best glide speed.

AIRPLANE WEIGHT	TARGET AIRSPEED	RATE OF DESCENT (FLAPS 0 DEGREES)
46,000 LB (20,865 KG)	211 KIAS	1,835 fpm
30,000 LB (13,607 KG)	173 KIAS	1,600 fpm

NOTE: Target glide speed provides a range of approximately 2.5 NM for each 1,000 feet altitude.



Challenger 605	ENG 05-21
	REV 49, Nov 19/18

Double Engine Failure (Cont'd)

(82) Descent and approach PLAN

NOTE: Recommended approach is a descending 360 degrees turn starting at 5,000 feet above the landing area.

(83) APU generator is available:

- ➡ **YES** – Go to (84)
- ➡ **NO** – Go to (88)

(84) APU generator is available:

- (85) ADG manual deploy handle STOW
- (86) ADG PWR TXFR OVERRIDE PRESS
- (87) Go to (89)

(88) APU generator is NOT available:

The following significant systems are not available when on emergency power only:

- Yaw damper 2 and AFCS 2,
- Ground spoilers,
- Flaps channel 2,
- Stabilizer trim channel 1,
- Mach trim, aileron and rudder trim,
- Hydraulic pumps 1B, 2B & 3A,
- DME, FMS 2,
- Copilot's PFD, MFD, navigation & communication systems,
- Right pack,
- Right probe heaters, right AOA vane heater and right ice detector,
- Nose wheel steering system, anti-skid system,
- Both windshield heaters, right window heater,
- Copilot's instrument lights, NAV lights, and L & R landing lights,
- SPS altitude compensation, and
- Auto throttle system.

(89) At 10,000 feet MSL or lowest safe altitude:

- (90) RAM-AIR PRESS IN
- (91) TAWS WARNING, FLAPS OFF PRESS IN
- (92) TAWS WARNING, RAAS OFF
(if installed) PRESS IN
- (93) Approach start point ESTABLISH
at 5,000 feet AGL over landing area.
- (94) Turn (20 to 30 degrees bank
angle) START
over landing area.
- (95) Target airspeed MAINTAIN



Double Engine Failure (Cont'd)

- (96) Downwind leg

ESTABLISH
at 2,500 feet AGL abeam landing area and 180
degrees from final approach heading.
- (97) Landing flaps

20 DEGREES
- (98) LDG GEAR

DN
- (99) Final approach

ESTABLISH
at 1,500 feet AGL.
- (100) Final approach speed

$V_{REF} + 14$
minimum.
- (101) Actual landing distance

INCREASE
as applicable by factor given below:

APU generator available	Emergency power only
1.55 (55%)	3.20 (220%)

When time and conditions permit:

- (102) Ditching and Forced Landing
Imminent

REVIEW
- (See Page F-LDG 08-2)

- COMPLETE -

Challenger 605	ENG 05-23
	REV 50, Mar 25/19

Engine Failure During Takeoff After Achieving V_1

Condition: Engine has failed and airplane is committed for takeoff.

Objective: Climb to a safe altitude.

- (1) TakeoffCONTINUE
- (2) AirplaneROTATE

When positive rate of climb is achieved:

- (3) Landing gearRETRACT
- (4) AirspeedMAINTAIN V_2

NOTE: If engine failure occurs above V_2 , maintain airspeed at current value (not more than $V_2 + 10$).

AFTER TAKEOFF CHECK:

→ If wing anti-ice system is on/required:

- 14TH STAGE ISOLPRESS IN

- I (5) Both THRUST REVERSERS OFF

At not less than 400 feet AGL or the engine-out level off height:

- (6) ThrustSET
to maximum continuous.
- (7) AirspeedACCELERATE
to V_{FTO} .
- (8) FLAPSUP
at a speed of not less than $V_{FTO} - 5$
- (9) ClimbINITIATE
- (10) CAS messages REVIEW

At a safe altitude:

- (11) In-Flight Engine
Failure/ShutdownACCOMPLISH

(See Page ENG 05-10)

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
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Challenger 605	ENG 05-24
	REV 58, Dec 08/20

Engine Failure in Climb During (V) ALTS CAP or (V) ALTV CAP

Condition: Engine has failed during level-off while still in (V) ALTS CAP or (V) ALTV CAP mode.

Objective: Maintain adequate speed protection.

- (1) Autopilot / FD.....DISCONNECT
 (2) Pitch attitude.....ADJUST
 to maintain the required single engine operating airspeed.

NOTE: Vertical FD commands may be used and autopilot re-engaged after FCC mode changes to (V) ALTS or (V) ALTV at desired altitude.

- (3) In-Flight Engine Failure/ShutdownACCOMPLISH
 (See Page ENG 05-10)

- COMPLETE -

Cross Bleed Relight/Starter Assisted

Condition: Engine is inoperative with no indication of severe damage.

Objective: Restart inoperative engine using bleed air.

NOTE: Use cross bleed for in-flight air starts whenever possible.

Affected engine:

- (1) Thrust lever CHECK SHUT OFF
- (2) Both BOOST PUMPs PRESS IN
- (3) APU LCV PRESS OUT

Operating engine:

- (4) 10TH STAGE bleed..... PRESS IN
- (5) Bleed air pressure60 PSI MINIMUM
Adjust operating engine’s thrust lever as required.
- (6) Relight envelope..... ESTABLISH
See table below.

STARTER-ASSISTED/CROSS BLEED RELIGHT ENVELOPE			
ALTITUDE	AIRSPEED	ITT	% N ₂
21,000 ft to 15,000 ft	200 KIAS up to 300 KIAS	90°C or less	0 to 55
15,000 ft to SL	140 KIAS up to 300 KIAS		

When ready to start:

- (7) CONT IGNITION PRESS IN

NOTE: If engine does not ignite in 25 seconds, retard thrust lever to SHUT OFF, press associated engine STOP switch/light, wait 20 seconds and repeat procedure.

Affected engine:

- (8) L (R) ENG START PRESS

When ITT is 90°C or colder and N₂ is 28%:

- (9) Thrust lever IDLE
- (10) Engine indications MONITOR carefully.

(11) Engine relights (within 25 seconds):

- ➡ YES – Go to (12)
- ➡ NO – Go to (24)



Challenger 605	ENG 05-26
	REV 49, Nov 19/18

Cross Bleed Relight/Starter Assisted (Cont'd)

(12) Engine relights (within 25 seconds):

- (13) Thrust levers AS REQUIRED
- (14) CONT IGNITION PRESS OUT
- (15) Applicable 10TH STAGE bleed PRESS IN
- (16) APU LCV AS REQUIRED
- (17) Both PACKs PRESS IN
- (18) COWL ANTI-ICE AS REQUIRED
- (19) 14TH STAGE ISOL PRESS OUT
- (20) Applicable GEN ON
- (21) Applicable HYDRAULIC pump AUTO
- (22) TAWS WARNING, FLAPS OFF PRESS OUT
- (23) Altimeters CROSS-CHECK

– COMPLETE –

(24) Engine does NOT relight:

- (25) CONT IGNITION PRESS OUT
- (26) Affected 10TH STAGE bleed PRESS OUT
- (27) APU LCV AS REQUIRED
- (28) Operative PACK PRESS IN
- (29) MFD or CDU, ATC/TCAS mode TA ONLY
- (30) Land at the nearest suitable airport.
- (31) Single Engine Approach and
Landing Procedure ACCOMPLISH

(See Page ENG 05–29)

– COMPLETE –

Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Engine
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Windmill Relight

Condition: Engine is inoperative with no indication of severe damage.

Objective: Restart inoperative engine using external air flow.

- NOTE:**
1. At thrust lever advance, N_2 should be stable or increasing.
 2. Windmill airstart efficiency is enhanced by attaining the highest practical airspeed and N_2 within the relight envelope.
 3. 300 KIAS is a target airspeed and may have to be exceeded to obtain the minimum required N_2 .

Affected engine:

- (1) Thrust lever CHECK SHUT OFF
- (2) Both BOOST PUMPs PRESS IN
- (3) Affected 10TH STAGE bleed PRESS OUT
- (4) 10TH STAGE ISOL PRESS OUT
- (5) APU LCV PRESS OUT
- (6) Relight envelope ESTABLISH
See table below.

WINDMILL RELIGHT ENVELOPE			
ALTITUDE	AIRSPPEED	ITT	% N_2
21,000 ft to 10,000 ft	300 KIAS up to 348 KIAS	90°C or less	12 to 55
10,000 ft to 8,000 ft			10 to 55
8,000 ft to SL	300 KIAS		

When ready to start:

- (7) CONT IGNITION PRESS IN

When ITT is 90°C or colder and N_2 is 12% or greater (10,000 feet or greater) or N_2 is 10% or greater (less than 10,000 feet):



Challenger 605	ENG 05-28
	REV 49, Nov 19/18

Windmill Relight (Cont'd)

Affected engine:

- (8) Thrust leverIDLE
when within start envelope.

NOTE: 1. If engine does not ignite in 25 seconds, retard thrust lever to SHUT OFF, maintain at least 300 KIAS for 30 seconds and repeat procedure. After ignition is achieved, monitor N_2 acceleration rate. N_2 acceleration should be uninterrupted and positive at all times. Stable idle must be achieved in 2 minutes of advancing thrust lever to IDLE.

2. During windmilling airstart, attempt to keep the flight control inputs to a minimum.

- (9) Engine indicationsMONITOR
carefully.

(10) Engine start is successful (within 25 seconds):

➡ **YES** – Go to (11)

➡ **NO** – Go to (23)

(11) Engine start is successful (within 25 seconds):

When engine stabilized at Flight Idle:

- (12) Thrust leversAS REQUIRED
(13) CONT IGNITIONPRESS OUT
(14) 10TH STAGE bleedAS REQUIRED
(15) APU LCVAS REQUIRED
(16) Applicable PACKPRESS IN
(17) COWL ANTI-ICEAS REQUIRED
(18) 14TH STAGE ISOLPRESS OUT
(19) Applicable GENON
(20) Applicable HYDRAULIC pumpAUTO
(21) TAWS WARNING, FLAPS OFFPRESS OUT
(22) AltimetersCROSS CHECK

– COMPLETE –

(23) Engine does NOT relight within 25 seconds:

- (24) CONT IGNITIONPRESS OUT
(25) Affected 10TH STAGE bleedPRESS OUT
(26) APU LCVAS REQUIRED
(27) Operative PACKPRESS IN
(28) MFD or CDU, ATC/TCAS modeTA ONLY



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
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Challenger 605	ENG 05-29
	REV 49, Nov 19/18

Windmill Relight (Cont'd)

- (29) Land at the nearest suitable airport.
- (30) Single Engine Approach and Landing ProcedureACCOMPLISH
- (See Page ENG 05–29)

– COMPLETE –

Single Engine Approach and Landing

Condition: Landing with one engine inoperative.

Objective: Ensure proper configuration for single landing.

DESCENT CHECK:

- (1) ATIS COPY
- (2) CAS messages..... REVIEW
- (3) TAWS WARNING, TERRAIN..... AS REQUIRED
- (4) TAWS WARNING, FLAPS OFF PRESS IN
- (5) TAWS WARNING, RAAS OFF (if installed) PRESS IN
- (6) Landing data.....SET
- (7) Approach speed $V_{REF} + 14$ minimum.
- (8) Actual landing distance (without thrust reverser) INCREASE by a factor of 1.55 (55%).
- (9) Approach briefingCOMPLETE
- (10) Shoulder harnessSECURE
- (11) Pressurization.....CHECK & SET
- (12) COWL ANTI-ICEAS REQUIRED
- (13) WING ANTI-ICEAS REQUIRED
- (14) L & R WSHLD/WIND HI OR LOW
- (15) Fuel quantity & balance CHECK

At transition level:

- (16) Altimeters SET/CROSS CHECK

IN RANGE CHECK:

- (17) APU START
- (18) PACK..... TRANSFER to APU bleed air.
- (19) Go-Around Thrust..... DETERMINE
Add 2% N_1 to the two engine takeoff N_1 .



Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Engine
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Single Engine Approach and Landing (Cont'd)

FAA Certified Airplanes:
 (20) Go-Around Thrust..... DETERMINE
 Normal takeoff N_1 .

(21) External lights..... AS REQUIRED

■ (22) Both PASS SIGNS ON

– COMPLETE –

BEFORE LANDING CHECK:

(23) CONT IGNITION AS REQUIRED

(24) WING ANTI-ICE AS REQUIRED

(25) COWL ANTI-ICE AS REQUIRED

(26) Cabin ADVISE

(27) LDG GEAR..... DN

(28) Brake and anti skid system check COMPLETE

(29) Operative THRUST REVERSER ARMED

(30) Landing flaps 20 DEGREES

(31) Approach speed MAINTAIN
 $V_{REF} + 14$.

(32) Actual landing distance
 (without thrust reverser) INCREASE
 by a factor of 1.55 (55%).

NOTE: If required, use remaining thrust reverser carefully upon landing.

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

Challenger 605	ENG 05-31
	REV 51, Apr 05/19

Single Engine Go-Around

Condition: Landing cannot be continued.

Objective: Perform safe single engine climb.

- (1) Single Engine Climb Thrust
(APR rating).....SET

FAA Certified Airplanes:

- (1) Normal Takeoff Thrust.....SET

- (2) TOGA PRESS

- I (3) FLIGHT SPOILER (if extended)RETRACT

- (4) Airplane ROTATE

- (5) FLAPS 20 DEGREES

When a positive rate of climb is achieved:

- (6) LDG GEAR.....UP

At a safe altitude (not less than 400 feet AGL):

- (7) FLAPS 0 DEGREES
at a speed of not less than $V_{FTO} - 5$.

- (8) Both THRUST REVERSERS OFF

- (9) **Go-around was due to engine failure during approach:**

➡ **YES** – Go to (10)

➡ **NO** – Go to (12)

- (10) **Go-around was due to engine failure during approach:**

- (11) In-Flight Engine
Failure/ShutdownACCOMPLISH

(See Page ENG 05–10)

– COMPLETE –

- (12) **Go-around was NOT due to engine failure during approach:**

- (13) Single Engine Approach and
Landing ProcedureACCOMPLISH

(See Page ENG 05–29)

– COMPLETE –

Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Engine
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Challenger 605	ENG 05-32
	REV 49, Nov 19/18

Engine Failure During Approach

Condition: Engine failure while on approach.

Objective: Reconfigure for single engine operation.

- (1) Autopilot DISCONNECT
- (2) ATS DISC PRESS

▲ CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (3) Operating engine thrust INCREASE
- (4) FLIGHT SPOILER RETRACT
- (5) Landing flaps 20 DEGREES
- (6) Airspeed INCREASE
to $V_{REF} + 14$.
- (7) Airplane RETRIM

(8) Decision to continue approach:

- ➡ **YES** – Go to (9)
- ➡ **NO** – Go to (17)

(9) Decision to continue approach:

- (10) PACKs TRANSFER
to APU bleed air.
- (11) Go-Around thrust DETERMINE
Add 2% N_1 to the two engine takeoff N_1 .

FAA Certified Airplanes:

- (11) Go-Around Thrust DETERMINE
Normal takeoff N_1 .

- (12) TAWS WARNING, FLAPS OFF PRESS IN
- (13) TAWS WARNING, RAAS OFF (if
installed) PRESS IN
- (14) Final approach speed MAINTAIN
 $V_{REF} + 14$
- (15) Actual landing distance
(without thrust reverser) INCREASE
by a factor of 1.55 (55%)

NOTE: Use remaining thrust reverser carefully upon landing.

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

After landing:



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
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Challenger 605	ENG 05-33
	REV 49, Nov 19/18

Engine Failure During Approach (Cont'd)

(16) Affected engine SHUTDOWN
 – COMPLETE –

(17) Decision to go-around:

(18) Single Engine Go-Around.....ACCOMPLISH
 (See Page ENG 05-31)

– COMPLETE –

ATS Failure

Condition: **FAIL** displayed in ATS Mode Status Display.

- (1) ATS DISC PRESS
 (2) Thrust levers..... POSITION MANUALLY

NOTE: 1. The steady **FAIL** message is displayed if the MSD is not receiving data from the ATS computer. Press the ATS DISC switch to confirm disengagement of the ATS. Do not re-engage the ATS.
 2. For an ATS flashing **FAIL** message, the ATS may be re-engaged by pressing the ATS push button switch on the ATS Control Panel. If the fault is still present, the ATS flashing **FAIL** message will re-appear, and ATS will not re-engage.

– COMPLETE –

ATS Low Speed

Condition: Flashing **AOA LIMIT** displayed in ATS Mode Status Display.

- (1) AirspeedSET
 speed bug higher than AOA limit.

– COMPLETE –

ATS Overspeed

Condition: Flashing **Vmo LIMIT** or **Mmo LIMIT** displayed in ATS Mode Status Display.

- (1) ATS DISC PRESS
 (2) Thrust levers..... POSITION MANUALLY

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
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Challenger 605	ENG 05-34
	REV 49, Nov 19/18

ATS Radio Altimeter Input Failure

Condition: RETARD mode engages too early or fails to engage at 50 feet.

- (1) ATS DISC PRESS
- (2) Thrust levers POSITION MANUALLY

– COMPLETE –

ATS Thrust Limited

Condition: Flashing **SPEED** displayed in ATS Mode Status Display.

- (1) ATS DISC PRESS
- (2) Thrust levers POSITION MANUALLY
and re-adjust attitude.

– COMPLETE –

ATS Uncommanded Disengage

- (1) ATS DISC PRESS
- (2) Thrust levers POSITION MANUALLY

NOTE: 1. The ATS will automatically disengage and illuminate the flashing **DISENG'D** message in the MSD if:

- an N_1 split of 13% or greater is detected,
- an N_1 greater than 98.4% is detected,
- if a Thrust Reverser unlocks.

2. After fault correction, the ATS can be re-engaged by depressing the ATS push button on the ATS control panel.

3. The ATS will disengage if the N_1 value is removed from the EICAS (no active selection present). This could happen when the bleeds are changed from ENGINES to APU during the IN RANGE check.

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
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Challenger 605	ENG 05-35
	REV 49, Nov 19/18

Dual ATS MSD Failure

- (1) ATS DISC PRESS
- (2) Thrust levers POSITION MANUALLY

NOTE: In the event of a dual MSD failure causing the displays to go blank, the crew must depress an ATS DISC switch to confirm ATS disengagement, and consider the system inoperative.

– COMPLETE –

Fluctuating Oil Pressure

Condition: Oil pressure fluctuations in excess of 5 psi and decreasing. May be associated with increasing oil temperature.

Objective: Reduce thrust and monitor.

- (1) **Choose applicable:**
 - ➡ Oil pressure is less than 25 psi – Go to (2)
 - ➡ Oil pressure is greater than 25 psi – Go to (4)
- (2) **Oil pressure is less than 25 psi:**
 - (3) Engine Oil Pressure Low Procedure ACCOMPLISH
(See Page ENG 05–5)

– COMPLETE –

- (4) **Oil pressure is greater than 25 psi:**
 - (5) ATS DISC PRESS

Affected engine:

- (6) Thrust lever CONFIRM AND IDLE
- (7) Descent (if required) INITIATE
to 29,000 feet or less, or until oil pressure stabilizes.

When affected engine oil pressure stabilizes:

- (8) Thrust lever ADJUST
as required.

NOTE:

1. If necessary, keep descending to a lower altitude to maintain a stable oil pressure, as cruise power setting is resumed.
2. If oil pressure fluctuations in excess of ± 5 psi resume, select affected thrust lever to IDLE and descend to a lower altitude.

– COMPLETE –

Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Engine
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Challenger 605	ENG 05-36
	REV 49, Nov 19/18

High Oil Temperature

82 OIL TEMP 155

..... Readout of 155 to 162°C (amber range)

82 OIL TEMP 163

..... Readout of 163°C (red range)

Condition: Oil temperature above normal range.

Objective: Reduce thrust and monitor.

(1) ATS DISC PRESS

Affected engine:

(2) Thrust lever ADJUST
to maintain oil temperature within normal range.

(3) Oil temperature stays high (red or amber):

➡ YES – Go to (4)

➡ NO – Go to (6)

(4) Oil temperature stays high (red or amber):

(5) In-Flight Failure/Shutdown ACCOMPLISH
(See Page ENG 05-10)

– COMPLETE –

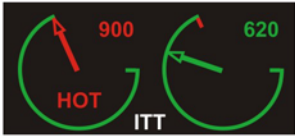
(6) Oil temperature in normal range (green):

(7) Engine indications MONITOR

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
--	--------

Hot Start



Condition: HOT icon on ITT gauge and/or readout indicates 900°C or hotter.

Objective: Discontinue engine start.

Affected engine:

- (1) Thrust leverSHUT OFF
- (2) IGNITIONPRESS OUT

If one engine is operating:

- (3) Inoperative engine BOOST PUMPPRESS OUT
- (4) Dry Motor.....ACCOMPLISH
until ITT is reduced below 120°C, or starter limit.

- COMPLETE -

Interrupted Start

Objective: Discontinue engine start.

If a start is interrupted:

- (1) Thrust leverSHUT OFF
- (2) IGNITIONPRESS OUT

If one engine is operating:

- (3) Inoperative engine BOOST PUMPPRESS OUT
- (4) Dry Motor..... WITHIN STARTER LIMITS

DRY MOTORING CYCLE		
START	MAXIMUM TIME ON	FOLLOWED BY
1	90 seconds	5 minutes off
2 and subsequent	30 seconds	5 minutes off

- COMPLETE -

Challenger 605	ENG 05-38
	REV 49, Nov 19/18

Low Idling

Condition: Idle N_2 split greater than 2% following engine start.

Objective: Notify maintenance personnel.

- NOTE:**
1. Engine to engine N_2 split at ground idle power should not be greater than 2% N_2 . If greater, notify maintenance personnel.
 2. Do not advance thrust lever above 70% N_2 until N_2 is stabilized within normal limits and within 2% of each other.
 3. N_2 should be around 63% at an outside air temperature of 15°C. Idle N_2 speed changes approximately 1% for every 10°C OAT.

– COMPLETE –

N_1 Fan Vibration

0.1 FAN VIB 2.7

..... FAN VIB readout equal to or greater than 2.7 MILS (amber)

Condition: Fan vibration exceeds limits.

Objective: Attempt to correct condition. Shutdown if necessary.

⚠CAUTION: It is not recommended that an engine be shut down unless there is another indication of a severe engine abnormality, i.e., high oil temperature, high oil pressure, abnormal increase in ITT, ITT exceedance, or abnormal engine vibration is felt through the airframe.

- NOTE:**
1. If engine was started within 2 hours of shutdown, allow 10 minutes for vibration levels to return to normal.
 2. An ITT increase of more than 10°C from stable, matched N_1 power is an abnormal increase.

(1) ATS DISC..... PRESS

(2) Airplane in icing conditions:

➡ YES – Go to (3)

➡ NO – Go to (7)

(3) Airplane in icing conditions:

Affected engine:



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
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Challenger 605	ENG 05-39
	REV 49, Nov 19/18

N₁ Fan Vibration (Cont'd)

- (4) Thrust leverADJUST
N₂ above 80% & N₁ above 70%.

Maintain thrust settings until vibration is reduced.

▲CAUTION: Do not exceed maximum continuous thrust setting.

- NOTE:**
1. Vibration may increase during the ice shedding process.
 2. Vibrations may momentarily indicate full scale during the ice shedding process.
 3. Thrust settings up to and including maximum continuous may be required to shed ice.

- (5) Engine indications MONITOR ITT

- (6) Go to (10)

(7) Airplane NOT in icing conditions:

Affected engine:

- (8) Thrust leverADJUST
to thrust that maintains vibration level within normal range.

- (9) EngineMONITOR

(10) Vibration continues, and severe engine abnormality indications are present:

- ➡ **YES** – Go to (11)
- ➡ **NO** – No further action required. **End of procedure.**

(11) Vibration continues, and severe engine abnormality indications are present:

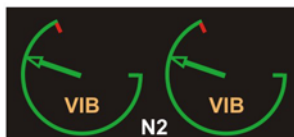
- (12) In-Flight Engine
Failure/ShutdownACCOMPLISH

(See Page ENG 05-10)

– COMPLETE –

Challenger 605	ENG 05-40
	REV 49, Nov 19/18

N₂ Core Vibration



Condition: N₂ vibration exceeds limits.

Objective: Attempt to correct condition. Shutdown if necessary.

▲ CAUTION: It is not recommended that an engine be shut down unless there is another indication of a severe engine abnormality, i.e., high oil temperature, high oil pressure, high ITT, or abnormal engine vibration is felt through the airframe.

(1) ATS DISC PRESS

Affected engine:

(2) Thrust lever ADJUST
to setting that maintains vibration level within normal range.

NOTE: If engine was started within 2 hours of shutdown, allow 10 minutes for vibration levels to return to normal.

(3) Engine indications MONITOR

(4) Vibration can be controlled or reduced:

➡ **YES** – No further action required. **End of procedure.**

➡ **NO** – Go to (5)

(5) Vibration can NOT be controlled or reduced:

(6) In-Flight Engine Failure/Shutdown ACCOMPLISH

(See Page ENG 05-10)

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
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Challenger 605	ENG 05-41
	REV 49, Nov 19/18

N₂ Stagnation

Condition: Slow N₂ acceleration in combination with rising ITT during start.

Objective: Discontinue engine start.

If during or after starting, N₂ stagnates in combination with rising ITT:

- (1) Thrust leverSHUT OFF
- (2) IGNITIONPRESS OUT

If one engine is operating:

- (3) Inoperative engine BOOST PUMPPRESS OUT
- (4) Dry Motor.....ACCOMPLISH
until ITT is reduced below 120°C or starter limit.

– COMPLETE –

No Lightoff

Condition: No ITT rise during start.

Objective: Discontinue engine start.

If an engine fails to light off:

- (1) Thrust leverSHUT OFF
- (2) IGNITIONPRESS OUT

If one engine is operating:

- (3) Inoperative engine BOOST PUMPPRESS OUT
- (4) Dry Motor..... WITHIN STARTER LIMITS

DRY MOTORING CYCLE		
START	MAXIMUM TIME ON	FOLLOWED BY
1	90 seconds	5 minutes off
2 and subsequent	30 seconds	5 minutes off

– COMPLETE –

Challenger 605	ENG 05-42
	REV 51, Apr 05/19

Post-Shutdown Engine Tail Pipe Fire

Condition: Engine ITT exceeds 350°C and rising rapidly, or flame reported at tail pipe.

Objective: Dry motor engine. Evacuate if necessary.

Affected engine:

- (1) Dry motor.....ACCOMPLISH
until ITT is reduced below 150°C or starter limit.

➡ FROM Quick Reference Card ⬅

NOTE: The APU or ground air source (if available) may be used to dry motor the engine.

(2) Tail pipe fire indications persist:

- ➡ **YES** – Go to (3)
- ➡ **NO** – No further action required. **End of procedure.**

(3) Tail pipe fire indications persist:

- (4) ENG FIRE PUSHPRESS IN
- (5) Emergency Evacuation Procedure.....ACCOMPLISH
(See Page F-LDG 08-1)
- (6) Control tower INFORM OF CONDITION
and request fire-fighting assistance.

– COMPLETE –

Challenger 605	ENG 05-43
	REV 51, Apr 05/19

Thrust Lever Inoperative

Condition: No engine response when moving thrust lever or thrust lever jammed.

Objective: Secure affected engine before commencing approach and review penalties.

At a safe altitude, before commencing the approach:

Affected engine:

- I (1) ENG FIRE PUSHPRESS IN
to shutdown affected engine.

NOTE: Ground spoilers may not be available upon landing.

- (2) Actual landing distance INCREASE
by a factor of 1.60 (60%), for a flaps 20 degrees landing
without the use of reverse thrust.

▲CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

Affected engine:

- (3) In-Flight Engine
Failure/ShutdownACCOMPLISH
(See Page ENG 05-10)

– COMPLETE –

Uncommanded Acceleration

Condition: Engine accelerates to a higher thrust than selected.

Objective: Attempt to control. Shutdown if necessary.

(1) ATS DISC PRESS

On ground, affected engine:

(2) Thrust lever CONFIRM AND IDLE
then SHUT OFF.

In flight, affected engine:

(3) Thrust lever CONFIRM AND IDLE
Check engine response.

➡ FROM Quick Reference Card ⬅

⚠ CAUTION: The ATS in **GREEN** mode N₁ synchronization will attempt inappropriate deceleration/acceleration of the remaining engine, to maintain synchronization until disengaged.

(4) **Engine responds:**

➡ **YES** – Go to (5)

➡ **NO** – Go to (10)

(5) **Engine responds:**

(6) Both thrust levers RETARD
to 79% N₂ or less.

(7) Both ENG SPEED OFF

NOTE: The APR system is inoperative when the ENG SPEED switches are selected OFF.

(8) Both thrust levers ADVANCE

⚠ CAUTION: Do not change power settings abruptly as engine response is degraded.

(9) Engine indications MONITOR

NOTE: Exercise caution to prevent overspeed during thrust reverser use, when operating with ENG SPEED switches OFF.

– COMPLETE –

(10) **Engine does NOT respond as expected:**

(11) In-Flight Engine
Failure/Shutdown ACCOMPLISH

(See Page ENG 05-10)

– COMPLETE –

Challenger 605	ENG 05-45
	REV 49, Nov 19/18

Uncommanded Deceleration or Abnormal Engine Oscillation/Fluctuation

Condition: Engine thrust does not agree with thrust lever selection.

Objective: Attempt to control. Shutdown if necessary.

NOTE: To prevent engine overspeed when selecting ENG SPEED switches OFF, N₂ indications should be referenced instead of N₁.

(1) ATS DISC PRESS

▲ CAUTION: The ATS in **SPEED** mode N₁ synchronization will attempt inappropriate deceleration/acceleration of the remaining engine to maintain synchronization until disengaged.

Affected engine:

(2) Thrust leverADJUST
to below 79% N₂ or IDLE (whichever occurs first).

(3) ENG SPEED OFF

(4) Thrust lever ADVANCE

▲ CAUTION: Do not change power settings abruptly as engine response is degraded.

(5) Engine indicationsMONITOR

(6) **Engine responds:**

➡ **YES** – Go to (7)

➡ **NO** – Go to (8)

(7) **Engine responds:**

NOTE: 1. To minimize asymmetric thrust, both ENG SPEED switches should be selected OFF prior to approach and landing.

2. Exercise caution, to prevent overspeed during thrust reverser use, when operating with ENG SPEED switches OFF.

3. The APR system is inoperative when the ENG SPEED switches are selected OFF.

– COMPLETE –

(8) **Engine does NOT respond as expected:**

(9) In-Flight Engine
Failure/ShutdownACCOMPLISH

(See Page ENG 05–10)

– COMPLETE –

Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Engine
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Challenger 605	ENG 05-46
	REV 49, Nov 19/18

Advisory Messages

ENGINE ADVISORY MESSAGES	
APR ARM	APR system is armed.
APR TEST 1 (2) OK	APR test successful at DCU 1 (2).
IGNITION A (B)	Ignition system A (B) is activated.
IGNITION A/B	Ignition systems A and B are on during continuous ignition conditions.
L (R) REV ARMED	L (R) thrust reverser armed and ready.

Status Messages

ENGINE STATUS MESSAGES	
DCU 1 (2) APR FAIL	APR relay did not energize during the test.
L (R) ENG ECU FAIL	APR relay has energized, but APR circuit not triggered.
L (R) ENG OIL SENSOR	Indication of oil pressure switch failure.

Challenger 605	ENG 05-47
	REV 49, Nov 19/18

Auto Throttle Mode Status Display (MSD)

MSD MESSAGE	DESCRIPTION
AOA LIMIT	ATS protecting AOA limit (low speed protection). Refer to ATS Low Speed procedure See ENG 05-33
DISENG'D	ATS disengaged (press DISC to clear). Refer to ATS Uncommanded Disengage procedure See ENG 05-34
FAIL (flashing)	ATS failed due to input error (i.e. no N_1 thrust limit from FMS...). Refer to ATS Failure procedure See ENG 05-33
FAIL (steady)	Power lost to the ATS computer. Refer to ATS Failure procedure See ENG 05-33
SPEED	In speed mode, thrust as reached the N_1 limit and airspeed is 20 knots (or 0.06 Mach) below target speed for longer than 10 seconds. Refer to ATS Thrust Limited procedure See ENG 05-34
SPEED Mmo LIMIT	ATS protecting M_{MO} limit with speed mode activated (overspeed protection). Refer to ATS Overspeed procedure See ENG 05-33
SPEED Vmo LIMIT	ATS protecting V_{MO} limit with speed mode activated (overspeed protection). Refer to ATS Overspeed procedure See ENG 05-33

Mode Status Display Error Messages

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Engine
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Challenger 605	ENG 05-48
	REV 51, Apr 05/19

Power Supply and Circuit Breaker Summary

CB NAME	BUS	CB LOCATION
APR	BATT BUS	CBP5-A02
AUTO THROTTLE	DC BUS 1	CBP1-G07
ENG IGN A	AC ESS BUS	CBP3-C07
ENG IGN A	BATT BUS	CBP1-L07
ENG IGN B	BATT BUS	CBP1-L08
ENG IGN B	BATT BUS	CBP5-A01
ENG OIL POWER	APU BATT DIR	CBP5-B04
ENG START L	BATT BUS	CBP1-L10
ENG START R	BATT BUS	CBP1-L09
ENG VIB MON	AC BUS 1	CBP1-C07
OIL BYPASS IND	APU BATT DIR	CBP5-B03
OIL PRESS TRANS L	BATT BUS	CBP1-L06
OIL PRESS TRANS R	DC ESS BUS	CBP4-B06
THRUST REV 1	DC ESS BUS	CBP4-A06
THRUST REV 2	DC ESS BUS	CBP4-A07
THRUST REV AUTO STOW 1	DC ESS BUS	CBP4-A04
THRUST REV AUTO STOW 2	DC ESS BUS	CBP4-A05

Challenger 605	ENVR TOC-1
	REV 50, Mar 25/19

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Challenger 605	ENVR TOC-2
	REV 49, Nov 19/18

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TRAINING PURPOSES ONLY

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Environmental / Bleed Air
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Challenger 605	ENVR 06-1
	REV 50, Mar 25/19

L 10TH DUCT



“BLEED AIR DUCT”

10TH STAGE



L

Condition: Bleed air leak or overheat detected in the left 10th stage duct.

Objective: Isolate the left duct and prevent further use.

(1) L 10TH STAGE bleed.....PRESS OUT

(2) APU LCVPRESS OUT

(3) 10TH STAGE ISOLPRESS OUT

→ If APU was providing bleed air:

• R 10TH STAGE bleedPRESS IN

→ If above 40,000 feet:

• L COWL ANTI-ICEPRESS IN

(4) L PACK.....PRESS OUT

(5) **L 10TH DUCT** stays on:

➔ YES – Go to (6)

➔ NO – No further action required. **End of procedure.**

(6) **L 10TH DUCT** stays on:

(7) ATS DISC.....PRESS

(8) Left thrust lever.....RETARD
to eliminate warning.

(9) **L 10TH DUCT** stays on (after 30 seconds):

➔ YES – Go to (10)

➔ NO – No further action required. **End of procedure.**

(10) **L 10TH DUCT** stays on (after 30 seconds):

(11) In-Flight Engine
Failure/ShutdownACCOMPLISH

(See Page ENG 05–10)

– COMPLETE –

Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Environmental / Bleed Air
--	---------------------------

R 10TH DUCT

“BLEED AIR DUCT”

10TH STAGE



R

Condition: Bleed air leak or overheat detected in the right 10th stage duct.

Objective: Isolate the right duct and prevent further use.

(1) R 10TH STAGE bleedPRESS OUT

(2) 10TH STAGE ISOLPRESS OUT

→ If above 40,000 feet:

• R COWL ANTI-ICEPRESS IN

(3) R PACKPRESS OUT

(4) **R 10TH DUCT** stays on:

➔ YES – Go to (5)

➔ NO – No further action required. **End of procedure.**

(5) **R 10TH DUCT** stays on:

(6) ATS DISCPRESS

(7) Right thrust leverRETARD
to eliminate warning.

(8) **R 10TH DUCT** stays on (after 30 seconds):

➔ YES – Go to (9)

➔ NO – No further action required. **End of procedure.**

(9) **R 10TH DUCT** stays on (after 30 seconds):

(10) In-Flight Engine
Failure/ShutdownACCOMPLISH

(See Page ENG 05–10)

– COMPLETE –

L (R) 14TH DUCT**“BLEED AIR DUCT”****14TH STAGE****L or R**

Condition: Bleed air leak or overheat detected in the indicated 14th duct.

Objective: Isolate the affected duct and prevent further use.

Affected engine:

(1) 14TH STAGE bleed.....PRESS OUT

- NOTE:**
1. Avoid icing conditions.
 2. Cowl anti-icing will not be available on affected engine.
 3. Thrust reverser is inoperative on affected engine.

→ If wing anti-ice is required:

- 14TH STAGE ISOLPRESS IN

(2) **L (R) 14TH DUCT** stays on:

- ➔ **YES** – Go to (3)
- ➔ **NO** – No further action required. **End of procedure.**

(3) **L (R) 14TH DUCT** stays on:

(4) ATS DISC.....PRESS

Affected engine:

(5) Thrust leverRETARD
to eliminate warning.

(6) **L (R) 14TH DUCT** stays on (after 30 seconds):

- ➔ **YES** – Go to (7)
- ➔ **NO** – No further action required. **End of procedure.**

(7) **L (R) 14TH DUCT** stays on (after 30 seconds):

(8) In-Flight Engine
Failure/ShutdownACCOMPLISH
(See Page ENG 05–10)

– COMPLETE –

CABIN ALT



“CABIN PRESSURE”

Condition: Excessive cabin altitude.

Objective: Descend to a safe altitude and attempt to regain control of cabin pressure.

- (1) Oxygen masks.....DON
Set 100%.
- (2) Crew communication.....ESTABLISH
- (3) Both PASS SIGNS ON
- (4) Passenger oxygen.....DEPLOY
- (5) Descent INITIATE
to 10,000 feet or lowest safe altitude, whichever is higher.
- (6) Thrust levers.....IDLE
- (7) ATS DISC.....PRESS
- (8) FLIGHT SPOILER.....MAX

➡ FROM Quick Reference Card ◀

- (9) Airspeed V_{MO}/M_{MO}
maximum.

CAUTION: If structural damage is suspected, do not exceed the airspeed at which the damage occurred, and minimize maneuvering loads.

WARNING: Operation above 10,000 feet cabin altitude may result in hypoxia.

At a safe altitude:

- (10) Oxygen and masks.....AS REQUIRED

NOTE:

1. Closing the doors on the mask stowage compartments and pressing RESET will stop the flow of oxygen to the masks.
2. If supplemental crew oxygen is still required, setting masks to normal (N) will reduce consumption.

- (11) Passenger oxygen.....CLOSED
then NORMAL to reset

(12) **CABIN ALT** accompanied by any door/hatch message or unsafe condition:

- ➡ **YES** – Go to (13)
- ➡ **NO** – Go to (16)



Challenger 605	ENVR 06-5
	REV 49, Nov 19/18

CABIN ALT (Cont'd)

- (13) **CABIN ALT** accompanied by any door/hatch message or unsafe condition:
- (14) Do not attempt to repressurize the airplane.
- (15) Applicable door procedureACCOMPLISH
when at a safe cabin altitude.
- (See Page DOOR 03-1)

- COMPLETE -

- (16) **No door/hatch message or unsafe condition present:**
Restore control of cabin pressurization manually:
- (17) PRESS CONTMANUAL
- (18) MAN ALTHOLD DN
to decrease cabin altitude.
- (19) MAN RATEMAX INCR

- (20) **Cabin pressurization control restored:**
- ➡ **YES** – Go to (21)
 - ➡ **NO** – Go to (23)
- (21) **Cabin pressurization control restored:**
- (22) Manual Cabin Pressurization Control ProcedureCONTINUE
- (See Page ENVR 06-13)

- COMPLETE -

- (23) **No control over cabin pressurization:**
- (24) Unpressurized Flight ProcedureACCOMPLISH
if required.
- (See Page ENVR 06-14)

- COMPLETE -

DIFF PRESS

- Condition:** Cabin differential pressure exceeds limits.
- Objective:** Reduce cabin differential pressure and manually control cabin pressure.
- (1) PRESS CONTMANUAL
- (2) MAN ALTHOLD UP
- (3) MAN RATEAS REQUIRED
to increase cabin altitude and reduce differential pressure.



Challenger 605	ENVR 06-6
	REV 49, Nov 19/18

DIFF PRESS (Cont'd)

(4) DIFF PRESS stays on:

➡ **YES** – Go to (5)

➡ **NO** – Go to (16)

(5) DIFF PRESS stays on:

(6) Descent INITIATE
to 10,000 feet or lowest safe altitude, whichever is higher.

(7) Left or Right PACK PRESS OUT

(8) DIFF PRESS stays on:

➡ **YES** – Go to (9)

➡ **NO** – Go to (16)

(9) DIFF PRESS stays on:

(10) Both PACKs PRESS OUT

(11) DIFF PRESS stays on:

➡ **YES** – Go to (12)

➡ **NO** – Go to (16)

(12) DIFF PRESS stays on:

(13) EMER DEPRESS PRESS IN

(14) Descent INITIATE
to lowest possible altitude.

(15) Unpressurized Flight Procedure ACCOMPLISH

(See Page ENVR 06-14)

– COMPLETE –

(16) DIFF PRESS goes out:

(17) Cabin altitude/differential pressure MONITOR

(18) Manual Cabin Pressurization
Control Procedure CONTINUE

(See Page ENVR 06-13)

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Environmental / Bleed Air
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Challenger 605	ENVR 06-7
	REV 51, Apr 05/19

AUTO PRESS

┌PRESS CONT┐



Condition: Automatic pressurization mode has failed.

- (1) Manual Cabin Pressurization
Control ProcedureACCOMPLISH
(See Page ENVR 06–13)

NOTE: Do not attempt to reselect auto mode.

– COMPLETE –

CABIN ALT

Condition: Cabin altitude exceeds caution limits.

Objective: Restore control of cabin pressurization.

- (1) 10TH STAGE bleeds CHECK PRESSED IN
- (2) Both PACKs CHECK PRESSED IN
- (3) Thrust INCREASE
- (4) EMER DEPRESS CHECK PRESSED OUT
- (5) MAN ALT MID POS/CENTRE
- (6) **Automatic pressurization is restored:**
- ➡ **YES** – No further action required. **End of procedure.**
- ➡ **NO** – Go to (7)
- (7) **Automatic pressurization is NOT restored:**
- (8) Manual Cabin Pressurization
Control ProcedureACCOMPLISH
(See Page ENVR 06–13)

– COMPLETE –

Challenger 605	ENVR 06-8
	REV 49, Nov 19/18

CPAM FAIL

Condition: Both cabin pressure acquisition modules have failed.
EICAS pressurization information is inoperative.

Objective: Descent to a safe altitude and continue unpressurized.

- (1) Descent INITIATE
to 10,000 feet or lowest safe altitude, whichever is higher.
- (2) Unpressurized Flight Procedure ACCOMPLISH
(See Page ENVR 06–14)

NOTE: EICAS pressurization system readouts (e.g., cabin altitude, rate of change and differential pressure) are not available. EICAS cabin pressure warning and caution messages are also not available.

– COMPLETE –

OXY LO PRESS

Condition: Oxygen pressure is low.

Objective: Ensure adequate oxygen supply. Descend if necessary.

- (1) **OXY pressure readout is less than 825 psi:**
 - ➡ **YES** – Go to (2)
 - ➡ **NO** – Go to (4)
- (2) **OXY pressure readout is less than 825 psi:**
- (3) Descent INITIATE
to a safe altitude.

– COMPLETE –

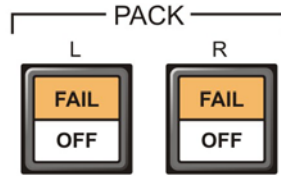
- (4) **OXY pressure readout is greater than 825 psi:**
- (5) Crew oxygen masks TEST
and check pressure readout.
- (6) Oxygen quantity MONITOR
- (7) **OXY pressure readout drops to 0 psi:**
 - ➡ **YES** – Go to (8)
 - ➡ **NO** – No further action required. **End of procedure.**
- (8) **OXY pressure readout drops to 0 psi:**
- (9) Descent INITIATE
to a safe altitude.

– COMPLETE –

Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Environmental / Bleed Air
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Challenger 605	ENVR 06-9
	REV 49, Nov 19/18

L (R) PACK HI PRESS



Condition: Over-pressurization and shutdown of the indicated pack.

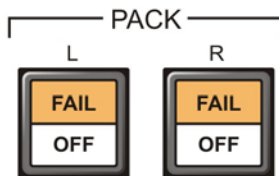
Objective: Attempt to reset.

NOTE: Indicated pack valve, 10th stage bleed valve and isolation valve will close automatically during an overpressure condition.

→ **If above 40,000 feet:**

- Associated COWL ANTI-ICE PRESS IN
- (1) Affected PACK PRESS OUT
- (2) Affected 10TH STAGE bleed PRESS OUT
- (3) Affected 10TH STAGE bleed PRESS IN
- (4) Bleed air pressure CONFIRM
less than 60 psi.
- (5) Affected PACK PRESS IN
- (6) **L (R) PACK HI PRESS stays on:**
 - ➡ **YES** – Go to (7)
 - ➡ **NO** – No further action required. **End of procedure.**
- (7) **L (R) PACK HI PRESS stays on:**
 - (8) Affected PACK PRESS OUT
 - (9) Affected 10TH STAGE bleed PRESS OUT

– COMPLETE –

L (R) PACK HI TEMP

Condition: Overtemperature and shutdown of the indicated pack.

Objective: Attempt to reset.

NOTE: Indicated pack valve will close automatically during an overtemperature condition.

→ **If above 40,000 feet:**

- Associated COWL ANTI-ICE PRESS IN
- (1) Affected PACK PRESS OUT
- (2) Affected 10TH STAGE bleed PRESS OUT

NOTE: If temperature control is in MAN with full HOT selected, decrease temperature before resetting pack.

When L (R) PACK HI TEMP goes out:

- (3) Affected 10TH STAGE bleed PRESS IN
- (4) Affected PACK PRESS IN

(5) L (R) PACK HI TEMP comes on again:

- ➡ **YES** – Go to (6)
- ➡ **NO** – No further action required. **End of procedure.**

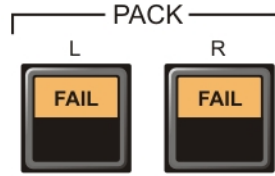
(6) L (R) PACK HI TEMP comes on again:

- (7) Affected PACK PRESS OUT
- (8) Affected 10TH STAGE bleed PRESS OUT

– COMPLETE –

Challenger 605	ENVR 06-11
	REV 49, Nov 19/18

L (R) PACK NOT OFF



Condition: Indicated pack continues to operate when commanded off.

Objective: Attempt to secure the indicated pack.

(1) Affected PACKPRESS IN

→ If above 40,000 feet:

- Associated COWL ANTI-ICEPRESS IN

(2) Affected PACKPRESS OUT

(3) **L (R) PACK NOT OFF** stays on:

- ➡ YES – Go to (4)
- ➡ NO – No further action required. **End of procedure.**

(4) **L (R) PACK NOT OFF** stays on:

(5) Affected 10TH STAGE bleedPRESS OUT

(6) 10TH STAGE ISOLPRESS OUT

→ If APU is operating and **L PACK NOT OFF** is on:

- APU LCVCHECK PRESSED OUT

– COMPLETE –

PAX OXY ON

Condition: Passenger oxygen system is active (AUTO or manual).

(1) Passenger oxygen.....CHECK STATUS

– COMPLETE –

Emergency Descent

Condition: Situation requires rapid descent to a lower altitude.

Objective: Rapidly bring the airplane to a safe altitude.

- | | |
|-----------------------------|--|
| (1) Oxygen masks..... | DON
Set 100%. |
| (2) Crew communication..... | ESTABLISH |
| (3) Both PASS SIGNS | ON |
| (4) Passenger oxygen..... | DEPLOY |
| (5) Descent | INITIATE
to 10,000 feet or lowest safe altitude, whichever is higher. |
| (6) Thrust levers..... | IDLE |
| (7) ATS DISC | PRESS |
| (8) FLIGHT SPOILER | MAX |

➡ **FROM Quick Reference Card** ⬅

- | | |
|--------------------|-----------------------------|
| (9) Airspeed | V_{MO}/M_{MO}
maximum. |
|--------------------|-----------------------------|

⚠ CAUTION: If structural damage is suspected, do not exceed the airspeed at which the damage occurred, and minimize maneuvering loads.

⚠ WARNING: Operation above 10,000 feet cabin altitude may result in hypoxia.

At a safe altitude:

- | | |
|----------------------------|-------------|
| (10) Oxygen and masks..... | AS REQUIRED |
|----------------------------|-------------|

NOTE:

1. Closing the doors on the mask stowage compartments and pressing RESET will stop the flow of oxygen to the masks.
2. If supplemental crew oxygen is still required, setting masks to normal (N) will reduce consumption.

- | | |
|----------------------------|--------------------------------|
| (11) Passenger oxygen..... | CLOSED
then NORMAL to reset |
|----------------------------|--------------------------------|

– COMPLETE –

Manual Cabin Pressurization Control

┌PRESS CONT┐



ALT	MAN	3200
RATE		↑ 200
ΔP		-0.5

Condition: Automatic pressurization system has failed or is selected OFF.

Objective: Ensure the cabin altitude / differential pressure is adequate for cruise flight level / altitude.

- (1) MAN ALTMID POS/CENTRE
 (2) PRESS CONTMANUAL

▲CAUTION: Do not exceed the maximum normal cabin differential pressure of 8.8 psid.

(3) **Choose applicable:**

- ➡ **To increase cabin altitude** – Go to (4)
- ➡ **To decrease cabin altitude** – Go to (7)
- ➡ **To maintain cabin altitude** – Go to (10)

(4) **To increase cabin altitude:**

- (5) MAN ALTHOLD UP
 (6) MAN RATEAS REQUIRED

(7) **To decrease cabin altitude:**

- (8) MAN ALT HOLD DN
 (9) MAN RATEAS REQUIRED

(10) **To maintain cabin altitude:**

When reaching target cabin altitude as per the table below:

- (11) MAN ALTMID POS/CENTRE

Cruise Flight Level	180	200	220	240	260	280	300	350	400	410
Target Cabin Altitude	2100	2400	2700	3000	3400	3800	4200	5200	6600	6900

Before landing:

- (12) Cabin altitudeADJUST
 to landing field elevation.

NOTE: Do not set cabin altitude below landing field elevation.

If differential pressure is not zero upon landing:

- (13) MAN ALTHOLD UP



Challenger 605	ENVR 06-14
	REV 51, Apr 05/19

Manual Cabin Pressurization Control (Cont'd)

(14) MAN RATE MAX INCR

– COMPLETE –

Unpressurized Flight

Condition: Normal or manual cabin pressurization is unavailable.

Objective: Provide a sufficient airflow in the cabin.

(1) Airplane altitude MAX 10,000 FEET
or lowest safe altitude, whichever is higher.

(2) **PACKs are OFF:**

➡ **YES** – Go to (3)

➡ **NO** – Go to (9)

(3) **PACKs are OFF:**

(4) RAM-AIR PRESS IN

(5) Airspeed 250 KIAS
recommended during cruise to provide sufficient airflow to
cabin.

(6) Both PACKs PRESS OUT

(7) 10TH STAGE bleeds PRESS OUT

(8) Go to (10)

(9) **PACKs are ON:**

(10) EMER DEPRESS PRESS IN

– COMPLETE –

Challenger 605	ENVR 06-15
	REV 49, Nov 19/18

Advisory Messages

ENVIRONMENTAL / BLEED AIR ADVISORY MESSAGES	
CABIN ALT WARN HI	Verify the LDG ALT selector.
DUCT TEST OK	10th and 14th stages, bleed leak detection test successful.
LOOP A (B) TEST OK	Duct overheat loop circuit test successful.

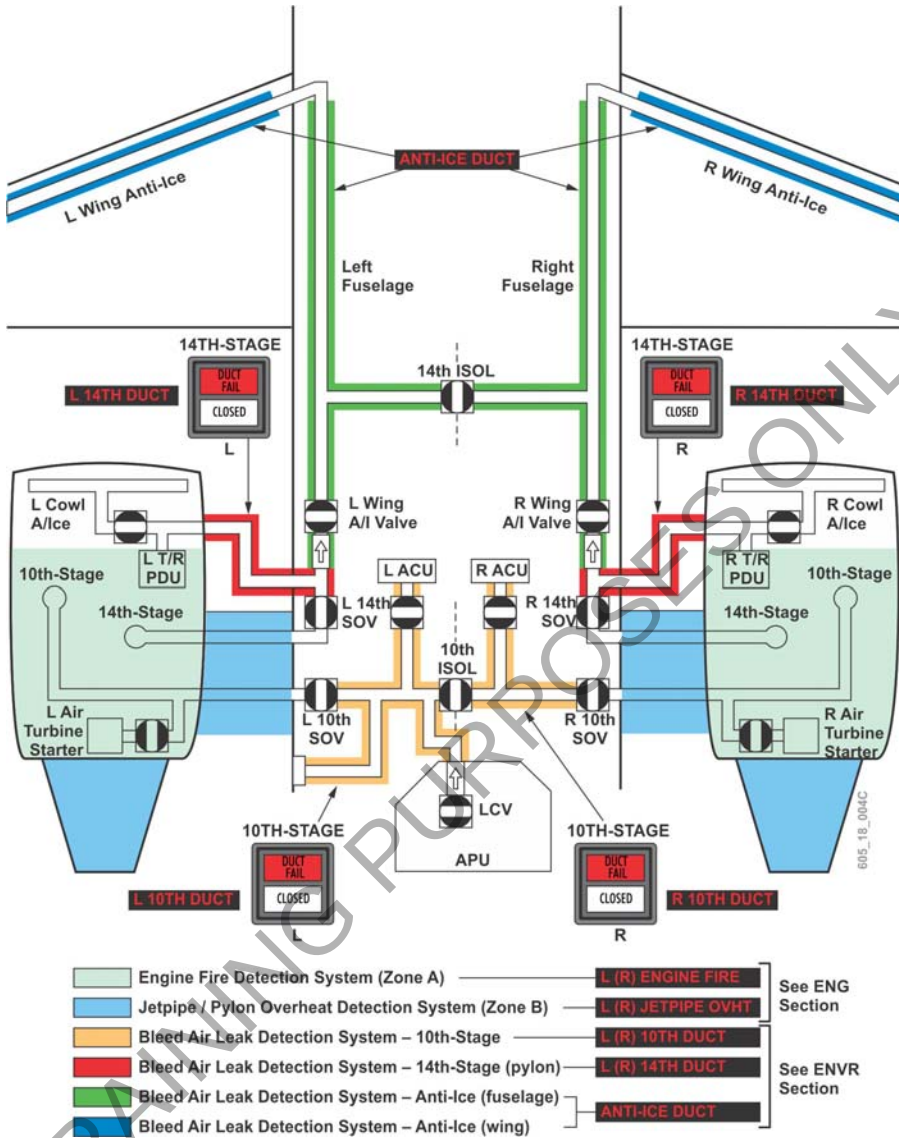
Status Messages

ENVIRONMENTAL / BLEED AIR STATUS MESSAGES	
CPAM FAULT	One of two CPAM is inoperative. Pressurization system monitor.
DISPLAY FAN FAIL	Avionics cooling fan failure.

TRAINING PURPOSES ONLY

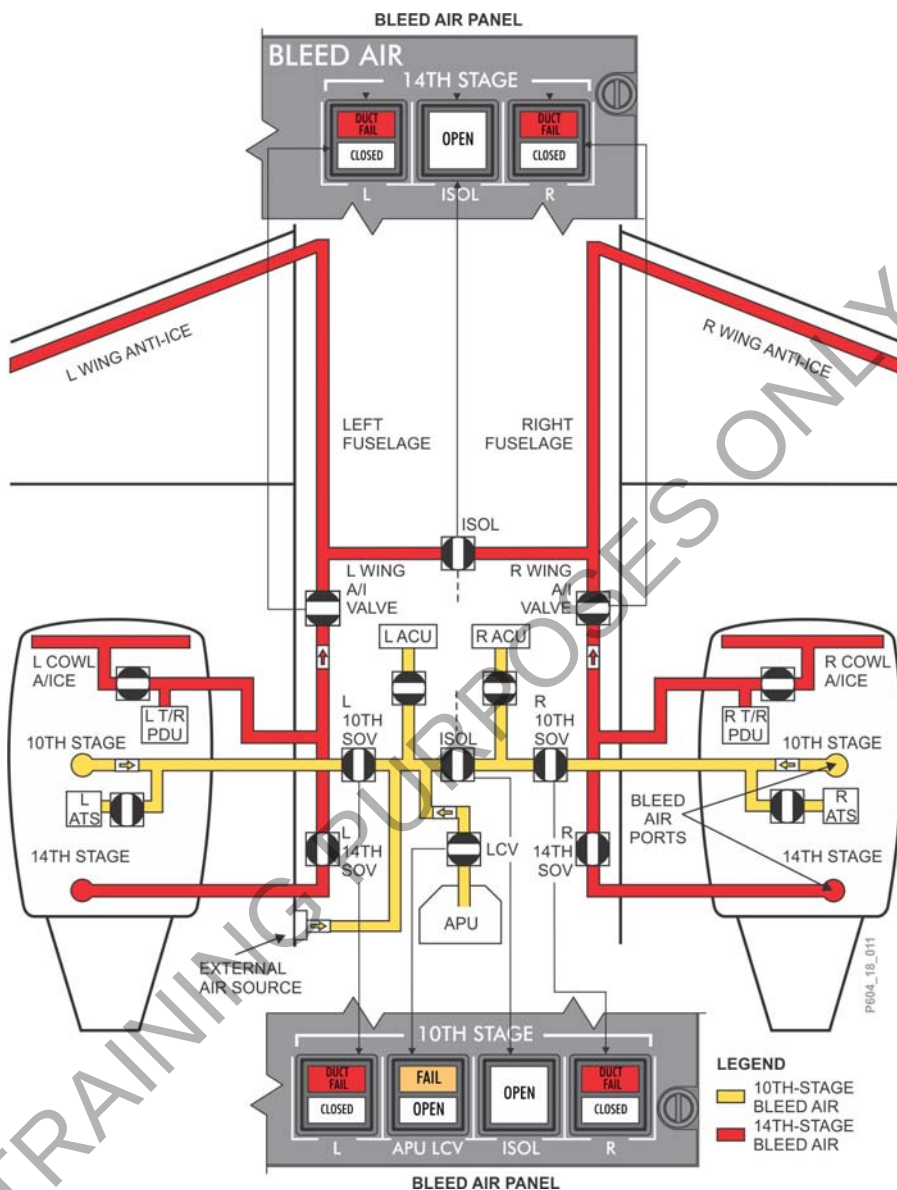
SCHEMATICS

Bleed Air Leak Detection System



SCHEMATICS (CONT'D)

Bleed Air System



Challenger 605	ENVR 06-18
	REV 49, Nov 19/18

Power Supply and Circuit Breaker Summary

CB NAME	BUS	CB LOCATION
10 ST B/AIR ISOL	DC ESS BUS	CBP4-B11
10 ST B/AIR SOV L	DC ESS BUS	CBP4-C11
10 ST B/AIR SOV R	DC BUS 2	CBP2-F11
14 ST B/AIR ISOL SOV	DC BUS 1	CBP1-F13
14 ST B/AIR SOV L	DC BUS 1	CBP1-F12
14 ST B/AIR SOV R	DC BUS 2	CBP2-F12
AIR COND CONT L	DC ESS BUS	CBP4-B09
AIR COND CONT R	DC BUS 2	CBP2-F09
B/LEAK L DET	AC ESS BUS	CBP3-B01
B/LEAK R DET	AC ESS BUS	CBP3-B02
B/LEAK TEST	DC ESS BUS	CBP4-B12
CABIN PRESS CONT	BATT BUS	CBP2-N05
CABIN TEMP CONT AUTO	DC BUS 2	CBP2-E15
CABIN TEMP CONT LO LIMIT	DC BUS 2	CBP2-E13
CABIN TEMP CONT MAN	DC BUS 2	CBP2-E14
CABIN TEMP SENSE	AC BUS 2	CBP2-B13
CKPT TEMP CONT AUTO	DC BUS 1	CBP1-E15
CKPT TEMP CONT LO LIMIT	DC ESS BUS	CBP4-C09
CKPT TEMP CONT MAN	DC ESS BUS	CBP4-C10
CKPT TEMP SENSE	AC BUS 1	CBP1-C09
CPAM1 / EMER DERESS	DC ESS BUS	CBP4-B05
CPAM 2	DC BUS 2	CBP2-G05
FOOT WARMER	DC UTILITY BUS 1	CBP1-E01
RAM AIR SOV	BATT BUS	CBP2-P12

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Environmental / Bleed Air
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Challenger 605	FCTL TOC-1
	REV 63, Mar 14/22

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FLAPS FAIL	FCTL 07-5
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POWER SUPPLY AND CIRCUIT BREAKER

SUMMARY

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Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Flight Controls
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Challenger 605	FCTL TOC-2
	REV 49, Nov 19/18

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TRAINING PURPOSES ONLY

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Flight Controls
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Challenger 605	FCTL 07-1
	REV 65, Jun 24/22

CONFIG FLAPS or CONFIG SPOILERS or CONFIG STAB



“CONFIG FLAPS”
“CONFIG SPOILERS”
“CONFIG TRIM”

Condition: Takeoff attempted with misconfigured flight controls.

Objective: Discontinue takeoff.

(1) TakeoffREJECT

(2) Airplane configuration.....CONFIRM

Check the position of the following items:

- Flaps,
- Spoilers,
- Stabilizer.

(3) **Erroneous configuration messages and aural occur:**

➡ **YES** – Go to (4)

➡ **NO** – No further action required. **End of procedure.**

(4) **Erroneous configuration messages and aural occur:**

(5) **WOW OUTPUT** ProcedureACCOMPLISH

(See Page GEAR 10–12)

– COMPLETE –

AILERON PCU

Condition: One or more jammed aileron power control unit.

Objective: Assess controllability, review landing configuration and penalties.

NOTE: It may be necessary to reduce airspeed to improve airplane handling.

(1) AutopilotDISCONNECT

If necessary:

(2) Aileron controls.....OVERPOWER

(3) **Difficult to maintain desired flight path:**

➡ **YES** – Go to (4)

➡ **NO** – Go to (8)

(4) **Difficult to maintain desired flight path:**

At a safe altitude:



Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Flight Controls
---	------------------------

AILERON PCU (Cont'd)

- (5) Aileron controls..... RELEASE PRESSURE
- (6) ROLL DISC PULL AND TURN

NOTE:

1. The pilot with the operative aileron will be required to maintain a roll control input to offset the effects of the aileron PCU malfunction.
2. If the right aileron control circuit is inoperative, the autopilot system is inoperative.

- (7) Go to (9)

(8) Fight path control is adequate:

- (9) Land at the nearest suitable airport.

NOTE: Controllability is reduced. Select longest runway available with minimum turbulence and crosswind.

Prior to landing:

- (10) TAWS WARNING, FLAPS OFF PRESS IN
- (11) TAWS WARNING, RAAS OFF (if installed) PRESS IN
- (12) Landing flaps 20 DEGREES
- (13) Approach speed $V_{REF} + 14$ minimum.
- (14) ATS DISC PRESS prior to 100 feet AGL.

▲ CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (15) Actual landing distance INCREASE as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

Challenger 605	FCTL 07-3
	REV 49, Nov 19/18

ALT COMP FAIL

Condition: Stall Protection Computer altitude compensation function has failed.

Objective: Review penalties.

NOTE: The stick pusher may be inoperative. One or both channels of the stick shaker will remain operative.

Prior to landing:

- (1) Approach speed $V_{REF} + 10$ minimum.
- (2) Actual landing distance INCREASE
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.40 (40%)	1.30 (30%)

– COMPLETE –

ELEVATOR SPLIT

Condition: Elevator surface position difference exceeds allowable tolerance.

Objective: Assess controllability, review landing configuration and penalties.

NOTE: It may be necessary to reduce airspeed to improve airplane handling.

- (1) Autopilot DISCONNECT
- (2) Elevator controls MOVE SLOWLY
and monitor position on EICAS to identify malfunctioning elevator surface.

If necessary:

- (3) Elevator controls OVERPOWER

(4) Difficult to maintain desired flight path:

➡ **YES** – Go to (5)

➡ **NO** – Go to (14)

(5) Difficult to maintain desired flight path:

At a safe altitude:

- (6) Elevator controls RELEASE PRESSURE
- (7) PITCH DISC PULL AND TURN



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Flight Controls
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ELEVATOR SPLIT (Cont'd)

(8) Determine jammed side:

- ➡ **Left side is jammed** – Go to (9)
- ➡ **Right side is jammed** – Go to (11)

(9) Left side is jammed:

NOTE: Do not re-engage autopilot.

(10) Go to (15)

(11) Right side is jammed:

(12) STALL PROT, PUSHER (either left or right) OFF

NOTE: Stick shaker on operative control column, operates at a lower intensity.

(13) Go to (15)

(14) Fight path control is adequate:

(15) Land at the nearest suitable airport.

NOTE: Controllability is reduced. Select the longest runway available with minimum turbulence and crosswind.

Prior to landing:

(16) TAWS WARNING, FLAPS OFF PRESS IN

(17) TAWS WARNING, RAAS OFF (if installed) PRESS IN

(18) Landing flaps 20 DEGREES

(19) Approach speed $V_{REF} + 14$ minimum.

(20) ATS DISC PRESS prior to 100 feet AGL.

▲ CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

(21) Actual landing distance INCREASE as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

Challenger 605	FCTL 07-5
	REV 49, Nov 19/18

FLAPS FAIL

Condition: Flaps failed to reach commanded position.

Objective: Review landing configuration and penalties.

(1) Choose applicable:

- ➡ **Flap position is less than 8 degrees** – Go to (2)
- ➡ **Flap position is 8 degrees or greater, but less than 45 degrees** – Go to (5)
- ➡ **Flap position is 45 degrees** – Go to (15)

(2) Flap position is less than 8 degrees:

(3) HYDRAULIC pumps 1B, 2B and 3B ON

(4) Go to (6)

(5) Flap position is 8 degrees or greater, but less than 45 degrees:

(6) FLAPSSELECT
to closest detent position less than actual flaps position,
and do not attempt to operate flaps any further.

(7) Maximum enroute airspeed V_{FE}
for next greater flap setting from failed position.

NOTE: If flaps are confirmed retracted (0 degrees), reduction of cruise airspeed is not required.

⚠CAUTION: If possible, avoid prolonged flight in icing. If icing is encountered for an extended period of time, aerodynamic buffet may be experienced.

AR Certified Airplanes:

- (8) If flap position is great than 5 degrees:
Land at the nearest suitable airport.

Prior to landing:

→ **If flap position is less than 20 degrees:**

- Fuel balance CHECK
- (9) TAWS WARNING, FLAPS OFF PRESS IN
- (10) TAWS WARNING, RAAS OFF (if installed) PRESS IN
- (11) Final approach speed $V_{REF} + \Delta V_{REF}$
for next smaller flap setting failed position.



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Flight Controls
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FLAPS FAIL (Cont'd)

(12) ATS DISC PRESS
prior to 100 feet AGL.

▲ CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

(13) Thrust levers POSITION MANUALLY

(14) Actual landing distance INCREASE
as applicable by factor given below:

Flaps (Degrees)	ΔV_{REF} (KTS)	ALD Factor	
		Without Thrust Reversers	With Thrust Reversers
0	30	1.80 (80%)	1.75 (75%)
20	14	1.55 (55%)	1.50 (50%)
30	7	1.40 (40%)	1.35 (35%)

NOTE: Accomplish a normal descent typically with a vertical speed of not more than 400 fpm at touchdown.

▲ CAUTION: 1. Improper landing technique during a flaps failure can generate nose gear loads sufficient to cause structural damage.

To prevent damage:

- After main gear touchdown, gently lower the nose to the runway.
- Apply brakes only after nose wheel touchdown.

2. With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

– COMPLETE –

(15) Flap position is 45 degrees:

NOTE: If aerodynamic buffet is encountered after flying in icing conditions, reduce buffet by increasing the minimum approach speed to V_{REF} (Flaps 45 degrees) + 5 KIAS..

(16) No further action required.

– COMPLETE –

I	Challenger 605	FCTL 07-7
		REV 61, Aug 27/21

FLT SPLRS

On ground:

- (1) Do not takeoff.

– COMPLETE –

FLT SPLRS DEPLOY

Condition: Flight spoilers are deployed at low altitude or when thrust at high power.

- (1) FLIGHT SPOILERRETRACT

– COMPLETE –

GND SPLRS

Condition: Ground spoilers have failed.

Objective: Review landing configuration and penalties.

(1) Identify phase of flight:

- ➡ On ground – Go to (2)
- ➡ In flight – Go to (5)

(2) On ground:

NOTE: **GND SPLRS** may come on at taxi speeds above 16 knots.

- (3) AirplaneSTOP
- (4) Ground spoilers pre-flight check.....ACCOMPLISH
Ground spoilers are unserviceable if message stays on.

– COMPLETE –

(5) In flight:

- (6) GND SPOILERSDISARM

(7) **GND SPLRS** stays on:

- ➡ YES – Go to (8)
- ➡ NO – Go to (17)

(8) **GND SPLRS** stays on:

Prior to landing:

- (9) TAWS WARNING, FLAPS OFFPRESS IN
- (10) TAWS WARNING, RAAS OFF (if installed)PRESS IN



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Flight Controls
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Challenger 605	FCTL 07-8
	REV 63, Mar 14/22

GND SPLRS (Cont'd)

- (11) Landing flaps 20 DEGREES
- (12) Approach speed $V_{REF} + 14$
minimum prior to 100 feet AGL.
- (13) ATS DISC PRESS
prior to 100 feet AGL.

▲ CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (14) Actual landing distance INCREASE
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.60 (60%)	1.55 (55%)

NOTE: Landing distance factors for failed ground spoilers are based upon serviceable flight spoilers.

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

After touchdown:

- (15) GND SPOILERS ARM
- (16) FLIGHT SPOILER MAX

– COMPLETE –

- (17) **GND SPLRS** goes out:

Prior to landing:

- (18) Actual landing distance INCREASE
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.30 (30%)	1.25 (25%)

After touchdown:

- (19) GND SPOILERS ARM
- (20) FLIGHT SPOILER MAX

– COMPLETE –

Challenger 605	FCTL 07-9
	REV 63, Mar 14/22

GND SPLRS DEPLOY

Condition: Ground spoilers deployed with ratio altitude greater than 10 feet.

Objective: Review landing configuration and penalties.

(1) GND SPOILERSDISARM

Prior to landing:

(2) TAWS WARNING, FLAPS OFFPRESS IN

(3) TAWS WARNING, RAAS OFF (if installed)PRESS IN

(4) Landing flaps 20 DEGREES

(5) Approach speed $V_{REF} + 14$ minimum.

(6) ATS DISC PRESS prior to 100 feet AGL.

▲ CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

(7) Actual landing distance INCREASE as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.60 (60%)	1.55 (55%)

NOTE: Landing distance factors for failed ground spoilers are based upon serviceable flight spoilers.

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

After touchdown:

(8) GND SPOILERSARM to deploy ground spoilers.

(9) FLIGHT SPOILERMAX

– COMPLETE –

GND SPLRS NOT ARMED

Condition: Ground spoilers are not armed with airplane in takeoff or in landing mode.

Objective: Review penalties.

(1) GND SPOILERSARM



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Flight Controls
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Challenger 605	FCTL 07-10
	REV 63, Mar 14/22

GND SPLRS NOT ARMED (Cont'd)

- (2) GND SPLRS NOT ARMED stays on:
- ➡ YES – Go to (3)
- ➡ NO – No further action required. End of procedure.
- (3) GND SPLRS NOT ARMED stays on:
- (4) Actual landing distance INCREASE
as applicable by factor given below:
- | WITHOUT THRUST REVERSERS | WITH THRUST REVERSERS |
|--------------------------|-----------------------|
| 1.30 (30%) | 1.25 (25%) |
- NOTE: Landing distance factors for failed ground spoilers are based upon serviceable flight spoilers.
- After touchdown:
- (5) FLIGHT SPOILERMAX

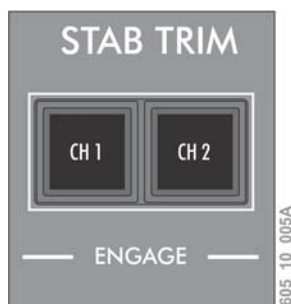
– COMPLETE –

MACH TRIM



- Condition: Mach trim has failed.
- Objective: Review penalties.
- With at least one STAB TRIM channel engaged:
- (1) MACH TRIM PRESS TO ENGAGE
- NOTE: With a Mach trim failure, do not exceed 250 KIAS (0.7M), unless the autopilot is engaged and functioning normally.

– COMPLETE –

STAB TRIM

Condition: Both pitch trim channels are off or failed. Will also disengage Mach trim.

Objective: Attempt to reset at least one stab channel.

NOTE: For stabilizer trim runaway, (See Page FCTL 07-21)

(1) STAB TRIM DISC.....PRESS, HOLD AND RELEASE

(2) STAB TRIM CH 1 and CH 2.....PRESS TO ENGAGE

(3) **STAB TRIM** stays on:

➡ **YES** – Go to (4)

➡ **NO** – Go to (9)

(4) **STAB TRIM** stays on:

(5) STAB TRIM, CH 2.....PRESS TO ENGAGE

(6) **STAB TRIM** stays on:

➡ **YES** – Go to (7)

➡ **NO** – Go to (9)

(7) **STAB TRIM** stays on:

NOTE: 1. Minimize changes to airspeed and configuration to minimize control forces or autopilot servo loads.

2. Disregard **MACH TRIM**.

(8) Go to (11)

(9) **STAB TRIM** goes out:

(10) MACH TRIM.....PRESS TO ENGAGE

– COMPLETE –

(11) Autopilot is engaged:

➡ **YES** – Go to (12)

➡ **NO** – Go to (23)



STAB TRIM (Cont'd)**(12) Autopilot is engaged:**

(13) Autopilot LEAVE ENGAGED
as long as needed to reduce workload.

(14) AP HOLDING NOSE DOWN or NOSE UP is on:➡ **YES** – Go to (15)➡ **NO** – Go to (23)**(15) AP HOLDING NOSE DOWN or NOSE UP is on:**

(16) AirspeedADJUST
as required to keep autopilot engaged as long as possible.

NOTE: 1. If **AP HOLDING NOSE DOWN** is on, decrease
airspeed. If **AP HOLDING NOSE UP** is on, increase
airspeed.

2. **AP HOLDING NOSE DOWN** or **NOSE UP** will go out
if the airspeed is adjusted.

(17) AP HOLDING NOSE DOWN or NOSE UP stays on:➡ **YES** – Go to (18)➡ **NO** – Go to (21)**(18) AP HOLDING NOSE DOWN or NOSE UP stays on:**

NOTE: Anticipate an out-of-trim condition with autopilot
disengagement.

(19) AutopilotDISCONNECT

(20) Go to (24)

(21) AP HOLDING NOSE DOWN or NOSE UP goes out:

(22) Go to (24)

(23) Autopilot is NOT engaged:**(24) Prior to landing, choose applicable:**➡ **Failure occurred with flaps at 45 degrees** – Go to
(25)➡ **Failure occurred with flaps less than 45 degrees** – Go
to (27)**(25) Failure occurred with flaps at 45 degrees:**

(26) Landing flapsLEAVE AT 45 DEGREES



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Challenger 605	FCTL 07-13
	REV 61, Aug 27/21

STAB TRIM (Cont'd)

- ▲ CAUTION:** 1. Do not dump fuel if trim runaway is in nose down direction.
- Fuel dumping will cause the CG to move forward requiring a larger pull force to counter and causing any push force to be reduced.
- Consider use of fuel dumping in the event of a nose-up trim runaway condition.
2. A landing may be made with fuel in the tail tank.

– COMPLETE –

(27) Failure occurred with flaps less than 45 degrees:

- (28) TAWS WARNING, FLAPS OFF PRESS IN
- (29) TAWS WARNING, RAAS OFF (if installed) PRESS IN
- (30) Landing flaps 20 DEGREES
- (31) Approach speed $V_{REF} + 14$ minimum.
- (32) ATS DISC PRESS prior to 100 feet AGL.

- ▲ CAUTION:** ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.
- (33) Actual landing distance INCREASE as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

- ▲ CAUTION:** 1. Do not dump fuel if trim runaway is in nose down direction.
- Fuel dumping will cause the CG to move forward requiring a larger pull force to counter and causing any push force to be reduced.
- Consider use of fuel dumping in the event of a nose-up trim runaway condition.
2. A landing may be made with fuel in the tail tank.
3. Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

STALL FAIL

Condition: One or both channels of the stall protection system have failed. Stick pusher is inoperative.

Objective: Review penalties.

- (1) STALL PROT, PUSHER
(either left or right) OFF
- (2) **Accompanied by nuisance shaker operation and/or warbler aural warning:**
 - ➔ **YES** – Go to (3)
 - ➔ **NO** – Go to (6)
- (3) **Accompanied by nuisance shaker operation and/or warbler aural warning:**
- (4) Affected STALL PROT, L CH or R
CH circuit breakers (1N5 or 4C5) PULL
- (5) Go to (7)
- (6) **NO nuisance shaker operation and/or warbler aural warning:**
- (7) **Prior to landing:**
- (8) Approach speed $V_{REF} + 10$
minimum, to maintain margin above stall.
- (9) Actual landing distance INCREASE
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.40 (40%)	1.30 (30%)

– COMPLETE –

Aileron or Rudder Trim Runaway

Condition: Uncommanded aileron or rudder trim movement.

Objective: Attempt to correct condition.

- (1) Aileron or rudder OVERPOWER
- (2) Affected trim NEUTRAL

NOTE: Use of autopilot may alleviate trim problem.

– COMPLETE –

Challenger 605	FCTL 07-15
	REV 61, Aug 27/21

Aileron System Jammed

Condition: Unable to move ailerons.

Objective: Assess controllability. Review landing configuration and penalties.

NOTE: With a jammed primary flight control, it may be necessary to reduce airspeed to improve airplane handling.

- (1) Autopilot DISCONNECT
- (2) Aileron controls..... OVERPOWER
Use aileron trim to alleviate forces on controls, if jam has occurred away from the neutral position.
- (3) **Jam continues:**
 - ➡ **YES** – Go to (4)
 - ➡ **NO** – Go to (9)
- (4) **Jam continues:**
- (5) Aileron controls..... RELEASE PRESSURE
- (6) ROLL DISC PULL AND TURN
- (7) Jammed side DETERMINE
If right side, do not re-engage autopilot.
- (8) Go to (10)
- (9) **Jam is cleared:**
- (10) Land at the nearest suitable airport.

NOTE: Controllability is reduced. Select longest runway available with minimum turbulence and crosswind.

Prior to landing:

- (11) TAWS WARNING, FLAPS OFF PRESS IN
- (12) TAWS WARNING, RAAS OFF (if installed) PRESS IN
- (13) Landing flaps 20 DEGREES
- (14) Approach speed $V_{REF} + 14$ minimum.
- (15) ATS DISC PRESS
prior to 100 feet AGL.

▲ CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Flight Controls
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Aileron System Jammed (Cont'd)

(16) Actual landing distance INCREASE
as applicable, by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

Elevator System Jammed

Condition: Unable to move elevators.

Objective: Assess controllability. Review landing configuration and penalties.

NOTE: With a jammed primary flight control, it may be necessary to reduce airspeed to improve airplane handling.

- (1) Autopilot DISCONNECT
- (2) Elevator controls..... OVERPOWER
Use stabilizer trim to compensate, if elevator jam has occurred away from the trimmed position.

- (3) **Jam continues:**
- ➡ **YES** – Go to (4)
 - ➡ **NO** – Go to (13)

- (4) **Jam continues:**
- (5) Elevator controls..... RELEASE PRESSURE
- (6) PITCH DISC PULL AND TURN

- (7) **Choose applicable:**
- ➡ **Right side is jammed** – Go to (8)
 - ➡ **Left side is jammed** – Go to (11)

- (8) **Right side jammed:**
- (9) STALL PROT, PUSHER (left or right) OFF

NOTE: Stick shaker on operative control column, operates at a lower intensity.



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Challenger 605	FCTL 07-17
	REV 61, Aug 27/21

Elevator System Jammed (Cont'd)

(10) Go to (14)

(11) Left side jammed:

NOTE: Do not re-engage autopilot.

(12) Go to (14)

(13) Jam is cleared:

(14) Land at the nearest suitable airport.

NOTE: Controllability is reduced. Select the longest runway available with minimum turbulence and crosswind.

Prior to landing:

(15) TAWS WARNING, FLAPS OFF PRESS IN

(16) TAWS WARNING, RAAS OFF (if installed) PRESS IN

(17) Landing flaps 20 DEGREES

(18) Approach speed $V_{REF} + 14$ minimum.

(19) ATS DISC PRESS prior to 100 feet AGL.

▲ CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

(20) Actual landing distance INCREASE as applicable, by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

Flight Spoiler Asymmetric Extension

Condition: Uncommanded roll during flight spoiler extension.

Objective: Review penalties.

- (1) FLIGHT SPOILERRETRACT
Do not reselect in flight.
- (2) RunwaySELECT
to minimize crosswind.
- (3) Actual landing distance INCREASE
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.30 (30%)	1.25 (25%)

– COMPLETE –

Flight Spoiler Asymmetric Retraction

Condition: Uncommanded roll during flight spoiler retraction.

Objective: Review penalties.

- (1) Lateral controlESTABLISH
using ailerons and rudder, if necessary.
- (2) FLIGHT SPOILERRETRACT/ADJUST
to eliminate roll.
- (3) RunwaySELECT
to minimize crosswind.
- (4) Approach speed $V_{REF} + 10$
minimum.
- (5) Actual landing distance INCREASE
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.50 (50%)	1.40 (40%)

– COMPLETE –

Flight Spoilers Failure to Deploy

Condition: Flight spoilers cannot be deployed.

Objective: Review penalties.

- (1) Actual landing distance INCREASE
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.30 (30%)	1.25 (25%)

– COMPLETE –

Flight Spoilers Lever Jam

Condition: Flight spoilers cannot be moved.

Objective: Review penalties.

- (1) Approach speed $V_{REF} + 10$
minimum.
- (2) Actual landing distance INCREASE
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.50 (50%)	1.40 (40%)

– COMPLETE –

Overspeed Warning



CLACKER

Condition: V_{MO}/M_{MO} has been exceeded.

Objective: Reduce speed.

- (1) Airspeed REDUCE
until less than V_{MO}/M_{MO} or V_{FE} .

– COMPLETE –

Challenger 605	FCTL 07-20
	REV 61, Aug 27/21

Rudder System Jammed

Condition: Unable to move rudder.

Objective: Assess controllability. Review landing configuration and penalties.

NOTE: With a jammed primary flight control, it may be necessary to reduce airspeed to improve airplane handling.

(1) Autopilot DISCONNECT

(2) YD DISC PRESS

(3) Rudder pedals OVERPOWER

(4) Jam continues:

➡ **YES** – Go to (5)

➡ **NO** – Go to (8)

(5) Jam continues:

(6) Straight flight MAINTAIN
using aileron and differential thrust, until touchdown.

NOTE: Approximately 20% N_1 differential thrust may be required to maintain wings-level flight using thrust alone. If available, rudder trim can be used to maintain directional control.

(7) Go to (9)

(8) Jam is cleared:

(9) Land at the nearest suitable airport.

NOTE: Select the longest runway available with minimum turbulence and crosswind.

Prior to landing:

(10) TAWS WARNING, FLAPS OFF PRESS IN

(11) TAWS WARNING, RAAS OFF (if installed) PRESS IN

(12) Landing flaps 20 DEGREES

(13) Approach speed $V_{REF} + 14$
minimum.

(14) ATS DISC PRESS
prior to 100 feet AGL.

▲CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

(15) Actual landing distance INCREASE
as applicable, by factor given below:



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Flight Controls
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Rudder System Jammed (Cont'd)

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

NOTE: After touchdown, use nose wheel steering and, if necessary, differential braking to maintain directional control, as the airplane will turn in the direction of the jammed rudder.

- COMPLETE -

Stabilizer Trim Malfunction

Condition: Unable to command stab trim to desired position with no apparent failure.

Objective: Use stabilizer trim channel 2 only.

- (1) STAB TRIM DISC..... PRESS, HOLD & RELEASE
- (2) STAB TRIM, CH 2 PRESS TO ENGAGE

NOTE: Do not engage STAB TRIM CH 1.

- (3) MACH TRIM PRESS TO ENGAGE

- COMPLETE -

Stabilizer Trim Runaway

Condition: Abnormal or uncommanded change in pitch trim and/or control forces. May be accompanied by a warning tone (drumroll).

Objective: Assume manual control. Review landing configuration and penalties.

- (1) Controls ASSUME MANUAL CONTROL and override runaway.
 - (2) Pilot or copilot's STAB TRIM DISC.....PRESS, HOLD AND RELEASE

FROM Quick Reference Card

Stabilizer Trim Runaway (Cont'd)

- NOTE:**
1. Flight spoilers will reduce airspeed and provide nose up pitch.
 2. Reducing thrust will reduce airspeed and provide nose up pitch.

(3) STAB TRIM message is out or uncommanded stabilizer motion continues:

- ➡ **YES** – Go to (4)
➡ **NO** – Go to (9)

(4) STAB TRIM message is out or uncommanded stabilizer motion continues:

- (5) STAB CH1 and CH2 HSTCU circuit breakers (2F8 and 4B8) **PULL**
(6) Airspeed **ADJUST**

- NOTE:**
1. If a pull force is required, increase airspeed. If a push force is required, decrease airspeed.
 2. Do not exceed V_{MO} / M_{MO} or minimum maneuvering speed as appropriate.
 3. Autopilot, stabilizer trim and Mach trim are not available.

- (7) Land immediately at the nearest suitable airport.
(8) Go to (12)

(9) STAB TRIM message is on or uncommanded stabilizer motion stops:

- (10) Airspeed **ADJUST**

- NOTE:**
1. If a pull force is required, increase airspeed. If a push force is required, decrease airspeed.
 2. Do not exceed V_{MO} / M_{MO} or minimum maneuvering speed as appropriate.
 3. Autopilot, stabilizer trim and Mach trim are not available.

- (11) Land immediately at the nearest suitable airport.

(12) Prior to landing:

- (13) TAWS WARNING, FLAPS OFF **PRESS IN**
(14) TAWS WARNING, RAAS OFF (if installed) **PRESS IN**



Stabilizer Trim Runaway (Cont'd)

- ▲ CAUTION:** 1. Selection of Flaps 20 will increase nose down pitching moment slightly, and therefore should be selected just prior to approach in the event of a full nose down stabilizer position. Selection of Flaps 20 significantly reduces landing field length compared to Flaps 0.
2. If control forces are excessive, some relief may be gained by adjusting airspeed or flap position. Reduced flap settings generally reduce pull forces. Increased flap settings generally reduce push forces.

- (15) Recommended landing flaps 20 DEGREES
- (16) Final approach speed $V_{REF} + \Delta V_{REF}$
for applicable flap setting.
- (17) ATS DISC PRESS
prior to 100 feet AGL.

▲ CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (18) Actual landing distance INCREASE
as applicable by factor below:

Flaps Setting (Degrees)	ΔV_{REF} (KTS)	ALD Factor	
		Without Thrust Reversers	With Thrust Reversers
0	30	1.80 (80%)	1.75 (75%)
20	14	1.55 (55%)	1.50 (50%)
30	7	1.40 (40%)	1.35 (35%)

- ▲ CAUTION:** 1. Do not dump fuel if trim runaway is in nose down direction.

Fuel dumping will cause the CG to move forward requiring a larger pull force to counter and causing any push force to be reduced.

Consider use of fuel dumping in the event of a nose-up trim runaway condition.

2. A landing may be made with fuel in the tail tank.

▲ WARNING: In the event of a go-around, increasing thrust will increase nose down forces. Increase thrust slowly to avoid excessive pitch force.

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

Challenger 605	FCTL 07-24
	REV 49, Nov 19/18

Advisory Messages

FLIGHT CONTROLS ADVISORY MESSAGES	
AILERON MON OK	All four aileron PCU failure detection sensors are functioning properly.
T/O CONFIG OK	Engines running with flaps, flight spoilers, parking brake and stabilizer trim correctly set for takeoff.

Status Messages

FLIGHT CONTROLS STATUS MESSAGES	
FLAPS MOTOR OVHT	One flap motor is hot and has shut down.
STAB CH 1 (2) INOP	Respective channel not engaged or failed; other channel engaged/not failed.

Challenger 605	FCTL 07-25
	REV 49, Nov 19/18

Power Supply and Circuit Breaker Summary

CB NAME	BUS	CB LOCATION
AIL TRIM	DC BUS 2	CBP2-F05
FLAP/TRIM IND	BATT BUS	CBP1-N06
FLAP CONT 1	DC ESS BUS	CBP4-A09
FLAP CONT 2	DC BUS 2	CBP2-F03
FLAP MOTOR 1	AC ESS BUS	CBP3-B11
FLAP MOTOR 2	AC BUS 2	CBP2-B05
GND SPLR	DC BUS 1	CBP1-F04
GND SPLR	DC BUS 2	CBP2-F04
STAB CH 1 HSTA	AC BUS 2	CBP2-C05
STAB CH 1 HSTCU	DC BUS 2	CBP2-F08
STAB CH 2 HSTA	AC ESS BUS	CBP3-A05
STAB CH 2 HSTCU	DC ESS BUS	CBP4-B08
STALL PROT L CH	BATT BUS	CBP1-N05
STALL PROT R CH	DC ESS BUS	CBP4-C05
STALL PROT STICK PUSH	BATT BUS	CBP1-N04
SURF POS IND L	AC ESS BUS	CBP3-C06
SURF POS IND R	AC BUS 2	CBP2-B11

Challenger 605	FCTL 07-26
	REV 49, Nov 19/18

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TRAINING PURPOSES ONLY

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Flight Controls
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Challenger 605	F-LDG TOC-1
	REV 49, Nov 19/18

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TRAINING PURPOSES ONLY

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Forced Landing / Evacuation
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Challenger 605	<i>F-LDG TOC-2</i>
	<i>REV 49, Nov 19/18</i>

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TRAINING PURPOSES ONLY

<i>Non-normal procedures</i> <i>PSP 605-15 – QRH-I Vol. 2</i>	<i>Forced Landing / Evacuation</i>
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Challenger 605	F-LDG 08-1
	REV 51, Apr 05/19

Emergency Evacuation

Condition: Continued occupation of the airplane is unsafe.

Objective: Properly secure and safely exit the airplane in a timely manner.

Pilot:

- (1) PARKING BRAKESET
- (2) Evacuation.....COMMAND
- (3) GND SPOILERSDISARM
- (4) FLIGHT SPOILER.....RETRACT
- (5) Thrust levers.....SHUT OFF
- (6) APU, LH and RH ENG FIRE
PUSHPRESS IN
- (7) BATT MASTER OFF
to prevent CVR erasure.

Copilot: On EVACUATION command

- (8) ATC NOTIFY
of condition and intention to evacuate.
- (9) Both PACKsPRESS OUT
- (10) EMER DEPRESSPRESS IN
- (11) EMER LTS ON

▶

FROM Quick Reference Card

◀

- (12) Exits..... OPEN
- (13) Passenger evacuation.....ASSIST
and direct passengers away from airplane.
- (14) AirplaneABANDON
by any suitable exit.

– COMPLETE –

Challenger 605	F-LDG 08-2
	REV 51, Apr 05/19

Ditching or Forced Landing Imminent

Condition: Unplanned off field landing at low altitude.

Objective: Complete critical steps and configuration prior to impact.

PRELIMINARY

- (1) Both PACKsPRESS OUT
(2) EMER DEPRESSPRESS IN
prior to 200 feet AGL.

FROM Quick Reference Card

JUST BEFORE CONTACT

- (3) Thrust levers.....SHUTOFF
(4) APU, LH and RH ENG FIRE
PUSHPRESS IN
(5) Water (or terrain) CONTACT
with minimum forward speed, but not less than stick shaker
speed, and at minimum sink rate.

AFTER CONTACT

When the airplane has stopped:

- (6) ENG and APU BOTTLE ARMED
(3)PRESS
and hold to discharge.
(7) Door and overwing exit..... OPEN
(8) Evacuation.....INITIATE
(9) BATT MASTER OFF
to prevent CVR erasure.
(10) AirplaneABANDON
with survival equipment.

– COMPLETE –

Forced Landing

Condition: Off field landing with altitude/time available.

Objective: Prepare cabin and configuration.

PRELIMINARY

- (1) Descent PLAN
(2) Crew ALERT & BRIEF
(3) ATC NOTIFY
(4) Transponder7700



Challenger 605	F-LDG 08-3
	REV 51, Apr 05/19

Forced Landing (Cont'd)

- (5) MFD or CDU/TCASSET
- (6) ELT (if installed) ON
- (7) Both PASS SIGNS ON
- (8) Loose equipment.....SECURE
- (9) TAWS circuit breaker (1B13).....PULL
- (10) AUDIO WARNING DCU 1 and 2 DISABLE
- (11) Survival equipment.....CHECK
- (12) Crew harness and belts..... ON & TIGHT
- (13) Shoulder harness lock.....LOCK
- (14) LDG GEAR.....AS REQUIRED
- (15) FLAPSAS REQUIRED

APPROACH

At approximately 2,000 feet:

- (16) Landing area conditions and wind direction DETERMINE
- (17) Descent rate/approach speed ESTABLISH
- (18) Both PACKsPRESS OUT
- (19) PRESS CONTMANUAL
- (20) MAN ALTHOLD UP
- (21) MAN RATEMAX INCR

When completely depressurized:

- (22) L and R 10TH STAGE bleedsPRESS OUT
- (23) L and R BOOST PUMPsPRESS OUT
- (24) APU (if not essential)..... SHUTDOWN

At 500 feet (Radio Altitude):

- (25) PA.....ALERT
Ditching imminent.
- (26) Radio TRANSMIT
final position.
- (27) BRACE FOR IMPACTORDER
“BRACE – BRACE – BRACE”.
- (28) EMER LTS ON
- (29) LANDING LTS ON

JUST BEFORE CONTACT

- (30) Thrust levers.....SHUTOFF



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Forced Landing / Evacuation
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Challenger 605	F-LDG 08-4
	REV 51, Apr 05/19

Forced Landing (Cont'd)

- (31) APU, LH and RH ENG FIRE
PUSH PRESS IN
- (32) Terrain CONTACT
with minimum forward speed, but not less than stick shaker
speed, and at minimum sink rate.

AFTER CONTACT

When the airplane has stopped:

- (33) ENG and APU BOTTLE ARMED
(3) PRESS
and hold to discharge.
- (34) Door and overwing exit OPEN
- (35) Evacuation INITIATE
- (36) BATT MASTER OFF
to prevent CVR erasure.
- (37) Airplane ABANDON
with survival equipment.

– COMPLETE –

Planned Ditching

Condition: Off field water landing with altitude/time available.

Objective: Prepare cabin and configuration.

PRELIMINARY

- (1) Descent PLAN
- (2) Crew ALERT & BRIEF
- (3) ATC NOTIFY
- (4) Transponder 7700
- (5) ELT (if installed) ON
- (6) MFD or CDU/TCAS SET
- (7) Both PASS SIGNS ON
- (8) Loose equipment SECURE
- (9) TAWS circuit breaker (1B13) PULL
- (10) AUDIO WARNING DCU 1 and 2 DISABLE
- (11) Survival equipment CHECK
- (12) Life vest, harness and belts ON & TIGHT
- (13) Shoulder harness lock LOCK
- (14) LDG GEAR UP



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Forced Landing / Evacuation
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Challenger 605	F-LDG 08-5
	REV 51, Apr 05/19

Planned Ditching (Cont'd)

(15) FLAPSAS REQUIRED

APPROACH

At approximately 2,000 feet:

(16) Sea conditions and wind direction DETERMINE

(17) Ditching headingESTABLISH

- Wind speed less than 15 knots: contact parallel to swells.
- Wind speed 15 to 45 knots: compromise between wind and swell.
- Wind speed greater than 45 knots: ditch into wind.

(18) Descent rate/approach speed ESTABLISH

(19) Both PACKsPRESS OUT

(20) PRESS CONTMANUAL

(21) MAN ALTHOLD UP

(22) MAN RATEMAX INCR

When completely depressurized:

(23) L and R 10TH STAGE bleedsPRESS OUT

(24) MAN ALTHOLD DN
to close both outflow valves.

(25) L and R BOOST PUMPsPRESS OUT

(26) APU (if not essential) SHUTDOWN

At 500 feet (Radio Altitude):

(27) PAALERT
Ditching imminent.

(28) Radio TRANSMIT
final position.

(29) BRACE FOR IMPACTORDER
“BRACE – BRACE – BRACE”.

(30) EMER LTS ON

(31) LANDING LTS ON

JUST BEFORE CONTACT

(32) Thrust leversSHUTOFF

(33) APU, LH and RH ENG FIRE
PUSHPRESS IN

(34) Water CONTACT
with minimum forward speed, but not less than stick shaker
speed, and at minimum sink rate.

AFTER WATER CONTACT



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Forced Landing / Evacuation
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Challenger 605	F-LDG 08-6
	REV 51, Apr 05/19

Planned Ditching (Cont'd)

When the airplane has stopped:

- (35) ENG and APU BOTTLE ARMED
(3) PRESS
and hold to discharge.

(36) Door and overwing exit..... OPEN

(37) Evacuation..... INITIATE

(38) BATT MASTER OFF
to prevent CVR erasure.

(39) Airplane ABANDON
with survival equipment.

– COMPLETE –

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TRAINING PURPOSES ONLY

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Challenger 605	FUEL 09-1
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AUTO TAIL XFER

Condition: Tail tank automatic fuel transfer has failed.

Objective: Manually transfer tail tank fuel.

NOTE: If a level sensor fails during fuel transfer, tail tank to auxiliary tank fuel transfer may continue after the **AUTO TAIL XFER** is on.

(1) Identify phase of flight:

- ➡ **On ground** – Go to (2)
- ➡ **In flight** – Go to (6)

(2) On ground:

After AC power is applied:

(3) FUEL FSCU CH 1 and CH 2 cbs
(1M11 and 4A10)..... OPEN

After 3 minutes:

(4) FUEL FSCU CH 1 and CH 2 cbs
(1M11 and 4A10)..... CLOSE

(5) AUTO TAIL XFER stays on:

- ➡ **YES** – Report for maintenance. **End of procedure.**
- ➡ **NO** – No further action required. **End of procedure.**

(6) In flight:

(7) Tail tank fuel quantity CHECK

(8) Tail tank quantity is less than 90 lb and AUTO TAIL XFER goes out within 2 minutes:

- ➡ **YES** – No further action required. **End of procedure.**
- ➡ **NO** – Go to (9)

(9) Tail tank quantity is 90 lb or greater and AUTO TAIL XFER stays on for more than 2 minutes:

(10) TAIL TANK TRANSFER,
MANUAL OVRD MANUAL

(11) Tank quantities MONITOR
Check AUX tank quantity is less than 7,000 lb (3,175 kg) .

(12) PRI or SEC TAIL TANK
TRANSFER PRESS IN
to start applicable pump.

(13) Tank quantities MONITOR
Check that tail quantity is decreasing.

When tail tank is empty:



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AUTO TAIL XFER (Cont'd)

(14) PRI or SEC TAIL TANK
TRANSFERPRESS OUT

NOTE: Do not deselect manual transfer until after landing.

– COMPLETE –

AUTO TAIL XFER INHIB

Condition: Auto tail tank fuel transfer is inhibited.

Objective: Manually transfer tail tank fuel.

(1) **Engine failure is suspected:**

➡ **YES** – Go to (2)

➡ **NO** – Go to (5)

(2) **Engine failure is suspected:**

NOTE:

1. Fuel dumping must only be carried out with flaps at 0 degrees.
2. Fuel dumping in known lightning conditions is prohibited.
3. Tail tank contents will deplete from full to empty, in approximately 30 minutes.

(3) FUEL, DUMPPRESS IN

(4) Tank quantitiesMONITOR
Check that tail tank quantity is decreasing.

NOTE: Reduce tail tank fuel quantity to 1,400 lb (635 kg).
If auxiliary tank quantity is less than 3,000 lb (1,361 kg),
dump fuel until tail tank is empty.

– COMPLETE –

(5) **No engine failure condition:**

(6) TAIL TANK TRANSFER,
MANUAL OVRDMANUAL

(7) Tank quantitiesMONITOR
Check AUX tank quantity is less than 7,000 lb (3,175 kg).

(8) PRI or SEC TAIL TANK
TRANSFERPRESS IN

(9) Tank quantitiesMONITOR
Check that fuel transfers from tail tank to AUX tank.

When tail tank is empty:



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Challenger 605	FUEL 09-3
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AUTO TAIL XFER INHIB (Cont'd)

(10) PRI or SEC TAIL TANK
TRANSFERPRESS OUT

– COMPLETE –

AUX TANK HEAVY

Condition: Excessive AUX tank to tail tank fuel ratio.

Objective: Manually control tail tank fuel.

→ If fuel dumping is in progress:

- FUEL, DUMPPRESS OUT
- Tank quantitiesMONITOR
Check that tail tank quantity is not decreasing.

(1) **AUX TANK HEAVY** stays on:

- ➡ **YES** – Go to (2)
- ➡ **NO** – No further action required. **End of procedure.**

(2) **AUX TANK HEAVY** stays on:

(3) TAIL TANK TRANSFER,
MANUAL OVRDMANUAL

(4) Tank quantitiesMONITOR
Check AUX tank quantity is less than 7,000 lb (3,175 kg).

(5) PRI and SEC TAIL TANK
TRANSFERPRESS OUT

(6) Tank quantitiesMONITOR
Check that AUX tank quantity is decreasing, and tail tank
fuel quantity is not decreasing.

NOTE: If AUX tank is not decreasing, AUX and tail tank fuel is unusable.

(7) **AUX TANK HEAVY** is still on:

- ➡ **YES** – Go to (8)
- ➡ **NO** – No further action required. **End of procedure.**

(8) **AUX TANK HEAVY** is still on:

Prior to landing:

(9) TAWS WARNING, FLAPS OFFPRESS IN

(10) TAWS WARNING, RAAS OFF (if
installed)PRESS IN



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Fuel
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Challenger 605	FUEL 09-4
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AUX TANK HEAVY (Cont'd)

- (11) Landing flaps 20 DEGREES
- (12) Approach speed $V_{REF} + 14$ minimum.
- (13) ATS DISC PRESS prior to 100 feet AGL.

▲ CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (14) Actual landing distance INCREASE as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

BULK FUEL TEMP

Condition: Fuel temperature in the left main tank is colder than -37°C .

Objective: Operate in warmer air mass.

- (1) **Identify phase of flight:**

- ➡ **On ground** – Go to (2)
- ➡ **In flight** – Go to (5)

- (2) **On ground:**

- (3) APU/Engines DO NOT START if fuel temperature is colder than the specified limits for fuel type.

- (4) Airplane DO NOT TAKEOFF if anticipated air mass temperature is colder than the specified limit for fuel type.

– COMPLETE –

- (5) **In flight:**

- (6) Descend or deviate to warmer air mass, and/or increase airspeed as required.

– COMPLETE –

Challenger 605	FUEL 09-5
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DUMP VALVE OPEN

Condition: Fuel dump valve open more than 5 minutes.

- (1) Tank quantitiesMONITOR
- (2) FUEL, DUMPAS REQUIRED

NOTE: Tail tank contents will decrease from full to empty in approximately 30 minutes.

– COMPLETE –

L (R) ENG SOV

Condition: Indicated engine fuel shutoff valve is not in commanded position.

Objective: Complete engine shutdown as required.

(1) **Choose applicable:**

- ➡ **L (R) ENG SOV is on with associated ENG FIRE PUSH pressed in** – Go to (2)
- ➡ **L (R) ENG SOV is on with associated ENG FIRE PUSH pressed OUT** – Go to (5)

(2) **L (R) ENG SOV is on with associated ENG FIRE PUSH pressed in:**

- (3) Affected BOOST PUMP CONFIRM PRESSED OUT
- (4) Land at the nearest suitable airport.

– COMPLETE –

(5) **L (R) ENG SOV is on with associated ENG FIRE PUSH pressed OUT:**

- (6) Engine instruments CONFIRM ENGINE FAILURE
using N_1 , fuel flow, ITT & N_2 .

(7) **Engine failure is confirmed:**

- ➡ **YES** – Go to (8)
- ➡ **NO** – No further action required. **End of procedure.**

(8) **Engine failure is confirmed:**

- (9) In-Flight Engine Failure/ShutdownACCOMPLISH
(See Page ENG 05–10)

– COMPLETE –

Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Fuel
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Challenger 605	FUEL 09-6
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L (R) FUEL FILTER

Condition: Indicated fuel filter is clogged and imminent bypass.

(1) Affected engineMONITOR engine indication and performance.

– COMPLETE –

FUEL IMBALANCE

Condition: Fuel imbalance exists between left and right wing tanks.

Objective: Determine cause and correct fuel imbalance.

▲ CAUTION: If **FUEL IMBALANCE** is on:

- Do not taxi until imbalance is rectified.
- Do not land until imbalance is rectified, unless a greater emergency exists.

NOTE:

1. The maximum permissible fuel imbalance for taxi, takeoff and landing is 400 lb (182 kg);
2. The maximum permissible fuel imbalance in flight is 800 lb (363 kg).

(1) Fuel distributionMONITOR

Affected high tank:

(2) L (R) TO AUX XFLOWPRESS IN

(3) Tank quantitiesMONITOR

When tank quantities have equalized:

(4) L or R TO AUX XFLOWPRESS OUT

(5) Fuel imbalance persists:

- ➡ **YES** – Go to (6)
- ➡ **NO** – No further action required. **End of procedure.**

(6) Fuel imbalance persists:

(7) GRAVITY/XFLOWPRESS IN

NOTE:

1. A steady heading sideslip will accelerate gravity crossflow operation. Fly one wing low slip (on the side with less fuel) to ensure proper operation and to accelerate gravity crossflow.
2. During uncoordinated flight, fuel indications are inaccurate. Accurate fuel indications will be obtained after 30 seconds when level and stabilized in coordinated flight.



Challenger 605	FUEL 09-7
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FUEL IMBALANCE (Cont'd)

On the low quantity side:

- (8) Affected engineREDUCE THRUST
- (9) Tank quantitiesMONITOR

When main tank quantities have equalized:

- (10) GRAVITY/XFLOWPRESS OUT
- (11) Tank quantitiesMONITOR

(12) Imbalance persists (fuel leak from wing tank is suspected):

- ➡ **YES** – Go to (13)
- ➡ **NO** – No further action required. **End of procedure.**

(13) Imbalance persists (fuel leak from wing tank is suspected):

- (14) Fuel Leak ProcedureACCOMPLISH
(See Page FUEL 09–19)

– COMPLETE –

L (R) FUEL LO PRESS

Condition: Indicated fuel pressure is low.

Objective: Monitor possible fuel leaks.

(1) Associated L (R) MAIN EJECTOR FAIL and/or L (R) SCAV EJECTOR is on:

- ➡ **YES** – Go to (2)
- ➡ **NO** – Go to (13)

(2) Associated L (R) MAIN EJECTOR FAIL and/or L (R) SCAV EJECTOR is on:

- (3) Associated engine fuel flow indicationCHECK
- (4) Associated wing tank fuel quantityCHECK

(5) Affected fuel tank quantity is decreasing abnormally:

- ➡ **YES** – Go to (6)
- ➡ **NO** – Go to (11)

(6) Affected fuel tank quantity is decreasing abnormally:

- (7) Associated BOOST PUMPPRESS OUT



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L (R) FUEL LO PRESS (Cont'd)

(8) Fuel quantity continues to decrease abnormally:

➡ YES – Go to (9)

➡ NO – Go to (11)

(9) Fuel quantity continues to decrease abnormally:

(10) Fuel Leak Procedure ACCOMPLISH

(See Page FUEL 09–19)

– COMPLETE –

(11) Affected fuel tank quantity is decreasing normally:

(12) Affected engine instruments MONITOR

– COMPLETE –

(13) Associated L (R) MAIN EJECTOR FAIL and L (R) SCAV EJECTOR are out:

(14) Affected engine instruments MONITOR

– COMPLETE –

L (R) FUEL LO TEMP

Condition: Fuel temperature of the indicated engine is low.

(1) Identify phase of flight:

➡ On ground – Go to (2)

➡ In flight – Go to (4)

(2) On ground:

(3) Do not attempt takeoff.

– COMPLETE –

(4) In flight:

(5) Associated engine instruments MONITOR

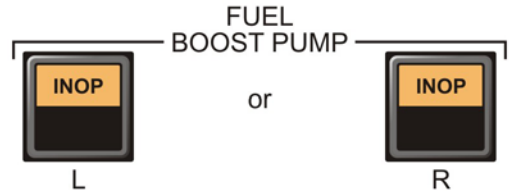
NOTE: Under these conditions, the fuel feed to the engine is not preheated. L (R) FUEL FILTER may come on, indicating an impending bypass.

– COMPLETE –

Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Fuel
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Challenger 605	FUEL 09-9
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L (R) FUEL PUMP



Condition: Indicated fuel pump has failed.

Objective: Correct condition.

- (1) Indicated BOOST PUMPPRESS OUT
then PRESS back IN.
- (2) Indicated **L (R) FUEL PUMP** and BOOST PUMP INOP light stay on:
 - ➡ **YES** – Go to (3)
 - ➡ **NO** – No further action required. **End of procedure.**
- (3) Indicated **L (R) FUEL PUMP** and BOOST PUMP INOP light stay on:
- (4) Indicated engine instrumentsMONITOR

– COMPLETE –

FUEL XFLOW SOV OPEN

Condition: Left or right to AUX crossflow valve is open while fuel balance is normal.

Objective: Monitor fuel balance.

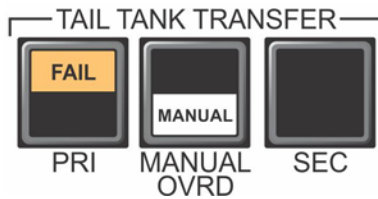
- (1) Tank quantitiesMONITOR

When main tanks are equalized:

- (1) L (R) TO AUX XFLOWPRESS OUT

– COMPLETE –

PRI TAIL XFER PUMP



Condition: Primary pump has failed during manual fuel transfer.

Objective: Reduce tail tank fuel quantity.

- (1) SEC TAIL TANK TRANSFERPRESS IN
- (2) PRI TAIL TANK TRANSFER.....PRESS OUT
- (3) Tank quantitiesMONITOR
Secondary transfer pump should maintain tail and auxiliary tank quantities normal (i.e., tail tank is decreasing).

(4) Transfer is inoperative:

- ➡ **YES** – Go to (5)
- ➡ **NO** – Go to (9)

(5) Transfer is inoperative:

- (6) Airplane attitudeLEVEL FLIGHT
if practical.

NOTE:

1. Fuel dumping must only be carried out with flaps at 0 degrees.
2. Fuel dumping in known lightning conditions is prohibited.
3. Tail tank contents will deplete from full to empty, in approximately 30 minutes.

- (7) FUEL, DUMP.....PRESS IN
- (8) Tank quantitiesMONITOR
Check that tail tank quantity is decreasing.

NOTE: Continue to dump fuel until level flight is achieved. In level flight, continue to dump fuel until tail tank quantity is 1,400 lb (635 kg). If auxiliary tank quantity is less than 3,000 lb (1,361kg), dump fuel until tail tank is empty.

– COMPLETE –

(9) Fuel is transferring:

When tail tank is empty:

- (10) SEC TAIL TANK TRANSFERPRESS OUT

– COMPLETE –

Challenger 605	FUEL 09-11
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L (R) SCAV EJECTOR

Condition: Indicated scavenge ejector output pressure is low.

Objective: Monitor for possible loss of motive flow or fuel leak.

(1) Associated L (R) MAIN EJECTOR FAIL message is on:

➡ **YES** – Go to (2)

➡ **NO** – Go to (4)

(2) Associated L (R) MAIN EJECTOR FAIL message is on:

(3) Motive Flow Failure Procedure.....ACCOMPLISH

(See Page FUEL 09–26)

– COMPLETE –

(4) Associated L (R) MAIN EJECTOR FAIL message is out:

(5) Fuel quantity in either wing tank is less than 500 lb (230 kg):

➡ **YES** – Go to (6)

➡ **NO** – No further action required. **End of procedure.**

(6) Fuel quantity in either wing tank is less than 500 lb (230 kg):

**(7) Airplane attitude5 DEGREES NOSE UP
maximum.**

(8) Tank quantitiesMONITOR

(9) Total fuel quantity is decreasing abnormally:

➡ **YES** – Go to (10)

➡ **NO** – No further action required. **End of procedure.**

(10) Total fuel quantity is decreasing abnormally:

(11) Affected BOOST PUMPPRESS OUT

(12) Fuel quantity continues to decrease abnormally:

➡ **YES** – Go to (13)

➡ **NO** – No further action required. **End of procedure.**

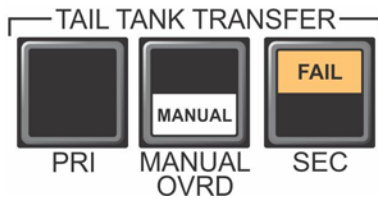
(13) Fuel quantity continues to decrease abnormally:

(14) Fuel Leak ProcedureACCOMPLISH

(See Page FUEL 09–19)

– COMPLETE –

SEC TAIL XFER PUMP



Condition: Secondary pump has failed during manual fuel transfer.

Objective: Reduce tail tank fuel quantity.

- (1) PRI TAIL TANK TRANSFER.....PRESS IN
- (2) SEC TAIL TANK TRANSFERPRESS OUT
- (3) Tank quantitiesMONITOR
Primary transfer pump should maintain tail and auxiliary tank quantities normal (i.e., tail tank is decreasing).

(4) Transfer is inoperative:

- ➡ **YES** – Go to (5)
- ➡ **NO** – Go to (9)

(5) Transfer is inoperative:

- NOTE:**
1. Fuel dumping must only be carried out with flaps at 0 degrees.
 2. Fuel dumping in known lightning conditions is prohibited.
 3. Tail tank contents will deplete from full to empty, in approximately 30 minutes.

- (6) Airplane attitudeLEVEL FLIGHT
- (7) FUEL, DUMPPRESS IN
- (8) Tank quantitiesMONITOR
Check that tail tank quantity is decreasing.

NOTE: Continue to dump fuel until level flight is achieved. In level flight, continue to dump fuel until tail tank quantity is 1,400 lb (635 kg). If auxiliary tank quantity is less than 3,000 lb (1,361kg), dump fuel until tail tank is empty.

– COMPLETE –

(9) Fuel is transferring:

When tail tank is empty:

- (10) PRI TAIL TANK TRANSFER.....PRESS OUT

– COMPLETE –

Challenger 605	FUEL 09-13
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TAIL DUMP SOV

Condition: Tail dump shutoff valve is not in commanded position.

Objective: Monitor for possible fuel leaks.

(1) **Choose applicable:**

- ➡ **FUEL DUMP is pressed in** – Go to (2)
- ➡ **FUEL DUMP is pressed OUT** – Go to (4)

(2) **FUEL DUMP is pressed in:**

(3) Tank quantitiesMONITOR
Check that tail tank quantity is decreasing.

→ **If tail tank quantity is not decreasing:**

- Land at the nearest suitable airport.

– COMPLETE –

(4) **FUEL DUMP is pressed OUT:**

(5) Tank quantitiesMONITOR
Check that tail tank is not leaking.

– COMPLETE –

TAIL TANK HEAVY

Condition: Excessive tail tank to AUX tank fuel ratio.

Objective: Reduce tail tank fuel quantity.

(1) **TAIL TANK TRANSFER AUTO INHIB light is out:**

- ➡ **YES** – Go to (2)
- ➡ **NO** – Go to (12)

(2) **TAIL TANK TRANSFER AUTO INHIB light is out:**

(3) TAIL TANK TRANSFER,
MANUAL OVRDMANUAL

(4) Tank quantitiesMONITOR
Check AUX tank quantity is less than 7,000 lb (3,175 kg).

(5) PRI or SEC TAIL TANK
TRANSFERPRESS IN

(6) Tank quantitiesMONITOR
Check that tail tank quantity is decreasing.



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TAIL TANK HEAVY (Cont'd)

(7) Tail tank fuel quantity is decreasing and **TAIL TANK HEAVY** goes out:

➡ **YES** – Go to (8)

➡ **NO** – Go to (10)

(8) Tail tank fuel quantity is decreasing and **TAIL TANK HEAVY** goes out:

When tail tank is empty:

(9) PRI or SEC TAIL TANK
TRANSFERPRESS OUT

– COMPLETE –

(10) Tail tank fuel quantity is not decreasing and **TAIL TANK HEAVY** stays on:

(11) Go to (13)

– COMPLETE –

(12) **TAIL TANK TRANSFER AUTO INHIB light is on:**

(13) FUEL, DUMPPRESS IN

- NOTE:**
1. Fuel dumping must only be carried out with flaps at 0 degrees.
 2. Fuel dumping in known lightning conditions is prohibited.
 3. Tail tank contents will deplete from full to empty, in approximately 30 minutes.

(14) Tank quantitiesMONITOR
Check that tail tank fuel quantity is decreasing.

NOTE: Reduce tail tank fuel quantity to 1,400 lb (635 kg).
If auxiliary tank quantity is less than 3,000 lb (1,361 kg),
dump fuel until tail tank is empty.

– COMPLETE –

TAIL XFER SOV

Condition: Tail transfer shutoff valve is not in commanded position.

Objective: Manually control tail tank fuel.

(1) Tank quantitiesMONITOR

(2) TAIL TANK TRANSFER,
MANUAL OVRDMANUAL



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TAIL XFER SOV (Cont'd)

(3) Tank quantitiesMONITOR
Check AUX tank quantity is less than 7,000 lb (3,175 kg).

(4) **AUX tank fuel quantity is greater than 7,000 lb (3,175 kg):**
 ➡ **YES** – Go to (5)
 ➡ **NO** – Go to (8)

(5) **AUX tank fuel quantity is greater than 7,000 lb (3,175 kg):**

(6) PRI and SEC TAIL TANK
TRANSFERPRESS OUT

(7) Tank quantitiesMONITOR

– COMPLETE –

(8) **AUX tank fuel quantity is less than 7,000 lb (3,175 kg):**

Transfer excessive tail tank fuel:

(9) PRI or SEC TAIL TANK
TRANSFERPRESS IN

(10) Tank quantitiesMONITOR

When tail tank is empty:

(11) PRI or SEC TAIL TANK
TRANSFERPRESS OUT

– COMPLETE –

TAIL XFER SOV CLSD

Condition: Tail transfer shutoff valve not fully opened while tail tank transfer pump pressure is normal.

Objective: Manually control tail tank fuel.

(1) Tank quantitiesMONITOR
Check that tail tank quantity is not decreasing.

(2) TAIL TANK TRANSFER,
MANUAL OVRDMANUAL

(3) Tank quantitiesMONITOR
Check AUX tank quantity is less than 7,000 lb (3,175 kg).

(4) PRI TAIL TANK TRANSFERPRESS IN

(5) **TAIL XFER SOV CLSD stays on:**

➡ **YES** – Go to (6)

➡ **NO** – Go to (17)



Challenger 605	FUEL 09-16
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TAIL XFER SOV CLSD (Cont'd)

(6) TAIL XFER SOV CLSD stays on:

- (7) SEC TAIL TANK TRANSFERPRESS IN
- (8) PRI TAIL TANK TRANSFER.....PRESS OUT
- (9) Tank quantitiesMONITOR
Check that tail tank fuel quantity is decreasing.

(10) Transfer is inoperative:

- ➡ YES – Go to (11)
- ➡ NO – Go to (15)

(11) Transfer is inoperative:

- (12) Airplane attitudeLEVEL FLIGHT
if practical.

NOTE:

1. Fuel dumping must only be carried out with flaps at 0 degrees.
2. Fuel dumping in known lightning conditions is prohibited.
3. Tail tank contents will deplete from full to empty, in approximately 30 minutes.

- (13) FUEL, DUMPPRESS IN
- (14) Tank quantitiesMONITOR
Check that tail tank quantity is decreasing.

NOTE: Continue to dump fuel until level flight is achieved. In level flight, continue to dump fuel until tail tank quantity is 1,400 lb (635 kg). If auxiliary tank quantity is less than 3,000 lb (1,361 kg), dump fuel until tail tank is empty.

– COMPLETE –

(15) Fuel transfer resumes:

When tail tank is empty:

- (16) PRI or SEC TAIL TANK
TRANSFERPRESS OUT

– COMPLETE –

(17) TAIL XFER SOV CLSD goes out:

When tail tank is empty:

- (18) PRI or SEC TAIL TANK
TRANSFERPRESS OUT

– COMPLETE –

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Challenger 605	FUEL 09-17
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TAIL XFER SOV OPEN

Condition: Tail transfer shutoff valve not fully closed while tail tank transfer pump pressure is low.

Objective: Manually control tail tank fuel.

- (1) Tank quantitiesMONITOR
- (2) TAIL TANK TRANSFER,
MANUAL OVRDMANUAL
- (3) Tank quantitiesMONITOR
Check AUX tank quantity is less than 7,000 lb (3,175 kg).
- (4) **AUX tank fuel is greater than 7,000 lb (3,175 kg):**
 - ➡ **YES** – Go to (5)
 - ➡ **NO** – Go to (8)
- (5) **AUX tank fuel is greater than 7,000 lb (3,175 kg):**
- (6) PRI and SEC TAIL TANK
TRANSFERPRESS OUT
- (7) Tank quantitiesMONITOR

– COMPLETE –

- (8) **AUX tank fuel is less than 7,000 lb (3,175 kg):**

Transfer excessive tail tank fuel quantity:

- (9) PRI or SEC TAIL TANK
TRANSFERPRESS IN
- (10) Tank quantitiesMONITOR

When tail tank is empty:

- (11) PRI or SEC TAIL TANK
TRANSFERPRESS OUT

– COMPLETE –

AUX Tank Contents Increasing Abnormally (during Left or Right TO AUX Crossflow)

Condition: Lateral fuel imbalance.

- (1) Tank quantitiesMONITOR
Verify that lateral imbalance is corrected.
- (2) L or R TO AUX XFLOWPRESS OUT

NOTE: Auxiliary tank content may increase during Left or Right TO AUX crossflow depending upon fuel level.

– COMPLETE –

Challenger 605	FUEL 09-18
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Fuel Crossflow Valve Fails to Close

Condition: Fuel crossflow continues after L or R TO AUX crossflow valve commanded closed.

Objective: On ground: correct condition before takeoff. In flight: control fuel balance.

(1) Identify phase of flight:

- ➡ **On ground** – Go to (2)
- ➡ **In flight** – Go to (4)

(2) On ground:

→ **If fuel imbalance is greater than 400 lb (182 kg):**

- Do not taxi.

(3) Report for maintenance.

– COMPLETE –

(4) In flight:

(5) Fuel imbalance is greater than 800 lb (363 kg):

- ➡ **YES** – Go to (6)
- ➡ **NO** – No further action required. **End of procedure.**

(6) Fuel imbalance is greater than 800 lb (363 kg):

- (7) Airplane attitude STRAIGHT & LEVEL
- (8) GRAVITY XFLOW PRESS IN
- (9) Tank quantities MONITOR

When main tanks have equalized:

- (10) GRAVITY XFLOW PRESS OUT
- (11) Tank quantities MONITOR

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Fuel
--	------

Challenger 605	FUEL 09-19
	REV 51, Apr 05/19

Fuel Leak

Condition: Fuel leak is evident, suspected or confirmed.

Objective: Attempt to minimize leak and land.

(1) Diversion may be required.

NOTE: If visibility permits, a visual check from the cabin may enable identification of the leak source.

Leak is confirmed or suspected:

(2) Land immediately at the nearest suitable airport.

- ▲CAUTION:** 1. Do not delay landing while attempting to determine location of leak.
2. Expedite landing if tank quantity gauge indications are amber.

NOTE: 1. The minimum fuel quantity for go-around is 500 lb (230 kg) per wing (with the airplane level), and assuming a maximum airplane climb attitude of 10 degrees nose up.

2. Do not action **FUEL IMBALANCE**, **L (R) FUEL LO PRESS**, **L (R) SCAV EJECTOR** or **L (R) MAIN EJECTOR FAIL**. Disregard fuel imbalance limitations.

(3) AutopilotMONITOR

NOTE: Anticipate out of trim situation when disconnecting autopilot.

(4) Fuel leak from engine is suspected:

- ➡ **YES** – Go to (5)
- ➡ **NO** – Go to (10)

(5) Fuel leak from engine is suspected:

On low quantity side:

- (6) Thrust lever CONFIRM AND IDLE
- (7) Thrust lever CONFIRM AND SHUT OFF
- I** (8) Affected ENG FIRE PUSH PRESS IN
- (9) In-Flight Engine Failure/Shutdown ACCOMPLISH
- (See Page ENG 05–10)

– COMPLETE –

(10) Fuel leak from wing tank is suspected:

(11) GRAVITY/XFLOW PRESS OUT



Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Fuel
--	------

Challenger 605	FUEL 09-20
	REV 49, Nov 19/18

Fuel Leak (Cont'd)

(12) Range is a consideration:

- ➡ **YES** – Go to (13)
- ➡ **NO** – Go to (29)

(13) Range is a consideration:

NOTE: Consider use of asymmetric thrust to use as much fuel as possible from the leaking tank.

(14) Choose applicable, leak confirmed in:

- ➡ **AUX tank** – Go to (15)
- ➡ **Tail tank** – Go to (19)
- ➡ **Wing tank** – Go to (23)
- ➡ **Unable to determine** – Go to (32)

(15) Leak confirmed in AUX tank:

(16) TAIL TANK TRANSFER,
MANUAL OVRDMANUAL

(17) Auxiliary tank quantityMONITOR

When auxiliary tank quantity has stabilized:

(18) PRI TAIL TANK TRANSFER.....PRESS IN
Ensure auxiliary and tail tank quantities remain within limits.

– COMPLETE –

(19) Leak confirmed in tail tank:

(20) TAIL TANK TRANSFER,
MANUAL OVRDMANUAL

(21) PRI and SEC TAIL TANK
TRANSFER.....PRESS IN
Ensure auxiliary and tail tank quantities remain within limits.

When **AUX TANK HEAVY** comes on:

(22) PRI and SEC TAIL TANK
TRANSFER.....PRESS OUT

– COMPLETE –



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Fuel
--	------

Challenger 605	FUEL 09-21
	REV 49, Nov 19/18

Fuel Leak (Cont'd)

(23) Leak confirmed in wing tank:

(24) Affected L (R) TO AUX XFLOWPRESS IN

NOTE: Anticipate increasing lateral trim and control forces.

When trim or control forces become excessive:

(25) L (R) TO AUX XFLOWPRESS OUT

(26) Repeat steps (24) and (25), as necessary to conserve as much fuel as possible from the leaking tank.

(27) Tank quantitiesMONITOR

NOTE: 1. Consider use of asymmetric thrust to maintain fuel imbalance within limits.

2. If using asymmetric thrust, disengage ATS.

(28) Go to (34)

(29) Range is NOT a consideration:

(30) Tank quantitiesMONITOR

NOTE: 1. Consider use of asymmetric thrust to maintain fuel imbalance within limits.

2. If using asymmetric thrust, disengage ATS.

(31) Go to (34)

– COMPLETE –

(32) Unable to determine leak source:

(33) Tank quantitiesMONITOR

NOTE: 1. Consider use of asymmetric thrust to maintain fuel imbalance within limits.

2. If using asymmetric thrust, disengage ATS.

(34) Landing with FUEL IMBALANCE on :

➡ **YES** – Go to (35)

➡ **NO** – No further action required. **End of procedure.**

(35) Landing with FUEL IMBALANCE on:

NOTE: Select the longest runway available with minimum turbulence and crosswind.

Prior to landing:

(36) TAWS WARNING, FLAPS OFFPRESS IN

(37) TAWS WARNING, RAAS OFF (if installed)PRESS IN



Fuel Leak (Cont'd)

- (38) Landing flaps 20 DEGREES
- (39) Approach speed $V_{REF} + 14$ minimum.
- (40) ATS DISC PRESS prior to 100 feet AGL.

▲ CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (41) Actual landing distance INCREASE as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

Loss of Fuel Quantity Indication

Condition: Fuel quantity indication dashed.

Objective: Manually control and/or calculate fuel quantity.

- (1) Choose applicable:
- ➡ **Loss of Auxiliary Tank Fuel Indication** – Go to (2)
 - ➡ **Loss of Tail Tank Fuel Indication** – Go to (11)
 - ➡ **Loss of Totalizer Fuel Indication** – Go to (16)
 - ➡ **Loss of Wing Tank Fuel Indication** – Go to (18)
- (2) **Loss of Auxiliary Tank Fuel Indication:**
- (3) TAIL TANK TRANSFER, MANUAL OVRD MANUAL
- (4) PRI or SEC TAIL TANK TRANSFER PRESS IN to start applicable pump.
- (5) Tank quantities MONITOR
Check that tail tank quantity is decreasing.



Challenger 605	FUEL 09-23
	REV 49, Nov 19/18

Loss of Fuel Quantity Indication (Cont'd)

(6) Tail tank quantity is decreasing:

- ➡ **YES** – Go to (7)
- ➡ **NO** – Go to (9)

(7) Tail tank quantity is decreasing:

When tail tank is empty:

(8) PRI or SEC TAIL TANK
TRANSFERPRESS OUT
– COMPLETE –

(9) Tail tank quantity is NOT decreasing:

NOTE: 1. Fuel dumping must only be carried out with flaps at 0 degrees.
2. Fuel dumping in known lightning conditions is prohibited.
3. Tail tank contents will decrease from full to empty, in approximately 30 minutes.

(10) FUEL, DUMPPRESS IN
– COMPLETE –

(11) Loss of Tail Tank Fuel Indication:

(12) TAIL TANK TRANSFER,
MANUAL OVRDMANUAL
(13) PRI and SEC TAIL TANK
TRANSFERPRESS IN
(14) Tank quantitiesMONITOR
Check AUX tank quantity is less than 7,000 lb (3,175 kg).

When tail tank is empty:

(15) PRI and SEC TAIL TANK
TRANSFERPRESS OUT
– COMPLETE –

(16) Loss of Totalizer Fuel Indication:

(17) Tank quantitiesMONITOR
Compare fuel burn to sum of tank quantities.
– COMPLETE –

(18) Loss of Wing Tank Fuel Indication:

(19) Both ENG SPEED ON



Challenger 605	FUEL 09-24
	REV 49, Nov 19/18

Loss of Fuel Quantity Indication (Cont'd)

- (20) Airplane attitudeSTRAIGHT & LEVEL
- (21) Engine fuel flow indicationsMATCH
- (22) PACKsTRANSFER
to engine bleeds.
- (23) APUSHUTDOWN
if not required.

NOTE: APU operation will increase fuel burn from right wing tank, by a rate of 100 to 300 lb/hr (45 to 136 kg/hr).

- (24) Airplane trimADJUST
- (25) Missing fuel tank quantityCALCULATE
using FMS fuel used page and displayed indications.
- (26) Fuel distributionMONITOR
for imbalance.

- (27) **Excessive imbalance exists (more than 25% of available trim required):**
 - ➡ **YES** – Go to (28)
 - ➡ **NO** – Go to (32)

- (28) **Excessive imbalance exists (more than 25% of available trim required):**

Affected high tank:

- (29) L (R) TO AUX XFLOWPRESS IN

NOTE: Cross flow transfers fuel at a rate of approximately 15 lb/min (7 kg/min) or greater.

When excessive imbalance is corrected:

- (30) Affected L (R) TO AUX XFLOWPRESS OUT
- (31) Go to (33)

(32) No excessive imbalance:

- (33) Fuel reserves.....INCREASE 5%

– COMPLETE –

Challenger 605	FUEL 09-25
	REV 49, Nov 19/18

Low Fuel Warning

Condition: Amber fuel quantity readouts indicating a low fuel condition.

Objective: Expedite landing.

- (1) Airplane attitude5 DEGREES NOSE UP maximum.
- (2) Fuel balance CHECK
- (3) Land at the nearest suitable airport.

NOTE: Do not attempt go-around. The minimum fuel quantity for go-around is 500 lb (230 kg) per wing (with the airplane level), and assuming a maximum airplane climb attitude of 10 degrees nose up.

- COMPLETE -

TRAINING PURPOSES ONLY

Challenger 605	FUEL 09-26
	REV 49, Nov 19/18

Motive Flow Failure

Condition: Loss of engine fuel pressure (same side **MAIN EJECTOR FAIL** and **SCAV EJECTOR** messages).

Objective: Attempt to correct condition and monitor for fuel leaks.

(1) Both BOOST PUMPS CHECK PRESSED IN

(2) **Choose applicable:**

➡ **Engine thrust is inadequate** – Go to (3)

➡ **Engine thrust is normal** – Go to (7)

(3) **Engine thrust is inadequate:**

(4) ATS DISC PRESS

(5) Unaffected engine ADJUST THRUST
to maintain flight path.

(6) Go to (8)

(7) **Engine thrust is normal:**

(8) Tank quantities MONITOR

(9) **Total fuel quantity is decreasing normally:**

➡ **YES** – No further action required. **End of procedure.**

➡ **NO** – Go to (10)

(10) **Total fuel quantity is decreasing abnormally:**

(11) Fuel Leak Procedure ACCOMPLISH

(See Page FUEL 09-19)

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Fuel
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Challenger 605	FUEL 09-27
	REV 49, Nov 19/18

Advisory Messages

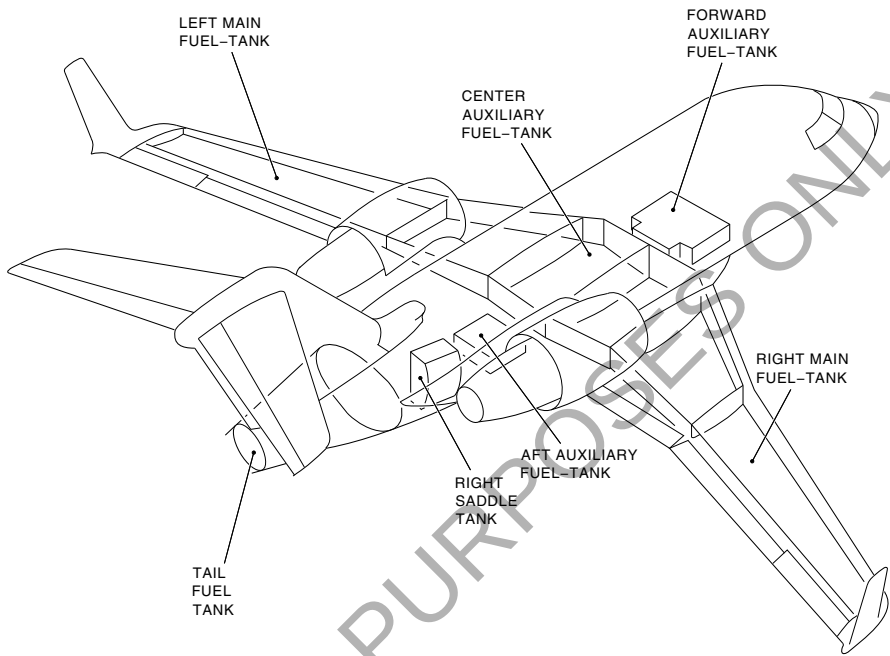
FUEL ADVISORY MESSAGES	
L (R) ENG SOV CLSD	L (R) engine SOV closed, after respective engine fire push switch is pushed.
L (R) FUEL PUMP ON	L (R) fuel pump on, with sufficient outlet fuel pressure.

Status Messages

FUEL STATUS MESSAGES	
FUEL CH 1 (2) FAIL	Respective channel in the fuel system computer has failed.
FUEL QTY ACCURACY	Fuel calibration failed on both channels.
FUEL QTY ATT CORR	Invalid signal failed on both channels.
L (R) MAIN EJECTOR FAIL	Low fuel pressure at respective main ejector, with respective engine running.
PRI TAIL XFER FAIL	Low pressure detected during automatic transfer.
SEC TAIL XFER FAIL	Low pressure detected during automatic transfer.
TAIL RFL SOV OPEN	Tail tank refuel SOV is open.

SCHEMATICS

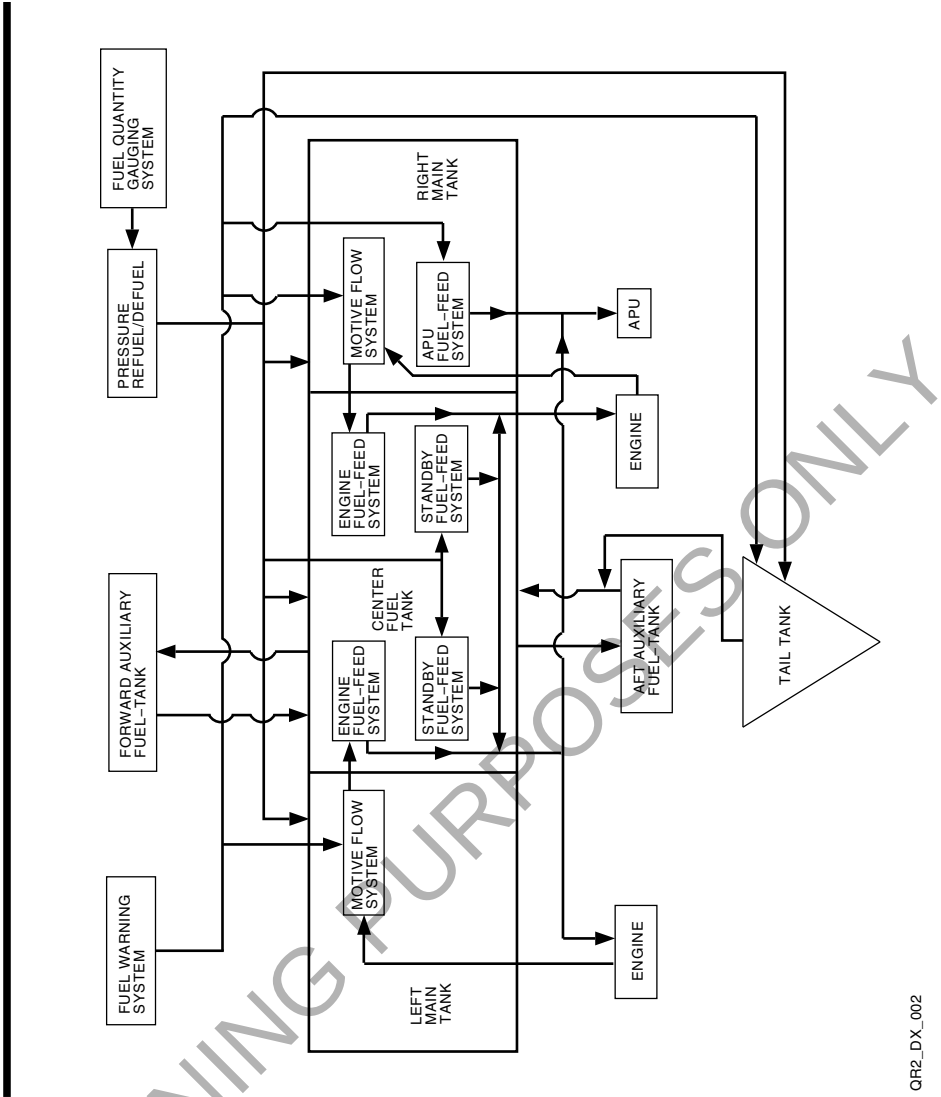
Fuel System Storage



QR2_DX_001

- COMPLETE -

SCHEMATICS (CONT'D)



QR2_DX_002

Power Supply and Circuit Breaker Summary

CB NAME	BUS	CB LOCATION
APU FUEL PUMP	BATT BUS	CBP2-P04
FSCU CH 2	DC ESS BUS	CBP4-A10
FSCU PRIM FUEL XFER CONT	DC BUS 1	CBP1-G09
FUEL DEFUEL – SOV POS	DC BUS 1	CBP1-G08
FUEL DEFUEL – T/T SOVS	APU BATT DIR	CBP5-B07
FUEL FSCU CH 1	BATT BUS	CBP1-M11
FUEL GRAV XFLOW	BATT BUS	CBP1-M12
FUEL PUMP L	BATT BUS	CBP1-M10
FUEL PUMP R	DC BUS 2	CBP2-G08
FUEL PUMP CONT L	BATT BUS	CBP1-M09
FUEL PUMP CONT R	DC BUS 2	CBP2-G07
FUEL PUMP PRIM XFER	AC BUS 1	CBP1-D02
FUEL PUMP SEC XFER	AC BUS 2	CBP2-C08
FUEL SOV APU	DC EMERG BUS	CBP1-S03
FUEL SOV L ENG	DC EMERG BUS	CBP1-S02
FUEL SOV R ENG	DC EMERG BUS	CBP1-S01
FUEL X FEED CONT	BATT BUS	CBP2-P05
FUEL/DEFUEL – FSCU	APU BATT DIR	CBP5-B05
FUEL/DEFUEL – L/R AUX SOV'S	APU BATT DIR	CBP5-B06
FUEL/DEFUEL – PANEL	APU BATT DIR	CBP5-B08
SEC FUEL XFER CONT	DC BUS 2	CBP2-G09
T/T DUMP SOV	BATT BUS	CBP2-P06
T/T DUMP SOV CONT	BATT BUS	CBP2-P07
T/T XFER SOV	DC ESS BUS	CBP4-A08
T/T XFER SOV CONT	BATT BUS	CBP1-M08

Challenger 605	GEAR TOC-1
	REV 63, Mar 14/22

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Challenger 605	GEAR TOC-2
	REV 49, Nov 19/18

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TRAINING PURPOSES ONLY

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Gear / Brakes
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Challenger 605	GEAR 10-1
	REV 64, Jun 02/22

GEAR DISAGREE



“GEAR DISAGREE”

Condition: One or more landing gear position indications displays red hash marks.

Objective: Manually lower landing gear.

(1) Airspeed REDUCE

- Flaps at 0 degrees – 181 KIAS
- Flaps at 20 degrees – 197 KIAS
- Flaps at 30 degrees – 197 KIAS
- Flaps at 45 degrees – 189 KIAS

▲ CAUTION: Leave hydraulic pump 3B OFF if pump was selected OFF due to a low or decreasing pressure, or a low quantity in system No. 3.

(2) HYDRAULIC pumps 2B and 3B ON

(3) **GEAR DISAGREE** goes out and three gear position indications agree:

- ➔ **YES** – No further action required. **End of procedure.**
- ➔ **NO** – Go to (4)

(4) **Gear disagree occurred when LDG GEAR lever was selected UP:**

- ➔ **YES** – Go to (11)
- ➔ **NO** – Go to (5)

(5) **Gear disagree occurred when LDG GEAR lever was selected DN:**

I (6) LDG GEAR leverUP

NOTE: If necessary, press DN LCK REL button to release landing gear lever.

Wait 30 seconds, then:

I (7) LDG GEAR lever DN

- NOTE:**
1. Steps (6) and (7) may be repeated as many times as desired.
 2. Quickly cycling the landing gear lever UP then DN may also provide a down and locked indication. For quick cycling, do not wait for the landing gear UP indications before re-selecting the landing gear lever down.



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Gear / Brakes
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Challenger 605	GEAR 10-2
	REV 64, Jun 02/22

GEAR DISAGREE (Cont'd)

(8) GEAR DISAGREE goes out and three DN indications are displayed:

- ➡ **YES** – No further action required. **End of procedure.**
- ➡ **NO** – Go to (9)

(9) GEAR DISAGREE stays on:

I (10) LDG GEAR leverUP

(11) Gear disagree persists:

▲CAUTION: Failure to pull the LANDING GEAR MANUAL RELEASE handle to its full extension may prevent successful landing gear extension.

NOTE: The force required to operate the LANDING GEAR MANUAL RELEASE handle is greater than 40 pounds. The flight crew may choose to reposition the seat prior to handle deployment. In safe flight conditions, the flight crew may also choose to stand up to pull the handle.

(12) LANDING GEAR MANUAL RELEASEPULL
to full extension.

NOTE: **NOSE DOOR OPEN** may be on when the LANDING GEAR MANUAL RELEASE handle is pulled. If the nose landing gear is in transition, this message will be inhibited.

I (13) LDG GEAR lever DN

(14) GEAR DISAGREE stays on:

- ➡ **YES** – Go to (15)
- ➡ **NO** – Go to (23)

(15) GEAR DISAGREE stays on:

(16) HYDRAULIC pumps 3A and 3B OFF

(17) Hydraulic pressureVERIFY
system 3 pressure is decreasing.

When hydraulic system 3 pressure is less than 200 psi:

(18) HYDRAULIC pumps 3A and 3B ON

(19) Landing gear indicationCHECK
for three DN indications.

(20) GEAR DISAGREE stays on:

- ➡ **YES** – Go to (21)
- ➡ **NO** – Go to (23)



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Gear / Brakes
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Challenger 605	GEAR 10-3
	REV 63, Mar 14/22

GEAR DISAGREE (Cont'd)

(21) GEAR DISAGREE stays on:

(22) Landing Gear Up/Unsafe Landing
 ProcedureACCOMPLISH

I (See Page GEAR 10–17)

– COMPLETE –

(23) GEAR DISAGREE goes out:

▲ CAUTION: Do not stow LANDING GEAR MANUAL RELEASE handle until gear locking pins are installed.

Nose wheel steering may not be available upon landing even if the **STEERING INOP** caution message is not displayed.

(24) No further action required.

– COMPLETE –

TRAINING PURPOSES ONLY

Challenger 605	GEAR 10-4
	REV 49, Nov 19/18

MLG BAY OVHT



Condition: Main landing gear bay overheat detected.

Objective: Cool brakes.

(1) Airspeed 197 KIAS maximum.

(2) LDG GEAR DN

➤

FROM Quick Reference Card

◀

(3) **MLG BAY OVHT** stays on:

- ➡ **YES** – Go to (4)
- ➡ **NO** – Go to (8)

(4) **MLG BAY OVHT** stays on:

- (5) LDG GEAR LEAVE DN
- (6) Airspeed 250 KIAS maximum.
- (7) Land at the nearest suitable airport.

– COMPLETE –

(8) **MLG BAY OVHT** goes out:

- (9) Flight CONTINUE with gear down for minimum of 5 minutes.
- (10) LDG GEAR UP

– COMPLETE –

Challenger 605	GEAR 10-5
	REV 49, Nov 19/18

NOSE DOOR OPEN



“NOSE DOOR”

Condition: Nose door still open after landing gear is locked in position, or at high speed.

Objective: Review penalties and land.

- (1) Airspeed REDUCE
 - Flaps at 0 degrees – 181 KIAS
 - Flaps at 20 degrees – 197 KIAS
 - Flaps at 30 degrees – 197 KIAS
 - Flaps at 45 degrees – 189 KIAS
- (2) LDG GEAR..... DN
- (3) Land at the nearest suitable airport.

NOTE: Use of LANDING GEAR MANUAL RELEASE will cause **NOSE DOOR OPEN** to be on.

– COMPLETE –

PARKING BRAKE



“CONFIG BRAKES”

Condition: Parking brake is on while at high thrust.

Objective: On ground: check configuration before takeoff. In flight: review penalties.

- (1) **Identify phase of flight:**
 - ➡ **On ground (both N_1 greater than 70%)** – Go to (2)
 - ➡ **In flight** – Go to (5)
- (2) **On ground (both N_1 greater than 70%):**
- (3) Thrust levers..... RETARD
- (4) PARKING BRAKE AS REQUIRED

– COMPLETE –

- (5) **In flight:**
- (6) PARKING BRAKE CHECK RELEASED



Challenger 605	GEAR 10-6
	REV 49, Nov 19/18

PARKING BRAKE (Cont'd)

(7) **PARKING BRAKE** stays on:

- ➡ **YES** – Go to (8)
- ➡ **NO** – No further action required. **End of procedure.**

(8) **PARKING BRAKE** stays on:

▲CAUTION: Inboard and outboard anti-skid systems are inoperative. Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

(9) Actual landing distance INCREASE
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.05 (105%)	1.65 (65%)

– COMPLETE –

A/SKID INBD

Condition: Inboard anti-skid is failed or not armed. May be accompanied with **PARK BRAKE SOV** if parking brake shutoff valve is failed.

Objective: Review penalties.

▲CAUTION: Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

(1) Actual landing distance INCREASE
as applicable by factor shown below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.05 (105%)	1.65 (65%)

If loss of directional control or braking is observed:

- (2) Brakes RELEASE
- (3) ANTI-SKID OFF
- (4) Brakes RE-APPLY

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Gear / Brakes
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Challenger 605	GEAR 10-7
	REV 49, Nov 19/18

A/SKID INBD and A/SKID OUTBD

Condition: Both anti-skid channels failed or not armed.

Objective: Review penalties.

(1) ANTI-SKID OFF

▲ CAUTION: Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

(2) Actual landing distance INCREASE
as applicable by factor shown below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.05 (105%)	1.65 (65%)

– COMPLETE –

A/SKID OUTBD

Condition: Outboard anti-skid is failed or not armed.

Objective: Review penalties.

▲ CAUTION: Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

(1) Actual landing distance INCREASE
as applicable by factor shown below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.05 (105%)	1.65 (65%)

If loss of directional control or braking is observed:

(2) Brakes RELEASE

(3) ANTI-SKID OFF

(4) Brakes RE-APPLY

– COMPLETE –

Challenger 605	GEAR 10-8
	REV 49, Nov 19/18

INBD (OUTBD) BRAKE PRESS

Condition: Low brake pressure on indicated brakes system.

Objective: Review penalties.

- (1) Hydraulic pressure and fluid quantityMONITOR
- (2) Actual landing distance INCREASE
as applicable by factor shown below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.70 (70%)	1.50 (50%)

NOTE: Maximize the use of reverse thrust.

– COMPLETE –

PARK BRAKE SOV

Condition: Parking brake shutoff valve is not in commanded position.

Objective: Review penalties.

NOTE: **PARK BRAKE SOV** indicates that the parking brake shutoff valve is failed or is closed with the parking brake not set, possibly affecting inboard anti-skid system performance.

▲ CAUTION: Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

- (1) Actual landing distance INCREASE
as applicable by factor shown below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.05 (105%)	1.65 (65%)

– COMPLETE –

Challenger 605	GEAR 10-9
	REV 54, Dec 18/19

PROX SYS

Condition: Failure in PSEU output circuitry. May be accompanied by **GND SPLRS**.

Objective: Review penalties.

- NOTE:**
1. EICAS landing gear position indications are inoperative.
 2. Normal landing gear extension/retraction is not available.
 3. Anti-skid, ground spoilers and nose wheel steering systems are inoperative.
 4. Nose door open warning system is not available.

(1) Airspeed **REDUCE**

- Flaps at 0 degrees – 181 KIAS
- Flaps at 20 degrees – 197 KIAS
- Flaps at 30 degrees – 197 KIAS
- Flaps at 45 degrees – 189 KIAS

(2) HYDRAULIC pumps 2B & 3B **ON**

(3) LDG GEAR **DN**

▲ CAUTION: Failure to pull the LANDING GEAR MANUAL RELEASE handle to its full extension may prevent successful landing gear extension.

NOTE: The force required to operate the LANDING GEAR MANUAL RELEASE handle is greater than 40 pounds. The flight crew may choose to reposition the seat prior to handle deployment. In safe flight conditions, the flight crew may also choose to stand up to pull the handle.

(4) LANDING GEAR MANUAL RELEASE **PULL**

(5) GND SPOILERS **DISARM**

(6) ANTI-SKID **OFF**

▲ CAUTION: Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

(7) NOSE STEER **OFF**

NOTE: If landing with the nosewheel steering inoperative, select the longest runway available with minimum turbulence and crosswind. Use differential braking, rudder and engine thrust as required during taxi.



PROX SYS (Cont'd)

- (8) Actual landing distance INCREASE
as applicable by factor shown below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.20 (120%)	1.75 (75%)

– COMPLETE –

STEERING INOP

Condition: Nosewheel steering has failed.

Objective: On ground: attempt to correct condition. In flight: review operational considerations.

- (1) **Identify phase of flight:**

➡ **On ground** – Go to (2)

➡ **In flight** – Go to (11)

- (2) **On ground:**

- (3) Nose wheel CENTER
Use differential braking/thrust/reverse thrust as required.

- (4) Nose wheel steering tiller ENSURE CENTERED

- (5) Rudder pedals ENSURE CENTERED

NOTE: NOSE STEER switch must be selected to ARMED to enable nosewheel steering circuit and centering of the nosewheel.

- (6) NOSE STEER CYCLE OFF AND ARMED

- (7) Nose wheel steering CHECK
Ensure normal operation.

NOTE: Check nose wheel steering responsiveness by taxiing forward and checking proper aircraft response to left/right tiller inputs and rudder pedal inputs.

- (8) **STEERING INOP stays on:**

➡ **YES** – Go to (9)

➡ **NO** – No further action required. **End of procedure.**

- (9) **STEERING INOP stays on:**

- (10) NOSE STEER OFF

– COMPLETE –



Challenger 605	GEAR 10-11
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STEERING INOP (Cont'd)

(11) In flight:

NOTE: Centering of nosewheel steering tiller, rudder pedals, and/or cycling switch from OFF to ARMED will not enable centering of nosewheel.

(12) NOSE STEER OFF

- NOTE:**
1. When landing with the nosewheel steering inoperative, select the longest runway available with minimum turbulence and crosswind.
 2. Use differential braking, rudder and engine thrust, as required, to assist directional control.

– COMPLETE –

WOW INPUT

Condition: PSEU is unable to determine if airplane is in flight mode or in ground mode.

(1) NOSE STEERENSURE ARMED

(2) GND SPOILERSDISARM

CAUTION: Do not override the landing gear downlock.

NOTE: If landing gear are down and locked, gear will not retract. If a go-around is required in this configuration, add 2.0 kts to the approach climb speed and reduce the approach climb gradient by 2.4%.

Prior to landing:

(3) Actual landing distance INCREASE
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.30 (30%)	1.25 (25%)

After touchdown:

(4) GND SPOILERSARM

(5) FLIGHT SPOILERMAX

– COMPLETE –

Challenger 605	GEAR 10-12
	REV 65, Jun 24/22

WOW OUTPUT

Condition: One or more PSEU critical drivers not in expected state.

Objective: On ground: do not takeoff. In flight: review operational considerations.

(1) Identify phase of flight:

- ➡ **On ground** – Go to (2)
- ➡ **In flight** – Go to (5)

(2) On ground:

- (3) Do not takeoff.
- (4) ADG DEPLOY CONT, AUTO
circuit breaker (2N6) **PULL**
to prevent inadvertent deployment.

– COMPLETE –

(5) In flight:

NOTE: After completing the Manual Cabin Pressurization Control Procedure, return to this checklist to finish the remaining steps.

- (6) Manual Cabin Pressurization
Control Procedure **ACCOMPLISH**
(See Page ENVR 06–13)
- (7) GND SPOILERS **DISARM**
- (8) NOSE STEER **OFF**

(9) STALL FAIL is on:

- ➡ **YES** – Go to (10)
- ➡ **NO** – Go to (12)

(10) STALL FAIL is on:

NOTE: After completing the **STALL FAIL** Procedure, return to this checklist to finish the remaining steps.

- (11) **STALL FAIL** Procedure **ACCOMPLISH**
(See Page FCTL 07–14)

(12) STALL FAIL is out:

Prior to landing:

- (13) Actual landing distance **INCREASE**
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.30 (30%)	1.25 (25%)



Challenger 605	GEAR 10-13
	REV 63, Mar 14/22

WOW OUTPUT (Cont'd)

After touchdown:

- (14) GND SPOILERSARM
- (15) FLIGHT SPOILERS.....MAX

If loss of braking occurs at less than 35 knots:

- (16) ANTI-SKID OFF

▲ CAUTION: Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

If nose wheel steering is required:

- (17) NOSE STEERARMED

After landing:

- (18) ADG DEPLOY CONT, AUTO
circuit breaker (2N6)PULL
to prevent inadvertent deployment.

– COMPLETE –

Asymmetric Braking / Loss of Braking

Objective: Attempt to brake with anti-skid OFF.

If loss of directional control or braking is observed:

- (1) Wheel brakes RELEASE momentarily.
- (2) ANTI-SKID OFF
- (3) Wheel brakes RE-APPLY as required.



FROM Quick Reference Card



CAUTION: Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

– COMPLETE –

Landing Gear Lever Jammed in the UP Position

Condition: Unable to select gear lever down.

Objective: Manually lower landing gear.

- (1) Airspeed REDUCE
 - Flaps at 0 degrees – 181 KIAS
 - Flaps at 20 degrees – 197 KIAS
 - Flaps at 30 degrees – 197 KIAS
 - Flaps at 45 degrees – 189 KIAS
- (2) HYDRAULIC pumps 2B and 3B ON

CAUTION: Failure to pull the LANDING GEAR MANUAL RELEASE handle to its full extension may prevent successful landing gear extension.

NOTE: The force required to operate the LANDING GEAR MANUAL RELEASE handle is greater than 40 pounds. The flight crew may choose to reposition the seat prior to handle deployment. In safe flight conditions, the flight crew may also choose to stand up to pull the handle.



Challenger 605	GEAR 10-15
	REV 65, Jun 24/22

Landing Gear Lever Jammed in the UP Position (Cont'd)

- (3) LANDING GEAR MANUAL RELEASEPULL
to full extension.

NOTE: 1. **NOSE DOOR OPEN** may be displayed when the LANDING GEAR MANUAL RELEASE handle is pulled. If the nose landing gear is in transition, this message will be inhibited.

2. **GEAR DISAGREE** will appear with the LDG GEAR lever jammed in the UP position, if any landing gear is not in its uplock.

- (4) Landing gear indicationCHECK
for three **DN** indications.

- (5) Three landing gear symbols indicate **DN**:

- ➡ **YES** – Go to (14)
- ➡ **NO** – Go to (6)

- (6) One or more landing gear symbols do **NOT** indicate **DN**:

- (7) HYDRAULIC pumps 3A and 3B OFF
- (8) Hydraulic pressureVERIFY
system 3 pressure is decreasing.

When hydraulic system 3 pressure is less than 200 psi:

- (9) HYDRAULIC pumps 3A and 3B ON
- (10) Landing gear indicationCHECK
for three DN indications.

- (11) Gear disagree persists:

- ➡ **YES** – Go to (12)
- ➡ **NO** – Go to (14)

- (12) Gear disagree persists:

- (13) Landing Gear Up/Unsafe procedureACCOMPLISH

I (See Page GEAR 10–17)

– COMPLETE –

- (14) Three landing gear symbols indicate **DN**:

Prior to landing:

- (15) NOSE STEER OFF



Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Gear / Brakes
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Challenger 605	GEAR 10-16
	REV 63, Mar 14/22

Landing Gear Lever Jammed in the UP Position (Cont'd)

- ▲ CAUTION:** 1. Do not stow LANDING GEAR MANUAL RELEASE handle until gear locking pins are installed.
2. Nose wheel steering will not be available upon landing.

(16) No further action required.

– COMPLETE –

Landing Gear Manual Extension

Condition: Unable to extend gear normally.

Objective: Manually lower landing gear.

- (1) Airspeed REDUCE
- Flaps at 0 degrees – 181 KIAS
 - Flaps at 20 degrees – 197 KIAS
 - Flaps at 30 degrees – 197 KIAS
 - Flaps at 45 degrees – 189 KIAS

▲ CAUTION: Leave hydraulic pump 3B OFF if pump was selected OFF due to a low or decreasing pressure, or a low quantity in system No. 3.

- (2) HYDRAULIC pumps 2B and 3B ON
- (3) LDG GEAR DN

▲ CAUTION: Failure to pull the LANDING GEAR MANUAL RELEASE handle to its full extension may prevent successful landing gear extension.

NOTE: The force required to operate the LANDING GEAR MANUAL RELEASE handle is greater than 40 pounds. The flight crew may choose to reposition the seat prior to handle deployment. In safe flight conditions, the flight crew may also choose to stand up to pull the handle.

- (4) LANDING GEAR MANUAL RELEASE PULL to full extension.

NOTE: **NOSE DOOR OPEN** may be on when the LANDING GEAR MANUAL RELEASE handle is pulled. If the nose landing gear is in transition, this message will be inhibited.

- (5) Landing gear indication CHECK for three **DN** indications.



Landing Gear Manual Extension (Cont'd)

(6) Three landing gear symbols indicate **DN**:

➡ **YES** – Go to (7)

➡ **NO** – Go to (9)

(7) Three landing gear symbols indicate **DN**:

▲ **CAUTION:** 1. Do not stow LANDING GEAR MANUAL RELEASE handle until gear locking pins are installed.

2. Nose wheel steering may not be available upon landing even if **STEERING INOP** is not displayed.

(8) No further action required.

– COMPLETE –

(9) One or more landing gear symbols do NOT indicate **DN**:

(10) **GEAR DISAGREE** Procedure.....ACCOMPLISH

(See Page GEAR 10-1)

– COMPLETE –

Landing Gear Up/Unsafe Landing

Condition: Landing with one or more gear up or unsafe.

Objective: Prepare the airplane for an unsafe landing.

- NOTE:**
1. **If one main landing gear is up or unsafe**, hold applicable wing up for as long as possible. Maintain directional control with rudder and nosewheel steering (if considered safe). When wing touches ground, apply asymmetrical braking for directional control.
 2. **If nose landing gear is up or unsafe**, relocate passengers (if possible) to obtain aft CG. Trim stabilizer nose-up after touchdown. Gently lower the nose before elevator effectiveness is lost.
 3. **If all wheels are up or unsafe**, perform a nose high attitude touchdown but do not reduce touchdown speed below stick shaker speed.
 4. **If both main landing gear cannot be locked down**, consideration should be given to landing with all wheels up.

PRELIMINARY

(1) Descent PLAN

PREPARATION



Landing Gear Up/Unsafe Landing (Cont'd)

- (2) Crew ALERT & BRIEF
 (3) ATC NOTIFY
 (4) Both PASS SIGNS ON
 (5) Loose equipment..... SECURE
 (6) TAWS circuit breaker (1B13)..... PULL
 (7) AUDIO WARNING DCU 1 and 2..... PRESS IN
 (8) Shoulder harness TIGHT & LOCKED

APPROACH

- (9) Both PACKs PRESS OUT
 (10) PRESS CONT MANUAL
 (11) MAN ALT HOLD UP
 (12) MAN RATE AS REQUIRED

When depressurized:

- (13) L and R 10TH STAGE bleeds PRESS OUT
 (14) Landing flaps 45 DEGREES
 (15) EMER LTS ON
 (16) Brace for impact ORDER
 "BRACE-BRACE-BRACE".

BEFORE IMPACT

- (17) Airplane pitch attitude MAINTAIN NOSE HIGH
 (18) Ground/Landing field CONTACT
 with minimum forward speed, but not less than stick shaker
 speed, and at minimum sink rate.

AFTER LANDING

- (19) Thrust levers SHUT OFF

→ If any gear extended:

- HYDRAULIC pumps 3A and 3B ON
 until landing gear is secured with locking pins.
- (20) Emergency Evacuation Procedure ACCOMPLISH
 (See Page F-LDG 08-1)

- COMPLETE -

I

Challenger 605	GEAR 10-19
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Advisory Messages

GEAR / BRAKES ADVISORY MESSAGES	
A/SKID IN TEST	Anti-skid system undergoing self test (with the parking brake off).
PARKING BRAKE ON	Parking brake set with one or both engines less than 70% N ₁ .

Status Messages

GEAR / BRAKES STATUS MESSAGES	
PROX SYS FAULT	Failure in the PSEU.
WOW OUTPUT FAIL	WOW output failure in a non-critical system.

TRAINING PURPOSES ONLY

Power Supply and Circuit Breaker Summary

CB NAME	BUS	CB LOCATION
ANTI SKID	DC BUS 1	CBP1-F07
ANTI SKID	DC BUS 2	CBP2-F07
BRAKE PRESS IND	DC BUS 2	CBP2-G12
HARD LDG IND	DC BUS 1	CBP1-H11
NOSE STEER	DC BUS 1	CBP1-F06
NOSE STEER	DC BUS 2	CBP2-F06
WOW / GEAR / SPLR 1	DC BUS 1	CBP1-F02
WOW / GEAR / SPLR 2	BATT BUS	CBP2-N02
WOW/GEAR/SPLR 1	BATT BUS	CBP1-N02
WOW/GEAR/SPLR 2	DC BUS 2	CBP2-F02

Challenger 605	HYD TOC-1
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POWER SUPPLY AND CIRCUIT BREAKER SUMMARY

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Challenger 605	HYD TOC-2
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TRAINING PURPOSES ONLY

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Hydraulics
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Challenger 605	HYD 11-1
	REV 63, Mar 14/22

HYD 1 (2) HI TEMP

Condition: Indicated hydraulic system fluid temperature is high.

Objective: Monitor hydraulic fluid temperature. Shutdown associated engine if necessary.

NOTE: Disregard **HYD 1 (2) HI TEMP** if one other hydraulic system has already failed (i.e., leave pumps on).

(1) Identify phase of flight:

- ➡ **In flight** – Go to (2)
- ➡ **On ground** – Go to (15)

(2) In flight:

(3) Choose applicable:

- ➡ **Temperature is colder than 107°C (225°F)** – Go to (4)
- ➡ **Temperature is between than 107°C (225°F) and 135°C (275°F)** – Go to (7)
- ➡ **Temperature is 135°C (275°F) or hotter** – Go to (11)

(4) Temperature is colder than 107°C (225°F):

- (5) Affected HYDRAULIC "B" pump..... OFF
- (6) Affected system temperatureMONITOR

– COMPLETE –

(7) Temperature is between than 107°C (225°F) and 135°C (275°F):

- (8) Affected HYDRAULIC "B" pump..... OFF
- (9) Affected system temperatureMONITOR
- (10) Land at the nearest suitable airport.

– COMPLETE –

(11) Temperature is 135°C (275°F) or hotter:

- (12) Affected HYDRAULIC "B" pump..... OFF

NOTE: – When shutting down an engine due to a hydraulic system high temperature condition, make sure to apply the actual landing distance correction factor given in the table below.

- After completing the In-Flight Engine Failure/Shutdown procedure, return to this checklist to complete the remaining steps.



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Hydraulics
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Challenger 605	HYD 11-2
	REV 63, Mar 14/22

HYD 1 (2) HI TEMP (Cont'd)

Affected engine:

- (13) In-Flight Engine
 Failure/ShutdownACCOMPLISH
 (See Page ENG 05-10)

- (14) Review inoperative systems and penalties:

	SYSTEM 1 FAILED	SYSTEM 2 FAILED
Inoperative systems	Ground Spoilers (left AND right)	• Outboard brakes (when system 2 accumulator pressure is depleted) • Main landing gear down lock assist actuator
ALD Factor	1.60 (60%)	2.25 (125%)

- COMPLETE -

- (15) On ground:
 (16) Do not takeoff. Report to maintenance.

- COMPLETE -

HYD 1 LO PRESS

Condition: Hydraulic system 1 pressure is low.

Objective: Attempt to restore. Review penalties if unable.

- (1) HYDRAULIC pump 1B ON
- (2) Hydraulic pressure and fluid quantityCHECK
- (3) **System 1 quantity is less than 5%, OR pressure is less than 1,800 psi, OR pressure is decreasing rapidly:**
- ➡ **YES** – Go to (4)
- ➡ **NO** – Go to (15)
- (4) **System 1 quantity is less than 5%, OR pressure is less than 1,800 psi, OR pressure is decreasing rapidly:**
- (5) HYDRAULIC pump 1B OFF
- (6) Land at the nearest suitable airport.



HYD 1 LO PRESS (Cont'd)

(7) Inoperative systems REVIEW

- Ground spoilers inoperative (left and right).

NOTE: The engine-driven pump has been operating without hydraulic fluid. Log length of time that engine-driven pump has been running dry. Monitor fluid temperature readouts.

(8) ANTI SKID LEAVE ARMED

(9) TAWS WARNING, FLAPS OFF PRESS IN

(10) TAWS WARNING, RAAS OFF (if installed) PRESS IN

(11) Landing flaps 20 DEGREES

(12) Approach speed $V_{REF} + 14$ minimum.

(13) ATS DISC PRESS prior to 100 feet AGL.

▲ CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

(14) Actual landing distance INCREASE as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.60 (60%)	1.55 (55%)

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

(15) **System 1 pressure and quantity are normal:**

(16) HYDRAULIC pump 1B LEAVE ON

(17) Hydraulic pressure and fluid quantity MONITOR

– COMPLETE –

Challenger 605	HYD 11-4
	REV 49, Nov 19/18

HYD 1 LO PRESS and HYD 2 LO PRESS

Condition: Pressure is low on BOTH hydraulic systems 1 AND 2.

Objective: Attempt to restore. Review penalties if unable.

- (1) All HYDRAULIC pumps ON
- (2) Hydraulic pressure and fluid quantity CHECK
- (3) **Systems 1 and 2 quantities are less than 5%, OR pressure is less than 1,800 psi, OR pressure is decreasing rapidly :**
 - ➡ **YES** – Go to (4)
 - ➡ **NO** – No further action required. **End of procedure.**
- (4) **Systems 1 and 2 quantities are less than 5%, OR pressure is less than 1,800 psi, OR pressure is decreasing rapidly:**
- (5) HYDRAULIC pumps 1B and 2B OFF
to prevent pump overheat.
- (6) Inoperative systems REVIEW
 - Left and right ground spoilers inoperative;
 - Left and right flight spoilers inoperative;
 - Gear down lock assist actuator inoperative;
 - Outboard brakes inoperative (when system 2 accumulator pressure is depleted).
- (7) ANTI-SKID LEAVE ARMED
- (8) Land at the nearest suitable airport.

Prior to landing:

- (9) TAWS WARNING, FLAPS OFF PRESS IN
- (10) TAWS WARNING, RAAS OFF (if installed) PRESS IN
- (11) Landing flaps 20 DEGREES
- (12) Approach speed $V_{REF} + 14$
minimum.
- (13) ATS DISC PRESS
prior to 100 feet AGL.

⚠CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (14) Actual landing distance INCREASE
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.55 (155%)	1.95 (95%)

NOTE: Maximize the use of reverse thrust.



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Hydraulics
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Challenger 605	HYD 11-5
	REV 49, Nov 19/18

HYD 1 LO PRESS and HYD 2 LO PRESS (Cont'd)

▲ CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

HYD 1 LO PRESS and HYD 3 LO PRESS

Condition: Pressure is low on BOTH hydraulic systems 1 AND 3.

Objective: Attempt to restore. Review penalties if unable.

- (1) Airspeed 200 KIAS maximum.
- (2) All HYDRAULIC pumps ON
- (3) Hydraulic pressure and fluid quantity CHECK
- (4) **Systems 1 and 3 quantities are less than 5%, OR pressure is less than 1,800 psi, OR pressure is decreasing rapidly:**
 - ➡ **YES** – Go to (5)
 - ➡ **NO** – No further action required. **End of procedure.**
- (5) **Systems 1 and 3 quantities are less than 5%, OR pressure is less than 1,800 psi, OR pressure is decreasing rapidly:**
- (6) HYDRAULIC pumps 1B, 3A and 3B OFF
- (7) Inoperative systems REVIEW
 - Left aileron and left elevator inoperative;
 - Left and right ground spoilers inoperative;
 - Normal landing gear extension and retraction inoperative;
 - Inboard brakes inoperative (when system 3 accumulator pressure is depleted);
 - Nose door will remain open;
 - Nose wheel steering inoperative (may have nose wheel shimmy);
 - Parking brake inoperative.
- (8) NOSE STEER OFF
- (9) ANTI-SKID LEAVE ARMED



Challenger 605	HYD 11-6
	REV 63, Mar 14/22

HYD 1 LO PRESS and HYD 3 LO PRESS (Cont'd)

(10) Land at the nearest suitable airport.

- NOTE:**
- Flight path control is limited with hydraulic systems 1 and 3 failed.
 - Select the longest runway available with minimal crosswind and turbulence.
 - Rudder control is adequate for normal flight and should be used in coordination with aileron during turns.
 - After completing the Landing Gear Manual Extension Procedure, return to this checklist to finish the remaining steps.

(11) Landing Gear Manual Extension Procedure.....ACCOMPLISH

(See Page GEAR 10–16)

Prior to landing:

(12) TAWS WARNING, FLAPS OFFPRESS IN

(13) TAWS WARNING, RAAS OFF (if installed)PRESS IN

(14) Landing flaps20 DEGREES

(15) Approach speed $V_{REF} + 14$ minimum.

(16) ATS DISCPRESS prior to 100 feet AGL.

▲CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

(17) Actual landing distanceINCREASE as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.40 (140%)	1.90 (90%)

NOTE: Maximize the use of reverse thrust.

▲CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

Challenger 605	HYD 11-7
	REV 49, Nov 19/18

HYD 2 LO PRESS

Condition: Hydraulic system 2 pressure is low.

Objective: Attempt to restore. Review penalties if unable.

- (1) HYDRAULIC pump 2B ON
- (2) Hydraulic pressure and fluid quantity CHECK
- (3) **System 2 quantity is less than 5%, OR pressure is less than 1,800 psi, OR pressure is decreasing rapidly:**
 - ➡ **YES** – Go to (4)
 - ➡ **NO** – Go to (10)
- (4) **System 2 quantity is less than 5%, OR pressure is less than 1,800 psi, OR pressure is decreasing rapidly:**
- (5) HYDRAULIC pump 2B OFF
- (6) Land at the nearest suitable airport.
- (7) Inoperative systems (see below) REVIEW
 - Outboard brakes inoperative (when system 2 accumulator pressure is depleted);
 - Main landing gear down lock assist actuator inoperative.

NOTE: The engine-driven pump has been operating without hydraulic fluid. Log length of time that engine-driven pump has been running dry. Monitor fluid temperature readouts.

- (8) ANTI-SKID LEAVE ARMED
- (9) Actual landing distance INCREASE
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.70 (70%)	1.50 (50%)

NOTE: Maximize the use of reverse thrust.

– COMPLETE –

(10) System 2 pressure and quantity are normal:

- (11) HYDRAULIC pump 2B LEAVE ON
- (12) Hydraulic pressure and fluid quantity MONITOR

– COMPLETE –

Challenger 605	HYD 11-8
	REV 49, Nov 19/18

HYD 2 LO PRESS and HYD 3 LO PRESS

Condition: Pressure is low on BOTH hydraulic systems 2 AND 3.

Objective: Attempt to restore. Review penalties if unable.

- (1) Airspeed 200 KIAS maximum.
- (2) All HYDRAULIC pumps ON
- (3) Hydraulic pressure and fluid quantity CHECK
- (4) **Systems 2 and 3 quantities are less than 5%, OR pressure is less than 1,800 psi, OR pressure is decreasing rapidly:**
 - ➡ **YES** – Go to (5)
 - ➡ **NO** – No further action required. **End of procedure.**
- (5) **Systems 2 and 3 quantities are less than 5%, OR pressure is less than 1,800 psi, OR pressure is decreasing rapidly:**
- (6) HYDRAULIC pumps 2B, 3A and 3B OFF to prevent pump overheat.
- (7) Inoperative systems REVIEW
 - Right aileron and right elevator inoperative;
 - Normal landing gear extension and retraction inoperative;
 - Main landing gear down lock assist actuator inoperative;
 - Inboard and outboard brakes inoperative (when systems 2 & 3 accumulator pressures are depleted);
 - Nose door will remain open;
 - Nose wheel steering inoperative (may have nose wheel shimmy);
 - Parking brake inoperative.
- (8) Land at the nearest suitable airport.

- NOTE:**
1. Flight path control is limited with hydraulic systems 2 and 3 failed.
 2. Select the longest runway available with minimal crosswind and turbulence.
 3. Rudder control is adequate for normal flight and should be used in coordination with aileron, if necessary, during turns.
 4. After completing the Landing Gear Manual Extension Procedure, return to this checklist to finish the remaining steps.

Prior to landing:

- (9) NOSE STEER OFF
- (10) ANTI-SKID OFF



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Hydraulics
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Challenger 605	HYD 11-9
	REV 63, Mar 14/22

HYD 2 LO PRESS and HYD 3 LO PRESS (Cont'd)

NOTE: After completing the Landing Gear Manual Extension Procedure, return to this checklist to complete the remaining steps.

(11) Landing Gear Manual Extension ProcedureACCOMPLISH

I (See Page GEAR 10–16)

NOTE: 1. Unless it is possible to restore use of hydraulic system 2, landing gear extension relies upon gear free fall after use of gear manual extension control.

2. A sideslip, for each side, must be performed to achieve MLG down-lock. It may be required to increase airspeed to ensure down-lock during sideslip.

(12) TAWS WARNING, FLAPS OFFPRESS IN

(13) TAWS WARNING, RAAS OFF (if installed)PRESS IN

(14) Landing flaps20 DEGREES

(15) Approach speed $V_{REF} + 14$ minimum.

NOTE: The above double hydraulic system failure has an extremely remote probability. If it occurs, maximum use of reverse thrust must be used upon landing.

(16) ATS DISCPRESS prior to 100 feet AGL.

▲CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

(17) Actual landing distanceINCREASE by a factor of 3.50 (250%).

▲CAUTION: 1. Extreme caution is required during braking to avoid tire damage. Maximize use of reverse thrust.

2. Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

Upon landing:

(18) Both enginesAPPLY maximum reverse thrust.

(19) BrakesAPPLY light to moderate braking until safe taxi speed is maintained.



Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Hydraulics
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Challenger 605	HYD 11-10
	REV 49, Nov 19/18

HYD 2 LO PRESS and HYD 3 LO PRESS (Cont'd)

(20) Brake pressureMONITOR
 Limited braking is possible using brake accumulator pressure.

NOTE: **INBD** and **OUTBD BRAKE PRESS** come on at less than 1,015 psi.

– COMPLETE –

HYD 3 HI TEMP

Condition: Hydraulic system 3 fluid temperature is high.

Objective: Limit the use of hydraulic system 3.

- (1) **Identify phase of flight:**
- ➡ **In flight** – Go to (2)
 - ➡ **On ground** – Go to (6)
- (2) **In flight:**
- (3) HYDRAULIC pumps 3A and 3B OFF
- (4) System No. 3 temperatureMONITOR

→ **If temperature is hotter than 107°C (225°F):**

- Land at the nearest suitable airport.

Prior to landing:

- (5) HYDRAULIC pumps 3A and 3B ON

– COMPLETE –

- (6) **On ground:**
- (7) Do not takeoff. Report to maintenance.

– COMPLETE –

HYD 3 LO PRESS

Condition: Hydraulic system 3 pressure is low.

Objective: Attempt to restore. Review penalties if unable.

- (1) HYDRAULIC pump 3B ON
- (2) Hydraulic pressure and fluid quantity CHECK



Challenger 605	HYD 11-11
	REV 63, Mar 14/22

HYD 3 LO PRESS (Cont'd)

(3) **System 3 quantity is less than 5%, OR pressure is less than 1,800 psi, OR pressure is decreasing rapidly:**

➡ **YES** – Go to (4)

➡ **NO** – Go to (11)

(4) **System 3 quantity is less than 5%, OR pressure is less than 1,800 psi, OR pressure is decreasing rapidly:**

(5) HYDRAULIC pumps 3A and 3B OFF

(6) Land at the nearest suitable airport.

(7) Inoperative systems (see below) REVIEW

- Inboard brakes inoperative (when system 3 accumulator pressure is depleted);
- Normal landing gear extension and retraction inoperative;
- Nose door will remain open;
- Nose wheel steering inoperative (may have nose wheel shimmy);
- Parking brake inoperative.

(8) ANTI-SKID LEAVE ARMED

NOTE: After completing the Landing Gear Manual Extension Procedure, make sure to apply the actual landing distance correction factor given in the table below.

(9) Landing Gear Manual Extension Procedure ACCOMPLISH

I (See Page GEAR 10–16)

(10) Actual landing distance INCREASE
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.70 (70%)	1.50 (50%)

NOTE: Maximize the use of reverse thrust.

– COMPLETE –

(11) **System 3 pressure and quantity are normal:**

(12) HYDRAULIC pump 3B LEAVE ON

(13) Hydraulic pressure and fluid quantity MONITOR

– COMPLETE –

Challenger 605	HYD 11-12
	REV 49, Nov 19/18

HYD EDP 1A (2A)

Condition: Indicated engine driven hydraulic pump pressure is low.

Objective: Attempt to maintain associated hydraulic system pressure normal.

- (1) Associated HYDRAULIC pump 1B or 2B ON
- (2) Hydraulic pressure and fluid quantity CHECK
- (3) **System quantity low:**
 - ➡ **YES** – Go to (4)
 - ➡ **NO** – No further action required. **End of procedure.**
- (4) **System quantity low:**
- (5) Associated HYDRAULIC pump 1B or 2B OFF

NOTE: The engine-driven pump has been operating without hydraulic fluid. Log length of time that engine-driven pump has been running dry. Monitor fluid temperature readouts.

– COMPLETE –

HYD PUMP 1B

Condition: Hydraulic electric pump 1B pressure is low.

Objective: Attempt to restore.

- (1) HYDRAULIC pump 1B ON
- (2) **HYD PUMP 1B stays on:**
 - ➡ **YES** – Go to (3)
 - ➡ **NO** – No further action required. **End of procedure.**
- (3) **HYD PUMP 1B stays on:**
- (4) HYDRAULIC pump 1B OFF

– COMPLETE –

Challenger 605	HYD 11-13
	REV 49, Nov 19/18

HYD PUMP 2B

Condition: Hydraulic electric pump 2B pressure is low.

Objective: Attempt to restore.

(1) HYDRAULIC pump 2B ON

(2) **HYD PUMP 2B stays on:**

➡ **YES** – Go to (3)

➡ **NO** – No further action required. **End of procedure.**

(3) **HYD PUMP 2B stays on:**

(4) HYDRAULIC pump 2B OFF

– COMPLETE –

HYD PUMP 3A (3B)

Condition: Indicated hydraulic system 3 electric pump pressure is low.

Objective: Attempt to restore.

(1) HYDRAULIC pumps 3A and 3B ON

(2) **HYD PUMP 3A (3B) stays on:**

➡ **YES** – Go to (3)

➡ **NO** – No further action required. **End of procedure.**

(3) **HYD PUMP 3A (3B) stays on**

(4) Associated HYDRAULIC pump OFF

– COMPLETE –

HYD SOV 1 (2)

Condition: Indicated hydraulic shutoff valve is not in commanded position.

(1) Land at the nearest suitable airport.

– COMPLETE –

Challenger 605	HYD 11-14
	REV 49, Nov 19/18

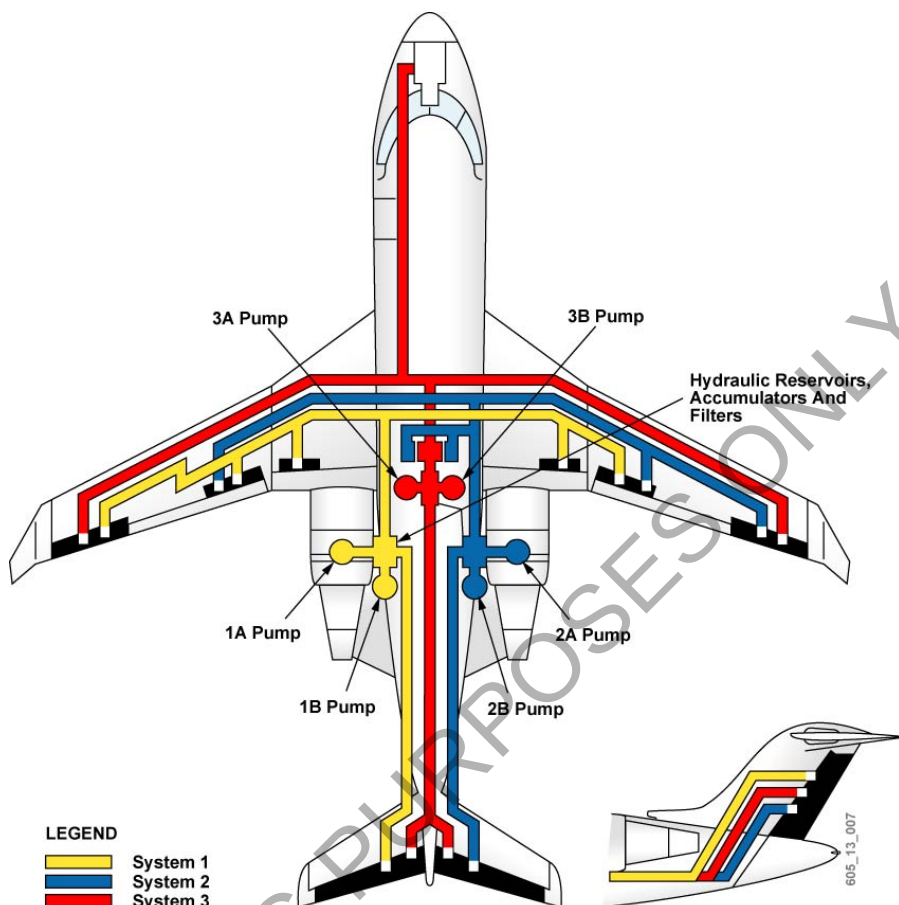
Advisory Messages

HYDRAULICS ADVISORY MESSAGES	
HYD SOV 1 (2) CLSD	Hydraulic SOV 1 (2) confirmed closed, after respective engine fire push switch is pushed.

TRAINING PURPOSES ONLY

SCHEMATICS

Hydraulics System Distribution



Challenger 605	HYD 11-16
	REV 49, Nov 19/18

Power Supply and Circuit Breaker Summary

CB NAME	BUS	CB LOCATION
ESS HYD PUMP SUPPLY	ADG BUS	CBP3-A11
HYD PUMP 1B	AC BUS 2	CBP2-A05
HYD PUMP 2B	AC BUS 1	CBP1-B11
HYD PUMP 3A	AC BUS 2	CBP2-A02
HYD PUMP 3B	AC BUS 1	CBP1-B08
HYD SOV L ENG	DC EMERG BUS	CBP1-S05
HYD SOV R ENG	DC EMERG BUS	CBP1-S04
HYD SYST AC PUMP CONT 1	DC BUS 2	CBP2-G02
HYD SYST AC PUMP CONT 2	DC BUS 1	CBP1-G02
HYD SYST AC PUMP CONT 3A	DC BUS 2	CBP2-G04
HYD SYST IND 1	DC BUS 2	CBP2-G03
HYD SYST IND 2	DC BUS 1	CBP1-G03
HYD SYST IND 3	BATT BUS	CBP2-P03
HYD SYST PUMP CONT 3B	BATT BUS	CBP2-P02

Challenger 605	ICE TOC-1
	REV 50, Mar 25/19

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<i>Non-normal procedures</i> PSP 605-15 – QRH-I Vol. 2	<i>Ice Protection</i>
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Challenger 605	ICE TOC-2
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TRAINING PURPOSES ONLY

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Ice Protection
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ANTI-ICE DUCT



“ANTI-ICE DUCT”

Condition: Leak detected in the fuselage or wing anti-ice ducting.

Objective: Leave icing condition and remove ice accumulation if required.

(1) WING ANTI-ICE OFF

(2) Leave icing conditions.

(3) **ANTI-ICE DUCT** stays on (after 30 seconds):

➡ **YES** – Go to (4)

➡ **NO** – Go to (11)

(4) **ANTI-ICE DUCT** stays on (after 30 seconds):

(5) 14TH STAGE bleeds.....PRESS OUT

NOTE: With both 14th stage bleed air shutoff valves closed, cowl anti-icing, wing anti-icing and thrust reversers are not available.

(6) Both COWL ANTI-ICESPRESS OUT

After leaving icing conditions:

(7) **Ice accumulation observed on the heated portion of the wing:**

➡ **YES** – Go to (8)

➡ **NO** – No further action required. **End of procedure.**

(8) **Ice accumulation observed on the heated portion of the wing:**

(9) Airspeed INCREASE
to V_{MO}/M_{MO} to disperse ice, if possible.

(10) **Ice still present on wing leading edge (unable to remove ice):**

➡ **YES** – Go to (16)

➡ **NO** – No further action required. **End of procedure.**

(11) **ANTI-ICE DUCT** goes out (after 30 seconds):

(12) WING ANTI-ICE STBY

When **ANTI-ICE DUCT comes on:**

(13) WING ANTI-ICE OFF

(14) Repeat steps (12) and (13), as required.



ANTI-ICE DUCT (Cont'd)

(15) Ice still present on wing leading edge (unable to remove ice):

- ➡ YES – Go to (16)
- ➡ NO – No further action required. **End of procedure.**

(16) Ice still present on wing leading edge (unable to remove ice):

Assure adequate stall margin:

(17) Maneuvering speed 200 KIAS minimum.

⚠WARNING: Even small accumulations of ice on the wing leading edge can change the stall characteristics, stall speed or the warning margins provided by the stall protection system.

Prior to landing:

(18) TAWS WARNING, FLAPS OFF PRESS IN

(19) TAWS WARNING, RAAS OFF (if installed) PRESS IN

(20) Landing flaps 20 DEGREES

(21) Approach speed $V_{REF} + 27$

(22) ATS DISC PRESS prior to 100 feet AGL.

⚠CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

(23) Actual landing distance INCREASE as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.80 (80%)	1.70 (70%)

At 50 feet AGL:

(24) Thrust levers IDLE and do not prolong the landing flare.

⚠CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

Challenger 605	ICE 12-3
	REV 50, Mar 25/19

WING OVHT



“WING OVERHEAT”

Condition: Overheat detected in left and/or right wing leading edge.

Objective: Control wing anti-ice using standby mode.

(1) WING ANTI-ICE OFF

(2) After 40 seconds, **WING OVHT** stays on:

➡ **YES** – Go to (3)

➡ **NO** – Go to (8)

(3) After 40 seconds, **WING OVHT** stays on:

(4) Both 14TH STAGE bleedsPRESS OUT

NOTE: With both 14th stage bleed air shutoff valves closed, cowl anti-icing, wing anti-icing and thrust reversers are not available.

(5) Both COWL ANTI-ICESPRESS OUT

(6) Leave icing conditions.

If required:

(7) Ice Dispersal ProcedureACCOMPLISH

I (See Page ICE 12-13)

– COMPLETE –

(8) After 40 seconds, **WING OVHT** goes out:

(9) WING ANTI-ICE STBY

(10) **L HEAT** or **R HEAT** light is **OUT** (and/or **L (R) WING A/ICE** is on):

➡ **YES** – Go to (11)

➡ **NO** – Go to (14)

(11) **L HEAT** or **R HEAT** light is **OUT** (and/or **L (R) WING A/ICE** is on):

(12) 14TH STAGE ISOLPRESS IN

(13) Go to (15)

(14) **L HEAT** or **R HEAT** light is **ON** (and/or **L (R) WING A/ICE** is on):

(15) **WING OVHT** comes on again:

➡ **YES** – Go to (16)

➡ **NO** – No further action required. **End of procedure.**



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Challenger 605	ICE 12-4
	REV 63, Mar 14/22

WING OVHT (Cont'd)

(16) WING OVHT comes on again:

(17) WING ANTI-ICE OFF

(18) Both 14TH STAGE bleedsPRESS OUT

NOTE: With both 14th stage bleed air shutoff valves closed, cowl anti-icing, wing anti-icing and thrust reversers are not available.

(19) Both COWL ANTI-ICEsPRESS OUT

(20) Leave icing conditions.

If required:

(21) Ice Dispersal ProcedureACCOMPLISH

(See Page ICE 12-13)

- COMPLETE -

L (R) AOA HEAT

Condition: Indicated angle of attack heater has failed.

Objective: Attempt to reset.

(1) ANTI-ICE PROBES OFF THEN ON

Do not rely on stall protection system or TAWS if the caution persists.

(2) No further action required.

- COMPLETE -

Challenger 605	ICE 12-5
	REV 65, Jun 24/22

L and R AOA HEAT

Condition: Both angle of attack heaters have failed.

Objective: Attempt to reset.

(1) ANTI-ICE PROBES OFF THEN ON

(2) **L and R AOA HEAT** stays on:

➡ **YES** – Go to (3)

➡ **NO** – No further action required. **End of procedure.**

(3) **L and R AOA HEAT** stays on:

(4) Avoid icing conditions.

(5) **STALL FAIL** Procedure ACCOMPLISH

I (See Page FCTL 07–14)

– COMPLETE –

AUX AOA CASE HEAT

Condition: Auxiliary angle of attack vane base heater has failed.

Objective: Attempt to reset.

(1) ANTI-ICE PROBES OFF THEN ON

Do not rely on low speed cue or Normalized AOA displayed on both PFDs if the caution persists.

(2) No further action required.

– COMPLETE –

AUX AOA VANE HEAT

Condition: Auxiliary angle of attack vane heater has failed.

Objective: Attempt to reset.

(1) ANTI-ICE PROBES OFF THEN ON

Do not rely on low speed cue or Normalized AOA displayed on both PFDs if the caution persists.

(2) No further action required.

– COMPLETE –

Challenger 605	ICE 12-6
	REV 52, Jun 20/19

L (R) COWL A/ICE

Condition: Indicated cowl anti-ice pressure not as expected for commanded position.

Objective: Review penalties. Leave icing conditions if required.

(1) Choose applicable:

- ➡ **Cowl anti-ice is pressed in** – Go to (2)
- ➡ **Cowl anti-ice is pressed out** – Go to (7)

(2) Cowl anti-ice is pressed in (L (R) COWL A/ICE may indicate low pressure):

- (3) Affected 14TH STAGE bleed CHECK PRESSED IN
- (4) ATS DISC PRESS
- (5) Engine thrust INCREASE N₂
to more than 78%.
- (6) CONT IGNITION PRESS IN

→ If wing anti-ice is on/required:

- 14TH STAGE ISOL PRESS IN
- Leave icing conditions.

→ If required:

- Ice Dispersal Procedure ACCOMPLISH
(See Page ICE 12-13)

– COMPLETE –

(7) Cowl anti-ice is pressed out (L (R) COWL A/ICE may indicate high pressure):

- (8) ATS DISC PRESS
- (9) Affected engine thrust REDUCE N₂
to less than 85%.
- (10) Affected 14TH STAGE bleed PRESS OUT

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Ice Protection
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I

Challenger 605

ICE 12-7

REV 50, Mar 25/19

ICE

Condition: Ice detected while wing and/or cowl anti-ice systems are not selected ON.

Objective: Activate anti-ice.

- (1) Thrust 78% N₂ minimum.
- (2) Both COWL ANTI-ICEs PRESS IN
- (3) WING ANTI-ICE NORM

If required:

- (4) L and R WSHLD/WIND ANTI-ICE HI
- (5) **ICE** stays on (**WING/COWL A/ICE ON** is out or **WING L HEAT** or **R HEAT** light is out):
 - ➔ **YES** – Go to (6)
 - ➔ **NO** – No further action required. **End of procedure.**
- (6) **ICE** stays on (**WING/COWL A/ICE ON** is out or **WING L HEAT** or **R HEAT** light is out):
- (7) Thrust INCREASE
- (8) **ICE** is still on (**WING/COWL A/ICE ON** is still out or **WING L HEAT** or **R HEAT** light is still out):
 - ➔ **YES** – Go to (9)
 - ➔ **NO** – No further action required. **End of procedure.**
- (9) **ICE** is still on (**WING/COWL A/ICE ON** is still out or **WING L HEAT** or **R HEAT** light is still out):
- (10) Leave icing conditions.

– COMPLETE –

ICE DETECTORS

Condition: Both ice detectors are failed.

(1) Icing conditions are present or are anticipated:

➡ **YES** – Go to (2)

➡ **NO** – No further action required. **End of procedure.**

(2) Icing conditions are present or are anticipated:

(3) Both COWL ANTI-ICES PRESS IN

(4) WING ANTI-ICE NORM

**(5) WING/COWL A/ICE ON is out or
WING L HEAT or R HEAT light is out:**

➡ **YES** – Go to (6)

➡ **NO** – No further action required. **End of procedure.**

**(6) WING/COWL A/ICE ON is out or
WING L HEAT or R HEAT light is out:**

(7) Thrust INCREASE

**(8) WING/COWL A/ICE ON is still out or
WING L HEAT or R HEAT light is still out:**

➡ **YES** – Go to (9)

➡ **NO** – No further action required. **End of procedure.**

**(9) WING/COWL A/ICE ON is still out or
WING L HEAT or R HEAT light is still out:**

(10) Leave icing conditions.

– COMPLETE –

L (R) PITOT BASE HEAT

Condition: Indicated pitot-static base heater has failed.

Objective: Attempt to reset.

(1) ANTI-ICE PROBES OFF THEN ON

(2) L (R) PITOT BASE HEAT is still on:

➡ **YES** – Go to (3)

➡ **NO** – No further action required. **End of procedure.**

(3) L (R) PITOT BASE HEAT is still on:

(4) Select ADC source to operative side.

– COMPLETE –

Challenger 605	ICE 12-9
	REV 63, Mar 14/22

L (R) PITOT TUBE HEAT

Condition: Indicated pitot-static heater has failed.

Objective: Attempt to reset.

(1) ANTI-ICE PROBES OFF THEN ON

I (2) L (R) PITOT TUBE HEAT is still on:

I ➡ YES – Go to (3)

I ➡ NO – No further action required. End of procedure.

I (3) L (R) PITOT TUBE HEAT is still on:

(4) Select ADC source to operative side.

– COMPLETE –

L (R) PROBE HEAT OFF

Condition: Airplane is on ground and either indicated PROBE switch is selected OFF or, all indicated side anti-ice heaters are failed.

Objective: Attempt to reset.

(1) ANTI-ICE PROBES OFF THEN ON

In flight – No further action required.

– COMPLETE –

L (R) STATIC HEAT

Condition: Indicated standby static port heater has failed.

Objective: Attempt to reset.

(1) ANTI-ICE PROBES OFF THEN ON

I (2) Do not rely on associated instruments if the caution persists:

- ISI for altitude and airspeed, and
- Cabin differential pressure on EICAS.

(3) No further action required.

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Ice Protection
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Challenger 605	ICE 12-10
	REV 63, Mar 14/22

STBY PITOT HEAT

Condition: Standby pitot probe heater has failed.

Objective: Attempt to reset.

- (1) ANTI-ICE PROBES OFF THEN ON
- (2) Do not rely on ISI for airspeed indications if the caution persists.
- (3) No further action required.

– COMPLETE –

TAT PROBE HEAT

Condition: True air temperature probe heater has failed.

Objective: Attempt to reset.

- (1) ANTI-ICE PROBES OFF THEN ON
- (2) Do not rely on air temperature indications if the caution persists.
- (3) No further action required.

– COMPLETE –

L (R) WINDOW HEAT

Condition: Indicated window heat system failed.

- (1) Affected ANTI-ICE, WSHLD/WIND RESET
then LO or HI, as applicable.

NOTE: Moving the COCKPIT and CABIN COLD-HOT knobs to HOT may help reduce fogging.

– COMPLETE –

Challenger 605	ICE 12-11
	REV 50, Mar 25/19

L (R) WING A/ICE

Condition: Insufficient pressure and temperature in indicated wing leading edge while wing anti-ice is selected ON.

Objective: Correct condition.

- (1) Affected 14TH STAGE bleed CHECK PRESSED IN
- (2) Engine thrust INCREASE
- (3) **L (R) WING A/ICE is still on:**

- ➡ **YES** – Go to (4)
- ➡ **NO** – No further action required. **End of procedure.**

- (4) **L (R) WING A/ICE is still on:**

- (5) 14TH STAGE ISOL PRESS IN

- (6) **L (R) WING A/ICE stays on after 30 seconds:**

- ➡ **YES** – Go to (7)
- ➡ **NO** – No further action required. **End of procedure.**

- (7) **L (R) WING A/ICE stays on after 30 seconds:**

- (8) WING ANTI-ICE STBY
- (9) Leave icing conditions.

If required:

- (10) Ice Dispersal Procedure ACCOMPLISH

I (See Page ICE 12-13)

– COMPLETE –

Challenger 605	ICE 12-12
	REV 50, Mar 25/19

WING A/ICE SNSR

Condition: Wing anti-ice control and/or overheat sensors failed.

Objective: Operate in standby mode.

- (1) WING ANTI-ICE STBY
- (2) **L (R) WING A/ICE comes on:**
 - ➡ **YES** – Go to (3)
 - ➡ **NO** – No further action required. **End of procedure.**
- (3) **L (R) WING A/ICE comes on:**
- (4) 14TH STAGE ISOL PRESS IN
- (5) WING ANTI-ICECHECK STBY
- (6) **L (R) WING A/ICE stays on:**
 - ➡ **YES** – Go to (7)
 - ➡ **NO** – No further action required. **End of procedure.**
- (7) **L (R) WING A/ICE stays on:**
- (8) Leave icing conditions.
- If required:**
- (9) Ice Dispersal ProcedureACCOMPLISH
- I (See Page ICE 12–13)

– COMPLETE –

I	Challenger 605	ICE 12-13
		REV 50, Mar 25/19

L (R) WSHLD HEAT

Condition: Indicated windshield heat is failed.

Objective: Attempt to reset.

- (1) Affected ANTI-ICE, WSHLD/WIND RESET then LO or HI, as applicable.

- (2) **L (R) WSHLD HEAT stays on:**

- ➡ **YES** – Go to (3)
- ➡ **NO** – No further action required.

NOTE: Moving the COCKPIT and CABIN COLD–HOT knobs to HOT may help reduce fogging.

– COMPLETE –

- (3) **L (R) WSHLD HEAT stays on:**

- (4) Leave icing conditions as soon as possible.

– COMPLETE –

Ice Dispersal

Condition: Ice accumulation observed on heated portion of the wing.

Objective: Attempt to remove ice accumulation. Review penalties if unable.

After leaving icing conditions, and if ice accumulation is observed on the heated portion of the wing:

- (1) Airspeed INCREASE to V_{MO}/M_{MO} , if possible, to disperse ice.

- (2) **Ice still present on wing leading edge (unable to remove ice):**

- ➡ **YES** – Go to (3)
- ➡ **NO** – No further action required. **End of procedure.**

- (3) **Ice still present on wing leading edge (unable to remove ice):**

Assure adequate stall margin:

- (4) Maneuvering airspeed 200 KIAS MINIMUM

⚠WARNING: Even small accumulations of ice on the wing leading edge can change the stall speed, stall characteristics or the warning margins provided by the stall protection system.



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Ice Dispersal (Cont'd)

Prior to landing:

- (5) TAWS WARNING, FLAPS OFFPRESS IN
- (6) TAWS WARNING, RAAS OFF (if installed)PRESS IN
- (7) Landing flaps 20 DEGREES
- (8) Approach speed $V_{REF} + 27$
- (9) ATS DISC PRESS prior to 100 feet AGL.

CAUTION: ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (10) Actual landing distance INCREASE as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.80 (80%)	1.70 (70%)

At 50 feet AGL:

- (11) Thrust leversIDLE and do not prolong the landing flare.

CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

– COMPLETE –

Insufficient Heat for SUPP GND WING ANTI-ICE



Condition: Not enough heat in the supplemental ground wing anti-ice system. Temp **FAIL LO** light is on.

Objective: Correct condition.

NOTE: Extended taxi with a thrust reverser deployed may cause **FAIL LO** light to come on.



Insufficient Heat for SUPP GND WING ANTI-ICE (Cont'd)

(1) Choose applicable:

- ➡ Thrust reverser stowed – Go to (2)
- ➡ Thrust reverser deployed – Go to (4)

(2) Thrust reverser stowed:

(3) Go to (8)

(4) Thrust reverser deployed:

(5) Thrust reverse lever(s)STOW

(6) FAIL LOFAIL LO light stays on after 30 seconds:

- ➡ YES – Go to (7)
- ➡ NO – No further action required. **End of procedure.**

(7) FAIL LO light stays on after 30 seconds:

(8) SUPP GND WING ANTI-ICE..... OFF

(9) Return to ramp and report for maintenance.

– COMPLETE –

Insufficient Heat for Wing Anti-Icing

Condition: Anti-ice selected ON but not indicating heat (no associated **A/ICE ON** message or green **HEAT** light).

Objective: Correct condition.

(1) Engine thrust..... INCREASE

(2) WING A/ICE ON or WING/COWL A/ICE ON is on or WING L HEAT or R HEAT light is on:

- ➡ YES – No further action required. **End of procedure.**
- ➡ NO – Go to (3)

(3) WING A/ICE ON or WING/COWL A/ICE ON is out or WING L HEAT or R HEAT light is out:

(4) Leave icing conditions.

– COMPLETE –

One Ice Detector Failed

Condition: One ice detector channel has failed. Associated **ICE DETECTOR 1 (2)** status message posted.

NOTE: Icing conditions exist in flight at a TAT of 10°C (50°F) or colder, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or colder.

When in icing conditions:

- (1) Both COWL ANTI-ICEsPRESS IN
- (2) WING ANTI-ICENORM
- (3) **ICE** stays on (**WING/COWL A/ICE ON** is out or **WING L HEAT** or **R HEAT** light is out):
 - ➔ **YES** – Go to (4)
 - ➔ **NO** – No further action required. **End of procedure.**
- (4) **ICE** stays on (**WING/COWL A/ICE ON** is out or **WING L HEAT** or **R HEAT** light is out):
- (5) Thrust INCREASE
- (6) **ICE** is still on (**WING/COWL A/ICE ON** is still out or **WING L HEAT** or **R HEAT** light is still out):
 - ➔ **YES** – Go to (7)
 - ➔ **NO** – No further action required. **End of procedure.**
- (7) **ICE** is still on (**WING/COWL A/ICE ON** is still out or **WING L HEAT** or **R HEAT** light is still out):
- (8) Leave icing conditions.

– COMPLETE –

I

Challenger 605	ICE 12-17
	REV 50, Mar 25/19

SUPP GND WING ANTI-ICE Overheat



Condition: Overheat detected in the supplemental ground wing anti-ice system. Temp **FAIL HI** light is on.

Objective: Correct condition.

- (1) SUPP GND WING ANTI-ICE..... OFF
- (2) Return to ramp and report for maintenance.

– COMPLETE –

TRAINING PURPOSES ONLY

Challenger 605	ICE 12-18
	REV 50, Mar 25/19

Advisory Messages

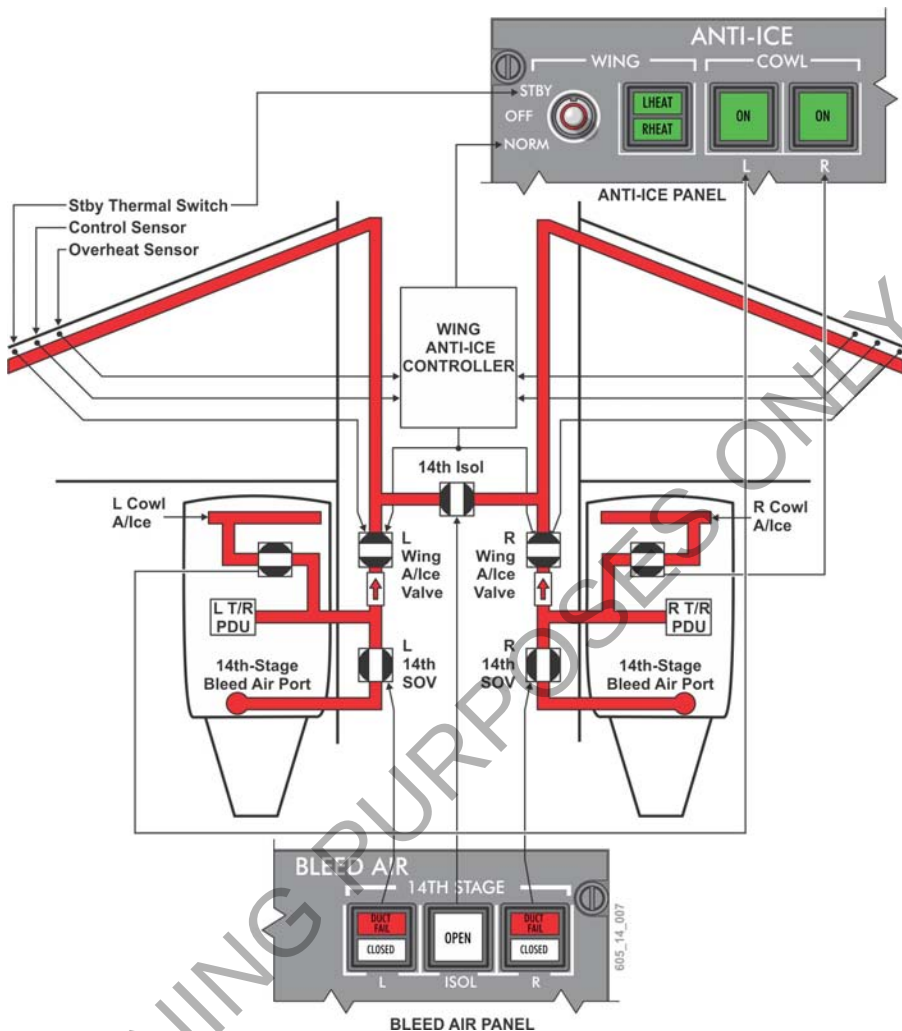
ICE PROTECTION ADVISORY MESSAGES	
COWL A/ICE ON	Both cowl anti-ice on with sufficient heat and with the engines running.
L (R) COWL A/ICE ON	L (R) cowl anti-ice on with sufficient heat and with respective engine running.
ICE	Icing is detected, but wing and cowl anti-ice are both on.
L (R) WINDOW HEAT OK	L (R) window heat test successful.
WING A/ICE ON	Wing anti-ice is on and sufficient heat is being supplied to the wing.
WING A/ICE TEST OK	Wing anti-ice test successful.
WING/COWL A/ICE ON	Both cowl and wing anti-ice are on with the engines running and sufficient heat is being supplied respectively.
L (R) WSHLD HEAT OK	L (R) windshield heat test successful.

Status Messages

ICE PROTECTION STATUS MESSAGES	
ICE DETECTOR 1 (2) FAIL	Respective ice detector failure. See Page ICE 12-16

SCHEMATICS

Wing/Cowl Anti-ice



Power Supply and Circuit Breaker Summary

CB NAME	BUS	CB LOCATION
A/ICE AUTO 1	DC BUS 1	CBP1-F10
A/ICE AUTO 2	DC BUS 2	CBP2-F10
A/ICE MAN 1	DC ESS BUS	CBP4-B10
A/ICE MAN 2	BATT BUS	CBP2-N10
A/ICE VALVE L ENG	BATT BUS	CBP2-N08
A/ICE VALVE R ENG	BATT BUS	CBP2-N09
ADS HTR L CONT	DC ESS BUS	CBP4-A11
ADS HTR R CONT	DC BUS 2	CBP2-G11
HEATERS – AOA L	AC ESS BUS	CBP3-B05
HEATERS – AOA R	AC BUS 2	CBP2-A13
HEATERS – AUX AOA	AC BUS 1	CBP1-C12
HEATERS – PITOT L	AC ESS BUS	CBP3-B04
HEATERS – PITOT R	AC BUS 2	CBP2-A14
HEATERS – PITOT STBY	AC ESS BUS	CBP3-B06
HEATERS – STATIC L	AC ESS BUS	CBP3-B07
HEATERS – STATIC R	AC BUS 2	CBP2-A15
HEATERS – TAT	AC BUS 1	CBP1-C11
HEATERS – WIND L	AC ESS BUS	CBP3-B03
HEATERS – WIND R	AC BUS 2	CBP2-A12
HEATERS – WSHLD L	AC BUS 1	CBP1-A10
HEATERS – WSHLD L	AC BUS 1	CBP1-A11
HEATERS – WSHLD R	AC BUS 2	CBP2-A10
HEATERS – WSHLD R	AC BUS 2	CBP2-A11
HTR CONT L WIND	DC BUS 1	CBP4-B13
HTR CONT R WIND	DC BUS 2	CBP2-F15
HTR CONT L WSHLD	DC BUS 1	CBP1-F14
HTR CONT R WSHLD	DC BUS 2	CBP2-F14
ICE DET 1	AC ESS BUS	CBP3-C08
ICE DET 2	AC BUS 2	CBP2-B12

Challenger 605	OTHER TOC-1
	REV 49, Nov 19/18

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NON-EICAS PROCEDURES

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POWER SUPPLY AND CIRCUIT BREAKER SUMMARY

Power Supply and Circuit Breaker Summary	OTHER 13-6
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TRAINING PURPOSES ONLY

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Other Procedures
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Challenger 605	OTHER TOC-2
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TRAINING PURPOSES ONLY

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Other Procedures
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Challenger 605	OTHER 13-1
	REV 49, Nov 19/18

EMER LTS OFF

Condition: Emergency lights are OFF (by manual selection or due to low power supply voltage).

(1) EMER LTSCHECK STATUS

– COMPLETE –

Lightning Strike

Condition: Suspected lightning strike.

Objective: Review penalties.

NOTE: May be associated with an unannounced stall system failure. Failure to the stall system must be assumed.

(1) STALL PROT, PUSHER
(either left or right) OFF

Prior to landing:

(2) Approach speed $V_{REF} + 10$
minimum, to maintain margin above stall.

(3) Actual landing distance INCREASE
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.40 (40%)	1.30 (30%)

– COMPLETE –

Pilot Incapacitation

In case of pilot incapacitation, the following are general guidelines:

- Ensure autopilot is engaged.
- Check that incapacitated pilot does not interfere with flight controls. It is preferable to have the incapacitated pilot removed from the flight compartment.
- If unable to remove the incapacitated pilot, lock shoulder harness of incapacitated pilot and move seat rearwards.
- If an immediate landing is imperative, obtain advice on most suitable airport where medical assistance can be readily rendered.
- Check on possibility of obtaining assistance from pilots who may be travelling as passengers on board the airplane.

– COMPLETE –

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Other Procedures
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Challenger 605	OTHER 13-2
	REV 49, Nov 19/18

Rejected Takeoff

Condition: Decision to stop the airplane.

Objective: Discontinue takeoff and stop airplane.

Simultaneously:

- (1) Takeoff DISCONTINUE IMMEDIATELY
- (2) Thrust levers IDLE
- (3) ATS DISC PRESS
- (4) Wheel brakes APPLY MAX BRAKING until safe stop.
- (5) FLIGHT SPOILER MAX
- (6) Operating thrust reverser(s) APPLY MAX consistent with directional control.



FROM Quick Reference Card



(7) Choose applicable, takeoff was rejected due to:

- ➡ Engine fire or severe damage – Go to (8)
- ➡ Engine failure – Go to (10)
- ➡ Other reasons – Go to (13)

(8) Takeoff was rejected due to engine fire or severe damage:

- (9) Engine Fire/Severe Damage Procedure (On Ground) ACCOMPLISH
(See Page ENG 05–3)

– COMPLETE –

(10) Takeoff was rejected due to engine failure:

- (11) Thrust lever of failed engine SHUT OFF
- (12) Go to (14)

– COMPLETE –

(13) Takeoff was rejected due to other reasons:

(14) Evacuation is required:

- ➡ YES – Go to (15)
- ➡ NO – Go to (17)

(15) Evacuation is required:

- (16) Emergency Evacuation Procedure ACCOMPLISH
(See Page F–LDG 08–1)

– COMPLETE –



Non-normal procedures PSP 605–15 – QRH–I Vol. 2	Other Procedures
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Challenger 605	OTHER 13-3
	REV 49, Nov 19/18

Rejected Takeoff (Cont'd)

(17) Evacuation is NOT required:

(18) Passengers ADVISE
to remain in their seats.

– COMPLETE –

Stall Recovery

Condition: Approaching stall. May be associated with buffeting, uncommanded roll, stick shaker activation and/or stall warbler aural warning.

Objective: Recover from stall.

- | | | |
|-----|----------------------|-------------|
| (1) | Autopilot | DISCONNECT |
| (2) | Pitch attitude | LOWER NOSE |
| (3) | ATS DISC | PRESS |
| (4) | Thrust levers | MAX POWER |
| (5) | Roll attitude | WINGS LEVEL |
| (6) | FLIGHT SPOILER | RETRACT |

➤ FROM Quick Reference Card ◀

After airspeed increases and stall warning is out:

- (7) Pitch attitudeADJUST
- (8) Thrust levers and airplane configurationADJUST

NOTE:

1. It is essential that the AOA be immediately reduced, even if this means a loss of altitude.
2. Avoid abrupt or aggressive pitch control inputs during recovery. Inappropriate recovery inputs can result in a secondary stall.
3. Height loss resulting from high AOA recovery, especially at cruise altitude and/or low initial thrust conditions, can be significant.

– COMPLETE –

Challenger 605	OTHER 13-4
	REV 49, Nov 19/18

Suspected External Damage (Ground Only)

Condition: Crew suspects external damage to airplane caused by impact from a bird or other animal, drone, ground equipment, etc.

Objective: Assess affected area to determine if dispatch is possible.

NOTE: 1. If the crew suspects that the airplane has external damage caused by foreign object impact, (bird or other animal, drone, ground equipment, maintenance vehicle, etc.) either during ground operation or from a previous flight, a thorough walkaround, must be performed.

- (1) Suspected External Damage
(Ground Only) ProcedureACCOMPLISH
Refer to Flight Crew Operating Manual Volume 1, OTHER PROCEDURES.

- COMPLETE -

Windshield Outer Face Ply Failure

Condition: Visible damage to the windshield external layer.

Objective: Avoid icing conditions.

- (1) Affected ANTI-ICE, WSHLD/WIND OFF
- (2) Minimize flight into visible moisture or icing conditions.

NOTE: Icing conditions exist in flight at a TAT of 10°C (50°F) or colder, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or colder.

- COMPLETE -

Challenger 605	OTHER 13-5
	REV 49, Nov 19/18

Status Messages

OTHER PROCEDURES STATUS MESSAGES	
EMER LTS ON	Emergency lights on.

TRAINING PURPOSES ONLY

Power Supply and Circuit Breaker Summary

CB NAME	BUS	CB LOCATION
AFT CABIN UPWASH LTS	DC BUS 2	CBP2-K05
BEACON LIGHTS	DC BUS 2	CBP2-G13
BOARD / STAIR LTS	MAIN BATT DIR	CBP6-A06
CLOCK	DC ESS BUS	CBP4-C07
CLOCK STANDBY	MAIN BATT DIR	CBP6-B03
CMS – MCE	DC UTILITY BUS 2	CBP2-E01
CMS – ZONE 1	DC UTILITY BUS 2	CBP2-E02
CMS – ZONE 2	DC UTILITY BUS 2	CBP2-E03
CMS – ZONE 3	DC UTILITY BUS 2	CBP2-E04
CMS MCE	DC BUS 2	CBP2-K01
ELT	BATT BUS	CBP1-M05
EXT SERV LTS	MAIN BATT DIR	CBP6-A05
FWD CABIN UPWASH LTS	DC BUS 1	CBP1-E14
INTG LTS PED & OVHD	BATT BUS	CBP1-P08
INTG LTS PLT C/PLT	DC ESS BUS	CBP4-C03
LDG/TAXI LTS NOSE	AC BUS 1	CBP1-C14
LDG/TAXI LTS L WING	AC BUS 1	CBP1-C15
LDG/TAXI LTS R NOSE	AC BUS 2	CBP2-C14
LDG/TAXI LTS R WING	AC BUS 2	CBP2-C15
LIGHTS CHRT HDLR	DC BUS 1	CBP1-G11
LIGHTS CKPT FLOOR	DC BUS 1	CBP1-G13
LIGHTS EMERG	DC ESS BUS	CBP4-A12
LIGHTS FLOOD	DC ESS BUS	CBP4-A13
LIGHTS REAR A/COLL	DC BUS 1	CBP1-G14
LIGHTS WING A/COLL	DC BUS 2	CBP2-G14
LIGHTS WING INSP	DC BUS 1	CBP1-G12
LOGO LIGHTS	AC UTILITY BUS 1	CBP1-D08
MAP LIGHT	BATT BUS	CBP1-P09
MAP LIGHT	BATT BUS	CBP2-N11

**Power Supply and Circuit Breaker Summary
(Cont'd)**

CB NAME	BUS	CB LOCATION
NAV LTS	AC BUS 1	CBP1-C13
OXYGEN MONITOR	BATT BUS	CBP2-P09
PASS WARN SIGNS	DC BUS 1	CBP1-F11
PULSE LDG LTS	DC BUS 1	CBP1-G10
SERV LTS	MAIN BATT DIR	CBP6-A07
WATER – HTR CONT	DC UTILITY BUS 1	CBP1-E04
WATER – SYSTEM	DC UTILITY BUS 1	CBP1-E05
WATER SYSTEM	APU BATT DIR	CBP5-B11
WTR HTRS – GALLEY	AC UTILITY BUS 2	CBP2-D07
WTR HTRS – LAV	AC UTILITY BUS 2	CBP2-D08
WTR HTRS – LINE	AC UTILITY BUS 2	CBP2-D06
WTR HTRS – TANK	AC UTILITY BUS 2	CBP2-D05

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TRAINING PURPOSES ONLY

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Other Procedures
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Challenger 605	SMK TOC-1
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ENG BOTTLE 1 (2) LO	SMK 14-10
L (R) FIRE FAIL	SMK 14-10
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SCHEMATICS

Bleed Air Leak Detection System	SMK 14-13
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POWER SUPPLY AND CIRCUIT BREAKER SUMMARY

Power Supply and Circuit Breaker Summary	SMK 14-14
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Challenger 605	SMK TOC-2
	REV 49, Nov 19/18

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TRAINING PURPOSES ONLY

Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Smoke / Fire
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Challenger 605	SMK 14-1
	REV 49, Nov 19/18

Smoke/Fire/Fumes

Condition: Smoke or fumes suspected or observed in the cabin / cockpit area.

Objective: Land immediately. Fight fire if time and conditions permit.

⚠WARNING: Time is critical during smoke/fire/fumes emergencies. The flight crew should consider an immediate landing anytime the situation cannot be controlled.

⚠CAUTION: Passenger oxygen masks should not be deployed when performing smoke or fire procedures.

(1) Oxygen masks/goggles
(if required)DON
Set to 100% and EMERGENCY.

(2) Crew communicationESTABLISH

(3) Land at the nearest suitable airport.

⚠CAUTION: Continuing to destination is not recommended. If the decision to continue is made, the flight crew must acknowledge that the threat has been positively identified, confirmed to be extinguished and the smoke or fumes have dissipated.

⚠WARNING: Anytime smoke or fumes becomes the greatest threat, accomplish the Smoke or Fumes Removal Procedure. (See Page SMK 14-7)

(4) **Time and conditions permit and source can be quickly identified:**

➡ **YES** – Go to (5)

➡ **NO** – Go to (7)

(5) **Time and conditions permit and source can be quickly identified:**

(6) **Choose applicable, source is:**

➡ **Air-conditioning** – Go to (9)

➡ **Electrical** – Go to (22)

➡ **Cabin** – Go to (52)



Challenger 605	SMK 14-2
	REV 49, Nov 19/18

Smoke/Fire/Fumes (Cont'd)

- (7) Time and conditions do NOT permit:
- (8) Smoke or Fumes
Removal ProcedureACCOMPLISH
(See Page SMK 14–7)
– COMPLETE –

- (9) Source is air-conditioning:

If APU is supplying bleed air to the packs:

- (10) PACKsTRANSFER
to engine bleeds.
- (11) L PACKPRESS OUT

Monitor & determine if smoke/fire/fumes continue.

- (12) Smoke/fire/fumes continue:

- ➡ YES – Go to (13)
- ➡ NO – Go to (20)

- (13) Smoke/fire/fumes continue:

- (14) L PACKPRESS IN
- (15) R PACKPRESS OUT

Monitor & determine if smoke/fire/fumes continue.

- (16) Smoke/fire/fumes continue:

- ➡ YES – Go to (17)
- ➡ NO – Go to (20)

- (17) Smoke/fire/fumes continue:

- (18) R PACKPRESS IN
- (19) Go to (21)

- (20) Smoke/fire/fumes stop:

If required:



Smoke/Fire/Fumes (Cont'd)

(21) Smoke or Fumes
Removal ProcedureACCOMPLISH
(See Page SMK 14–7)
– COMPLETE –

(22) Source is electrical:

(23) AC and DC electrical
loadsMONITOR

(24) Source / bus is positively identified:

- ➔ **YES** – Go to (25)
- ➔ **NO** – Go to (31)

(25) Source / bus is positively identified:

⚠WARNING: The table that follows provides guidance on how to isolate each electrical bus. Select only the steps required to isolate the positively identified bus. Do not use the table in an attempt to identify the source bus by trial and error.

(26) Affected electrical system ISOLATE
See table below:

IDENTIFIED BUS	APPLICABLE CIRCUIT BREAKER/SWITCH
AC UTLY 1 & 2 DC UTLY 1 & 2	AC/DC UTILITY OFF
AC BUS 1 and DC BUS 1	AC ESS XFER..... ALTN Left AUTOXFER..... OFF GEN 1 OFF
AC BUS 2 and DC BUS 2	Right AUTOXFER..... OFF GEN 2 OFF
DC BUS 1 CONTROL	FEED (1E8) PULL



Smoke/Fire/Fumes (Cont'd)

IDENTIFIED BUS	APPLICABLE CIRCUIT BREAKER/SWITCH
DC BUS 2 CONTROL	FEED (2E8)..... PULL
AC ESS BUS	115 VAC ESS FEED (1B5 & 2B8)..... PULL
DC ESS BUS	RCCB CONT (2M6), FEED 1 (2M8) and FEED 2 DC ESS (2M10)..... PULL
DC BATT BUS	RCCB CONT MAIN (2M4), APU BATT (2M5), FEED 1 (2M7) and FEED 2 BATT BUS (2M9) PULL
DC EMER BUS	APU BATT FEED (1S7) and EMERG BUS FEED (1M1) PULL

(27) Smoke/fire/fumes stop:

- ➔ **YES** – Go to (28)
- ➔ **NO** – Go to (31)

(28) Smoke/fire/fumes stop:

(29) BUS INOP LIST REVIEW
(See Page ELEC 04–15)

If required:

(30) Smoke or Fumes
Removal ProcedureACCOMPLISH
(See Page SMK 14–7)

– COMPLETE –
▼

Challenger 605	SMK 14-5
	REV 51, Apr 05/19

Smoke/Fire/Fumes (Cont'd)

(31) Source / bus is NOT positively identified:

- (32) ADG manual deploy
handlePULL
and leave extended.
- (33) AC ESS BUS CHECK POWERED

When ADG power is established:

- I** (34) STAB TRIM, CH 2 PRESS TO ENGAGE
- (35) R PACKPRESS OUT
- (36) GEN 1 and GEN 2 OFF/RESET
and leave off.
- (37) APU GEN..... OFF/RESET
and leave off.
- (38) Descent.....INITIATE
to 10,000 feet or lowest safe altitude,
whichever is higher.
- (39) Leave icing conditions (if possible).

(40) Smoke/fire/fumes continue:

- ➔ **YES** – Go to (41)
- ➔ **NO** – Go to (49)

(41) Smoke/fire/fumes continue:

- (42) GEN 1 and GEN 2 ON
- (43) ADG manual deploy
handleSTOW
- (44) ADG PWR TXFR
OVERRIDEPUSH
- (45) HYDRAULIC pump 3B..... AUTO/ON
- (46) CAS messages REVIEW
- (47) Affected systems RESTORE



Smoke/Fire/Fumes (Cont'd)

(48) Smoke or Fumes

Removal ProcedureACCOMPLISH

(See Page SMK 14–7)

– COMPLETE –

(49) Smoke/fire/fumes stop:

The following significant systems are not available when on emergency power only:

- Yaw damper 2 and AFCS 2,
- Ground spoilers,
- Flaps channel 2,
- Stabilizer trim channel 1,
- Mach trim, aileron and rudder trim,
- Hydraulic pumps 1B, 2B & 3A,
- DME, FMS 2,
- Copilot's PFD, MFD, navigation & communication systems,
- Right pack,
- Right probe heaters, right AOA vane heater and right ice detector,
- Nose wheel steering system, anti-skid system,
- Both windshield heaters, right window heater,
- Copilot's instrument lights, NAV lights, and L & R landing lights,
- SPS altitude compensation, and
- Auto throttle system.

NOTE: 1. With Mach trim not available, do not exceed 250 KIAS (0.7 M), unless the autopilot is engaged and functioning normally.

2. Without probe and AOA vane heaters, do not rely on stall protection system, ISI for altitude and airspeed, and cabin ΔP on EICAS.

3. Flaps will operate (extend or retract) at half speed.

(50) Actual landing distance INCREASE
as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.50 (150%)	1.85 (85%)



Challenger 605	SMK 14-7
	REV 49, Nov 19/18

Smoke/Fire/Fumes (Cont'd)

▲CAUTION: Before selecting reverse thrust, ensure that nose wheel has touched down, and nose down elevator is applied.

(51) Smoke or Fumes
Removal ProcedureACCOMPLISH
(See Page SMK 14–7)
– COMPLETE –

(52) Source is cabin:

(53) Designated crew member ADVISE
to isolate and extinguish source,
and secure area.

(54) AC/DC UTILITY OFF

(55) Smoke or Fumes
Removal ProcedureACCOMPLISH
(See Page SMK 14–7)
– COMPLETE –

Smoke or Fumes Removal

Condition: Smoke or fumes are a concern for the safe continuation of the flight.

Objective: Clear smoke from cockpit and cabin.

(1) Descent.....INITIATE
to 10,000 feet or lowest safe altitude,
whichever is higher.

(2) Passenger oxygenCLOSED

(3) PRESS CONTMANUAL

(4) MAN ALTHOLD UP

(5) MAN RATEMAX INCR



Challenger 605	SMK 14-8
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Smoke or Fumes Removal (Cont'd)

(6) EMER DEPRESSPRESS IN

NOTE: Ram air may be used to assist in smoke clearance. Maintain at least 250 KIAS to clear smoke using the RAM AIR valve.

(7) **Smoke/fire/fumes are under control:**

➔ **YES** – Go to (8)

➔ **NO** – Go to (10)

(8) **Smoke/fire/fumes are under control:**

If required:

(9) Smoke/Fire/Fumes

ProcedureCONTINUE

(See Page SMK 14–1)

– COMPLETE –

(10) **Smoke/fire/fumes are uncontrolled:**

(11) Land immediately at the nearest suitable airport.

▲**CAUTION:** 1. Depending upon the severity of the situation, the flight crew should expedite the landing.

2. The crew should also consider an overweight, tailwind, ditching or a forced off-airport landing.

(12) Emergency Evacuation

ProcedureACCOMPLISH

(See Page F-LDG 08–1)

– COMPLETE –

Non-normal procedures PSP 605–15 – QRH-I Vol. 2	Smoke / Fire
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Challenger 605	SMK 14-9
	REV 49, Nov 19/18

APU BOTTLE LO

Condition: APU firex bottle pressure is low.

Objective: Verify bottle status. Shutdown APU if required.

- (1) FIRE DETECTION TESTWARN
- (2) APU bottle light.....CHECK ON
to confirm that bottle is charged.
- (3) **APU bottle light is on:**
 - ➡ **YES** – No further action required. **End of procedure.**
 - ➡ **NO** – Go to (4)
- (4) **APU bottle light is out:**
- (5) PACKs.....TRANSFER
to engine bleeds.
- (6) APU SHUTDOWN

– COMPLETE –

APU FIRE FAIL

Condition: APU fire detection loop is failed.

Objective: Shutdown APU unless required for flight.

- (1) PACKs.....TRANSFER
to engine bleeds.
- (2) **APU generator is essential (both main generators failed):**
 - ➡ **YES** – Go to (3)
 - ➡ **NO** – Go to (5)
- (3) **APU generator is essential (both main generators failed):**
- (4) APU indicationsMONITOR

– COMPLETE –

- (5) **APU generator is NOT essential:**
- (6) APU SHUTDOWN

– COMPLETE –

Challenger 605	SMK 14-10
	REV 49, Nov 19/18

ENG BOTTLE 1 (2) LO

Condition: Indicated engine fire bottle pressure is low.

Objective: Verify bottle status.

- (1) FIRE DETECTION TESTWARN
- (2) Engine Bottle 1 and 2 lights.....CHECK ON
to confirm that engine bottles are charged.
- (3) No further action required.

NOTE: If any one bottle light is out, assume that the affected bottle is discharged.

– COMPLETE –

L (R) FIRE FAIL

Condition: Indicated engine fire detection loop is failed.

- (1) Affected engineMONITOR

– COMPLETE –

L (R) JETPIPE OVHT FAIL

Condition: Indicated jetpipe and pylone overheat detection loop is failed.

- (1) Affected engineMONITOR

– COMPLETE –

MLG OVHT FAIL

Condition: Gear bay overheat detection loop is failed.

Objective: Prevent potential high temperature in gear bay.

- (1) **Failure occurs within 30 minutes after takeoff:**

➡ **YES** – Go to (2)

➡ **NO** – No further action required. **End of procedure.**

- (2) **Failure occurs within 30 minutes after takeoff:**

- (3) Airspeed 197 KIAS
maximum.

- (4) Altitude 20,000 FEET OR LESS
but not below lowest safe altitude.



Non-normal procedures PSP 605-15 – QRH-I Vol. 2	Smoke / Fire
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MLG OVHT FAIL (Cont'd)

- (5) LDG GEAR DN
Leave DN for 5 minutes, then UP and resume climb.

– COMPLETE –

TRAINING PURPOSES ONLY

Challenger 605	SMK 14-12
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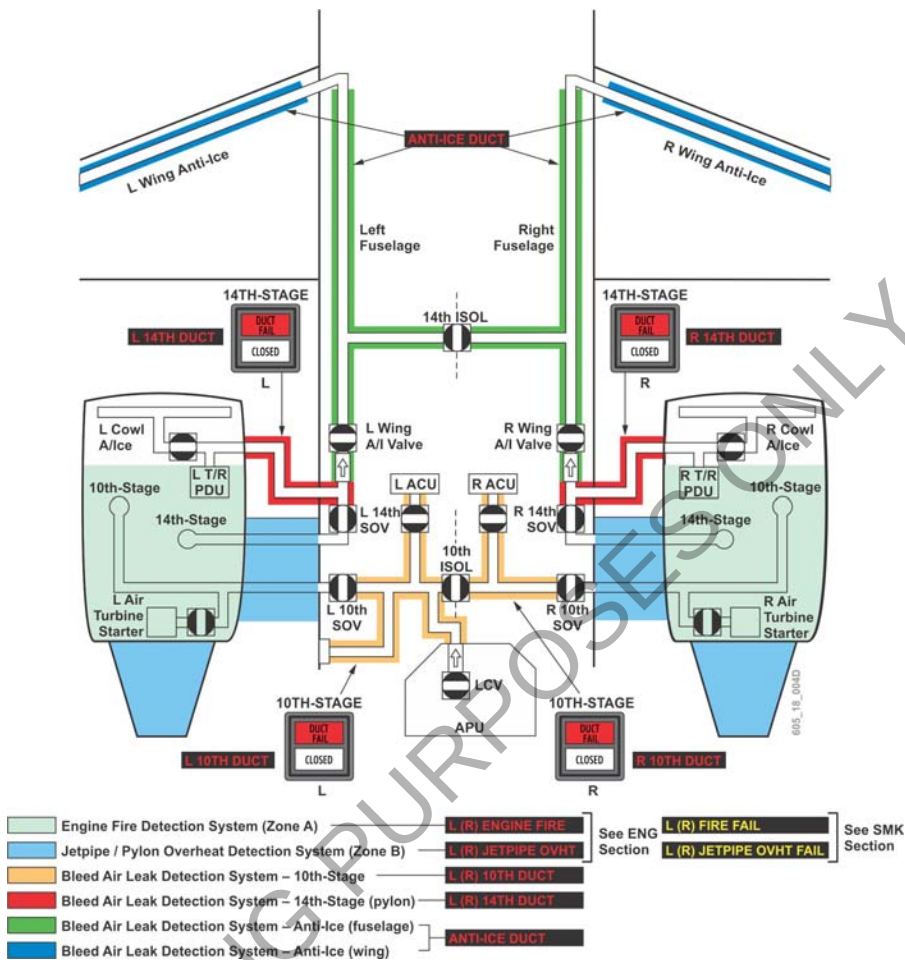
Advisory Messages

SMOKE / FIRE ADVISORY MESSAGES	
APU SQUIB 1 (2) OK	APU firex squib 1 (2) test successful.
L (R) ENG SQUIB 1 (2) OK	L (R) engine firex squib 1 (2) test successful.

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SCHEMATICS

Bleed Air Leak Detection System



Challenger 605	SMK 14-14
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Power Supply and Circuit Breaker Summary

CB NAME	BUS	CB LOCATION
APU FAULT ANN	BATT BUS	CBP1-L02
APU FIRE EXT	DC EMERG BUS	CBP1-R05
ENG JET PIPE OVHT L	BATT BUS	CBP1-N08
ENG JET PIPE OVHT R	BATT BUS	CBP1-N07
FIRE DETECTORS APU	BATT BUS	CBP1-N11
FIRE DETECTORS L ENG	BATT BUS	CBP1-N10
FIRE DETECTORS R ENG	BATT BUS	CBP1-N09
FIRE DETECTORS TEST	BATT BUS	CBP1-N12
FIRE EXT 1 L ENG	DC EMERG BUS	CBP1-R02
FIRE EXT 1 R ENG	DC EMERG BUS	CBP1-R01
FIRE EXT 2 L ENG	DC EMERG BUS	CBP1-R04
FIRE EXT 2 R ENG	DC EMERG BUS	CBP1-R03
MLG BAY OVHT DET	BATT BUS	CBP2-N04
SMOKE DETECT	DC BUS 2	CBP2-K04

GO / NO GO GUIDE

This Go / No Go Guide contains data for quick reference.

This guide in no way supersedes data outlined in the MMEL.

In case of conflict, the MMEL takes precedence.

AFTER PERFORMING THE APPLICABLE QRH PROCEDURE and the message persists, check dispatchability as follows:

WARNING MESSAGES (RED)	
EICAS MESSAGE	DISPATCH AND / OR MEL REF
AFCS MSGS FAIL	22-10-1
ANTI-ICE DUCT	NO GO
APU FIRE	NO GO
APU OVERSPEED	49-00-2
APU OVERTEMP	49-00-2
CABIN ALT	NO GO
CONFIG (AP, FLAPS, SPOILERS, STAB).	NO GO
DIFF PRESS	NO GO
EMER PWR ONLY	NO GO
ENG OIL PRESS L (R)	NO GO
ENGINE	NO GO
ENGINE FIRE L (R)	NO GO
ENGINE START SOV L (R)	NO GO
GEAR DISAGREE	NO GO
JETPIPE OVHT L (R)	NO GO
MLG BAY OVHT	NO GO
NOSE DOOR OPEN	NO GO
PARKING BRAKE	NO GO
PASSENGER DOOR	NO GO
WING OVHT	NO GO
10TH (14TH) DUCT L (R)	NO GO

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GO / NO GO GUIDE

CAUTION MESSAGES (AMBER)	
EICAS MESSAGE	DISPATCH AND / OR MEL REF
AC 1 (2) AUTOXFER	NO GO
AC BUS 1 (2)	NO GO
AC ESS BUS	NO GO
ADS-B OUT FAIL	34-54-1
ADS-B OUT FAULT	34-54-1
AILERON PCU	NO GO
ALT COMP FAIL	NO GO
AOA HEAT L (R)	30-31-1
AP PITCH TRIM	22-10-1
AP HOLDING LWD (RWD, NOSE DOWN, NOSE UP)	22-10-1
APR CMD SET	NO GO
APR INOP	76-10-1
APU BATTERY OFF	24-32-3
APU BOTTLE LO	26-10-1
APU PUMP	28-25-1
APU FIRE FAIL	26-10-1
APU GEN OFF	24-22-2
APU GEN OVLD	24-22-2
APU LCV FAIL	49-00-1. Return to ramp.
APU NEG G SOV	NO GO
APU OIL PRESS (TEMP)	49-00-2
APU SOV	NO GO
A/SKID INBD (OUTBD)	32-44-1. Return to ramp.
AUTO PRESS	21-31-1
AUTO TAIL XFER	28-13-2. Return to ramp.
AUTO TAIL XFER INHIB	28-13-2. Return to ramp.
AUX TANK HEAVY	NO GO
AUX AOA CASE (VANE) HEAT	30-31-1
BAGGAGE DOOR	52-70-4
BATTERY BUS	NO GO
BRAKE PRESS INBD (OUTBD)	NO GO
BULK FUEL TEMP	NO GO

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GO / NO GO GUIDE

CAUTION MESSAGES (AMBER)	
EICAS MESSAGE	DISPATCH AND / OR MEL REF
CABIN ALT	NO GO
COWL A/ICE L (R)	30-20-1
CPAM FAIL	21-33-1. Return to ramp.
DC BUS 1 (2)	NO GO
DC EMER (ESS) BUS	NO GO
DUMP VALVE OPEN	28-13-2. Return to ramp.
EFIS COMP INOP	NO GO
EFIS MISCOMP	NO GO
EICAS COMP INOP	NO GO
ELEVATOR SPLIT	NO GO
EMER DEPRESS	NO GO
EMER LTS OFF	NO GO
ENG BOTTLE 1 (2) LO	NO GO
ENG MISCOMP L (R)	NO GO
ENG SOV L (R)	NO GO
ENGINE L (R)	NO GO
FIRE FAIL L (R)	NO GO
FLAPS FAIL	NO GO
FLT SPLRS	NO GO
FLT SPLRS DEPLOY	NO GO
FUEL FILTER L (R)	NO GO
FUEL IMBALANCE	NO GO
FUEL LO PRESS L (R)	NO GO
FUEL LO TEMP L (R)	NO GO
FUEL PUMP L (R)	28-23-1
FUEL XFLOW SOV OPEN	NO GO
GEN 1 (2) OFF (OVLD) <DOT>	NO GO
GEN 1 (2) OFF (OVLD) <FAA>	24-21-1
GND SPLRS	27-67-1
GND SPLRS DEPLOY	NO GO
GND SPLRS NOT ARMED	NO GO
HYD 1 (2) (3) HI TEMP	NO GO

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GO / NO GO GUIDE

CAUTION MESSAGES (AMBER)	
EICAS MESSAGE	DISPATCH AND / OR MEL REF
HYD 1 (2) (3) LO PRESS	NO GO
HYD EDP 1A (2A)	NO GO
HYD PUMP 1B (2B)	29-11-1
HYD PUMP 3A	29-12-1
HYD PUMP 3B	NO GO
HYD SOV 1 (2)	NO GO
ICE	NO GO
ICE DETECTORS	30-80-1
JETPIPE OVHT FAIL L (R)	NO GO
MACH TRIM	22-20-1
MAIN BATT OFF	24-32-1
MLG OVHT FAIL	NO GO
OXY LO PRESS	NO GO
PACK HI PRESS L (R)	21-51-2
PACK HI TEMP L (R)	21-51-2
PACK NOT OFF L (R)	NO GO
PARK BRAKE SOV	NO GO
PAX DOOR LATCH	52-70-2
PAX DOOR OUT HNDL	52-70-2
PAX DOOR STOW	52-70-2
PAX OXY ON	NO GO
PFD X-TALK FAIL	NO GO
PITOT TUBE (BASE) HEAT L (R)	30-31-1
PRI (SEC) TAIL XFER PUMP	28-15-3. Return to ramp.
PROBE HEAT OFF L (R)	NO GO
PROX SYS	NO GO
REV UNLOCKED L (R)	78-30-1. Return to ramp.
REV UNSAFE L (R)	78-30-1. Return to ramp.
SCAV EJECTOR L (R)	NO GO
STAB TRIM	NO GO
STALL FAIL	NO GO
STATIC HEAT L (R)	30-31-1

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GO / NO GO GUIDE

CAUTION MESSAGES (AMBER)	
EICAS MESSAGE	DISPATCH AND / OR MEL REF
STBY PITOT HEAT	30-31-1
STEERING INOP <DOT>	32-50-1. Return to ramp.
STEERING INOP <FAA>	NO GO
TAIL DUMP SOV	28-13-2. Return to ramp.
TAIL TANK HEAVY	28-13-2. Return to ramp.
TAIL XFER SOV	28-13-2. Return to ramp.
TAIL XFER SOV CLSD (OPEN)	28-13-2. Return to ramp.
TAT PROBE HEAT	30-31-1
WINDOW HEAT (L)	NO GO
WINDOW HEAT (R)	30-41-1
WING A/ICE L (R)	30-12-1. Return to ramp.
WING A/ICE SNSR	30-12-1. Return to ramp.
WOW INPUT (OUTPUT)	NO GO
WSHLD HEAT L (R)	30-41-1
XPDR 1 (2) FAIL	34-54-1
XPDR 1 & 2 FAIL	34-54-1
YAW DAMPER	22-20-2

GO / NO GO GUIDE

STATUS MESSAGES (WHITE)	
EICAS MESSAGE	DISPATCH AND / OR MEL REF
AC ESS ALTN	NO GO
AFCS 1 (2) INOP	22-10-2
APU BATT CHGR FAIL	24-32-4
CAS MISCOMP	NO GO
CPAM FAULT	21-33-1
CVR FAIL	23-71-1
DATALINK FAIL	23-11-4
DCU 1 (2) APR FAIL	76-10-1
DCU 1 (2) AURAL INOP	NO GO
DCU 1 (2) INOP	NO GO
DISPLAY FAN FAIL	21-24-1
EMER LTS ON	NO GO
ENG ECU FAIL L (R)	76-00-1
ENG OIL SENSOR L (R)	NO GO
ESS TRU 1 (2) FAIL	NO GO
EXTERNAL DOOR OPEN <DOT>	NO GO
EXTERNAL DOOR OPEN <FAA>	52-70-14
FANS 1/A FAIL	23-11-4
FD 1 (2) FAIL	22-10-2
FDR FAIL	31-31-1
FLAPS MOTOR OVHT	27-50-1. Return to ramp.
FUEL CH 1 (CH 2) FAIL <DOT>	NO GO
FUEL CH 1 (CH 2) FAIL <FAA>	28-41-4
FUEL QTY ACCURACY (ATT CORR)	NO GO
HGS FAIL <DOT>	NO GO
HGS FAIL <FAA>	34-32-1
IAPS DEGRADED (OVERTEMP)	22-13-1. Return to ramp.
ICE DETECTOR 1 (2) FAIL	30-80-1
IRS 1 (2) (3) IN ATT	NO GO (with 3rd IRS, 34-23-2)
IRS 1 (2) (3) POWER FAULT	NO GO (with 3rd IRS, 34-23-2)
IRS 3 INOP	34-23-2
MAIN BATT CHGR FAIL	24-32-2. Return to ramp.

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GO / NO GO GUIDE

STATUS MESSAGES (WHITE)	
EICAS MESSAGE	DISPATCH AND / OR MEL REF
MAIN EJECTOR FAIL L (R)	NO GO
MFD X-TALK FAIL	31-61-2
PROX SYS FAULT	NO GO
RAAS FAIL	34-42-2
RAAS NOT AVAILABLE	34-42-2
RDC FAIL	31-31-1
STAB CH 1 (CH 2) INOP	NO GO
TAIL RFL SOV OPEN	NO GO
TAIL XFER FAIL SEC (PRI)	28-15-3. Return to ramp.
TAWS BASIC FAIL	34-42-2
TAWS FAIL	34-42-2
TAWS TERR FAIL	34-42-2
TAWS TERR NOT AVAIL	34-42-2
TAWS WINDSHEAR FAIL	34-42-2
TSS FAN FAIL	34-54-1
WOW OUTPUT FAIL	NO GO
XPDR 1 (2) FAIL	34-54-1
YD 1 (2) INOP	22-20-2

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